

Lexical functions

The naked truth

Igor Mel'čuk

Alain Polguère

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To Lidiya Iordanskaja, with love.

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Writing conventions

Abbreviations

$\mathcal{A}$	a set of arcs in a network / graph
ACC	accusative case
Adj	Adjective
Adv	Adverb
$A_{(poss)}$	possessive adjective (<i>my, your, ...</i>)
ART	article or other determiner – e.g. <i>to have</i> ART <i>dream</i>
Claus	Clausative
circumst	circumstantial surface-syntactic relation
DAT	dative case
determ	determinative surface-syntactic relation
dir-obj	direct-objectival surface-syntactic relation
DSynt-	deep-syntactic
DSyntA	deep-syntactic actant
DSyntS	deep-syntactic structure
f	A given lexical function
f(L)	Application of the lexical function f to its argument (lexical unit) L
fr-LN	French Lexical Network
$\mathcal{G}$	a given graph
$\mathcal{G}_{LS}$	a given Lexical System graph
GEN	genitive case
GER	gerund
INSTR	instrumental case
L	a given lexical unit, i.e. a lexeme or an idiom
	Note. In the body of the book, we allow ourselves to use the symbol “L” in the sense of ‘referent of lexical unit L’ – e.g., [...] <i>prototypically used to do L</i> in the description of S_{instr} (Chapter 4, [17], p. 99).
L_{GENER}	generic term (of a semantic field)

Writing conventions

$\mathcal{L}$	a given natural language
LF	lexical function
LN	lexical network
LOC	locative case
LDOCE	<i>Longman Dictionary of Contemporary English</i> (online)
modif	modificative surface-syntactic relation
n	a given node in a network / graph
$\mathcal{N}$	a set of nodes in a network / graph
N	Noun
NOM	nominative case
obl-obj	oblique-objectival surface-syntactic relation
PL	plural
PoS	part of speech
possess	possessive surface-syntactic relation
prepos	prepositional surface-syntactic relation
PREPOS	prepositional case
S	substantive [= ‘noun’; in the name of a nominal lexical function]
SAE	Standard Average European [language type; roughly, Germanic, Romance and Slavic languages]
Sem-	semantic
SemA	semantic actant
SemS	semantic structure
SG	singular
SSyntS	surface-syntactic structure
subj	subjectival surface-syntactic relation
subj-adnom	subjectival-adnominal surface-syntactic relation
V	Verb

Symbols

*A	A is ungrammatical
$A \approx A$	A and B are approximately equal; in the present volume, the symbol “ $\approx$ ” mostly stands for <i>approximately</i>
$A \equiv B$	A and B are equivalent
$A \cong B$	A and B are approximately equivalent
$s_1 \oplus s_2$	linguistic union of (components of) linguistic signs s_1 and s_2

$E C$	constraint C on the expression E – e.g., $\text{Loc}_{in}(\text{weather}) = \text{in } [\sim \text{ with a modifier}]$ (i.e., one says <i>in this / such weather, in rainy weather, in weather like this</i> , etc.)
$E_1 \langle E_2 \rangle$	expression E_2 is a variant of expression E_1
$A \Rightarrow B$	A implies B
$A \Leftrightarrow B$	linguistic entity A of the level n of representation corresponds [= is semantically equivalent] to linguistic entity B of level $n + 1$, and vice versa; for instance: ‘Joey $\leftarrow 1$ – sleep’ [SemS] $\Leftrightarrow$ JOEY $\leftarrow 1$ – SLEEP _(V) [DSyntS].
$f_{\supset}, f_{\subset}, f_{\cap}$	lexical function that is semantically richer ($\supset$) or poorer ($\subset$) than f , or has semantic intersection ($\cap$) with f
$L_1 \circ \leftarrow \dots \rightarrow \circ L_2$	in a syntactic representation, indicates that the nodes L_1 and L_2 are coreferential – i.e., they have the same referent
$L_1 \text{--Dep--} L_2$	L_2 semantically / syntactically depends on L_1 by semantic / syntactic dependency Dep
$L \vdash \text{LexRel} \rightarrow L'$	L' is the target of the lexical relation LexRel originating from L
$\ulcorner L_1 \dots L_n \urcorner$	idiom – e.g., ‘ <i>put the cart before the horse</i> ’
$\sim$	the lexical unit under description in a given formula – e.g., $\text{Real}_1(\text{bicycle}) = \text{ride}$ [ART $\sim$]
$ $	[in a Russian wordform] boundary between a morphological stem and a suffix – e.g., <i>sobak a</i> ‘dog-SG.NOM’
$—$	graphical connector for segments of one wordform that is traditionally spelled as several words – e.g. <i>at _last</i>
σ	a given meaning
σ^f	meaning of the lexical function f
$\underline{\sigma}$	semanteme that is communicatively dominant in a semantic structure
$_$	empty actant slot in a semantic structure
Ω	external participant of the situation denoted by L (i.e., a participant that is not expressed as one of L ’s semantic actants)
$\emptyset$	zero sign
In the list of an LF application values (<i>Val</i>):	
$Val_1; Val_2$	there is a notable semantic difference between Val_1 and Val_2
$Val_1 < Val_2$	Val_2 is semantically ‘more’ than Val_1 ; the left and right scopes of “<” can be limited by the “;” symbol
// <i>Val</i>	<i>Val</i> is a fused value element (for a syntagmatic LF)
<i>Val</i> [GP]	GP is the government pattern of <i>Val</i> – e.g., $\text{Oper}_1(\text{supremacy}) = \text{detain}$ [ART $\sim$]

Special font formats

SMALL CAPITALS	lexical unit supplied, when needed, with its lexicographic number – e.g., the lexeme STEP _(V) 1.1 and the idiom ‘BLOW THE WHISTLE’
monospace font	lexical function’s name – e.g., Syn, A ₁ , ...
<i>italicized sans serif font</i>	important notion, when first introduced – e.g., <i>idiom</i>
small bold sans serif font	usage label – e.g., archaic , child language , colloquial , ...

Russian examples are in international scientific transliteration; it is as follows:

Aa – Aa	Ёё – Ёё	Ja ja – Яя	Oo – Oo	Tt – Tt	Žž – Жж
Bb – Бб	Èè – Ээ	Ju ju – Юю	Pp – Пп	Uu – Уу	’ – Ьь
Cc – Цц	Ff – Фф	Kk – Кк	Rr – Рр	Vv – Вв	” – Ъъ
Čč – Чч	Gg – Гг	Ll – Лл	Ss – Сс	Xx – Хх	
Dd – Дд	Ii – Ии	Mm – Мм	Šš – Шш	Yy – Ыы	
Ee – Ее	Jj – Ъъ	Nn – Нн	Šč, šč – Щщ	Zz – Зз	

Simple standard lexical functions: alphabetical list

Numbers to the right of lexical function [LF] names indicate their order of presentation in Chapter 4 – paradigmatic LFs, cf. “P” – and Chapter 5 – syntagmatic LFs, cf. “S”. Boxing of the numbers indicates the most productive and important LFs.

A systemic presentation of standard LFs is given in Chapter 3, Section 3.8 – see, in particular, the classified list of LFs p. 67.



A ₀	[9]	P	p. 87	Gener	[4]	P	p. 80	Pred	12	P	p. 92
A _i	[21]	P	p. 102	Germ	31	S	p. 116	Prepar	52	S	p. 151
Able _i	[22]	P	p. 103	Imper	27	P	p. 110	Propt _i	44	S	p. 135
Adv ₀	[10]	P	p. 89	Imperf	29	P	p. 112	Prox	56	S	p. 156
Adv _i	24	P	p. 104	Incep	[53]	S	p. 154	Qual _i	23	P	p. 104
Anti; Non	[3]	P	p. 76	Instr	43	S	p. 134	Real _i	[49]	S	p. 147
Bon	[37]	S	p. 124	Involv	65	S	p. 166	Redun	[34]	S	p. 120
Cap	15	P	p. 96	Labor _{ij}	[48]	S	p. 143	Result _i	30	P	p. 113
Caus	[57]	S	p. 157	Labreal _{ij}	[51]	S	p. 150	S ₀	[7]	P	p. 85
Claus ₀	[11]	P	p. 90	Liqu	[58]	S	p. 159	S _i	[13]	P	p. 94
Cont	[55]	S	p. 155	Loc _{ab}	42	S	p. 131	S _{instr}	17	P	p. 99
Contr	6	P	p. 84	Loc _{ad}	41	S	p. 131	S _{loc}	16	P	p. 98
Conv _{ijk}	[2]	P	p. 74	Loc _{in}	40	S	p. 131	S _{med}	18	P	p. 100
Copul	45	S	p. 137	Magn	[35]	S	p. 121	S _{mod}	20	P	p. 101
Culm	32	S	p. 116	Manif	64	S	p. 165	S _{res}	19	P	p. 100
Epit	33	S	p. 118	Minus	39	S	p. 130	Sing	[25]	P	p. 108
Equip	14	P	p. 95	Mult	[26]	P	p. 109	Son	63	S	p. 163
Excess	62	S	p. 162	Obstr	60	S	p. 161	Stop	61	S	p. 161
Fact _i	[50]	S	p. 149	Oper _i	[46]	S	p. 139	Sympt _{ijk}	66	S	p. 167
Figur	5	P	p. 83	Perf	28	P	p. 111	Syn	[1]	P	p. 72
Fin	[54]	S	p. 154	Perm	[59]	S	p. 159	V ₀	[8]	P	p. 86
Func _i	[47]	S	p. 142	Plus	38	S	p. 128	Ver	[36]	S	p. 123

1 Introduction

1.1 Our object of study: lexical functions

This book is about *lexical functions*. Therefore, our very first task is to explain – at least, approximately, in simple terms and with the help of most trivial examples – what lexical functions are. Let’s do it!

Consider the phrases in (1) and (2):¹

- (1) a. *pay* tribute [to someone]
b. **give* tribute [to someone]
- (2) a. *give* recognition [to someone]
b. **pay* recognition [to someone]

The verbs *pay* and *give*, used in (1a) and (2a) with the nouns *tribute* and *recognition*, respectively, are not interchangeable in spite of the following facts:

- The two nouns are semantically close.
- The two verbs are, so to speak, pure verbalizers of their nominal objects: they express a very general, impoverished meaning, approximately ‘do’, and add no semantic content to the nouns.

These verbs are empty of meaning in the given contexts to such an extent that *pay tribute* is semantically reducible to a hypothetical verb **to tribute* [someone], and *give recognition*, to the verb *to recognize* [someone (during a ceremony)]. For this reason, verbs such as *pay* and *give* are called *support*, or *light*, verbs (see Chapter 5, Section 5.4.2).

¹As is customary, we use the asterisk “*” in front of an expression to indicate that this expression is ungrammatical. Let it be emphasized that for an expression to be ungrammatical does not imply that it cannot be found in speech. For instance, at the time of writing, the one-billion-word Corpus of Contemporary American English (Davies 2008) contained 1,676 occurrences of *pay tribute* and 13 occurrences of the ungrammatical phrase **give tribute* (for all inflected forms of these phrases). Additionally, the acceptability of word combinations can fluctuate significantly due to geographical variation. We are grateful to Alastair Butler for the suggestion to use (1) and (2) as illustrations.

The choice of the appropriate support verb by the Speaker is, as one sees in (1) and (2), contingent upon a specific nominal lexeme – the direct object of this verb: the meaning ‘do’ is expressed by the verb *pay* if you ‘do’ *tribute*, but by the verb *give* if you ‘do’ *recognition*. In other words, the meaning ‘do’ is expressed not freely, that is, not by any convenient verb the Speaker is able to think of. It is expressed in a constrained way: by the verb that the noun imposes. To borrow from the language of mathematics, the meaning ‘do’ is expressed *as function of* the direct object of the corresponding verb. This is where the noun *function* in the term *lexical function* comes from. The reason for the adjective *lexical* is obvious: such functions apply to lexical units and return lexical entities – on the notion of lexical entity, see Section 2.1 (p. 22).



Syntagmatic lexical function

What is illustrated in (1) and (2) is a *syntagmatic* lexical function called Oper_1 (see Chapter 5, [46], p. 139), which describes a particular pattern of constrained lexical unit combinations – that is, a particular type of a two-component phrase where one lexical unit, freely selected by the Speaker, determines the selection of the other lexical unit. In this specific case, borrowing again from mathematical formalization, we write:

$$\begin{aligned}\text{Oper}_1(\textit{tribute}) &= \textit{pay} [\textit{tribute}] \\ \text{Oper}_1(\textit{recognition}) &= \textit{give} [\textit{recognition}]\end{aligned}$$

Constrained lexical combinations described by syntagmatic lexical functions are termed *collocations* (see Chapter 3, Section 3.3.2). They are extremely varied – support-verb constructions (presented above), realization-verb constructions (*pay a debt*, *satisfy a desire*, *ring a bell*), (de-)intensifying modifier constructions (*generous / meager salary*, *remember clearly / vaguely*), etc. – and their description requires a wide range of syntagmatic lexical functions.

Lexical functions are not limited to constrained combinations of *lexical units* within *phrases*: they similarly deal with constrained combinations of *morphemes*

within *wordforms*. For instance, in English, the derivational means used to produce from a verb V the noun that means ‘person who does V’ is contingent upon V, as illustrated in (3):

- | | | | |
|-----|---|--|--------------------------------|
| (3) | <i>teach</i> | $\Rightarrow$ <i>teacher</i> | : -er suffixation |
| | <i>represent</i> [a social group] | $\Rightarrow$ <i>representative</i> _(N) | : -(at)ive suffixation |
| | <i>chair</i> _(V) [a meeting] | $\Rightarrow$ <i>chairperson</i> | : compounding |
| | <i>guide</i> _(V) | $\Rightarrow$ <i>guide</i> _(N) | : V $\Rightarrow$ N conversion |
| | <i>steal</i> _(V) | $\Rightarrow$ <i>thief</i> | : suppletive form |

In parallel with what has been said about the meaning ‘do’ for our first set of illustrations, we say that the meaning ‘person who does [V]’ is expressed as function of the verb V in order to return the corresponding noun.



Paradigmatic lexical function

The constrained combinations of morphemes illustrated in (3) are also representable by means of a lexical function; in this case, a *paradigmatic* lexical function called S_1 (see Chapter 4, [13], p. 94). Accordingly, we write:

$$\begin{aligned}
 S_1(\textit{teach}) &= \textit{teacher} \\
 S_1(\textit{represent}) &= \textit{representative}_{(N)} \\
 S_1(\textit{chair}_{(V)}) &= \textit{chairperson} \\
 S_1(\textit{guide}_{(V)}) &= \textit{guide}_{(N)} \\
 S_1(\textit{steal}_{(V)}) &= \textit{thief}
 \end{aligned}$$

Notation. As can be seen, we indicate in subscript the part of speech of a lexical unit if it has a homonym of a different part of speech.

Summing up, lexical functions are a formal tool designed to describe specific patterns of constrained combinations of linguistic signs – of lexical units as well as of morphemes.² As such, lexical functions account for an important part of natural languages’ phraseology:³ lexical as well as morphological phraseology (Mel’čuk 2021a).

²The morphemes in question are mostly derivational morphemes, that is, morphemes whose meanings are fairly close to lexical meanings.

³On the notions of phraseme and phraseology, see Section 2.1, p. 17.

1 Introduction

The set of lexical functions is organized around a rigorously structured system of *simple standard* lexical functions, whose number is under 70 (see Chapter 3, p. 67). One of their defining properties is that they are language-universal (see Section 3.2.1, p. 40).



To avoid misunderstanding, let us stress that lexical functions are by no means semantic units – a kind of elementary meanings. They do not exist on the semantic level of linguistic representation. They are special – namely, deep – lexical units, appearing in the deep-syntactic structures of utterances and participating in deep-syntactic paraphrasing (see Section 3.2.3, p. 42).⁴ One can say that a lexical function is, metaphorically speaking, a lexical unit “of the second order.” Like any lexical unit, it possesses a signified, a signifier and a syntactics – a set of combinatorial properties; the remarkable fact is that a lexical function’s signifier is a set of alternating lexical expressions. On the syntactics – third component of a linguistic sign, see Mel’čuk (2013b).

Lexical Functions: The Naked Truth presents the system of lexical functions as fully as possible, but with the following two restrictions.

First, our approach is exclusively synchronic, which means that we avoid the diachronic perspective on word formation or collocations (cf. Blanco 2018), as well as the micro-diachronic study of lexical dynamics, such as the transformation of idioms into collocations (cf. Pausé & Polguère 2020, Polguère To appear).

Second, our presentation is restricted to theoretical and descriptive considerations and we do not touch on domains of practical applications of lexical functions such as:

- pedagogical applications, that is, language teaching and learning;
- psycholinguistics, that is, Mental Lexicon issues such as language acquisition, access to and loss of lexical knowledge;
- Natural Language Processing – e.g., extraction / interpretation of phraseological expressions, training of Language Models for Artificial Intelligence, etc.

⁴Recall the equivalence mentioned above: $give_{\text{Oper}_1(\text{recognition})} \text{recognition} \equiv \text{recognize}$.

However, one essential application of lexical functions is systematically taken into consideration throughout the book: it is the lexicographic modeling of systems of lexical units, performed according to the theoretical and descriptive principles of Explanatory Combinatorial Lexicology (Mel'čuk et al. 1995, Mel'čuk 2013a: Ch. 11). This includes dictionary models of the lexicon, such as Explanatory Combinatorial Dictionaries (Mel'čuk & Zholkovsky 1984, 2016, Mel'čuk et al. 1984, 1988, 1992, 1999), lexical databases (Polguère 2000b, Alonso Ramos et al. 2010, Barrios Rodríguez & Boguslavsky 2019), as well as lexical network models, such as Lexical Systems (Polguère 2014).

1.2 Our aims

Lexical functions – which are hypothesized as being language-universal – operate at the core of the structuring and functioning of natural languages: organization of lexical knowledge, lexical access / selection, paraphrasing, etc. It is thus important for the notion of lexical function to be understood by as many people as possible, inside and outside the field of linguistic research. However, the hard fact is that lexical functions are familiar to only a limited circle of “insiders” and, when used, they are often characterized in a simplistic way, which makes these characterizations mostly ineffective. This may be the result of a certain lack of pedagogy and attractiveness of texts presenting lexical functions. Yet the core cause of this state of affairs is the fact that the notion of lexical function is inherently difficult to assimilate, for at least two reasons.

First, the mastering of lexical functions requires the mastering of a rather rich set of underlying linguistic notions, which are relevant to all structural modules of natural languages. They pertain to semantics, syntax and morphology, that is, to the lexicon as well as to the grammar of natural languages. For instance, it is impossible to master the above-mentioned syntagmatic lexical function $Oper_1$ without mastering Meaning-Text use of (dependency) syntax, the distinction between semantic and deep-syntactic actants and, of course, the principles of lexicographic definition. In other words, to really *understand* lexical functions, one needs to have a bird's-eye view of the organization of natural languages and not focus solely on specific linguistic modules one happens to have a specific interest in. Lexical functions are fully embedded in the Meaning-Text approach to language studies (Mel'čuk 1981, 2016b, Kahane 2003, Milićević 2006) and have to be acquired as such. Figure 1.1 visualizes the architecture of a Meaning-Text model of natural languages: levels of utterance representation and linguistic modules that establish correspondences ($\Leftrightarrow$) between those levels. Though correspon-

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dences are bidirectional, this figure has to be read from bottom to top – according to the Speaker’s perspective, i.e., in the perspective of speech production.

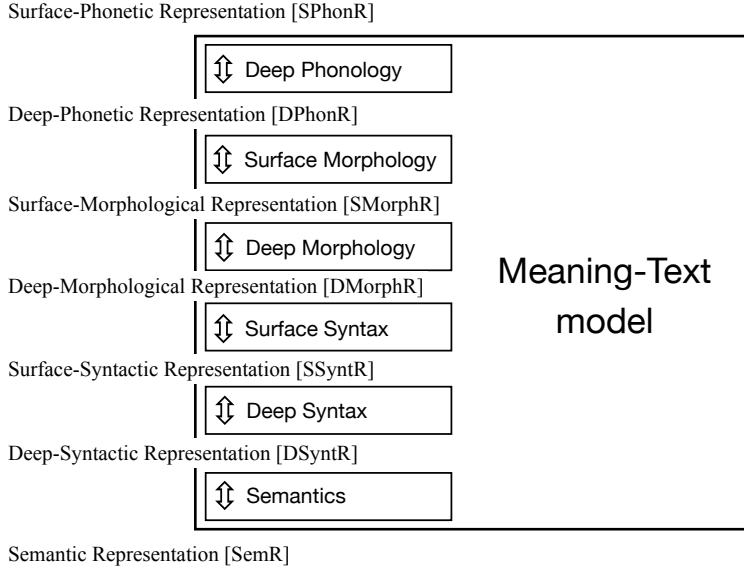


Figure 1.1: Architecture of a Meaning-Text model

Second, we have to acknowledge that the formal language of lexical functions is far from being streamlined and self-evident. It tries to reconcile two mutually incompatible imperatives: on the one hand, formal precision; on the other hand, legibility by humans and the possibility of easily applying the researcher’s linguistic intuition when reasoning on lexical function formulas. Consequently, the language of lexical functions is sometimes making use of unusual formal conventions – for instance, the use of actantial subscripts (to the names of lexical functions) with different interpretations depending on the context, which is not acceptable in a genuine formal language (Chapter 3, Sections 3.2.3.4 and 3.4.3).⁵

Faced with these difficulties, we decided to deal with the situation frontally, without cutting corners. However, because our ambition is, first of all, to be as much pedagogical as possible, we made a systematic effort at providing readers with all background and, even, insider knowledge needed to achieve an in-depth

⁵For an alternative (so-called *algebraic*) notation of lexical functions, see Kahane & Polguère (2001). Note that what is gained with this notation in terms of formal rigor and calculability is counterbalanced by a clear loss in terms of possible recourse to linguistic intuition when a lexicographer, a linguist, etc., wants to reason using lexical functions.

understanding of lexical functions. We hope therefore that the book can prove useful to a vast range of readers: linguists, lexicographers, researchers and developers in Natural Language Processing, language teachers, translators, etc. – in a nutshell, to everyone who deals with or is interested in Language.

1.3 How do we describe lexical functions?

As we just said, this book is intended as a fresh, more pedagogical presentation of the system of lexical functions, which has already been the topic of many publications – e.g., Alonso Ramos (1993), Mel'čuk (1996), Mel'čuk (2007), Mel'čuk & Milićević (2014: 210–223), Mel'čuk (2015b: Ch. 14), etc. So how are we proceeding to achieve this pedagogical goal?

Chapter organization. To start with, the chapter organization follows a progression that aims at a gradual acquisition of lexical functions:

- Chapter 2 introduces some basic lexicological, semantic and syntactic notions the knowledge of which is a prerequisite for the assimilation of lexical functions. The presentation of these preliminary notions encompasses important terminology and formal conventions used throughout the remainder of the book.
- Chapter 3 is an in-depth discussion of the notion of lexical function. It is intended to familiarize readers with all conceptual and formal aspects of lexical functions on which the description of individual lexical functions in the following chapters is based.
- Chapter 4 presents, in a logical order, all paradigmatic simple standard lexical functions.
- Chapter 5 does the same for syntagmatic simple standard lexical functions.
- Chapter 6 concludes the book with the discussion of the central role played by lexical functions in the structuring of natural language lexicons modeled as lexical networks.

While readers are expected to study Chapters 2 and 3 in a linear way, Chapters 4 and 5 are designed for random, or “opportunistic,” access to information on individual lexical functions. A profusion of cross-references helps establish connections between conceptually related lexical functions, thus allowing readers to browse a *system* of lexical functions, rather than following a one-way path through a *list*.

Detailed description template of lexical functions. The presentation of individual lexical functions follows a well-specified template that is explained in Chapter 3, Section 3.7. We have paid special attention to providing information required for *active* learning, where readers can not only understand what a given lexical function is, but also develop the ability to use it in various contexts (lexicographic work, language teaching, etc.).

As the book contains updates to the system of standard lexical functions that might be confusing with respect to information and practices found in earlier publications, *Historical remarks* have been added at the relevant points to help the readers in this respect.

Extensive linguistic exemplification. Though standard lexical functions are postulated to be language-universal, it would have been a colossal task to systematically illustrate each given lexical function with examples borrowed from a broad range of typologically different language families. However, to take into consideration the cross-linguistic relevance of lexical functions, our illustrations in the presentation of lexical functions are systematically done in English, French and Russian – the latter two being the mother tongues of the authors.

Facilitation of multiple access to information. We did not hesitate to introduce some redundancies in the volume in order to facilitate the different types of usage by readers: in-depth learning of the nature and use of lexical functions vs. punctual consultation of the description of individual lexical functions. In the same spirit, indexes (of notions and of lexical functions) are extensive, with some form of redundancy at times, in order to meet the multiple needs of individual readers.

Note that the full list of standard lexical functions is available in two forms: as an alphabetical list (p. 5); and as a “logical” list (Chapter 3, p. 67), where lexical functions are enumerated in the order they appear in Chapters 4 and 5.

Alternative approaches. As explained in Section 1.1, lexical functions serve to describe both collocations, embodying syntagmatic lexical relations, *and* derivations, embodying paradigmatic lexical relations. There exists a wealth of literature on these topics, which are generally treated as two non-related lexical phenomena; but the systematic presentation of alternative approaches to lexical functions is beyond the scope of the present book. We establish connections between the lexical function approach to the description of lexical relations and other well-known approaches when it is especially relevant to do so. In particular,

current approaches to the description of collocations in lexicographic models are examined at the end of Chapter 5, on syntagmatic lexical functions (Section 5.5, p. 171); semantic lexical relations for the structuring of so-called *-Net* models (cf. WordNet) are contrasted with the use of paradigmatic lexical functions in Chapter 6, on lexical networks (Section 6.1, p. 178).

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2 Preliminary linguistic notions

The description of lexical functions requires a preliminary set of linguistic notions that we need to make use of throughout the present volume. This chapter offers the reader a concise presentation of those notions, which are grouped into three classes: lexical notions (Section 2.1), semantic notions (Section 2.2) and syntactic notions (Section 2.3).

What follows is a series of definitions supplied with explanations and illustrations. Two types of definitions are used: analytic and enumerative definitions.

Analytic definitions constitute the basic type; they follow the Aristotelian definition pattern *per genus proximum et differentia specifica* ('by next kind and specific differences') – e.g., definitions for *lexeme* and *idiom* below. These definitions are of almost the same structure as lexicographic definitions described in Mel'čuk & Polguère (2018).¹ The first component of the definition is the *central* (or *generic*) *component*, and it corresponds to the Aristotelian *genus proximum*. It is followed by a series of *peripheral components*, which correspond to *differentia specifica*; each peripheral component is introduced by a bullet ("·").

Marginally, *enumerative definitions* are also used. They are structured as disjunctive lists of previously defined notions – e.g., the definition for *lexical unit* [= *lexeme* or *idiom*] below (p. 21).

2.1 Lexical notions

Lexical functions, as their name suggests, are primarily about the lexicon and the units that constitute it, i.e., words and phraseological expressions. Therefore, we begin with linguistic notions related to words, on the understanding that the word *word* is both too ambiguous and vague to belong to linguistic (in particular, lexicological) terminology. Let us start with the notion of *lexeme*.

¹For examples of actual lexicographic definitions, see the definitions of *snow*_(N) in Chapter 4 (Section 4.2, p. 82), *DEBT* in Chapter 5 (Section 5.4.3, p. 148) and *ANGER*_(N) in the Appendix (p. 209).



Notion definition: lexeme

A *lexeme* is a set of wordforms and analytic-form phrases such that

- their semantic, syntactic and morphological differences can be fully described in terms of inflectional significations.

Example. The lexeme AMAZE = { *amaze, amazes, amazed, amazing, have amazed, will amaze, am amazed, being amazed, will have amazed, ...* }.

A lexeme name is printed in SMALL CAPITALS, when needed accompanied by its part of speech and *lexicographic number* (in case of polysemy / homonymy) – e.g., PICTURE_(N) 1.1 ‘image’, PICTURE_(N) 1.2 ‘movie’, PICTURE_(N) II ‘mental representation’.² Lexicographic numbers used in this book are not borrowed from a particular lexicographic description, unless specified otherwise. They are, so to speak, conventional and serve only to distinguish word senses.

Lexemes are one of the two types of lexical units, the second one being idioms (see below). Lexemes are the primary type, as will be shown shortly. The above definition relies on the morphological notions of *wordform* and *inflection*, which have to be taken for granted here. A lexeme can be conceived of as a generalization over a set of wordforms and analytic-form phrases that all carry the same lexical meaning in their signified, while being related by shared signifiers.

The next notion to be defined is that of (lexemic) *phrase*. It is a core preliminary notion for two reasons:

1. Idioms (e.g., ‘PULL THE PLUG’) – the second type of lexical units that we need to define – are a particular class of phrasemes.
2. Lexical functions are especially relevant for the description of collocations (e.g., *wide awake* or *pay attention*), which are also a particular case of phrasemes (Chapter 3, Section 3.3.2, p. 48).

In the following definition, the term *Speaker*, written with a capital S, refers to the producer of the utterance under consideration (as opposed to a *speaker* of a language). The corresponding receiver of the utterance is referred to as the *Addressee*, with a capital A.

²On lexicographic numbers used in Explanatory Combinatorial Lexicology, see Mel’čuk (2013a: Ch. 13, Sec. 3.1).



Notion definition: (lexemic) phraseme

A (lexemic) *phraseme* is a complex lexemic expression such that

- the selection of at least one of its lexemes by the Speaker is not free – i.e., it is constrained by another lexeme of this expression.

Example. The phraseme *wide awake*, where the Speaker's choice of *wide* to express 'fully' is constrained by the adjective *awake*; *wide awake* can be contrasted with the non-idiomaticity of **wide dressed* 'fully dressed', **wide ready* 'fully ready', **wide known* 'fully known' (cf. idiomatic phrase *widely known*), etc.

The presence of phrasemes is overwhelming in natural languages; the set of all phrasemes of a given natural language is called its *phraseology*. In this volume, we concentrate on lexemic phrasemes (in particular, idioms and collocations) and disregard other families of phrasemes, such as syntactic and morphemic phrasemes (Mel'čuk 2015b: Ch. 16, 2021a, 2023). Therefore, the modifier *lexemic* will be omitted.

Equipped with the definition of phraseme, we can define the second type of lexical unit (along with lexemes): *idiom*.



Notion definition: idiom

An *idiom* is a phraseme such that

- it is semantically non-compositional.

Example. The idiom 'GREEN THUMB' 'great ability to take care of plants'.

An idiom's name is enclosed between two top corners: 'GREEN THUMB', 'ROCK THE BOAT', 'BY AND LARGE', ...

In the above definition, the phrase *is semantically non-compositional* refers to the fact that the meaning of an idiom is not a regular linguistic union of

the meanings of its lexical components,³ so that a lexicographic definition is required to describe an idiom's meaning. Three types of idioms are distinguished as regards their meaning: strong, semi- and weak idioms.

- A *strong idiom* has a meaning that does not contain the meaning of any of its components – e.g., 'SHOOT THE BREEZE' 'talk leisurely'.
- A *semi-idiom* has a meaning that contains the meanings of some of its lexical components – e.g., 'MORNING SICKNESS' 'nauseous feeling experienced by a pregnant woman typically in the *morning*'.
- A *weak idiom* has a meaning that contains the meanings of all its lexical components plus additional semantic material – e.g., 'BACON AND EGGS' 'dish made of fried *eggs* and fried *bacon*'.



Semantic non-compositionality is often mistaken for opacity or transparency, which are relevant to the perspective of the Addressee, i.e., to understanding of speech. Non-compositionality, however, is about the Speaker, who has to produce the idiom out of the semantic content the Speaker seeks to express. Idioms, especially semi- and weak idioms, are often easily interpretable by Addressees who do not know them. Even strong idioms can be based on metaphors that are quite transparent, at least in the contexts of use; such is the case of 'HAVE ONE'S HANDS FULL' 'be very busy', 'IN A NUTSHELL' 'concisely', etc. The semantic (non-)compositionality of an expression should therefore be estimated *strictly* by contrasting its meaning as a whole with that of the lexical (and grammatical) material it is formally made up from, without recourse to hypothetical understanding processes.

(Non-)compositionality is relevant in the perspective of speech production, while transparency is strictly about speech understanding. Since this book takes the Meaning-Text approach, it is exclusively (non-)compositionality that we consider in order to characterize idioms.

³On the operation of linguistic union, see Chapter 3, Section 3.4, p. 52.

Now that both notions of lexeme and idiom have been defined and clarified, we can offer an enumerative definition of *lexical unit*.



Notion definition: lexical unit

A *lexical unit* is a lexeme or an idiom.

Note that there is another, marginal type of lexical unit, which we do not mention in the above definition in order to simplify the presentation: *compounding elements*, also called *combining forms* (Prčić 2005, 2008) – e.g., ANTHROPO- (in *anthropomorphic*, ...), NEURO- (*neuroscience*, ...), or -PHILE (*francophile*, ...), -PHOBIA (*arachnophobia*, ...). They are defective lexemes, in the sense that they cannot be used independently: they function only in compounding.

A lexical unit is the unit of the linguistic description of a language's lexicon. To put it differently, it is a unit of lexicographic description. That is, a lexical unit is described by a *lexical entry*, of which it is the *headword*.

In conformity with the principles of the description of linguistic signs (Mel'čuk 2013a: Ch. 11, Sec. 2), the lexical entry for a given lexical unit L accounts for three major components of L:⁴

1. L's *signified* – described in the *semantic zone* of the lexical entry, which is centered around L's lexicographic definition; other possible elements of this zone are, for instance, *lexical connotations* (Iordanskaja & Mel'čuk 2009).
2. L's *signifier* – described in the *phonological / graphematic zone* of the entry.
3. L's *syntactics*, or *restricted cooccurrence* (*cooccurrence*, for short) – described in the *cooccurrence zone* of the entry.

L's restricted cooccurrence is its capacity to combine in utterances with other linguistic units, according to the following two facets:

- L as a particular syntactic unit belonging to a part of speech, characterized by a set of syntactic features and having a government pattern (see “Active syntactic valence,” Section 2.3, p. 31); this is L's *syntactic cooccurrence*;

⁴Strictly speaking, what follows is a characterization of the lexical entry of a lexeme. We do not indicate the particularities of the lexical entries of idioms, which are formally more complex.

2 Preliminary linguistic notions

- L as a particular lexical unit; this is L's *lexical cooccurrence*.

The expression *lexical cooccurrence of L* covers two different linguistic phenomena, both described by lexical functions:

- derivation, which involves the cooccurrence of (the stem of) lexical unit L with particular word formation means (Chapter 3, Section 3.3.1);
- collocation, which involves the cooccurrence of lexical unit L with particular lexical units (Chapter 3, Section 3.3.2).

Due to the existence of *polysemy*, lexical units are grouped within *vocables*.



Notion definition: **vocable**

A *vocable* is a set of lexical units such that

- all of them share a common signifier;
- their signifieds share important enough semantic components.

Examples. The vocable PICTURE_(N) contains the lexemes PICTURE_(N) 1.1 'image', PICTURE_(N) 1.2 'movie [= series of moving images that ...]' and PICTURE_(N) II 'mental image'; the vocable 「COME A CROPPER」 contains the idioms 「COME A CROPPER」I 'have a bad fall' and 「COME A CROPPER」II 'fail badly, 「as if」 「coming a cropper」I'.⁵

Finally, we give the definition of the notion of *lexical entity*. This notion covers a much larger spectrum of linguistic elements than the notion of lexical unit; it is necessary to encompass all possible values of lexical functions.⁶ It includes:

- lexical units – lexemes and idioms;
- semantically compositional phrasemes – collocations and linguistic clichés, such as *Have a nice day* (Mel'čuk 2020).

⁵The top corners enclosing semantemes indicate that they are meanings of idioms, in this case of the idioms 「AS IF」 and 「COME A CROPPER」I.

⁶On values of lexical functions, see Chapter 3, Section 3.1.1.



Notion definition: lexical entity

A *lexical entity* is a lexeme or one of the two other linguistic entities semantically similar to a lexeme: an idiom – i.e., non-compositional phraseme – or a compositional phraseme (a collocation or a linguistic cliché).

All concrete signs that constitute these entities – such as the wordforms and analytic-form phrases – are also considered to be lexical entities.



Further reading

On morphological notions (wordform, inflection, etc.), see Mel'čuk (2006); on phraseology and types of phrasemes, see Mel'čuk (2023).

2.2 Semantic notions

This section introduces several notions that pertain to the domain of linguistic meanings. We start with *semantemes*, which are main semantic entities.



Notion definition: semanteme

A *semanteme* of language $\mathcal{L}$ is a meaning such that

- it is expressed by a lexical unit of $\mathcal{L}$.

Examples. In English, the meaning 'not deep' (as of a body of water) is a configuration of two semantemes equivalent to a single semanteme 'shallow'; by contrast, in French, the corresponding configuration 'peu profond' lit. 'not_very deep' has no equivalent semanteme – French does not have an adjective corresponding to SHALLOW_(Adj).

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Semantemes come in two main families as regards their combinatorial characteristics.

A semanteme of the first family, called *semantic name*, is self-contained: it does not have open slots expected to be filled with other semantemes – in order to make it informationally complete. A semantic name denotes an *entity* (object, substance, being, ...) and, in terms of part of speech, the lexical unit expressing it is necessarily nominal – e.g., $\text{STONE}_{(N)}$, $\text{SAND}_{(N)}$, TIGER , ALEXANDER , VESUVIUS , MONTREAL , etc. The majority of semantic names are proper names.

A semanteme of the second family is binding rather than being self-contained: it is *predicative*, i.e., it controls open slots to be filled in by other semantemes. These latter are called *semantic actants* [*SemAs*] of the semanteme in question, and the slots themselves are called *actant slots*; they are represented by variables ‘X’, ‘Y’, ‘Z’, ... A semantic actant of a predicative semanteme ‘P’ is, in logical parlance, an *argument* of the predicate ‘P’. Predicative semantemes are further subdivided:

- A predicative semanteme that denotes a *fact* (an event, an action, a state, a characteristic, ...) is a *semantic predicate* – e.g., ‘X collides with Y’ = ‘collision of X with Y’, ‘war between X and Y for Z’, ‘X kisses Y on Z’, ‘happy X’, ‘tall X’, ‘X under Y’, etc. As can be seen, the lexical expressions of semantic predicates can be of any part of speech.
- A predicative semanteme that denotes an entity is a *semantic quasi-predicate* – e.g., ‘X’s bed’, ‘X’s blood’, ‘X, mother of Y’, etc. Like semantic names, quasi-predicates are expressed by nouns.

A predicative semanteme denotes an extralinguistic *situation* (or an entity implicated in a situation, for quasi-predicates), which is characterized by a set of required *participants* – as one can see from the examples above. A participant of an extralinguistic situation corresponds to a semantic actant of the semanteme that denotes this situation.

Semantic valence and external participant ‘ Ω ’. The cooccurrence of any given lexical unit whose corresponding semanteme is predicative – e.g., the lexeme $\text{KISS}_{(V)}$ [‘X kisses Y on Z’] – is largely determined by the set of actant slots it controls, called its *semantic valence*. However, speaking of a particular lexical unit L (mainly, about its definition and its lexical functions), it can be necessary to refer to an *external participant* of the situation denoted by L, i.e., a participant that is not part of L’s active semantic valence: it is not expressed as one of L’s *SemAs*. This participant is designated with the symbol Ω – see, for instance, the

presentation of Real_Ω in Chapter 5 ([49], “Realization verbs,” p. 148) illustrated with collocations such as $[\text{Tony}_\Omega]$ *climbs / scales the mountain*.⁷

The typology of semantemes that we have introduced is recapitulated below, based on two parameters of classification: 1) predicativity and 2) type of denotation – fact vs. entity. This gives four classes of semantemes, of which one is not present in natural languages: non-predicative semantemes denoting facts are logically impossible since, by definition, a fact is denoted by a predicative semanteme.



Classification of semantemes

	Predicative (binding)	Non-predicative (self-contained)
Denote facts	<i>semantic predicates:</i> 'X snores', 'doubt of X about Y', 'strange X', ...	_____
Denote entities	<i>semantic quasi-predicates:</i> 'nose of X', '[X is a] friend of Y', 'hat of X', ...	<i>semantic names:</i> 'sand', 'giraffe', 'Montreal', ...

The relation between a predicative semanteme ' σ ' and its SemA ' σ_{actant} ' is a *semantic dependency*, which is represented – as all linguistic dependencies are – by an arrow oriented from ' σ ' to ' σ_{actant} ': ' σ ' **–sem→** ' σ_{actant} '. Since a predicative semanteme can control more than one actant slot – e.g. 'X kisses Y on Z', the different SemAs must be distinguished. This is done by semantic actant numbers **1, 2, 3, ...**, as illustrated below with sentence (1).

(1) Judy kissed Ben on the cheek.

The configuration of semantemes seen in Figure 2.1 is the *semantic structure* [*SemS*] of sentence (1) and of all its paraphrases.⁸ It calls for the following three comments.

⁷Note that MOUNTAIN, the base of the collocation, is a semantic name: it does not control any semantic actant slot.

⁸On paraphrasing, see Chapter 3, Section 3.2.3.3 (p. 43).

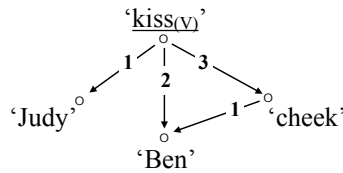


Figure 2.1: Semantic structure [SemS] of the sentence *Judy kissed Ben on the cheek*.

Note. The SemS in Figure 2.1 does not encode inflectional meanings, that is, *grammemes* expressed in (1): the indicative and the past tense of the verb KISS(V), as well as the singular of the noun CHEEK. For simplicity's sake, we follow the same policy in all SemSs in this volume.

1) Communicatively dominant semanteme. The sentence *Judy kissed Ben on the cheek* is about an act of kissing; therefore, the semanteme 'kiss' is *communicatively dominant*, which is indicated by underscoring. By contrast, the noun phrase *Ben's cheek that was kissed by Judy* is about the cheek (of Ben) and possesses an identical semantic structure, except for the semanteme 'cheek', which is *communicatively dominant* (i.e., underscored in the semantic structure of this phrase).

Communicative dominance is an essential component of the *communicative structure* of utterances, as theorized in Polguère (1990) and Mel'čuk (2001). For the sake of simplicity, other components of communicative structures – such as the Rheme vs. Theme opposition – are not encoded in utterance representations in this volume.

2) Semantic actants' numbering. Numbering of SemAs of a predicative meaning depends on its internal structure, that is, on its semantic decomposition. Kissing someone is to act upon someone, and the Agent of an action is taken to be SemA 1, the Patient undergoing an action is SemA 2, and the Target of the kiss (part of the Patient's body) is SemA 3.

3) Semantic structure as a network. Figure 2.1 shows that the SemA 3 of 'kiss' is a quasi-predicate that controls one actant slot: 'cheek of X'. This is due to the fact that 'cheek' is 'part of the body of X [such that...]'; all names of body parts necessarily express quasi-predicative semantemes. In this case, the SemA 1 of the quasi-predicate 'cheek' is 'Ben', and 'Ben' is itself the SemA 2 of 'kiss'. This configuration of semantic dependencies forms a *semantic network*.



Further reading

On *semantemes*, see Mel'čuk & Polguère (2008), Polguère (2012c) and Mel'čuk (2012: Part II, Sec. 2.2, 193–205); on *communicative dominance*, see Mel'čuk (2001: Part I, Sec. 2.3.1.1); on *SemA numbering*, see Mel'čuk (2015b: Ch. 12, Sec. 3.4.2.3) and Mel'čuk (2012: Ch. 4, Sec. 3); on *Meaning-Text semantic networks*, see Polguère (1997).

2.3 Syntactic notions

From the viewpoint of syntactic structures, two levels of representation are distinguished, which can be approximately characterized as follows.

The *deep-syntactic structure* [DSyntS] is the most direct lexical and syntactic expression of the corresponding SemS. It contains only the semantically full lexical units of the sentence to be produced, connected by labeled *deep-syntactic dependencies*, called *deep-syntactic relations*. The resulting structure is a *dependency tree*.

The *surface-syntactic structure* [SSyntS] is geared to the actual sentence in the sense that it serves to determine word order and morphological dependencies between lexemes, if any. It contains all and only the lexemes of the sentence to be produced, connected by labeled surface-syntactic dependencies, called *surface-syntactic relations*. The resulting structure is, of course, also a dependency tree.

Let us illustrate the *correspondences* between the SemS, the DSyntS and the SSyntS of a sentence, using sentence (2) as example.

(2) Judy shoots the breeze with Ben for hours.

“ $A \Leftrightarrow B$ ” means ‘linguistic entity A of the level n of representation corresponds, i.e. is semantically equivalent, to linguistic entity B of level $n + 1$; for instance, Figure 2.2 shows the SemS $\Leftrightarrow$ DSyntS correspondence for *Joey sleeps*.

In what follows we focus on two particular correspondences between utterance structures: SemS $\Leftrightarrow$ DSyntS and DSyntS $\Leftrightarrow$ SSyntS.

The SemS has been introduced and explained in Section 2.2. Here we need to explain the DSyntS and the SSyntS.

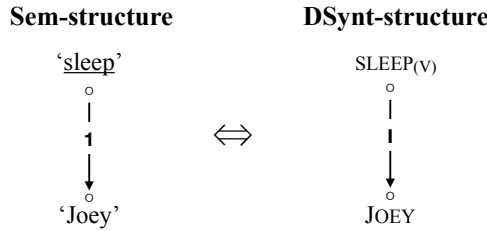


Figure 2.2: Example of a SemS ⇔ DSyntS correspondence

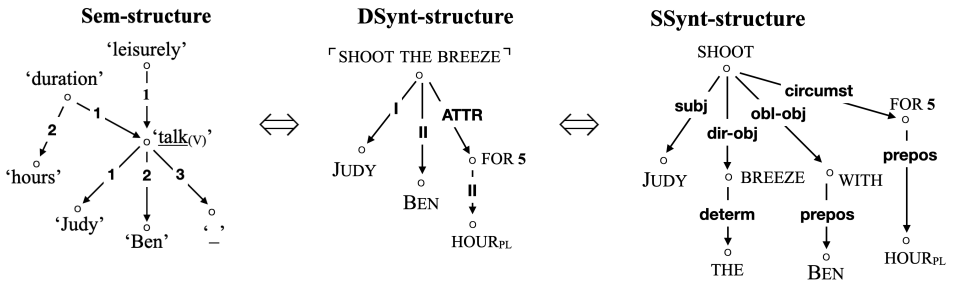


Figure 2.3: SemS ⇔ DSyntS ⇔ SSyntS correspondences for sentence (2): *Judy shoots the breeze with Ben for hours.*

Deep-syntactic structure [DSyntS] of sentence (2). The nodes of this tree are labeled with *deep lexical units*, i.e., semantically full lexical units, which express semantemes of the SemS. The idiom $\ulcorner$ SHOOT THE BREEZE $\urcorner$ expresses the semantic configuration $\text{'leisurely } -1 \rightarrow \text{talk}_{(V)}$, and the meaningful preposition FOR 5, the semanteme 'duration' .⁹ The other nodes are self-explanatory.

The arcs (branches) of this tree are labeled with DSynt-relations of two types:

- **I** and **II** are *actantial DSynt-relations*: the dependent members of these relations – JUDY and BEN – are DSynt-actants of their governor, the idiom $\ulcorner$ SHOOT THE BREEZE $\urcorner$ ($\text{'X } \ulcorner$ shoots the breeze $\urcorner$ with Y'). The notion of DSynt-actant will be defined shortly (p. 30).
- **ATTR** is a non-actantial DSynt-relation: it is not foreseen by the meaning of its governor and represents a dependency inversion – called *head-switching* – with respect to the corresponding semantic relation; cf. Figure 2.4.

⁹The lexicographic number of FOR 5 comes from the online *Longman Dictionary of Contemporary English* (LDOCE 2025), henceforth LDOCE.

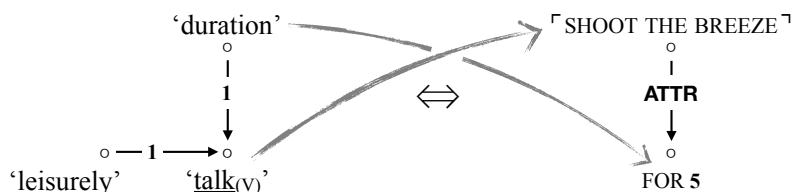


Figure 2.4: SemS ↔ DSyntS dependency inversion [= head-switching] involving the DSynt-relation **ATTR**

At the DSynt-level several other DSynt-relations are used – they are listed below. These relations are language-universal. In the present volume only actantial and **ATTR**(ibutive) DSynt-relations are relevant.



Language-universal DSynt-relations

I	1 st deep-syntactic relation / 1 st deep-syntactic actant
II	2 nd deep-syntactic relation / 2 nd deep-syntactic actant [Joey ←I– loves –II→ food]
...	
APPEND	appenditive; it links an structurally extraneous element, such as a parenthetical or an interjection, to its governor – the syntactic head of the clause [Sorry, ← APPEND –[I]– am late]
ADDRESS	addressative; it links a direct address to its governor, which is the syntactic head of the clause [Yes, – ADDRESS → Sir]
ATTR	attributive; it links a restrictive modifier / circumstantial to its governor [curious ← ATTR – student; arrived – ATTR → yesterday]
ATTR_{qual}	qualificative attributive; it links a qualificative modifier / circumstantial to its governor [students, – ATTR_{qual} → hungry for knowledge]
COORD	coordinative; it links the dependent member of a syntactic coordination to its governor [Ben – COORD → and Judy]
PSEUDO-COORD	pseudo-coordinative; it links the dependent member of elaboration or clarification to its governor [She came last year, – PSEUDO-COORD → in the spring]

A notion playing a key role in the presentation of lexical functions is that of *deep-syntactic actant* [DSyntA], mentioned above. It is defined as follows.



Notion definition: deep-syntactic actant [DSyntA]

A *deep-syntactic actant* I, II, ... of lexical unit L is a DSynt-dependent of L such that

- it expresses, in a prototypical case, a SemA of L and it is governed by L via an actantial DSynt-relation I, II, ...

There is also a frequent enough possibility of a DSyntA of L coming, or being raised, to L from one of its DSyntAs. Such *syntactic raising* is illustrated in Figure 2.5. This figure presents one of the support verbs mentioned in Chapter 1 (Section 1.1), namely the lexical function Oper₁ (Chapter 5, [46], p. 139).

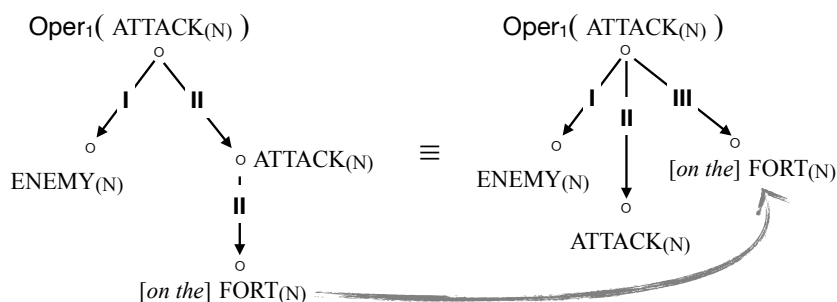
Oper₁(*attack*_(N)) is expressed in the surface-syntactic structure (see SSyntS below) by the lexical units 'CARRY OUT' or CONDUCT_(V). The *actant transfer* illustrated in Figure 2.5 is common with verbal syntagmatic lexical functions (support and realization verbs) – see Chapter 5, “Support [= light] verbs,” p. 141.

Remark. The second sentence (with raising) in Figure 2.5 is dubious: speakers seem to accept it only with a special prosody, namely, pauses before and after *on the fort*: *The enemy carried out – on the fort – a massive attack*. Other languages, such as French and Russian, allow for this construction without problem, as shown in (3a-b).

- (3) a. French L'ennemi a mené contre le fort une attaque massive.
b. Russian Vrag provël na fort massirovannuju ataku.

Surface-syntactic structure [SSyntS] of sentence (2). The nodes of the SSynt-tree for (2) are labeled with all lexemes of the sentence, including lexical components of the idiom (SHOOT, BREEZE and THE) and a governed preposition (WITH).

The arcs of this tree are labeled with SSynt-relations, which are language-specific – i.e., non-universal; there are several dozens of SSynt-relations in a language. We list below SSynt-relations featured in Figure 2.3 (p. 28) and in the whole volume.



The enemy $\ulcorner$ carried out $\urcorner_{\text{Oper}_1}$ a (massive) attack on the fort.

$\equiv$

The enemy $\ulcorner$ carried out $\urcorner_{\text{Oper}_1}$ on the fort a (massive) attack.

Figure 2.5: Syntactic raising of a DSyntA



Some SSynt-relations of English

circumstantial	<i>talked</i> --circumst-- <i>for</i> [hours]
determinative	<i>the</i> --determ-- <i>breeze</i> ; <i>my</i> --determ-- <i>daughter</i>
direct-objectival	<i>shoot</i> --dir-obj-- <i>breeze</i> ; <i>Marry</i> --dir-obj-- <i>me</i> ; <i>want</i> --dir-obj-- <i>to</i> [go]; <i>declare</i> --dir-obj-- <i>that</i> [it's OK]
modificative	<i>big</i> --modif-- <i>deal</i>
oblique-objectival	<i>shoot</i> [breeze] --obl-obj-- <i>with</i> [Ben]; <i>rely</i> --obl-obj-- <i>on</i> [you]
possessive	<i>tomorrow's</i> --possess-- <i>event</i>
prepositional	<i>with</i> --prepos-- <i>Ben</i> ; <i>for</i> --prepos-- <i>hours</i>
subjectival	<i>Judy</i> --subj-- <i>shoots</i> [the breeze]; <i>it</i> --subj-- <i>is</i> [late]
subjectival-adnominal	<i>speech</i> --subj-adnom-- <i>of</i> [the President]

Active syntactic valence. SemS $\Leftrightarrow$ DSyntS $\Leftrightarrow$ SSyntS correspondences are essentially based on the actantial structures – i.e., sets of actants – of lexical units. More precisely, these correspondences are controlled by the *active syntactic valence* of each lexical unit L they involve: how L's SemAs are expressed by means of DSyntAs, and how the latter are expressed at the surface-syntactic and mor-

phological levels.¹⁰

Consequently, to implement $\text{SemS} \Leftrightarrow \text{DSyntS} \Leftrightarrow \text{SSyntS}$ correspondences, the information about active syntactic valence of lexical units is needed. It is specified in their lexical entries by means of a table, called the *government pattern*. As an illustration, we give in (4) the government pattern of the verb $\text{KISS}_{(V)}$, appearing earlier in sentence (1) – Section 2.2, p. 25.¹¹

(4)			
	'X' $\Leftrightarrow$ I	'Y' $\Leftrightarrow$ II	'Z' $\Leftrightarrow$ III
	1. –subj→ N	1. –dir-obj→ N	1. –obl-obj→ on N

Another example of government pattern, for the noun $\text{ANGER}_{(N)}$, is found in the Appendix, p. 209.

The government pattern of a given lexical unit L describes L's restricted *syntactic* cooccurrence in L's lexical entry. By contrast, lexical functions – the topic of the present volume – describe restricted *lexical* cooccurrence of lexical units.



Further reading

On the notion of DSyntA, see Mel'čuk (2015b: Ch. 12); on the correspondence between the SemS and the DSyntS, see Polguère (1990) and Mel'čuk (2013a: Ch. 10); on SSynt-relations, see Mel'čuk (2015a, 2016a, 2021b: Ch. 2); on government patterns, see Miličević (2009) and Mel'čuk (2015b: Ch. 13).

¹⁰Unlike the situation with semantic valence (cf. Section 2.2, p. 24 above), the adjective *active* is required since active syntactic valence contrasts with passive syntactic valence: see Chapter 3, Section 3.5.1.

¹¹A lexical unit may have more than one government pattern. Thus, $\text{KISS}_{(V)}$ has another government pattern that describes utterances where the verb's SemA 3 ('Z') is expressed as its direct object (cf. *Judy kissed Ben's cheek*):

'X' $\Leftrightarrow$ I	'Z' $\Leftrightarrow$ II
1. –subj→ N	1. –dir-obj→ N

3 Notion of lexical function [LF]

This chapter presents the notion of lexical function – hereafter, LF – in nine steps:

- What is an LF? (Section 3.1)
- Simple standard LFs (Section 3.2)
- Two major classes of simple standard LFs (Section 3.3)
- Simple standard LF combinations (Section 3.4)
- Parts of speech of simple standard LFs (Section 3.5)
- Regularities in standard LFs (Section 3.6)
- Description template for simple standard LFs (Section 3.7)
- System of simple standard LFs (Section 3.8)
- Descriptive reliability of LFs (Section 3.9)

3.1 What is an LF?

3.1.1 Definition of the notion of LF

A *lexical function* is similar to a *function* in the mathematical sense of the term. Recall that a *mathematical* function f is a relation between two sets A and B such that:

- f applies to each individual element a_i of A , called f 's *argument* – this application is denoted as $f(a_i)$;
- $f(a_i)$ returns, in constant proportion, an element b_i of B , called $f(a_i)$'s *value*:
 $f(a_i) = b_i$;
- by the application $f(a_i)$, each a_i is associated to one and only one b_i .

3 Notion of lexical function [LF]

For instance, arithmetical function f_1 specified by the formula $f_1(x) = x + 3$ is a relation that associates ($\mapsto$) to any positive integer x another positive integer that is greater by three: $0 \mapsto 3$, $1 \mapsto 4$, $2 \mapsto 5$, ...

In the case of *lexical* functions, the arguments and the values are, of course, not numbers:

- the *argument* of an LF f is a particular lexical unit L of the language under description;
- the *value* of its application to L – i.e., $f(L)$ – is a set of (quasi-)synonymous lexical entities that represent alternative choices for the expression of a given meaning.¹

The lexical entities that constitute the value of an LF application can be lexemes, idioms, collocations, compounding elements or derivational affixes. The LF application (1) is an illustration with the LF A_1 : adjective that typically characterizes the DSyntA I of its argument L (see Chapter 4, [21], p. 102); in this case, L is $\text{DOUBT}_{(V)}$ and its DSyntA I the person who doubts.

$$(1) \quad A_1(\text{doubt}_{(V)}) = \text{doubtful}, \text{ in doubt}_{(N)}$$

The equality (1) tells us that the LF A_1 establishes a relation between the lexical unit $\text{DOUBT}_{(V)}$ and the set of near-synonymous lexical entities: the lexeme DOUBTFUL and the collocation *in doubt*.²



We use the term *value* (of an LF) either to refer to a set of alternative lexical entities or, as a shortcut, to one of these lexical entities. For instance, we can say that the lexeme DOUBTFUL is a value of $A_1(\text{doubt}_{(V)})$, where *a value* is an abbreviation for *an element of the value*. Similarly, we write formulas such as “ $L \equiv \text{Syn}(L)$ ” (cf. Section 3.3.1, p. 47) to state that the lexical unit L is equivalent to one of its synonyms (and not, of course, to the set of its synonyms).

Let us now propose a definition of lexical functions.

¹The notions of lexical units and lexical entities have been defined in Chapter 2, Section 2.1.

²The phrase *in doubt* is a deep-syntactic adjective (on deep parts of speech, see Section 3.5.1, p. 57).



Notion definition: lexical function

A *lexical function* f is a function such that

- it expresses a particular meaning ' σ^f ';
- it applies to a lexical unit L as its argument – this application being denoted as $f(L)$;
- it returns as value a set of (quasi-)synonymous lexical entities each of which expresses ' σ^f ' as a function of L .

This definition entails that LFs model such selection of lexical expressions for a given meaning ' σ^f ' that is contingent upon the lexical expression of another meaning with which ' σ^f ' is combined. Let us explain this.

Linguistic meanings can belong to two major classes from the viewpoint of their lexical expression: to an open class of *free* meanings or to a closed class of *constrained* meanings.



Free vs. constrained meanings and lexical functions

A *free meaning* is directly expressible by a lexical means with no regard for other lexical choices made in the production of the same sentence – e.g., 'not expected' can be freely expressed as *unexpected*. No matter what exactly is not expected – *birth*, *thunderstorm* or *behavior*, it can be characterized as *unexpected*. This is, so to speak, the default situation.

A *constrained meaning*, by contrast, is lexically expressed depending on another lexical choice. Thus, the meaning 'intense [X]' has to be lexically expressed as a function of the lexical expression of 'X': 'intense desire' can be expressed as *burning desire*, and 'intense despair' can be expressed as *deep despair* (while **deep desire* and **burning despair* are both ill-formed).

It is for constrained meanings that LFs come into play. Each LF f is associated to a constrained meaning – written ' σ^f ': an application $f(L)$ computes the lexical expression of ' σ^f ' as a function of the lexical unit L .

3 Notion of lexical function [LF]

The expression “meaning ‘ σ^f ’ associated to an FL” is an approximation. In fact, the semantic content ‘ σ^f ’ is not necessarily a genuine meaning. Two cases can be distinguished.³

First, ‘ σ^f ’ is a genuine meaning – e.g., ‘the one who does [L]’ for the LF S_1 (Chapter 4, [13], p. 94), as in $S_1(\textit{build}) = \textit{builder}$. Alternatively, the LF can correspond to a disjunction of genuine meanings – e.g., Magn (Chapter 5, [35], p. 121) corresponds to ‘intense’ (*mad* [love]), to ‘of big magnitude’ (*heavy* [weight]), to ‘to a high degree’ ([*stubborn*] ‘as a mule’), etc. Note that ‘ σ^f ’ can be a zero meaning: cf. support verbs, such as Oper_i (Chapter 5, [46], p. 139).

Second, ‘ σ^f ’ is, strictly speaking, not a meaning, but an indication of meaning identity of L and $f(L)$:

- with preservation of L’s part of speech; see, for instance, the LF Syn (Chapter 4, [1], p. 72): $\text{Syn}(\textit{buttocks}) = \textit{butt}$;
- with change of the part of speech; see, for instance, the LF S_0 (Chapter 4, [7], p. 85): $S_0(\textit{pursue}) = \textit{pursuit}$.

To conclude this introductory presentation of the notion of LF, let us stress that the meaning ‘ σ^f ’ of an LF f determines its *domain of application*, i.e. the set of its possible arguments: the meanings of the latter must be compatible with ‘ σ^f ’. Thus, the above-mentioned LF A_1 takes as arguments only lexical units with predicative meanings (Chapter 2, Section 2.2). As a rule, two situations can present themselves:

- application of an LF to a lexical unit that does not belong to its domain – i.e., to an illegitimate argument; for instance: $A_1(\textit{poppy})$, since the noun *POPPY* is not predicative;
- application of an LF to a legitimate argument that returns an empty value⁴ in the given language (no lexical entity); for instance, $A_1(\textit{swim}_{(V)})$ returns an empty value, since there is no special adjective in English to characterize somebody who is swimming.⁵

³Kahane & Polguère (2001) proposes a formal modeling of the meaning expressed by each LF, as well as of *combinations* of LFs presented in Section 3.4 below.

⁴More precisely, the value is the empty set {}.

⁵The expression *swimming* is an inflectional form (present participle) of the verb $\text{SWIM}_{(V)}$, whose use is accounted for by the grammar.

LFs and lexical networks. A natural language lexicon is essentially a huge set of lexical entities connected by a set of relations: in other words, it is a *lexical network*. LFs participate in the structuring of the lexicon in a fundamental way: each LF application $f(L)$ is equivalent to an elementary subnetwork. This is illustrated in Figure 3.1 with two LF applications involving the lexeme $PROMISE_{(N)}$ as argument.

Standard lexicographic representation:

$Syn_n(promise_{(N)})$	=	<i>commitment, engagement; oath, pledge_(N), vow_(N)</i>
$Real_1(promise_{(N)})$	=	<i>‘carry out’, fulfill, keep_(V) [ART ~]; honor_(V) [ART ~]</i>

Network representation:

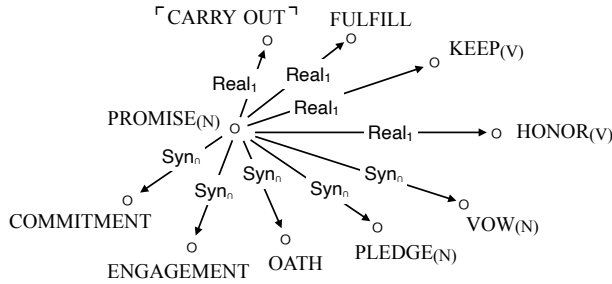


Figure 3.1: Lexical network representing a set of LF applications

The important issue of the structuring of lexical networks by LFs is considered in Chapter 6.

3.1.2 Notation for LF applications

Throughout this book, we use a set of notations (writing conventions and symbols) to present examples of LF applications. The general format of presentation for an LF application is as follows:

$$(2) \quad S_0(\textit{leave}_{(V)}) = \textit{departure}$$

The argument and the elements of the value of the LF are written in *italics* in the above formula – though, technically, they are lexemes and should be written using the standard convention for lexical units: $LEAVE_{(V)}$ and $DEPARTURE$. However, this violation of conventions allows for easier legibility; additionally, values might not be lexical units (e.g., they can be collocations) and it is easier to use a uniform presentation for values, disregarding their lexical status. Note that lists

3 Notion of lexical function [LF]

of values in our examples are not necessarily complete; we often limit ourselves to only one value element.

As illustrated in example (3), three symbols are used to separate the elements of an LF value: “,” for no or minimal semantic difference, “<” for the difference of degree, and “;” for a significant semantic difference.

- (3) $\text{Magn}(\text{salary}) = \text{big, generous, hefty, large, substantial} < \text{enormous, huge, massive; exorbitant}$
[The value *exorbitant* expresses an additional semanteme: ‘too much’.]

The “//” symbol used in (4) below indicates a fused value element (see Section 3.3.3.1, p. 51).

- (4) $\text{Magn}(\text{amaze}) = \text{utterly, // } \text{stun}_{(V)} [\equiv \text{utterly amaze}]$

Finally, two indications can follow a given value element L' .

1. The government pattern⁶ of L' , written between square brackets, specifies the syntactic configuration that L' can govern. This configuration is presented in simplified linear form; for instance in (5) below, where “~” stands for the LF’s argument ($\text{kiss}_{(N)}$) and “ART” stands for a determiner:

- (5) $\text{Oper}_1(\text{kiss}_{(N)}) = \text{give} [\text{ART} \sim \text{to } N_Y]$

2. Semantic or grammatical constraints associated with L' are introduced by a vertical bar “|”; for instance:

- (6) $\text{Oper}_1(\text{visit}_{(N)}) = \text{pay} [\text{ART} \sim \text{to } N_Y] \mid N_Y \text{ denotes a person}$

3.2 Simple standard LFs

The LFs as defined in Section 3.1 constitute a system built around a core of about seventy LFs, called *simple standard LFs*; they are referred to just as *LFs* in the subsequent discussion. Let us present the fundamental properties of these LFs.

The set of LFs has been empirically identified over the last 60 years in the context of the study of typologically varied languages (first publication on LFs: Zholkovsky & Mel’čuk 1965). The work on Russian and French Explanatory Combinatorial Dictionaries has allowed the researchers to sharpen the LF system and to consolidate the very notion of LF, under the elaboration of lexical entries,

⁶On government patterns, see Chapter 2, Section 2.3, p. 32.

where LFs are systematically used (Mel'čuk & Zholkovsky 1984, 2016, Mel'čuk et al. 1984, 1988, 1992, 1999, Mel'čuk & Polguère 2007). At the same time, Meaning-Text research on *syntactic paraphrasing* (Milićević 2007, Mel'čuk 2013a: Ch. 9) and dependency syntax (Mel'čuk & Pertsov 1987, Mel'čuk 1988, 2009, 2015a, 2016a, 2021b) have heavily contributed to the maturation of the notion.

Three topics are discussed below: fundamental properties of LFs (Section 3.2.1), non-standard LFs (Section 3.2.2) and LFs in deep syntax (Section 3.2.3).

3.2.1 Properties of simple standard LFs

Simple standard LFs have two fundamental properties: a vast domain of application (Section 3.2.1.1) and a vast range of possible values (Section 3.2.1.2). These properties follow from the character of the constrained meanings ' σ^f ' that simple standard LFs express.⁷

The first property implies two other more specific properties: potential morphological expression of ' σ^f ' in natural languages (Section 3.2.1.3) and universality of simple standard LFs (Section 3.2.1.4).

3.2.1.1 Vast domain of application

A given LF f takes, in a given natural language, a large number of arguments: it has a vast domain of application. To put it differently, the meaning ' σ^f ' must be sufficiently vague (i.e., impoverished) to be compatible with a large set of lexical meanings to which it is supposed to be united / associated.

Consequently, an f involves a significant subset of a language's lexicon, and this subset is semantically rather heterogenous. Thus, for instance, one may wish to postulate a simple standard LF with meaning 'young of L ' to describe the following *lexical relations*:

- (7) $cat \vdash \text{young of} \rightarrow kitten$
 $dog \vdash \text{young of} \rightarrow puppy$
 $pig \vdash \text{young of} \rightarrow piglet$

This, however, would not be correct since this LF's meaning specifies a very constrained *semantic class* of potential arguments: that of animate beings. Such a lexical relation can be described by an LF, but by a *non-standard* LF (see Section 3.2.2, p. 41).

⁷On constrained meanings, see Section 3.1.1, p. 35.



A lexical relation LexRel – of any type – between lexical units L and L' is denoted as $L \vdash \text{LexRel} \rightarrow L'$.

3.2.1.2 Vast range of possible values

The application $f(L)$ returns a large number of different values for different arguments: f is, therefore, not a trivial relation. An LF that would express the meaning 'of white color' would have a trivial value: $\text{WHITE}_{(\text{Adj})}$ for any lexical unit L. Such trivial cases must not be considered when identifying standard LFs.

3.2.1.3 Potential morphological expression of ' σ^f '

The first above property (vast domain of application) entails another significant property of standard LFs. Given their frequency in the language and their semantic regularity, one can expect the expression of ' σ^f ' by morphological means. This can happen, in a given language, either massively or sporadically; for instance:

- to implement the LF S_1 (Chapter 4, [13], p. 94), English uses the very productive derivational suffix -ER (*driv+er*, *advise+er*, ...);
- by contrast, one finds in French the unproductive nominal derivational suffix -AILLON that expresses the complex LF AntiBon (Chapter 5, [37], p. 124): French *avocat+ailлон* 'bad lawyer' (from *avocat* 'lawyer'), *écriv+ailлон* 'bad writer' (from *écrivain* 'writer'), and a few other nouns.

3.2.1.4 Universality of simple standard LFs

LFs that have the above properties are linguistically *universal*; by this, we mean two things.

Firstly, each LF is supposed to be found in all or most world languages. We say *most* because an LF can be trivial in a given language: it is always expressed there by the same linguistic means – a given lexical unit or a given affix.

Secondly, the system of LFs proposed in the present volume is presumed complete. It is believed (until proven wrong) that one cannot find in a language a lexical relation with the above properties of simple standard lexical functions that would not be included in the proposed system.

Given LFs' linguistic universality, it seems natural that their names come from Latin, so that they are not linked to any specific contemporary language.⁸ To keep LF formulas relatively short, Latin sources are truncated to a maximum of four or five characters – e.g., Syn, from Latin *synonymum*. In a few cases, pseudo-Latin terms are used, preceded by an asterisk – e.g., Conv_{ijk}, from Latin **conversivum*.

3.2.2 Non-standard LFs

Any natural language possesses an open set of constrained meanings ' σ^f ' such that:

- Each one of them is much richer than meanings of standard LFs.
- Because of this, a 'σ^f' can apply only to a semantically limited set of arguments; in point of fact, its domain of application is a clearly identifiable semantic field or a handful of lexical units. Quite frequently, its domain is even reduced to one lexical unit.
- There is not much variability in its expression – most of the time, just one lexical unit.

Such meanings are still expressed as a function of their argument and therefore they can be described by LFs, but by LFs of a completely different nature than standard LFs: these are *non-standard LFs*.

Three examples are given in (8): young of, mentioned earlier in (7), which only applies to the semantic class of lexical units denoting higher-order animals; over-protective of their children, which applies only to PARENT_(N), MOTHER_(N) and FATHER_(N) (and a few of their synonyms); of 366 days, which applies only to the lexical unit YEAR.

- (8) young of (*cat*_(N)) = *kitten*_(N)
 overprotective of her child (*mother*_(N)) = *helicopter*_(N) [~]
 of 366 days (*year*) = *leap*_(N) [~]

Non-standard LFs differ from standard LFs in the following aspects:

- They are not universal, but language-specific.
- Their number in a given language is unlimited, and they cannot be theoretically foreseen.

⁸The LF Equip (Chapter 4, [14], p. 95), from French *équipage* ‘crew’, is an exception.

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- They are so diverse that it is difficult to normalize them, and they have to be specified by the lexicographic definition of their value.
- Prototypically, they are not deep lexical units: they do not appear in deep-syntactic structures and do not participate in universal paraphrasing, unlike standard LFs – see Section 3.2.3 below. Their only job is to ensure correct lexical selection in the transition from the semantic to the deep-syntactic structure.



Further reading

On standard vs. non-standard LFs, see Polguère (2007) and Mel'čuk (2023: Appendix).

3.2.3 Standard LFs in deep syntax

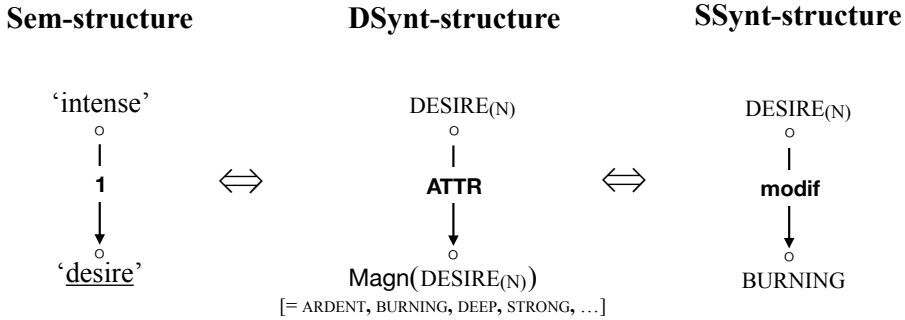
3.2.3.1 LFs as deep lexical units

An application $f(L)$ specifies a set of alternative lexical units: it is, so to speak, a meta-unit. It appears in the deep-syntactic structure [DSyntS] of a sentence, to be replaced by a specific lexical unit selected for the surface-syntactic structure [SSyntS]. As such, an application $f(L)$ is a deep lexical unit. This is illustrated by Figure 3.2.

Reminder. For the semantic and syntactic notions used in the present section – as well as for the representation of semantic and syntactic structures, see Chapter 2, Sections 2.2 and 2.3.

Figure 3.2 shows as well that, in linguistic representations, LFs appear only in DSyntSs. Being a lexical element in DSyntS, an LF has its own combinatorial properties, in particular, a deep part of speech (see Section 3.5, p. 56).

LFs fulfill two tasks in deep syntax: correct lexical selection (Section 3.2.3.2) and DSynt-paraphrasing (Section 3.2.3.3).

Figure 3.2: Magn in the DSyntS of the phrase *burning desire*

3.2.3.2 LFs and lexical selection

Lexical selection is an important part of what is known as *lexicalization* in speech production – e.g., in psycholinguistics research (Kempen & Huijbers 1983) and in computer text generation (Stede 1994).

Under lexical selection, LFs ensure correct choices in case a lexical unit has to be selected contingent upon another lexical unit in the same utterance.

Thus, Figure 3.2 above represents the following steps in lexicalization:

1. At the Sem-level, the semanteme 'intense' is identified as a constrained meaning (Section 3.1.1, p. 35), to be lexically expressed as function of the expression of the semanteme 'desire'.
2. At the DSynt-level, the semanteme 'intense' is expressed by (the name of) the LF Magn. This LF is applied to the lexeme DESIRE_(N): the application Magn(DESI_RE_(N)) represents a constrained lexical choice, more precisely, a set of the usable syntactic modifiers of DESIRE_(N), which express the semanteme 'intense'.
3. At the SSynt-level, the lexical selection is completed: the lexeme BURNING is introduced as a modifier of DESIRE_(N); this is done on the basis of the lexical entry of DESIRE_(N), where the value of the application Magn(DESI_RE_(N)) is specified.

3.2.3.3 LFs and paraphrasing

A vital feature of natural languages is their ability to express, in a general case, a given meaning by a large set of quasi-synonymous utterances, i.e., *paraphrases*, as shown in Figure 3.3.

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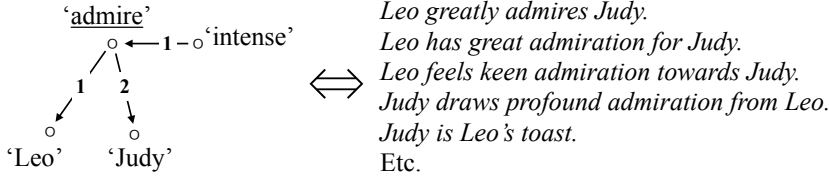


Figure 3.3: Paraphrasing potential of a natural language

This is but a modest illustration of what a natural language can do with regard to paraphrasing: a natural language is a paraphrasing device (Mel'čuk 2012: Ch. 2). Paraphrasing as such is not often used in actual speech production – only in specific situations: editing a text, rephrasing for better explanation, etc. However, the paraphrasing power of a language offers, at any given stage of speech production, multiple alternatives that enable the Speaker to speak fluently. When speaking a second language, a person generally doesn't master the same paraphrasing wealth and therefore production can be slowed down or outright blocked.

The system of standard LFs plays a core role in the use of language, because it is in terms of LFs that *deep-syntactic paraphrasing rules* are formulated. Since LFs are language-universal (see above, p. 40), the resulting system of paraphrasing rules is also universal. Such a paraphrasing system is described in Mel'čuk (1974: 141–176), Milićević (2007: Ch. 6) and Mel'čuk (2013a: Ch. 9). Here, we limit ourselves to citing two paraphrasing rules: $\mathbf{R}_{\text{Mult}}$ and $\mathbf{R}_{\text{Oper}_1}$.

$$\mathbf{R}_{\text{Mult}}: L_{(\text{N})\text{PL}} \equiv \text{Mult}(L_{(\text{N})})_{\text{SG}}$$

This rule describes the universal semantic equivalence between a plural noun $L_{(\text{N})\text{PL}}$ and the corresponding collective noun $\text{Mult}(L)$ – Chapter 4, [26], p. 109; for instance:

- (9) *workers* $\cong$ *workforce*
lies $\cong$ *bunch of lies*

Note that such equivalences hold for a specific meaning of the nominal plural: 'collective of [Ns]'. $\text{Mult}(L)$ itself is in the singular, as stated in the rule $\mathbf{R}_{\text{Mult}}$.

$$\mathbf{R}_{\text{Oper}_1}: L_{(\text{V})} \equiv \text{Oper}_1(S_0(L_{(\text{V})})) \text{ --II--> } S_0(L_{(\text{V})})$$

This rule describes the universal semantic equivalence between a verb $L_{(\text{V})}$ and the verbal collocation consisting of (i) the S_0 nominalization of $L_{(\text{V})}$ (Chapter 4,

[7], p. 85) and (ii) an Oper₁ support verb of this nominalization (Chapter 5, [46], p. 139). For instance:

(10) Leo *admires* Judy. $\equiv$ Leo *has admiration* for Judy.

3.2.3.4 Actantial subscripts in simple standard LFs

Actantial subscripts to LF names have frequently appeared in the preceding text. An explanation of their interpretation is supplied here for the names of simple standard LFs. The corresponding explanation concerning the names of LFs in combinations is provided in Section 3.4.3 (p. 55).

In the general case of an application $f_i(L)$ where f_i is a simple LF, the subscript “_{*i*}” refers to the DSyntA *i* of *L*. Here are two illustrations, with LFs S₂ (Chapter 4, [13], p. 94) and Func₃ (Chapter 5, [47], p. 142).

- (11) a. S₂(*eat* [*‘X eats Y’*]) = *food*
 b. Func₃(*ultimatum* [*‘X’s ultimatum to Y to do Z before time T’*]) = *call*_(V) [*for N_Z*]
 Ex.: *The ultimatum calls*_{Func₃(ultimatum)} *for the release*_Z *of prisoners.*

However, there are at least three special cases of LFs where the interpretation of actantial subscripts is not straightforward:

- Adv_(i), which stands for absolutive adverbials (such as *in the presence of*) – see Chapter 4, [24], Comment 2, p. 107;
- Sympt_{ijk}, which describes somatic phrasemes of psychological or physiological states – see Chapter 5, [66], p. 167;
- Caus, Liqu and Perm, which stand for verbs of causing – see Chapter 5, Section 5.4.5.

3.3 Two major classes of simple standard LFs

LFs are divided in two classes, based on the two functional axes of natural languages (Saussure 1972[1916]: Part 2, Ch. V):

- the *paradigmatic axis*, on which the Speaker performs the *selection* of linguistic units to be used;
- the *syntagmatic axis*, on which the Speaker performs the *combination* of selected linguistic units into higher-level units.

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These two axes and the corresponding linguistic operations (selection vs. combination) determine the two classes of LFs: paradigmatic LFs (Section 3.3.1) and syntagmatic LFs (Section 3.3.2).

3.3.1 Paradigmatic LFs

The application of a *paradigmatic LF* to a lexical unit L returns as value a set of lexical entities of which each element L' stands in a particular paradigmatic relation to L – i.e., L' is a possible substitute for L in the text. More precisely, L' is a *semantic derivative* of L.



Notion definition: semantic derivative

A *semantic derivative* of lexical unit L is a lexical entity L' whose meaning 'L'' includes 'L' such that

- the semantic difference ' σ ' = 'L' – L' is found in a large enough number of lexical pairs of the language – i.e., ' σ ' is regular;
- the language has dedicated means for expressing ' σ ' – e.g., morphological means.

L is called the *base of the derivative* L'.

As anticipated by the above definition, there is a particular case of semantic derivative: *morphological derivative*.



Notion definition: morphological derivative

A *morphological derivative* of lexical unit L is a semantic derivative L' of L such that

- the semantic difference 'L' – L' is expressed by a morphological means.

A typical case of derivatives is that of *Agent noun*, or more precisely, the name for the first semantic actant [SemA 1] 'X' of L. Here are a few illustrations:

3.3 Two major classes of simple standard LFs

- (12) *support*_(V) ⊢ Agent noun → *supporter* ‘X who supports’
agression ⊢ Agent noun → *agressor* ‘X who commits an aggression’
congress ⊢ Agent noun → *congressman* / *congresswoman* ‘X who is a member of a country’s congress’
boring ⊢ Agent noun → *bore*_(N) ‘X who is boring’ [*He is an awful bore.*]
*account*_(N) ⊢ Agent noun → *account holder* ‘X who has a bank account’
*steal*_(V) ⊢ Agent noun → *thief* ‘X who steals’
annoying ⊢ Agent noun → *vulgar* ‘*pain in the ass*’ ‘X that is annoying’
heal ⊢ Agent noun → *remedy*_(N) ‘X that heals’

Among these semantic derivatives, the first four are also morphological derivatives.

Semantic derivation is described by paradigmatic LFs in the following way:

- a given paradigmatic LF f specifies a particular family of semantic derivatives that express the meaning ‘ σ^f ’;
- it takes as arguments the bases of derivatives;
- it returns as values the derivatives themselves.

For instance, the semantic derivation of Agent nouns mentioned above is described by the paradigmatic LF S_1 with the meaning ‘somebody who / something that ...’ (Chapter 4, [13], p. 94). For the illustrations (12), it is encoded as in (13):

- (13) $S_1(\textit{support}_{(V)}) = \textit{supporter}$
 $S_1(\textit{agression}) = \textit{agressor}$
 $S_1(\textit{congress}) = \textit{congressman}, \textit{congresswoman}$
 etc.

Remark on the LF Syn. The LF Syn ‘synonym’ (Chapter 4, [1], p. 72) occupies a special place among paradigmatic LFs. It models a particular case of the most important relation underlying the functioning of natural languages: *synonymy in the broad sense*, i.e., semantic equivalence between linguistic expressions. Consequently, Syn underlies the simplest paraphrasing rule:

$$\boxed{\mathbf{R}_{\text{Syn}}: L \equiv \text{Syn}(L)}$$

Note that all paraphrasing rules – e.g., rules $\mathbf{R}_{\text{Mult}}$ and $\mathbf{R}_{\text{Oper}_1}$ (Section 3.2.3.3, p. 44) – establish the synonymy relations between linguistic expressions. Even more importantly, the Meaning-Text approach is built on the notion of *Meaning* (written with a capital *M*), which is the invariant of a set of paraphrases, i.e., synonymous expressions (Mel’čuk 1974: 10–11, 2012: 44–45, Mel’čuk & Milićević 2014: 102–103).

3.3.2 Syntagmatic LFs

The application of a *syntagmatic LF* to a lexical unit L_1 returns as value a set of lexical entities of which each element L_2 stands in a particular syntagmatic (i.e., cooccurrence) relation with L_1 . More precisely, L_2 forms a *collocation* with L_1 .⁹



Notion definition: collocation

A *collocation* is a compositional phraseme consisting, prototypically, of two lexical units L_1 and L_2 such that

- L_1 is selected by the Speaker freely, i.e., independently of any other lexical choice;
- L_2 is selected by the Speaker to express a particular meaning ‘ σ^f ’ in combination with L_1 as a function of L_1 .

L_1 is called the *base of the collocation* and L_2 , its *collocate*.

Examples. In the collocation *deep feeling*, *feeling* is the base and *deep*, the collocate; in *pay a visit*, *visit* is the base and *pay*, the collocate.

An emblematic case of collocation is that of intensification; here are three examples where collocates are italicized:

- (14) a. *nagging* doubt
b. Thanks *a million*!
c. drink ‘*like a sponge*’

Syntagmatic LFs take as arguments lexical units that are bases of collocations and return as values lists of corresponding collocates. For instance, intensification collocations are accounted for by the syntagmatic LF Magn (Chapter 5, [35], p. 121), so that collocations given in (14) are described as (15):

⁹The following definition would necessitate fine-tuning. In practice, instances of some sort of double-shot collocations can be found, where the collocate L_2 is not a lexical unit but is itself a collocation. For instance, the phraseme *under the effect of alcohol* has all appearances of being a double-shot collocation, where ALCOHOL is the base and *under the effect* is itself a collocation with EFFECT_(N) as its base. For a description of *under the effect* in terms of the LF A_2 , see Chapter 4, [21], p. 102; for a description of *under the effect of alcohol* in terms of the LF Propt₁, see Chapter 5, [44], p. 136. Note that the LF Sympt_{ijk} is often used to encode phrasemes that seem like double-shot collocations – see Chapter 5, [66], p. 167.

- (15) $\text{Magn}(\text{doubt}_{(N)}) = \text{strong} < \text{gnawing, nagging; lingering}$
 $\text{Magn}(\text{Thanks!}) = a \text{ lot} \mid \text{postposed}, a \text{ million} \mid \text{postposed}$
 $\text{Magn}(\text{drink}_{(V)}) = \text{heavily}, \text{'like a fish'}, \text{'like a sponge'}$



Further reading

For more on the notion of collocation, see Hausmann (1979), Wanner (1996), Mel'čuk (1998) and Mel'čuk (2023: Ch. 6).

Semantic target of a syntagmatic LF. For a full [= semantically loaded] syntagmatic LF such as Magn ,¹⁰ the meaning of $t(L)$, i.e. the meaning of L's collocate that is returned as value, necessarily applies to a component of L's lexicographic definition (Iordanskaja & Polguère 2005). Let us call such a component the (semantic) *target* of the LF.

In most cases, the LF target in L's definition is its central component and need not be explicitly indicated in the encoding of the LF. If, however, the target is not the central component, it must be specified as a subscript to the LF name; for instance, for Magn :

- (16) $\text{Magn}(\text{earthquake}) = \text{powerful, strong} < \text{major, massive}$
 $\text{Magn}_{\text{damage}}(\text{earthquake}) = \text{catastrophic, devastating; deadly}$
 Note. The approximate definition of EARTHQUAKE is 'sudden shaking of the Earth's surface, able to cause serious *damage* around location Y'.

Follows an illustration for the verbal syntagmatic LF Real_i (see Chapter 5, [49], p. 147):

- (17) $\text{Real}_{1, \text{'hostile action'}}(\text{anger}_{(N)}) : \text{vent} [A_{(\text{poss})}X \sim \text{on } N_Y]$
 $\text{Real}_{1, \text{'loss of self-control'}}(\text{anger}_{(N)}) : \text{'be beside oneself'} [with \sim]$

The lexical entry for $\text{ANGER}_{(N)}$ in the Appendix (p. 209) presents more examples of the same kind.

¹⁰By contrast, support verb LFs (see Chapter 5, Section 5.4.2) are examples of semantically empty syntagmatic LFs.

3.3.3 Paradigmatic and syntagmatic LFs in contact

Both classes of LFs – paradigmatic and syntagmatic ones – are designed to fulfill the same role, namely, to specify lexical expressions of constrained meanings (Section 3.1.1, p. 35). Therefore, they are expected to be frequently intertwined; and in fact, they are. Their interactions are of two kinds: with respect to the values they return (Section 3.3.3.1) and with respect to their use in linguistic description (Section 3.3.3.2).

3.3.3.1 Quid pro quo in LF values

The paradigmatic LFs express derivational meanings and are supposed to return semantic derivatives of their arguments; the syntagmatic LFs express collocational meanings and are supposed to return collocates of their arguments.



However, there is no airtight partition between the two classes of paradigmatic vs. syntagmatic LFs with respect to the lexical nature of their values.

This happens because both types of LFs do essentially the same job: they supply an appropriate expression means for a meaning to be added to the meaning of the base (of a derivative or of a collocation). The fundamental difference between the two types of LFs in this respect is as follows: for paradigmatic LFs, these means tend to be affixal – at least, in *Standard Average European* [SAE]¹¹ languages; for the syntagmatic LFs, the expressive means are normally separate lexical units. As a result, the values of the paradigmatic LFs are rather synthetic, and those of the syntagmatic LFs, rather analytic. But, as is always the case with natural languages, this distinction is not implemented 100% systematically: there are analytic values of paradigmatic LFs and synthetic values of syntagmatic LFs.

Analytic values of paradigmatic LFs. A paradigmatic LF *f* applied to a lexical unit *L* can return (instead of a one-piece semantic derivative) a collocation of *L*, which is, so to speak, an analytic semantic derivative of *L*. Such is the case of two previously mentioned examples: the analytic value *account holder* for *S*₁(*account*_(N)) and *in doubt* for *A*₁(*doubt*_(N)); see (18):

¹¹Roughly, SAE languages are Germanic, Romance and Slavic languages (Whorf 1941).

- (18) $S_1(\text{account}_{(N)}) = \sim \text{holder}$
 $A_1(\text{doubt}_{(N)}) = \text{doubtful, in } \sim$

This phenomenon is quite typical of some paradigmatic LFs, in particular of the collective Mult (Chapter 4, [26], p. 109):

- (19) $\text{Mult}(\text{fish}_{(N)}) = \text{school of } \sim$
 $\text{Mult}(\text{attack}_{(N)}) = \text{spate of } \sim\text{s}$
 $\text{Mult}(\text{activity}) = \text{flurry of } \sim$

Synthetic [= fused] values of syntagmatic LFs. A syntagmatic LF f applied to L can return – instead of a collocate of L – a semantic derivative of L : a lexical unit that synthetically expresses the linguistic union of the meanings ‘ L ’ and ‘ σ^f ’. For instance, in the case of the attenuation LF AntiMagn (antonym of Magn):

- (20) $\text{AntiMagn}(\text{rain}_{(N)}) = \text{light, // drizzle}_{(N)} [\cong \text{light rain}]$

The noun $\text{DRIZZLE}_{(N)}$ is a semantic derivative of $\text{RAIN}_{(N)}$, namely, a richer synonym $\text{Syn}_{\supset}$ rather than a genuine collocate, such as $\text{LIGHT}_{(\text{Adj})}$.

The phenomenon under consideration is called *fusion*, and the corresponding value is a *fused value*. Fusion is indicated, as shown above, by the symbol “//”.

3.3.3.2 Interaction of paradigmatic and syntagmatic LFs

Cohabitation of LFs in paraphrasing rules. Syntactic paraphrasing rules (cf. Section 3.2.3.3), which are crucial for the functioning of language, actively exploit both families of LFs, often making them collaborate within the same rule. This latter case is illustrated by rule $\mathbf{R}_{\text{Oper}_1}$ (Section 3.2.3.3, p. 44), which simultaneously uses the paradigmatic S_0 and the syntagmatic Oper_1 .

Semantic connections. A paradigmatic LF and a syntagmatic LF can show interesting semantic connections; here are two examples.

1. The paradigmatic LF S_1 applied to a verb L means ‘somebody / something that does L ’ – L is, so to speak, what S_1 is supposed to do; and the syntagmatic LF Fact_0 applied to a noun L' means ‘ $[L']$ does what it is supposed to do’. From this it follows:

- (21) $S_1(L_{(\text{'action'})}) = L'$ implies $\text{Fact}_0(L') = L_{(\text{'action'})}$
Examples:
 $S_1(\text{ski}_{(V)}) = \text{skier}$ implies $\text{Fact}_0(\text{skier}) = \text{ski}_{(V)}$
 $S_1(\text{sell}_{(V)}) = \text{seller}$ implies $\text{Fact}_0(\text{seller}) = \text{sell}_{(V)}$

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2. A modifying syntagmatic LF – Magn, Ver or Bon – in case of fusion adds to L a meaning such that the resulting L' can be considered a richer synonym $\text{Syn}_{\supset}$ of L, $\text{Syn}_{\supset}$ being a typical paradigmatic LF:

$$(22) \quad \text{Magn}(L) \quad = \parallel L' \quad \text{implies} \quad \text{Syn}_{\supset}(L) \quad = L'$$

Examples:

$$\begin{aligned} \text{Magn}(\text{tired}) &= \parallel \text{exhausted} & \text{implies} & \text{Syn}_{\supset}(\text{tired}) = \text{exhausted} \\ \text{Bon}(\text{musician}) &= \parallel \text{virtuoso}_{(N)} & \text{implies} & \text{Syn}_{\supset}(\text{musician}) = \text{virtuoso}_{(N)} \end{aligned}$$

3.4 Simple standard LF combinations

Simple LFs are combinable with each other; let us explain when such combinations are necessary.

A given constrained meaning ' σ ' can be representable as the *linguistic union* of two constrained meanings ' σ_1^f ' and ' σ_2^f ', which correspond to two standard LFs f_1 and f_2 . This situation is described by the following expression, where the symbol " $\oplus$ " stands for the operation of linguistic union: ' $\sigma = \sigma_1^f \oplus \sigma_2^f$ '.¹²

In such a case, ' σ ' has to be associated with an LF constructed by the combination of f_1 and f_2 . There are two types of such combinations: complex LFs (Section 3.4.1) and configurational LFs (Section 3.4.2).

3.4.1 Complex LFs

An LF f_1 can take another LF f_2 as its argument to form a specific combination of LFs, called a *complex LF* and denoted as $f_1 f_2$. This complex LF applies to L and returns L' expressing ' $\sigma_1^f \rightarrow \sigma_2^f$ '.

Here is an illustration where ' σ_1^f ' is 'begin' – expressed by the LF Incep – and ' σ_2^f ' is 'utilize' – expressed by the LF Real₁:

$$(23) \quad \text{IncepReal}_1(\text{road}) = \text{take}_{[\text{ART } \sim]} \\ [take \ a \ road \ \cong \ begin_{\text{Incep}} \ using_{\text{Real}_1} \ a \ road]$$

The name of a complex LF $f_1 f_2$ is an abbreviation for $f_1(f_2)$. This means that f_1 applies to f_2 to produce a complex LF $f_1(f_2)$, which then applies as a whole to L – i.e., $[f_1(f_2)](L)$.

¹²The operation of linguistic union is defined as follows in Mel'čuk (2006: Ch. 7, Sec. 2, def. 7.3, p. 386): "[...] an operation applicable to pairs of linguistic units (of language $\mathcal{L}$) – in particular, to signs – which unites them, according to their syntactics and/or the general rules of (the grammar of) $\mathcal{L}$."

This situation is quite different from a *composition* of LFs (in the mathematical sense). In a composition of LFs f_1 and f_2 , there is no actual production of an LF $f_1(f_2)$: the LF f_2 applies to L , then f_1 applies to the application $f_2(L)$, which is written as $f_1(f_2(L))$. The contrast between a complex LF $A_2\text{Real}_1$ and the corresponding composition of the same LFs is shown in (24):

- (24) $\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'}) = \text{use} [\text{ART } \sim]$
 $A_2(\text{use}_{(V)}) = \text{used}$
 $A_2\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'}) = \text{occupied}$
 $A_2(\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'})) = \text{used}$

As one can see, in this case, the composition of A_2 and Real_1 fails to return the idiomatic adjective expressing the meaning '[toilet] that is being utilized': namely, *occupied*, which will actually appear on a locked toilet door. (For an interesting case of LF composition with the LF *Son*, see Chapter 5, [63], p. 163.)

Remark. While complex LFs are overwhelmingly present in the lexicon, composition of LFs is very seldom relevant in lexicological modeling. It is mentioned here only because complex LFs are frequently mistaken for cases of LF composition.

As regards the two major classes of LFs – paradigmatic vs. syntagmatic (Section 3.3), there are four possible cases of complex LFs:

1. A paradigmatic LF applies to a paradigmatic LF
 $S_{\text{loc}}\text{Mult}(\text{senator}) = \text{senate } 2 \text{ [building]}^{13}$
2. A syntagmatic LF applies to a paradigmatic LF
 $\text{MagnMult}(\text{gunshot}) = \text{hail of } \sim s$
3. A paradigmatic LF applies to a syntagmatic LF
 $S_{\text{loc}}\text{Real}_1(\text{'racing car'}) = \text{'racing track'}$
4. A syntagmatic LF applies to a syntagmatic LF
 $\text{MagnSon}(\text{engine}) = \text{roar}_{(V)}$
 $\text{IncepSon}(\text{phone}_{(N)}) = \text{'go off'}$

A complex LF f_1f_2 encodes a deep-syntactic structure [DSyntS] (Chapter 2, Section 2.3) that is determined by the nature of f_1 and f_2 . Thus, for the case of *MagnSon* vs. *IncepSon*, we have the two DSyntSs visualized in Figure 3.4.

¹³The connection between *SENATOR* and *SENATE 2* can also be established in two steps:
 $\text{Mult}(\text{senator}) = \text{senate } 1$ and $S_{\text{loc}}(\text{senate } 1) = \text{senate } 2$.

3 Notion of lexical function [LF]

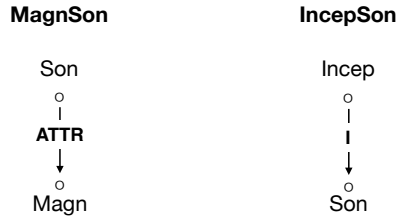


Figure 3.4: DSyntSs of complex LFs MagnSon and IncepSon



Part of speech of a complex LF

The head [= top node] of the DSyntS of a complex LF gives its part of speech to the value elements of this LF. For instance, the value of $S_{loc}Real_1$ – whose DSyntS is $S_{loc} \rightarrow Real_1$ – is a noun (as S_{loc}), and the value of MagnSon – whose DSyntS is Magn $\leftarrow ATTR$ Son – is a verb (as Son).

The part of speech of simple standard LFs is presented in detail in Section 3.5 (p. 56).

A complex LF can consist of more than two simple LFs. For instance, Figure 3.5 shows the DSyntSs of such complex LFs exemplified in (25):

- (25) a. $AntiBonReal_1(chair_{(N)}) = slouch_{(V)}, slump_{(V)} [in ART \sim]$
 b. $CausPlusFact_0(screw_{(N)}) = tighten [ART]$

Note. $CausPlusFact_0$ is an abbreviation of $CausPlusMagnFact_0$ – see Chapter 5, [38], “Adjectival / adverbial collocates,” p. 130.

3.4.2 Configurational LFs

The combination of two LFs f_1 and f_2 can be such that both LFs apply to L simultaneously, but independently of each other and return a single value expressing jointly both LFs. This is represented by the following formula: $f_1+f_2(L)$.

An f_1+f_2 configuration is called a *configurational LF*. Compare the following pair of LFs:

- (26) $A_1(sweat_{(V)}) = sweaty$ ‘who is sweating’
 $Magn+A_1(sweat_{(V)}) = drenched in sweat_{(N)}$ ‘who is sweating a lot’

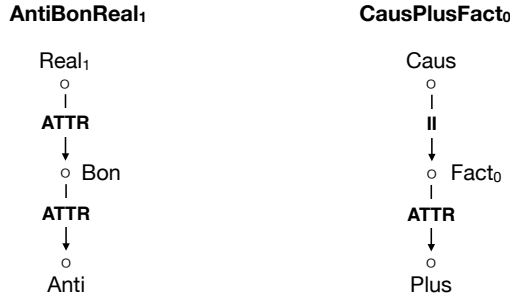


Figure 3.5: DSyntSs of complex LFs AntiBonReal₁ and CausPlusFact₀

In a configurational LF f_1+f_2 , the order of components is relevant: f_2 – the rightmost LF – is dominant, both syntactically and informationally. In the configuration f_1+f_2 where f_1 and f_2 are of different parts of speech, the component f_2 imposes its part of speech on the value of the configurational LF, i.e., it dominates f_1 syntactically. For instance, in the example below, the value is a verb, in accordance to the Oper₁ component of the configurational LF:

(27) Magn+Oper₁(*desire*_(N)) = *simmer* [with ~]

If f_1 and f_2 are of the same part of speech – e.g., both are modifying adjectives – the meaning of f_2 constitutes the informationally dominant component in the meaning of the value. For instance, the contrast between Bon+Magn and Magn+Bon is illustrated in (28):

(28) Bon+Magn(*meal*) = *hearty* ‘copious, and also good’
 Magn+Bon(*meal*) = *somptuous* ‘good, and also copious’

3.4.3 Actantial subscripts in LF combinations

3.4.3.1 Actantial subscripts in complex LFs

Two cases are to be distinguished for the actantial subscript to the name of a simple LF f_i used as a component of a complex LF (except for the rightmost one in the formula, which is interpretable in the standard way).

If f_i in a complex formula $f_i g$ is not a verb of causing (Chapter 5, Section 5.4.5), its argument is the next simple LF g , and its actantial subscript “_{*i*}” refers to the DSyntA_{*i*} of g . For instance, in the complex LF A₁Real₂, illustrated in (29), “₁” refers to the DSyntA₁ of Real₂ – that is, to the sick person taking the medication.

(29) A₁Real₂(*medication* [prescribed by *X* to *Y* against *Z*]) = *medicated*, on ~

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The actantial subscripts of the LF $A_1\text{Real}_2$ used in (29) are interpretable as follows:

- The lexical unit *MEDICATION* controls three DSyntAs: DSyntA I is the doctor X; DSyntA II, the sick person Y; DSyntA III, the illness Z.
- $\text{Real}_2(\text{medication}) - [Y] \text{ takes } [a \text{ medication}]$ – controls, by definition of the LF Real_i (Chapter 5, [49], p. 147), two DSyntAs:
 - DSyntA I, the sick person Y – cf. “₂” in Real_2 (since Y is DSyntA II of *MEDICATION*);
 - DSyntA II, the noun *MEDICATION*.
- The subscript “₁” of A_1 in the formula $A_1\text{Real}_2$ refers to the DSyntA I of its argument Real_2 , that is, to the sick person Y (and not to the DSyntA I of *MEDICATION*, that is, to the doctor X).

If f_i is a verb of causing, its actantial subscript “_i” refers to the corresponding DSyntA of L, just as it does when this LF is used alone. For instance, in the complex LF $\text{Caus}_2\text{Oper}_2$ illustrated in (30), the subscript “₂” of Caus_2 refers to the DSyntA II of $\text{trust}_{(N)}$, that is, to the person who is trusted:

- (30) $\text{Caus}_2\text{Oper}_2(\text{trust}_{(N)} [\text{of } X \text{ in } Y]) = \text{win}_{(V)} [\text{ART } \sim]$
Ex.: $\text{Joey}_Y \text{ won}_{\text{Caus}_2\text{Oper}_2(\text{trust}_{(N)})} \text{ the trust of Chandler}_X$.

3.4.3.2 Actantial subscripts in configurational LFs

There is nothing particular to be said about actantial subscripts in configurational LFs: the components of these LFs are either simple LFs, or complex LFs, and the explanations about actantial subscripts in these two types of LFs have already been provided.

3.5 Parts of speech of simple standard LFs

Since an LF is a deep lexical unit (cf. 3.2.3.1, p. 42), it belongs to a part of speech. More specifically, as we will see, it has only a deep-syntactic part of speech.

3.5.1 Notion of part of speech [PoS]

A *part of speech* [PoS] is a major class of lexical units that feature, at the syntactic level, the same *passive syntactic valence*. By passive syntactic valence of a lexical unit, we mean its ability to participate in particular syntactic constructions as the dependent member (Mel'čuk 2015b: 109):¹⁴

- a verb – in a finite form – is typically the syntactic head of a clause and need not be dependent;
- a noun is typically a syntactic actant;
- an adjective is typically a modifier of a noun;
- etc.

Two types of PoS are distinguished: deep PoS and surface PoS. There are five *deep parts of speech*:

- *Verb* [V],
- *Noun* [N],
- *Adjective* [Adj],
- *Adverb* [Adv],
- *Clausative* [Claus].

V, N, Adj and Adv do not call for further remarks. Claus, on the other hand, needs an explanation: a clausative is a lexical expression that is syntactically equivalent to an independent clause and is not a verb.¹⁵ Clausatives form a very heterogeneous lexical class, which includes emotional interjections (Wow!, Yuck!) and onomatopoeic interjections (MEOW!, BANG!), as well as interjection-like lexemes (WELL) and fixed phrases (‘BLESS YOU’ [to someone who just sneezed])¹⁶.

Deep PoS are linguistically universal in the sense that they are necessary and sufficient for describing deep-syntactic structures of sentences in any language.

¹⁴Passive syntactic valence is to be contrasted with active syntactic valence, introduced in Chapter 2, Section 2.3, p. 31.

¹⁵Tesnière (1959: Ch. 45) calls such expressions *mots phrases* or *phrasillons*.

¹⁶Even if BLESS is inherently a verb, the phrase *Bless you!* is not a verbal phrase: the form *bless* in it obviously cannot be considered an imperative, and neither can it be taken to be a finite form, since it has no subject. This form is “frozen” not only semantically, but also syntactically.

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In contrast to deep PoS, *surface PoS* are not linguistically universal: each language features its own set of surface PoS. Thus, English has prepositions, while Hungarian has only postpositions; English has determiners, while Russian does not. A given surface PoS is a “particular case” of a deep PoS: prepositions and conjunctions are subtypes of deep adverbs (Adv);¹⁷ determiners are subtypes of deep adjectives (Adj); etc. On deep vs. surface PoS, see Mel’čuk (2013a: Ch. 7, Sec. 2.1.3, 49–51).

In what follows, only deep PoS are considered, so that the adjective *deep* can be omitted.

To conclude, note that the PoS of a lexical unit L is a core element of L’s cooccurrence; as such, it can be expected to be the first piece of information given in L’s lexical entry – see e.g., the lexical entry of ANGER_(N) in the Appendix (p. 209).

3.5.2 LF’s inherent PoS

Most simple standard LFs possess an inherent PoS: e.g., $S_i(L)$ is always a noun, and $Oper_i(L)$ is always a verb. However, several paradigmatic LFs borrow their PoS from their argument. Such LFs are:

- the three “meta”-LFs, which underlie the whole LF system: Syn, Conv_{ijk} and Anti (with Contr) – Chapter 4, Section 4.1, p. 71;
- the “semantic pronoun”: Gener – Chapter 4, [4], p. 80;
- two LFs whose meanings are close to those of grammemes: the singulative Sing and the collective Mult – Chapter 4, Section 4.8, p. 108.

For the PoS of complex LFs, see Section 3.4.1, p. 53.

Part of speech tables for LF applications. An LF f calls for two syntactic characteristics as regards PoS: (i) the PoS of its possible arguments L and (ii) the corresponding PoS of its value – a deep lexical unit $f(L)$.

This correspondence is shown in a *PoS table* for each LF. Take as example the PoS table of Magn, intensification modifier:

¹⁷More precisely, prepositions and conjunctions form a subclass of adverbs that have an obligatory DSyntA II (on DSyntA, see Chapter 2, Section 2.3, p. 30).

Magn: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Magn(L)	Adv	Adj	Adv	Adv	Adj / Adv

The table header indicates that Magn is a modifier that functions as an adjective or an adverb. The body of the table specifies the correspondence between the PoS of L and that of the deep lexical unit Magn(L) – i.e., the PoS of Magn itself, for short.

When a PoS correspondence is theoretically possible, but not attested to our knowledge, the symbol “[?]” is added. The PoS table of Gener illustrates this situation.

Gener: borrows L’s PoS					
L	V	N	Adj	Adv	Claus
Gener(L)	V	N	Adj	Adv	Claus[?]

In the header of this table, the statement “borrows L’s PoS” indicates that Gener is an LF without inherent PoS.

3.6 Regularities in standard LFs

In spite of the basically idiosyncratic character of LFs, in many cases a given LF has the same values for quite a few different arguments: you *take a bath*, *a shower*, *a rest*, ...; you *are in despair*, *love*, *a rage*, ...; you *follow the rules*, *prescriptions*, *recommendations*, *instructions*, ... The reason is, of course, semantic proximity of these LF’s arguments: semantically related lexical units tend to feature the same values for a given LF.

In other words, there exist *regularities* among LFs of any natural language. No doubt that such regularities play an important role in the acquisition of LFs by speakers, as the former allow the latter to make predictions about LFs. These regularities should be exploited in such domains as language teaching, translation, etc. It is therefore important to extract them, and formulate *generalizations*, which should be stored in models of the lexicon in a way that supports the *inheritance* of regularities (Mel’čuk & Wanner 1996).

The elements of the values of an LF *f* shared by several argument lexical units can be generalized by extracting them from individual lexical entries of these

3 Notion of lexical function [LF]

units and storing them together: either on the argument side (for a group of lexical units) or on the LF side (attaching them to the LF itself). In practice, both methods can be combined in different proportions. We propose below four types of generalizations, that is, we indicate types of regularities with associated means of accounting for them: generalizations at vocable's level (Section 3.6.1), generalizations associated with a generic term (Section 3.6.2), joker values associated with given LFs (Section 3.6.3) and lexical entries for collocative lexical units (Section 3.6.4).

3.6.1 Generalizations at vocable's level

If several lexical units of a given vocable or of several semantically-related vocables share LF values, these values can be given under one of these units, with a cross-reference in the lexical entries of the others. For instance, we have for the verb $\text{ESCAPE}_{(V)}$:

- (31) S_1 : *escaper*
 $S_1\text{Perf}$: *escapee, fugitive_{(N)}, runaway_{(N)}*
 $S_1\text{Able}_1$: 'flight risk'
 S_2 : 'place of detention'
 A_1 : *fugitive_{(Adj)}, runaway_{(Adj)}*
 Able_2 : *escapable*

As the same LF relations hold for the corresponding noun $\text{ESCAPE}_{(N)}$, which is the S_0 (nominalization) of the verb, we can simply indicate the following cross-reference in its lexical entry:

- (32) $S_1, S_1\text{Perf}, S_1\text{Able}_1, S_2, A_1, \text{Able}_2$: $\uparrow \text{ESCAPE}_{(V)}$

Note that (32), which concerns two specific lexical units ($\text{ESCAPE}_{(V)}$ and $\text{ESCAPE}_{(N)}$), is in point of fact a consequence of a series of more general statements concerning the entire system of LFs itself, namely:

- (33) $S_i(L_{(V)}) = S_i(S_0(L_{(V)}))$
...
 $A_i(L_{(V)}) = A_i(S_0(L_{(V)}))$
...

3.6.2 Generalizations associated with a generic term

There are semantic classes of lexical units such that:

1. they all belong to the same semantic field;

2. they possess the same generic term L_{GENER} ¹⁸ – for instance, CAR, TRUCK_(N), VAN, BUS_(N), MOTORBIKE_(N), ... form a semantic class united under the same L_{GENER} VEHICLE in the sense of ‘motor vehicle’;
3. the meaning of L_{GENER} is sufficiently rich so that it belongs to the same semantic field as these lexical units – cf., VEHICLE belongs to the same semantic field as CAR, TRUCK_(N), etc.

Elements of such semantic classes may share the same values for some LFs. (This is only a tendency, with many exceptions, yet in some cases this tendency is quite clear.) If such is the case, generalizations can be made by storing all common values of LFs supplied with necessary constraints under a special *public zone* of L_{GENER} ’s lexical entry. For instance, if the following is stored in the public zone of the lexical entry for VEHICLE, it does not need to be repeated in the lexical entry for all names of vehicles:

VEHICLE, noun
 ...
Public
 ...
 S_1 : *driver; rider* | for two-wheel vehicle
 S_2 : *passenger*
 ...
 Real_1 [i.e., ‘use ~ as driver’] : *drive*_(V) [ART ~]; *ride*_(V)1 [ART ~] | for two-wheel vehicle
 Real_2 [i.e., ‘use ~ as passenger’] : *ride*_(V)2 [(in / on) ART ~], go [by ~]
 ...

A rather straightforward particular case of the above type generalization is the use of LFs with proper names. If the argument of an LF *f* happens to be a proper name $N_{(\text{prop})}$, the value of *f* must be computed substituting for $N_{(\text{prop})}$ the lexical unit *L* that denotes the entity named by $N_{(\text{prop})}$. In other words, the values of LFs for RHINE, GANGES, VOLGA, THAMES, MISSISSIPPI, etc. are those of the LFs for RIVER (or WATERCOURSE):

- (34) a. The Rhine ⟨This river⟩ *flows* from Switzerland to the North Sea.
 b. The Ganges ⟨This river⟩ *empties* into the Bay of Bengal.
 c. The Oka ⟨This river⟩ is a *tributary* of the Volga.
 d. The Thames ⟨This river⟩ was *swollen* and *flooding* the riverbanks.

¹⁸On generic terms, see the LF Gener, Chapter 4, [4], p. 80.

3.6.3 Joker values associated with given LFs

The level of generality of LF values might be such that it becomes feasible to associate with a given LF some general default elements of its value: we refer to such default elements as *joker values*. Thus, a generally valid expression in English of the LF S_0 for a verb $L_{(V)}$ is the phrase *the fact of V-ing* – or just simply *V-ing*; e.g.:

- (35) $\text{joker } S_0(\text{chew}_{(V)}) = \text{the fact of chewing, the chewing}$
 $\text{joker } S_0(\text{ponder}) = \text{the fact of pondering, the pondering}$

For the support verbs $\text{Oper}_1(L)$, the joker value elements are DO, MAKE, GIVE, HAVE, TAKE, ‘‘CARRY OUT’’ and maybe some more. Of course, for each of those, semantic restrictions on L have to be specified. For $\text{Func}_0(L)$ – another type of support verbs, joker value elements are, among others, ‘‘TAKE PLACE’’, HAPPEN (for L denoting a social event), etc. For intensifiers $\text{Magn}(L)$, when they have nominal arguments, such general elements are: $\text{BIG}_{(\text{Adj})}$, $\text{HIGH}_{(\text{Adj})}$ (for L denoting non-linear physical parameters such as $\text{PRESSURE}_{(N)}$, $\text{SPEED}_{(N)}$, $\text{TEMPERATURE}_{(N)}$), $\text{UTMOST}_{(\text{Adj})}$ / $\text{UTTER}_{(\text{Adj})}$ (for feelings), etc. An obvious Magn joker value element for adjectival lexemes as arguments is VERY.

The generalizations considered here are directly associated with LFs and their description presupposes the existence of special lexical entries dedicated to LFs.

3.6.4 Lexical entries for collocative lexical units

Collocative lexical units are ‘‘designed’’ to be selected by the Speaker as values of syntagmatic LFs: they have virtually no autonomy and are essentially used as collocates within collocations.

To ensure a sufficiently systematic description of collocative lexical units shared by many collocation bases, one has to compile for them special lexical entries, where semantic and syntactic conditions on their use are specified. Thus, the support verb *deliver* is an element of the value of the LF Oper_1 for many nouns of public activities (*deliver a speech / a lecture / a performance*). It receives a lexical entry of its own in the vocable DELIVER; accordingly, it need not be lexicographically described in all lexical entries for nouns it combines with but can be selected and used following indications found in its own entry.

For an example of a dictionary of collocations with entries for collocates, see Bosque (2010).

To round up the topic, still another perspective of possible generalizations for LFs is opened by accounting for the links between the meaning of a lexical unit L and the LFs applicable to it; for instance:

3.7 Description template for simple standard LFs

- L that is predicative expects actantial paradigmatic derivatives – cf. LFs S_i , A_i , ..., and syntagmatic verbal collocates such as support verbs ($Oper_i$, ...) and possibly realization verbs ($Real_i$, ...).
- L that denotes an artefact expects realization verbs such as $Real_1$, which expresses ‘make use of L’.
- L that denotes something producing sounds expects values for the LF Son .
- L that denotes an inner state of a human expects values for the LFs $Manif$ (‘[L] manifests itself in (part of the body of) X’) and $Sympt_{ijk}$.
- L that denotes a place expects values for the LF $Loc_{in/ad/ab}$.

The identification and proper modeling of such generalizations is an open research question.



Further reading

On semantic regularities in standard LF applications, see Apresjan (2009); on lexical inheritance of LF regularities, see Mel’čuk & Wanner (1996), Barrios Rodríguez (2009); on accounting for regularities in lexical entries for collocates, see Reuther (1996); on the lexicographic description of collocates, see Alonso Ramos & Mantha (1996), Alonso Ramos (2003); on extracting regularities – polysemy and lexicographic definitions – in lexical networks with the LF Gener, see Polguère (2022).

3.7 Description template for simple standard LFs

In Chapters 4 and 5, we follow a three-part template (see below, Parts A, B and C) for the presentation of each individual LF. This template is illustrated with the LF S_0 (Chapter 4, [7], p. 85).

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Part A. Identification number between square brackets and name of the LF together with its Latin source also between square brackets (cf. the remark on the naming of LFs on p. 41), followed by a rough characterization.

[7] S_0 [Latin *substantivum*]: nominalization

Part B. Semantic-syntactic characterization of the LF, followed by its PoS table.

S_0 applies to a non-nominal lexical unit L and returns a noun having the same meaning as L.

S_0 : nominal				
L	V	Adj	Adv	Claus
$S_0(L)$	N	N	N	N

If a series of LFs have an identical PoS table, it is extracted and mentioned just once before the presentation of these LFs – see, e.g., the shared PoS table for Syn, Conv_{ijk} and Anti (Chapter 4, Section 4.1, p. 72).

Part C. Illustration of the LF in English, French and Russian.

English

$S_0(\textit{present}_{(V)})$ = *presentation*

$S_0(\textit{leave}_{(V)})$ = *departure*

...

French

$S_0(\textit{présenter})$ = *présentation*

$S_0(\textit{partir})$ = *départ*

...

Russian

$S_0(\textit{predstavljat'})$ = *predstavlenie*

$S_0(\textit{uezžat'})$ = *ot'ezd*

...

As mentioned at the beginning of this chapter (Section 3.1.2, p. 37), our illustrations do not provide exhaustive lists for LF values; in most cases, only one value element is indicated.

Active syntactic valence of L. The use of an LF f that carries actantial subscripts (Conv_{12} , S_1 , Oper_2 , etc.) presupposes a preliminary identification of L 's active syntactic valence (Chapter 2, Section 2.3, p. 31). Thus, to understand (36) below, one must first establish that $\text{FEAR}_{(N)}$ is a lexeme that (i) denotes a type of feeling and (ii) controls two SemAs expressible as DSyntAs: $X'_{[1 \Leftrightarrow I]}$ *fear of* $Y_{[2 \Leftrightarrow II]}$.

- (36) $\text{Able}_1(\text{fear}_{(N)}) = \text{fearful}$
 $\text{Able}_2(\text{fear}_{(N)}) = \text{scary}$

Note. LFs being deep lexical units, subscripts “ $_1$ ” and “ $_2$ ” refer to $\text{FEAR}_{(N)}$'s DSyntAs I and II (on Able_i , see Chapter 4, [22], p. 103); these DSyntAs correspond respectively to SemAs 1 and 2 – cf. Chapter 2, Section 2.3, p. 30.

It would be tedious to supply each $f(L)$ illustration with the actantial structure of L ; readers are expected to establish L 's actantial structure themselves. This is, by the way, an excellent training in Explanatory Combinatorial Lexicology.

3.8 System of simple standard LFs

The next two chapters introduce LFs in an enumerative form and constitute a repository of paradigmatic LFs (Chapter 4) and syntagmatic LFs (Chapter 5). Though LFs are listed linearly in these chapters, it should be stressed that, in reality, the set of LFs constitutes a *system* where all elements are interconnected.¹⁹ The box named “System of simple standard lexical functions” (p. 67) lists all simple standard LFs following a logic that takes into account their systemic organization.

Three strategies are used to account for this organization: identification of central LFs (Section 3.8.1), subclassification of paradigmatic and syntagmatic LFs (Section 3.8.2) and cross-referencing between LFs' descriptions (Section 3.8.3).

3.8.1 Identification of central LFs

Not all LFs are of equal importance and some can be considered as being truly at the heart of the system of simple standard LFs. At least two features determine whether a given LF is central:

- its massive presence in the lexical network of the language – some LFs (e.g., Syn , Magn , Oper_i , ...) apply to many different arguments and return

¹⁹On the systemic organization of LFs and its computational modeling, see Jousse (2010) and Fonseca et al. (2016).

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many elements of their values; others (Figur, S_{mod} , Obstr, ...) apply much more marginally;

- its implication in paraphrasing and, more specifically, in the universal system of syntactic paraphrasing rules (Section 3.2.3.3, p. 43), which demonstrates its importance in the semantics $\Leftrightarrow$ syntax correspondence.

Mastering of the system of LFs has to develop starting from the most central LFs. They are signaled on p. 67 below by their boxed number.

3.8.2 Subclassification of paradigmatic and syntagmatic LFs

The ordering of LFs in the synthetic table below – cf. identification numbers – is logically motivated, starting with LFs whose meaning is closer to that of their argument. Thus, paradigmatic LFs (semantic derivatives) are listed before syntagmatic LFs (collocates). Subclassifications are established within each of the two main classes of LFs; they are as follows:

Paradigmatic LFs (Chaper 4)

- Three pillars of the LF system: Syn, $Conv_{ijk}$, Anti (Section 4.1)
- Three LFs related to Syn and Anti: Gener, Figur, Contr (Section 4.2)
- Structural, i.e. syntactically motivated, derivatives (Section 4.3)
- Predicative derivatives (Section 4.4)
- Nominal derivatives (Section 4.5)
 - Nominal actantial derivatives (Section 4.5.1)
 - Nominal circumstantial derivatives (Section 4.5.2)
- Adjectival actantial derivatives (Section 4.6)
- Adverbial actantial derivatives (Section 4.7)
- Singulative, collective and other inflection-like derivatives (Section 4.8)

Syntagmatic LFs (Chaper 5)

- Nominal collocates (Section 5.1)
- Adjectival / adverbial collocates (Section 5.2)
- Prepositional collocates (Section 5.3)
- Verbal collocates, distributed into seven subclasses corresponding to their increasing semantic load (Section 5.4):
 - Copula verb (Section 5.4.1)
 - Support [= light] verbs (Section 5.4.2)
 - Realization verbs (Section 5.4.3)
 - Phasic verbs (Section 5.4.4)

- Verbs of causing (Section 5.4.5)
- Verbs of mode of functioning (Section 5.4.6)
- Other verbal collocates (Section 5.4.7)



System of simple standard lexical functions [LFs]

Paradigmatic LFs (Chapter 4)				Syntagmatic LFs (Chapter 5)			
[1]	Syn	19	S _{res}	31	Germ	[49]	Real _i
[2]	Conv _{ijk}	20	S _{mod}	32	Culm	[50]	Fact _i
[3]	Anti; Non	[21]	A _i	33	Epit	[51]	Labreal _{ij}
[4]	Gener	[22]	Able _i	[34]	Redun	52	Prepar
5	Figur	23	Qual _i	[35]	Magn	[53]	Incep
6	Contr	24	Adv _i	[36]	Ver	[54]	Fin
[7]	S ₀	[25]	Sing	[37]	Bon	[55]	Cont
[8]	V ₀	[26]	Mult	38	Plus	56	Prox
[9]	A ₀	27	Imper	39	Minus	[57]	Caus
[10]	Adv ₀	28	Perf	40	Loc _{in}	[58]	Liqu
[11]	Claus ₀	29	Imperf	41	Loc _{ad}	[59]	Perm
12	Pred	30	Result _i	42	Loc _{ab}	60	Obstr
[13]	S _i			43	Instr	61	Stop
14	Equip			44	Propt _i	62	Excess
15	Cap			45	Copul	63	Son
16	S _{loc}			[46]	Oper _i	64	Manif
17	S _{instr}			[47]	Func _i	65	Involv
18	S _{med}			[48]	Labor _{ij}	66	Sympt _{ijk}

Tip. In the electronic version of the present volume, LF identification numbers are clickable hyperlinks leading to the description of corresponding LFs.

LFs are listed above in a revised order in comparison to the one used in previous Meaning-Text publications. For instance, Conv_{ijk} is introduced after Anti in

Mel'čuk (1996: 48–50), a reference text on LFs. More importantly, LFs' classification has been revised in some instances – e.g., Germ and Culm, now listed among syntagmatic LFs, were formerly classified as paradigmatic LFs.

3.8.3 Cross-referencing between LFs' descriptions

A profusion of cross-references in Chapters 4 and 5 make explicit the interconnections between individual (or families of) LFs. Their aim – together with the classification we propose – is to help the reader assimilate the set of simple standard LFs as a system, rather than a simple list of theoretical entities.

3.9 Descriptive reliability of LFs

Before we proceed with the detailed presentation of paradigmatic and syntagmatic LFs, let us answer a legitimate question: how trustworthy are LF-based descriptions with respect to formalization? In particular, can it be expected that any given lexical relation $L \vdash \text{LexRel} \rightarrow L'$ will be modeled with the same LF application $f(L) = L'$ by different individuals?

For seasoned lexicographers, with extensive experience of the use of LFs in the construction of lexical resources, this is a non-issue. We do know that the system of LFs is a reliable formal language for collaborative work on modeling paradigmatic and syntagmatic relations. However, it can be expected from our readers – faced with the theoretical and formal complexity of LFs – to question the actual applicability of the approach.

As with any formal descriptive tool, disagreements between researchers do exist in the context of encoding LF relations. There can be three reasons for this.

1. Lexicographers being human, they make mistakes, and those mistakes simply have to be fixed.
2. More interestingly, some lexical relations can give rise to alternative encodings that are formally equivalent – cf. example (22) of Section 3.3.3.2 (p. 52), fused values for Magn that entail a corresponding Syn₅ relations.
3. Finally, tricky cases do exist. A disagreement on the semantic content of L and/or L' involved in $L \vdash \text{LexRel} \rightarrow L'$ can entail different choices of a corresponding LF. The triplet of adjectival / adverbial LFs Magn, Ver and Bon is a notorious source of problems in this respect – see discussion in Chapter 5, Section 5.2, p. 126.

In spite of such unavoidable situations, the LF system proves to be reliable in many contexts of application, and in lexicography in particular. An argument for the robustness and non-aleatory nature of LF modeling can be found in the fact that it is possible to implement the automatic diagnosis of LF relations. In particular, LF-based descriptions can be used to train *machine learning models* in Natural Language Processing with the aim to perform the automatic identification of LF relations – for a synthesis, see Wanner (2022). Furthermore, Fisas & Wanner (2026) explores the feasibility of “teaching” a *Generative Artificial Intelligence* model a representative set of syntagmatic LFs in order to automatically build LF-encoded collocation resources.

4 Paradigmatic lexical functions

Paradigmatic LFs fall into eight major classes; they are presented here in the same order in which they appear in the cooccurrence zone of a lexical entry in an Explanatory Combinatorial Dictionary (Mel'čuk & Zholkovsky 1984, 2016, Mel'čuk et al. 1984, 1988, 1992, 1999):

- Three pillars of the LF system: Syn, Conv_{ijk}, Anti (Section 4.1)
- Three LFs related to Syn and Anti: Gener, Figur, Contr (Section 4.2)
- Structural, i.e. syntactically motivated, derivatives (Section 4.3)
- Predicative derivatives (Section 4.4)
- Nominal derivatives (Section 4.5)
 - Nominal actantial derivatives (Section 4.5.1)
 - Nominal circumstantial derivatives (Section 4.5.2)
- Adjectival actantial derivatives (Section 4.6)
- Adverbial actantial derivatives (Section 4.7)
- Singulative, collective and other inflection-like derivatives (Section 4.8)

The presentation order of paradigmatic LFs corresponds to the importance of their semantic contribution: from Syn, whose semantic import is null, to LFs that carry relatively rich semantic load.

4.1 Three pillars of the LF system: Syn, Conv_{ijk}, Anti

The lexical function Syn (synonym) is the most important in the system of LFs and in the structuring of the lexical stock of languages, given its intimate relation with paraphrasing – cf. the remark on rule **R_{Syn}**, Chapter 3, Section 3.3.1 (p. 47).

Syn expresses the identity of meanings, and the identity of meanings underlies the LF Conv_{ijk} (conversive), which expresses the identity of meanings with

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the permutation of the DSyntAs; Anti (antonym) expresses the contrariness of meanings.

We call Syn, Conv_{ijk} and Anti pillars of the LF system because these LFs are extensively used to express the semantic connections between LFs themselves.

These three LFs do not have their own inherent part of speech [PoS]: they are neither nominal, nor verbal, etc., but borrow the PoS of their argument. This is reflected in their PoS table, where L's PoS and that of f(L) are always identical. All three LFs possess the same PoS table, namely:

Syn, Conv _{ijk} , Anti: borrow L's PoS					
L	V	N	Adj	Adv	Claus
Syn / Conv _{ijk} / Anti(L)	V	N	Adj	Adv	Claus

[1] Syn [Latin *synonymum*]: synonym

Syn(L) has the same meaning and PoS as L and is, therefore, a possible substitute for L in the syntactic structure of the utterance with preservation of its meaning. L', a given value element of Syn(L), is a special case of L's semantic derivatives (Chapter 3, Section 3.3.1, p. 46): one where the semantic difference between 'L' and 'L'' (i.e., 'σ^{Syn}') is zero – see Chapter 3, "Paradigmatic LFs," p. 47.

As illustrated below, synonyms of L can belong to different registers and have different stylistic properties; what counts for synonymy is just identity of meanings.

English

Syn(<i>urinate</i>)	=	informal <i>pee</i> _(V) , colloquial <i>piss</i> _(V)
Syn(<i>car</i>)	=	<i>auto</i> _(N) , <i>automobile</i> _(N)
Syn _⊃ (<i>murder</i> _(N))	=	<i>assassination</i>
Syn _⊂ (<i>assassination</i>)	=	<i>murder</i> _(N)
Syn _⊂ (<i>murder</i> _(N))	=	<i>killing</i>
Syn _∩ (<i>murder</i> _(N))	=	<i>massacre</i> _(N) ; <i>slaughter</i> _(N)
Syn _⊂ (colloquial 'like a bat out of hell')	=	<i>fast</i> _(Adv)
Syn _⊃ (<i>fast</i> _(Adv))	=	colloquial 'like a bat out of hell'
Syn(<i>Ouch!</i>)	=	<i>Ow!</i>

The three set-theoretical subscripts "⊃", "⊂" and "∩" are used in the following sense:

- $\text{Syn}_{\supset}$ means ‘richer [= narrower] synonym’ – the meaning of $\text{Syn}_{\supset}(\text{L})$ includes that of L;
- $\text{Syn}_{\subset}$ means ‘less rich [= poorer, broader] synonym’ – the meaning of $\text{Syn}_{\subset}(\text{L})$ is included into that of L;
- $\text{Syn}_{\cap}$ means ‘intersecting synonym’ – the meanings of $\text{Syn}_{\cap}(\text{L})$ and L have an intersection, while each possesses a distinctive component.

These subscripts can accompany the names of other LFs as well – always with the same meaning.

French

$\text{Syn}(\text{uriner})$	=	colloquial <i>pisser</i>
$\text{Syn}(\text{voiture})$	=	<i>auto, automobile, colloquial bagnole</i>
$\text{Syn}(\text{‘doigt de pied’})$	=	<i>orteil</i>
$\text{Syn}_{\supset}(\text{meurtre})$	=	<i>assassinat</i>
$\text{Syn}_{\subset}(\text{assassinat})$	=	<i>meurtre</i>
$\text{Syn}_{\cap}(\text{meurtre})$	=	‘ <i>mise à mort</i> ’; <i>carnage, massacre</i>
$\text{Syn}_{\subset}(\text{‘à fond de train’})$	=	<i>vite</i>
$\text{Syn}_{\supset}(\text{vite})$	=	‘ <i>à fond de train</i> ’
$\text{Syn}(\text{Aïe !})$	=	<i>Ouille !</i>

Russian

$\text{Syn}(\text{močit'sja})$	=	colloquial <i>pisat'</i>
$\text{Syn}(\text{mašina})$	=	<i>avtomobil'</i> , dated <i>avto</i> , slang <i>tačka</i>
$\text{Syn}_{\supset}(\text{prestuplenie})$	=	<i>ubijstvo</i>
$\text{Syn}_{\subset}(\text{ubijstvo})$	=	<i>prestuplenie</i>
$\text{Syn}_{\cap}(\text{ubijstvo})$	=	formal <i>umerščvlenie</i> ; <i>bojnja, reznja</i>
$\text{Syn}_{\subset}(\text{‘kak strela’})$	=	<i>bystro</i>
$\text{Syn}_{\supset}(\text{bystro})$	=	‘ <i>kak strela</i> ’
$\text{Syn}(\text{Aj!})$	=	<i>Oj!</i>

Remark on reversibility. As can easily be seen in the above illustrations, the LF *Syn* is reversible:

$$\text{if } \text{Syn}(\text{L}) = \text{L}', \text{ then } \text{Syn}(\text{L}') = \text{L}.$$

Of course, in case poorer/richer synonyms are involved, the reversibility is as follows:

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if $\text{Syn}_{\subseteq}(L) = L'$, then $\text{Syn}_{\supset}(L') = L$.

The intersecting synonyms are reversible:

if $\text{Syn}_{\cap}(L) = L'$, then $\text{Syn}_{\cap}(L') = L$.

The same remark on reversibility applies to Syn's fellow LFs Conv_{ijk} and Anti.

[2] **Conv_{ijk}** [Latin **conversivum*]: **conversive**

$\text{Conv}_{ijk}(L)$ has the same meaning and PoS as L and is, therefore, a possible substitute for L in the deep-syntactic structure of the utterance, with preservation of meaning, but with the necessary permutation of L 's DSyntAs. This relation of paraphrase between L and its conversive is accounted for by the $\mathbf{R}_{\text{Conv}_{ijk}}$ paraphrasing rule:

$$\mathbf{R}_{\text{Conv}_{ijk}}: L \cong \text{Conv}_{ijk}(L)$$

The pair of sentences (1a–b) illustrates the semantic quasi-equivalence of two conversive lexical units:

- (1) a. Lightning *precedes* thunder.
 $\cong$
 b. Thunder *follows*_{Conv₂₁(precede)} lightning.

The equivalence between sentences (1a) and (1b) is approximate because they have different communicative structures: in (1a), the *Theme* is *lightning*, and in (1b), *thunder*. As (1a–b) shows, conversive substitutions allow the Speaker to express relevant communicative information.

English

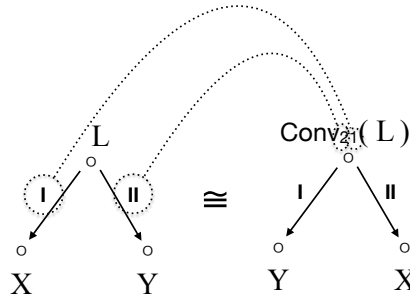
$\text{Conv}_{21}(\text{include } [X \text{ is a set}])$	=	<i>belong</i>
$\text{Conv}_{3214}(\text{lend})$	=	<i>borrow</i>
$[N_X \text{ lends } N_Y \text{ to } N_Z \text{ for period } N_W]$		
$\text{Conv}_{3214\cap}(\text{buy}_{(V)})$	=	<i>sell</i> _(V)
$[N_X \text{ buys } N_Y \text{ from } N_Z \text{ for } N_W]$		
$\text{Conv}_{3214\cap}(\text{sell}_{(V)})$	=	<i>buy</i> _(V)
$\text{Conv}_{1432\supset}(\text{buy}_{(V)})$	=	<i>pay</i> _(V)
$\text{Conv}_{21}([be] \text{ parent}_{(N)})$	=	<i>[be] child</i>
$\text{Conv}_{213}([be] \text{ teacher})$	=	<i>[be] pupil</i> [Prof. Wanner's pupil in mathematics]



The LF Conv_{ijk}, as in fact the majority of LFs, is based, in quite an essential way, on the notion of DSyntA, presented in Chapter 2, Section 2.3.

The symbol Conv is supplied with a string of actantial subscripts; this string encodes the relation between the active syntactic valence of L and that of Conv_{ijk}(L). Each number in the string refers to a DSyntA of L, and this number's position in the string refers to the corresponding DSyntA of Conv_{ijk}(L). Thus, the number 2 in the *first* position of the numerical subscript of Conv₂₁ means that DSyntA II of L is expressed by DSyntA I of Conv₂₁(L); the number 1 in the *second* position of the subscript indicates that L's DSyntA I corresponds to DSyntA II of Conv₂₁(L).

Figure 4.1 visualizes the correspondence between the DSyntAs of L and those of Conv₂₁(L) under the semantic quasi-equivalence of the two deep-syntactic structures: that constituted by L with its DSyntAs and that constituted by L's Conv₂₁-converse with its DSyntAs.



For instance: $X \text{ follows } Y. \cong Y \text{ precedes}_{\text{Conv}_{21}(\text{follow})} X.$

Figure 4.1: Correspondence between L's DSyntAs and those of Conv₂₁(L)

French

Conv₂₁(inclure [X is a set]) = appartenir, «faire partie»

Conv₃₂₁₄(prêter) = emprunter

[N_X prête N_Y à N_Z pour la période N_W]

Conv₃₂₁₄∩(acheter) = vendre

[N_X achète N_Y à N_Z pour N_W]

Conv₃₂₁₄∩(vendre) = acheter

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Conv ₁₄₃₂ (<i>acheter</i>)	=	<i>payer</i>
Conv ₂₁ ([être] <i>parent</i>)	=	[être] <i>enfant</i>
Conv ₂₁₃ ([être] <i>maître</i>)	=	[être] <i>élève</i> [du prof. Ménard en math]
Conv ₂₁ (<i>meilleur</i>)	=	<i>pire</i>

Russian

Conv ₂₁ (<i>soderžat'</i> [X is a set])	=	<i>prinadležat'</i>
Conv ₃₂₁₄ (<i>odolžit'</i> [N _{DAT}])	=	<i>odolžit'</i> [u N _{GEN}]
[N _{X-NOM} <i>odolžil</i> N _{Y-ACC} N _{Z-DAT} <i>na vremja</i> N _{W-ACC}]		
Conv ₃₂₁₄ (<i>kupit'</i>)	=	<i>prodat'</i>
[N _{X-NOM} <i>kupil</i> N _{Y-ACC} <i>u</i> N _{Z-GEN} <i>za</i> N _{W-ACC}]		
Conv ₃₂₁₄ (<i>prodat'</i>)	=	<i>kupit'</i>
Conv ₁₄₃₂ (<i>kupit'</i>)	=	<i>zaplatit'</i>
Conv ₂₁ ([byt'] <i>roditelem</i>)	=	[byt'] <i>reběnkom</i>
Conv ₂₁₃ ([byt'] <i>učitelem</i>)	=	[byt'] <i>učenikom</i> [prof. Menara po matematike]
Conv ₂₁ (<i>tjaželee</i>)	=	<i>legče</i>

[3] Anti [Latin **antonymum*]: antonym

Anti(L) has a meaning opposite to that of L and is, therefore, a possible substitute for L in the syntactic structure of the utterance with preservation of the meaning, but under the condition that a negation is added to Anti(L) as illustrated in (2a–b).

- (2) a. Chris was *absent* from the meeting.
 ≡
 b. Chris was *not present*_{Anti(absent)} at the meeting.

However, such equivalence is valid only for contradictory antonyms (see type 1 of antonyms after the illustrations below); with other types of antonyms, the semantic equivalence is only approximate or does not obtain at all. The paraphrastic relation holding between a lexical unit L and Anti(L) + Negation is too complex to be summarized here with a single paraphrasing rule, and it is considered below separately for each of the four types of antonyms.

English

Anti(<i>respect</i> _(N))	=	<i>disrespect</i> _(N)
Anti(<i>small</i> _(Adj))	=	<i>big</i> _(Adj)
Anti(<i>hot</i>)	=	<i>cold</i> _(Adj)
Anti(<i>with</i>)	=	<i>without</i>
Anti(<i>dress</i> _(V))	=	<i>undress</i> _(V)

Anti _∩ (<i>build</i> _(V))	=	<i>destroy</i>
Anti(<i>Yummy!</i>)	=	<i>Yuck!</i>

French

Anti(<i>respect</i>)	=	<i>irrespect</i>
Anti _⊃ (<i>mentionner</i>)	=	‘ <i>passer sous silence</i> ’
Anti(<i>petit</i> _(Adj))	=	<i>grand</i> _(Adj)
Anti(<i>chaud</i> _(Adj))	=	<i>froid</i> _(Adj)
Anti(<i>avec</i>)	=	<i>sans</i>
Anti(<i>habiller</i>)	=	<i>déshabiller</i>
Anti _∩ (<i>construire</i>)	=	<i>détruire</i>
Anti(<i>Miam !</i>)	=	<i>Beurk !</i>

Russian

Anti(<i>uvaženie</i>)	=	<i>neuvaženie</i>
Anti _⊃ (<i>upomjanut’</i>)	=	‘ <i>obojti molčaniem</i> ’
Anti(<i>malen’kij</i>)	=	<i>bol’šoj</i>
Anti(<i>gorjačij</i>)	=	<i>xolodnyj</i>
Anti(<i>s</i>)	=	<i>bez</i>
Anti(<i>odet’</i>)	=	<i>razdet’</i>
Anti _∩ (<i>stroit’</i>)	=	<i>razrušat’</i>
Anti(<i>Ux_ty!</i>) ¹	=	<i>Fu!</i>

Antonyms come in four types: contradictory, contrary, scalar and positioning antonyms.

1. *Contradictory (or complementary) antonyms.* A contradictory antonym is opposed to its “partner” by a negation of the highest level of its meaning, i.e., that applies to the central component of its lexicographic definition (Mel’čuk & Polguère 2018: Sec. 2.4.1): ‘Anti(L)’ = ‘non L’. Two mutually contradictory antonyms represent the two values of a binary category. For instance, PRESENT_(Adj) [*in a room*] is a contradictory antonym of ABSENT_(Adj) [*from a room*]:

$$‘[X] \text{ absent from } Y’ = ‘[X] \text{ not present in } Y’$$

Contradictory antonyms do not (or hardly) allow for gradation: **more/less present in a room*. In contrast, the remaining three types of antonyms are gradable.

¹The grey underscoring of the space between two strings of letters, as in *UX TY*, shows that, linguistically speaking, this expression is a single wordform, which is traditionally spelled in two words (cf. *AT ALL*, *AT LAST* or *BY FAR*).

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2. *Contrary antonyms*. Two mutually contrary antonyms are opposed by a negation that is deeply embedded in the lexicographic definition of one of them, i.e., that applies to a peripheral component. For instance, $\text{CLOSE}_{(V)}$ [*a door*] is a contrary antonym of $\text{OPEN}_{(V)}$ [*a door*]:

‘X opens Y’ = ‘X causes that Y becomes $\text{open}_{(\text{Adj})}$ ’
‘X closes Y’ = ‘X causes that Y becomes *not* $\text{open}_{(\text{Adj})}$ ’

These semantic decompositions show that ‘ $\text{close}_{(V)}$ ’ $\neq$ ‘ $\text{not open}_{(V)}$ ’ and, therefore, that $\text{CLOSE}_{(V)}$ and $\text{OPEN}_{(V)}$ are not contradictory antonyms.

3. *Scalar antonyms*. They are opposed by the components ‘more’ $\sim$ ‘less’ in their lexicographic definitions: ‘more’ / ‘less’ indicates the position on a gradable scale, or on an axis. For instance, $\text{HEAVY}_{(\text{Adj})}$ and $\text{LIGHT}_{(\text{Adj})}$ are scalar antonyms:

‘X is heavy’ = ‘X’s weight is *higher* [= ‘*more* high’] than the normal weight of Xs’
‘X is light’ = ‘X’s weight is *lower* [= ‘*less* high’] than the normal weight of Xs’

The opposition between ‘more’ and ‘less’ is reducible to the opposition by negation (Mel’čuk 2015b: 190). Note that if two adjectives are scalar antonyms, their comparative degrees are conversives as well; thus, we have ‘X is heavier than Y’ = ‘Y is lighter than X’.

4. *Positioning antonyms*. These antonyms denote two opposite spatial or temporal positions (in a literal or metaphorical sense). For instance, $\text{TOP}_{(N)}$ [*the top of the shelves*] and $\text{BOTTOM}_{(N)}$ [*the bottom of the shelves*] are positioning antonyms:

‘top of X’ = ‘part of X that is the *furthest* [= ‘*least* close’] to the surface on which X is located’
‘bottom of X’ = ‘part of X that is the *closest* [= ‘*most* close’] to the surface on which X is located’

As is seen, all types of antonyms are based on negation. It is the position of negation in the lexicographic definition of lexical units that are Anti(L) that determines the type of antonymy. In the enumeration of the four antonym types above, we started with the negation occupying the uppermost position in the definition and concluded with the negation that is the most deeply embedded.

A lexical unit can have antonyms of different types. Such is, for instance, the lexeme DESTROY , appearing in (3):

- (3) They destroyed the bridge.
‘They caused that the bridge ceased to exist’ = ‘They caused the bridge to not[exist]’

DESTROY has two antonyms: a contradictory one – SPARE_(V), cf. (4), and a contrary one – BUILD_(V), cf. (5).

- (4) They spared the bridge.
‘They not[caused that the bridge ceased to exist]’
- (5) They built the bridge.
‘They caused that the bridge not[not[exist]]’ = ‘They caused that the bridge exists’
(‘not[not[exist]]’ = ‘exist’)

For all four types of antonyms only one LF is used: Anti. Theoretically speaking, it is easy to distinguish the four Anti LFs by means of super- or subscripts – e.g. Anti^{contrad}, Anti^{contrary}, etc. This is, however, superfluous, since each antonym should have its own lexicographic description – in particular, its definition – in a lexicographic model, which allows the user of the model to see the meaning relation between L and Anti(L).

Note that the cases where a lexical unit has antonyms of different types are infrequent enough.

Sometimes the negation underlying the antonymy has to be encoded as such in LF formulas; then the LF Non ‘not’ is used. Thus, Non – rather than Anti – combines with semantically empty LFs, such as those that represent support verbs (Oper_i, Func_i and Labor_{ij}, in Chapter 5, Section 5.4.2). And, from a logical viewpoint, a semantically empty verb cannot have an antonym: an antonym of L is supposed to have an opposite meaning with respect to that of L, while a light verb has no meaning. Therefore, we have:

- (6) Oper₁(consent_(N)) = give, grant [ART ~]
NonOper₁(consent_(N)) = refuse [ART ~]



Historical remark

The previous presentations of LFs, e.g. Mel’čuk et al. (1999: 75–81), did not introduce the LF Non explicitly. In these publications the LF Non (as Non or non) was simply smuggled into complex LF formulas.

4.2 Three LFs related to Syn and Anti: Gener, Figur, Contr

[4] Gener [Latin *genus*]: generic term

Gener(L) – a generic term for L – stands for the name of one of the semantic classes to which L's meaning may belong. It applies to L of any PoS and, like Syn, it borrows L's PoS:

Gener: borrows L's PoS					
L	V	N	Adj	Adv	Claus
Gener(L)	V	N	Adj	Adv	Claus[?]

English

Gener(*love*_(N)) = *feeling* [of ~]; *emotion* [of ~]

Gener(*gas*_(N)) = [*gaseous*] *substance*

Gener(*see*_(V)) = *perceive* [*visually*]

French

Gener(*amour*) = *sentiment* [*d'~*]; *émotion*

Gener(*gaz*) = *substance* [*gazeuse*]

Gener(*voir*) = *percevoir* [*visuellement*]

Russian

Gener(*ljub|ov'*) = *čuvstvo* [*~vi*]; *émocija*

Gener(*gaz*) = [*gazoobraznoe*] *veščestvo*

Gener(*videt'*) = *vosprinimat'* [*zritel'no*]

Identifying Gener(L). For a lexical unit to belong to the value of Gener(L) it is necessary and sufficient to appear in the following constructions:

- for nouns: L, L', L'', ... and other Gener(L)

(7) *melons, raspberries, mangoes, ... and other fruits*_{Gener(*melon*)}

- for verbs, adjectives and adverbs: L, L', L'', ... and Gener(L) in other ways

(8) a. *see, smell, ... and perceive*_{Gener(*see*)} in other ways

b. *fried, boiled, ... and cooked*_{Gener(*fried*)} in other ways

These examples show that Gener is a powerful classifier: it identifies numerically important semantic classes of lexical units. For instance:

- the noun DOG is Gener of BASSET, BULLDOG, COLLIE, DACHSHUND, DALMATIAN, 「GERMAN SHEPHERD」, HUSKY, LABRADOR, POODLE, SETTER, ...
- the noun FEELING is Gener of ADORATION, DESPAIR_(N), DREAD_(N), EXCITEMENT, FEAR_(N), GRATITUDE, LOVE_(N), REMORSE, ...

The compatibility of a Gener with the above constructions allows for the distinction between Gener and Syn_C. Thus, DREAM_(N) is a Syn_C of NIGHTMARE, rather than a Gener, because English does not have an important enough class of lexical units denoting different types of dreams.

Gener-based paraphrasing. The lexical stock and the lexical cooccurrence of the language allowing, a given Gener(L) can be used in the two following equivalences grouped under paraphrasing rule **R_{Gener}**.²

R_{Gener}: a) $L_{(N)} \equiv \text{Gener}(L_{(N)}) -\mathbf{R} \rightarrow L_{(N)}$
*anger*_(N) $\equiv$ *feeling*_{Gener(anger_(N))} *of anger*
submission $\equiv$ *attitude*_{Gener(submission)} *of submission*
*smell*_(N) $\equiv$ *sense*_{Gener(smell_(N))} *of smell*

b) $L \equiv \text{Gener}(L) -\mathbf{R} \rightarrow A_0/\text{Adv}_0/\text{Adv}_1(L)$
For $L_{(N)}$: *anger*_(N) $\equiv$ *angry*_{A₀(anger_(N))} *feeling*_{Gener(anger_(N))}
For $L_{(V)}$: *see* $\equiv$ *perceive*_{Gener(see)} *visually*_{Adv₁(see)}
For $L_{(\text{Adj})}$: *visual* $\equiv$ *visually*_{Adv₀(visual)} *perceptible*_{Gener(visual)}

See below for the description of the LFs A₀ (Section 4.3, [9], p. 87), Adv₀ (Section 4.3, [10], p. 87) and Adv₁ (Section 4.7, [24], p. 104).

In subrules a) and b) above, the deep-syntactic relation **R** on the right-hand side of the equivalence remains unspecified, as it varies according to the semantic valence and government pattern of each particular Gener(L); for instance:

- (9) a. COLOR_(N, 'color of X that is Y'): *the color*_{Gener(purple_(N))} **-II** $\rightarrow$ *purple*
b. FEELING_(‘feeling of X towards Y that is Z’): *the feeling*_{Gener(love_(N))} **-III** $\rightarrow$ [of] *love*
c. NUMBER_(N): *the number*_{Gener(nine)} **-ATTR** $\rightarrow$ *nine*

²These equivalences are described by two distinct paraphrasing rules (Rules 4 and 5) in Mel’čuk (2013a: p. 152).

Multiplicity of the values of Gener(L). A lexical unit L can have multiple elements of the value for Gener(L). This happens in two cases.

The first case is trivial. As for any other LF, there can be multiple elements of the value of $\text{Gener}(L)$ that are (quasi-)synonyms; for instance:

- (10) Gener(*fear*_(N)) = *feeling [of ~]; emotion [of ~]*
[Syn_N(*feeling*) = *emotion*]
Gener(*bin*_(N)) = *container; receptacle*
[Syn_N(*container*) = *receptacle*]

The second case is more remarkable as it is conditioned by the special character of L's meaning. Two subcases are distinguished:

- L denotes more than one kind of thing and, therefore, L belongs to more than one semantic class. This phenomenon corresponds to L's *semantic ambivalence* (Milićević & Polguère 2010, Smirnova & Tolochin 2018), that is, to the presence of a disjunction in L's definition.³
- L's definition contains a classifying component – called *semantic-class marking component* (Mel'čuk & Polguère 2018: Sec. 2.4.1.2) – that underlies a value of Gener(L). If, at the same time, the central component of L's definition underlies also a value of Gener(L), which is a prototypical case, L obtains more than one type of Gener(L).

Let us take an example: the definition of the lexeme $\text{snow}_{(N)}$, which illustrates both situations.

- | | |
|------|--|
| (11) | <i>snow on Y</i> : frozen water that is falling or has fallen from the sky on
geographical place Y
• in the form of white flakes
or
this falling
• which is a weather phenomenon |
|------|--|

In the above definition, the disjunction in the central component ('or') and the semantic-class marking component ('which is a weather phenomenon') combine to produce two types of $\text{Gener}(\text{snow}_{(N)})$:

- (12) Gener(*snow*_(N)) = (i) *substance*;
(ii) *precipitation*, 'weather condition'

³See also the related notions of *facet/subsense* (Cruse 1995) and *logical polysemy* (Pustejovsky 1995).

Formula (12) presupposes that we are dealing with one single sense of $\text{SNOW}_{(N)}$, that is, with one lexeme, and not with a case of polysemy, that is, with two lexemes – one denoting a substance and one denoting the event of its falling. This can be demonstrated using the *Unifying cooccurrence* criterion presented in Mel'čuk (2013a: Ch. 11, Sec. 3.2.2). Indeed, a given occurrence of $\text{snow}_{(N)}$ in an utterance can be used to refer to both substance and event simultaneously:

- (13) The snow that *started* [snow as precipitation] this morning
is beautifully *fluffy* [snow as substance].

For this reason, both types of denotation of $\text{SNOW}_{(N)}$ (substance and meteorological event) have to be considered as belonging to one lexical unit, with a disjunctive central component in its definition.

The classifying power of Gener makes this LF particularly important in the structuring of natural language lexical networks and in the description of lexical regularities – see Chapter 3, Section 3.6.2.



Further reading

For more on generic terms and lexicographic definitions, see Polguère (2022: Sec. 3) and Goddard (2017).

[5] **Figur** [Latin *figuraliter* 'figuratively']: figurative term

$\text{Figur}(L)$ is a metaphorical expression having the same meaning as the noun L plus a somehow artistic representation of L ; it is either a lexical unit or a collocation whose base is L and the collocater, a noun that takes L as its DSyntA 1. The relation of paraphrase between L and $\text{Figur}(L)$ is accounted for by the $\mathbf{R}_{\text{Figur}}$ paraphrasing rule:

$$\mathbf{R}_{\text{Figur}}: L \cong \text{Figur}(L)$$

The equivalence established by this rule is approximate because the use of $\text{Figur}(L)$ necessarily adds to the meaning of L a rhetorical content, as illustrated by (14a–b):

4 Paradigmatic lexical functions

- (14) a. He never experienced *passion*.
 $\cong$
 b. He never experienced the *flame of passion*_{Figur(*passion*)}.

Figur: nominal	
L	N
Figur(L)	N

Being a figurative *substitute* explains the status of Figur as a paradigmatic (rather than syntagmatic) LF. It is used for stylistic effects, and for this reason it is close to the syntagmatic LF Epit – Chapter 5, [33], p. 118. See also the remark on the syntactic similarity of Figur to two other syntagmatic LFs: Germ and Culm – Chapter 5, [31–32], p. 118.

English

Figur([bubonic] <i>plague</i> _(N))	= 'Black Death'
Figur(<i>power</i> _(N))	= <i>reins of ~</i>
Figur(<i>passion</i>)	= <i>flame of ~</i>
Figur(<i>fog</i> _(N))	= <i>curtain of ~</i>

French

Figur(<i>pétrole</i>)	= 'or noir'
Figur(<i>pouvoir</i>)	= <i>rênes du ~</i>
Figur(<i>passion</i>)	= <i>flamme de la ~</i>
Figur(<i>brouillard</i>)	= <i>rideau de ~</i>

Russian

Figur(<i>vlast'</i>)	= 'brazdy pravlenija'
Figur(<i>strast'</i>)	= <i>plamja ~i</i>
Figur(<i>tuman</i>)	= <i>pelena ~a</i>

[6] Contr [Latin *contrarium*]: contrastive term

Contr(L) is a lexical entity that expresses a meaning that is in contrastive opposition with that of L, while not being an Anti of the latter.

The contrastive opposition of L and the value of Contr(L) is not just conceptual, but truly lexical – i.e., language-dependent. In particular, it has to be manifested in phrasemes of the language, such as the italicized idiom in (15) below for ICE_(N) and its Contr FIRE_(N).

4.3 Structural, i.e. syntactically motivated, derivatives

(15) They are ‘like ice and fire’, but friends nonetheless.

Contr: borrows L’s PoS					
L	V	N	Adj	Adv	Claus
Contr(L)	V	N	Adj	Adv	Claus

English

Contr(<i>sea</i>)	=	<i>land</i> _(N)
Contr(<i>ice</i> _(N))	=	<i>fire</i> _(N)
Contr(<i>black</i> _(Adj))	=	<i>white</i> _(Adj)
Contr(<i>here</i>)	=	<i>there</i>

French

Contr(<i>manger</i> _(V))	=	<i>boire</i> _(V)
Contr(<i>terre</i>)	=	<i>mer</i>
Contr(<i>eau</i>)	=	<i>feu</i>
Contr(<i>nous</i>)	=	<i>eux</i>
Contr(<i>noir</i> _(Adj))	=	<i>blanc</i> _(Adj)
Contr(<i>ici</i>)	=	<i>là</i>

Russian

Contr(<i>more</i>)	=	<i>suša</i>
Contr(<i>lěd</i>)	=	<i>plamen’</i>
Contr(<i>my</i>)	=	<i>oni</i>
Contr(<i>čěrnij</i>)	=	<i>belyj</i>
Contr(<i>zdes’</i>)	=	<i>tam</i>

4.3 Structural, i.e. syntactically motivated, derivatives

The following five LFs – S_0 , V_0 , A_0 , Adv_0 and $Claus_0$ – represent the five possible derivations that modify L’s deep part of speech, but do not entail semantic increments to L: nominalization, verbalization, adjectivization, adverbialization and clausativization. The “₀” subscript in the name of these LFs indicates that they are structural (purely syntactic) derivatives.

[7] **S_0** [Latin *substantivum*]: **nominalization**

S_0 applies to a non-nominal lexical unit L and returns a noun having the same meaning as L.

4 Paradigmatic lexical functions

S ₀ : nominal				
L	V	Adj	Adv	Claus
S ₀ (L)	N	N	N	N

English

S ₀ (<i>present</i> _(V))	=	<i>presentation</i>
S ₀ (<i>leave</i> _(V))	=	<i>departure</i>
S ₀ (<i>fall</i> _(V))	=	<i>fall</i> _(N)
S ₀ (<i>close</i> _(Adj))	=	<i>proximity</i>
S ₀ (<i>Bang!</i>)	=	<i>gunshot</i>

French

S ₀ (<i>présenter</i>)	=	<i>présentation</i>
S ₀ (<i>partir</i>)	=	<i>départ</i>
S ₀ (<i>tomber</i>)	=	<i>chute</i>
S ₀ (<i>proche</i>)	=	<i>proximité</i>
S ₀ (<i>près</i>)	=	<i>proximité</i>
S ₀ (<i>Pan !</i>)	=	「 <i>coup de feu</i> 」

Russian

S ₀ (<i>predstavljat'</i>)	=	<i>predstavlenie</i>
S ₀ (<i>uezžat'</i>)	=	<i>ot'ezd</i>
S ₀ (<i>padat'</i>)	=	<i>padenie</i>
S ₀ (<i>blizkij</i>)	=	<i>blizost'</i>
S ₀ (<i>rjadom</i>)	=	<i>blizost'</i>
S ₀ (<i>child language Pif-paf!</i>)	=	<i>vystrel</i>

[8] V₀ [Latin *verbum*]: verbalization

V₀ applies to a non-verbal lexical unit L and returns a verb having the same meaning as L.

V ₀ : verbal				
L	N	Adj	Adv	Claus
V ₀ (L)	V	V	V	V

Obviously, only lexical units being genuine predicates – i.e., denoting facts – can have a V₀.

4.3 Structural, i.e. syntactically motivated, derivatives

English

$V_0(\text{presentation})$	=	<i>present</i> _(V)
$V_0(\text{oath})$	=	<i>swear</i> _(V)
$V_0(\text{compliment}_{(N)})$	=	<i>compliment</i> _(V)
$V_0(\text{age}_{(N)})$	=	<i>be</i> [NUM + N] _Y <i>old</i> N denotes a time unit – e.g., MONTH, YEAR, CENTURY [<i>She is 33 years old.</i>]
$V_0(\text{stinking}_{(Adj)})$	=	<i>stink</i> _(V)
$V_0(\text{energetic})$	=	colloquial <i>‘be full of beans’</i>
$V_0(\text{impatient})$	=	colloquial <i>‘champ at the bit’</i>
$V_0(\text{Bang!})$	=	<i>fire</i> _(V)
$V_0(\text{Wow!})$	=	<i>be impressed</i>

French

$V_0(\text{présentation})$	=	<i>présenter</i>
$V_0(\text{serment})$	=	<i>jurer</i>
$V_0(\text{erreur})$	=	<i>se tromper</i>
$V_0(\text{âge})$	=	<i>avoir</i> [NUM + N] _Y N denotes a time unit – e.g., MOIS, AN, SIÈCLE [<i>Elle a 33 ans.</i>]
$V_0(\text{colloquial puant})$	=	colloquial <i>puer</i>
$V_0(\text{énergique})$	=	colloquial <i>‘avoir bouffé du lion’</i>
$V_0(\text{impatient})$	=	<i>s’impatienter</i> ; <i>‘prendre le mors aux dents’</i> , <i>‘ronger son frein’</i>
$V_0(\text{Pan !})$	=	<i>‘faire feu’</i>
$V_0(\text{Waouh !})$	=	<i>être en admiration</i>

Russian

$V_0(\text{predstavlenie})$	=	<i>predstavljat’</i>
$V_0(\text{kljatva})$	=	<i>pokljast’sja</i>
$V_0(\text{ošibka})$	=	<i>ošibit’sja</i>
$V_0(\text{vozrast})$	=	[N _{X-DAT}] <i>byt’</i> [NUM + N] _Y N denotes a time unit – e.g., MESJAC, GOD, VEK [<i>Ej 33 goda.</i>]
$V_0(\text{vonjučij})$	=	<i>vonjat’</i>
$V_0(\text{sčastlivyj})$	=	<i>‘byt’na sed’mom nebe’</i>
$V_0(\text{child language Pif-paf!})$	=	<i>vystrelit’</i>
$V_0(\text{Ux_ty!})$	=	<i>byt’ v vostorge</i>

[9] **A₀** [Latin *adjectivum*]: adjectivization

A₀ applies to a non-adjectival lexical unit L and returns an adjective having the same meaning as L.

4 Paradigmatic lexical functions

A ₀ : adjectival				
L	V	N	Adv	Claus
A ₀ (L)	Adj	Adj	Adj	Adj

English

A ₀ (<i>compare</i>)	=	<i>comparative</i>
A ₀ (<i>time</i> _(N))	=	<i>temporal</i>
A ₀ (<i>school</i> _(N))	=	<i>scholar</i>
A ₀ (<i>rotation</i>)	=	<i>gyratory</i>
A ₀ (<i>rapidly</i>)	=	<i>rapid</i> _(Adj)
A ₀ (<i>closely</i>)	=	<i>close</i> _(Adj)

French

A ₀ (<i>comparer</i>)	=	<i>comparatif</i>
A ₀ (<i>temps</i>)	=	<i>temporel</i>
A ₀ (<i>port</i>)	=	<i>portuaire</i>
A ₀ (<i>rotation</i>)	=	<i>giratoire</i>
A ₀ (<i>rapidement</i>)	=	<i>rapide</i> _(Adj)
A ₀ (<i>près</i>)	=	<i>proche</i>

Russian

A ₀ (<i>sravnivat'</i>)	=	<i>sravnitel'nyj</i>
A ₀ (<i>vremja</i>)	=	<i>vremennoj</i>
A ₀ (<i>port</i>)	=	<i>portovyj</i>
A ₀ (<i>vraščenje</i>)	=	<i>vrašatel'nyj</i>
A ₀ (<i>bystro</i>)	=	<i>bystryj</i>
A ₀ (<i>rjadom</i>)	=	<i>blizkij</i>

Contrary to S₀(L) and V₀(L), the LF application A₀(L) returns an adjective that does not have *exactly* the same meaning as L. This is so for the following reason. A₀(L) is used as adjectival modifier of a noun, for instance *the temporal coordinate* means ‘the coordinate relative to time’,⁴ and the syntactic relation of modification (deep-syntactic relation **ATTR**) between the lexemes COORDINATE and TEMPORAL [= A₀(*time*)] expresses the semanteme ‘relative [to]’, which connects ‘coordinate’ and ‘time’, as shown in Figure 4.2.

The semanteme ‘relative [to]’ is always included in the meaning of A₀. Given this semantic increment, it is not the case that TIME_(N) is S₀ of TEMPORAL; the S₀

⁴On the lexicographic definition of A₀ lexical units, see Mel'čuk & Polguère (2018: Sec. 4.1.1).

4.3 Structural, i.e. syntactically motivated, derivatives

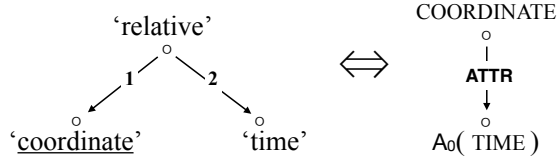


Figure 4.2: Semantic content carried by an A_0 syntactic modifier

of TEMPORAL is TEMPORALITY. The correspondences between these three lexemes are illustrated in paraphrases (16a–c).

- (16) a. We measure the dimension of existence relative to *time*.
 b. We measure the *temporal* dimension of existence.
 c. We measure the dimension of existence relative to its *temporality*.

The LF that associates TEMPORAL to TEMPORALITY is called A_1 – an adjective characterizing the first deep-syntactic actant of L (that is, of TEMPORALITY): $A_1(\text{TEMPORALITY}) = \text{TEMPORAL}$. The LF A_1 is presented below (Section 4.6, [21], p. 102).

The lexical triplet $\text{TIME}_{(N)}$, TEMPORAL and TEMPORALITY forms a configuration of lexical links that is visualized in Figure 4.3. This type of configuration is highly recurrent in the lexical network of English and other languages.

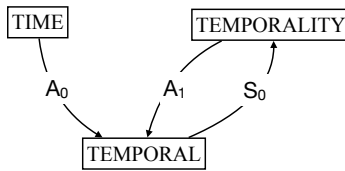


Figure 4.3: Typical configuration of lexical links for $A_0 \sim S_0 \sim A_1$

[10] Adv_0 [Latin *adverbium*]: adverbialization

Adv_0 applies to a non-adverbial lexical unit L and returns an adverb having the same meaning as L.

Adv_0 : adverbial				
L	V	N	Adj	Claus
$\text{Adv}_0(\text{L})$	Adv	Adv	Adv	Adv

4 Paradigmatic lexical functions

English

$\text{Adv}_0(\text{see}_{(V)})$	=	<i>visually</i> [<i>horrible to see</i> $\equiv$ <i>visually horrible</i>]
$\text{Adv}_0(\text{vision})$	=	<i>visually</i>
$\text{Adv}_0(\text{alphabet})$	=	<i>alphabetically</i>
$\text{Adv}_0(\text{rapid}_{(Adj)})$	=	<i>rapidly</i>
$\text{Adv}_0(\text{close}_{(Adj)})$	=	<i>closely</i>

French

$\text{Adv}_0(\text{voir})$	=	<i>visuellement</i> [<i>insupportable à voir</i> $\equiv$ <i>visuellement insupportable</i>]
$\text{Adv}_0(\text{vue})$	=	<i>visuellement</i>
$\text{Adv}_0(\text{alphabet})$	=	<i>alphabétiquement</i>
$\text{Adv}_0(\text{rapide}_{(Adj)})$	=	<i>rapidement</i>
$\text{Adv}_0(\text{proche})$	=	<i>près</i>

Russian

$\text{Adv}_0(\text{videt'})$	=	「 <i>na vid'</i> 」 [<i>protivno videt</i> $\equiv$ <i>protivnyj</i> 「 <i>na vid'</i> 」]
$\text{Adv}_0(\text{zrenie})$	=	<i>zritel'no</i>
$\text{Adv}_0(\text{alfavit})$	=	<i>po alfavitu</i>
$\text{Adv}_0(\text{bystryj})$	=	<i>bystro</i>
$\text{Adv}_0(\text{blizkij})$	=	<i>blizko</i>

Adv_0 should not be mistaken for another adverbialization LF, namely Adv_i , which is examined below (Section 4.7, [24], p. 104):

- (17) English $\text{Adv}_1(\text{see } [N_Y])$ = *at the sight* [*of N_Y*] ($\equiv$ *seeing* [*N_Y*])
 French $\text{Adv}_1(\text{voir } [N_Y])$ = *à la vue* [*de N_Y*]
 Russian $\text{Adv}_1(\text{videt'} [N_{Y-ACC}])$ = *pri vide* [*N_{Y-GEN}*]

[11] Claus_0 [Latin **clausativum*]: clausativization

Claus_0 applies to a non-clausative lexical unit L and returns a clausative having the same meaning as L.

$\text{Claus}_0(L)$ is a clausative lexical entity – that is, a lexical entity syntactically equivalent to a sentence (cf. Chapter 3, Section 3.5.1, p. 57). Claus_0 turns the meaning of its lexical argument L into the meaning of an utterance communicatively dominated by the semantic component ‘I signal [that L ...]’. In this respect, Claus_0 is not a pure structural derivation, just like A_0 (see above, p. 88); it adds a piece of communicative information (‘I signal [that L ...]’) to the derivative.



To signal vs. to communicate

To signal is a term (Mel'čuk 2001: 249–250): in contrast to *to communicate*, it denotes an enunciation mode that precludes negation and interrogation. A signaled expression does not state something about the world, but manifests an internal (mental, emotional or physiological) state of the Speaker.

Claus ₀ : clausative				
L	V	N	Adj	Adv
Claus₀(L)	Claus	Claus	Claus	Claus

Claus₀ should not be mistaken for the Imper LF (Section 4.8, [27], p. 110), whose values are syntactically also clausatives, but which carry an additional meaning: ‘I signal that *I want you to do* [L]’.

English

Claus₀(*congratulations*) = *Congratulations!, Congrats!*

Claus₀(*meow*_(V)) = *Meow!*

Claus₀Son(*cat*) = *Meow!*

Note. For the LF Son, see Chapter 5, [63], p. 163.

Claus₀(*explosion*) = *Boom!*

Claus₀(*gunshot*) = *Bang!, Pow!*

Claus₀(*informal yucky*) = *Yuck!*

French

Claus₀(*féliciter*) = *(Mes) Félicitations !; Chapeau !*

Claus₀(*miauler*) = *Miaou !*

Claus₀Son(*chat*) = *Miaou !*

Claus₀Son(*chute*) = *Badaboum !*

Claus₀(*explosion*) = *Boum !*

Claus₀(*« coup de feu »*) = *Pan !*

Claus₀(*dégoûtant*) = *Beurk !*

Russian

Claus₀(*pozdravlenije*) = *Pozdravljaju!*

4 Paradigmatic lexical functions

Claus ₀ (<i>mjaukat'</i>)	=	<i>Mjau!</i>
Claus ₀ Son(<i>koška</i>)	=	<i>Mjau!</i>
Claus ₀ Son(<i>padenie</i>)	=	<i>Bax!</i>
Claus ₀ (<i>vystrel</i>)	=	child language <i>Pif-paf!</i>
Claus ₀ (<i>protivnyj</i>)	=	<i>Fi!; Fu!</i>



Historical remark

The Claus₀ LF was introduced in Mel'čuk & Polguère (2021) in order to better dovetail the set of structural derivative LFs with that of deep parts of speech. Claus₀ cancels the neological LF Enun (Latin *ēnuntiātus*), which was used in the *DiCo* lexicographic project (Polguère 2000b).

4.4 Predicative derivatives

[12] Pred [Latin **praedicātum* '[a] predicate']: predicativization

Pred(L) is a verbal lexical unit that means 'be_(copul) L'; it serves to turn a predicative noun, an adjective or an adverb L into a verb. Thus, Pred plays the role of the copula; it is the paradigmatic counterpart of the syntagmatic LF Copul (Chapter 5, [45], p. 137).

Note that for adjectives and adverbs, the LF Pred is semantically equivalent to V₀ (see above, [8], p. 86):

$$(18) \text{ Pred}(\textit{unexpected}) = \text{V}_0(\textit{unexpected}) = \text{'come as a bolt from the blue'}$$

For quasi-predicative nouns denoting Agents, Pred is semantically equivalent to Fact₀ (Chapter 5, [50], p. 150):

$$(19) \text{ Pred}(\textit{cook}_{(N)}) = \text{Fact}_0(\textit{cook}_{(N)}) = \textit{cook}_{(V)} \text{'be a cook'}$$

Pred: verbal			
L	N _(predic)	Adj	Adv
Pred(L)	V	V	V

English

Pred(<i>beggar</i>)	=	<i>beg</i> ‘be a beggar’ [<i>He begs for a living.</i>]
Pred(<i>cook</i> _(N))	=	<i>cook</i> _(V) ‘be a cook’ [<i>He cooks in a posh restaurant.</i>]
Pred(<i>unexpected</i>)	=	‘come as a bolt from the blue’
Pred(<i>inseparable</i>)	=	‘be joined at the hip’

French

Pred(<i>mendiant</i>)	=	<i>mendier</i>
Pred(<i>cuisinier</i>)	=	<i>cuisiner</i>
Pred(<i>aisé</i> [financièrement])	=	‘avoir les moyens’
Pred _⊃ (<i>riche</i> _(Adj))	=	colloquial ‘péter dans la soie’
Pred(<i>inséparables</i> _(Adj))	=	‘être comme les (cinq) doigts de la main’, ‘être comme les deux doigts de la main’

Russian

Pred(<i>niščij</i>)	=	<i>niščenstvovat’</i>
Pred(<i>direktor</i>)	=	<i>direktorstvovat’</i>
Pred _⊃ (<i>bogatyj</i>)	=	‘katat’sja kak syr v masle’
Pred(<i>neožidannyj</i>)	=	literary ‘pojavit’sja kak čert iz tabakerki’
Pred(<i>nerazlučny</i>)	=	colloquial ‘[N _{X-ACC}] vodoj ne razol’šč’

Pred in configurational / complex LFs. Pred is mostly combined with adjectival / adverbial LFs Magn, Ver and Bon (Chapter 5, Section 5.2, [35 / 36 / 37], p. 121) to form configurational or complex LFs.

- (20) English Magn+Pred(*angry*) = *fume*_(V)
 English Magn+Pred(*hungry*) = *starve*
 French Magn+Pred(*chauve*_(N) ‘bald person’) = vulgar ‘avoir une perruque en peau de fesse’ lit. ‘have a buttock-skin wig’
- English PredMagn(*animosity*) =
 Magn+Func₀(*animosity*) = ‘run rampant’
 English PredVer(*dress*_(N)) = *fit*_(V); ‘sit well’ [with N]

Note that cases of the configurational *paradigmatic* LF Magn+Pred were mistakenly described as cases of the complex *syntagmatic* LF PredMagn in Mel’čuk (2015b: 216–217).

For more on the use of Pred in complex LFs, see the description of the syntagmatic LF Plus (Chapter 5, [38], p. 128).

4.5 Nominal derivatives

Nominal derivatives fall into two classes: actantial (Section 4.5.1) vs. circumstantial (Section 4.5.2). Note that adjectival (Section 4.6) and adverbial (Section 4.7) derivatives are only actantial.

4.5.1 Nominal actantial derivatives

This section presents, in the first place, the LFs $S_1, S_2, S_3, \dots$ – for short, S_i . S_i applies to a predicative lexical unit – predicate or quasi-predicate – and returns the name for its DSyntA *i*, *ii*, *iii*, ...

Also included in this section are two more specific LFs intimately related to S_i : Equip and Cap.

[13] S_i : typical name for L's DSyntA *i*

$S_i(L)$ is a nominal lexical entity that is the name for L's DSyntA *i*, *ii*, *iii*, ...

S_i : nominal					
L	V	N	Adj	Adv	Claus
$S_i(L)$	N	N	N	N	N

English

$S_1(\textit{sell}_{(V)}) = \textit{seller}$
[N_X sells N_Y to N_Z for N_W]
 $S_2(\textit{sell}_{(V)}) = \textit{merchandise}$
 $S_3(\textit{sell}_{(V)}) = \textit{buyer}$
 $S_4(\textit{sell}_{(V)}) = \textit{price}$
 $S_1(\textit{chemistry}) = \textit{chemist}$
[chemistry done by N_X]
 $S_1(\textit{mute}_{(Adj)}) = \textit{mute}_{(N)}$
[N_X is mute]
 $S_1(\textit{lazy}) = \textit{colloquial lazybones}$
[N_X is lazy]

French

$S_1(\textit{vendre}) = \textit{vendeur}$
 $S_2(\textit{vendre}) = \textit{marchandise}$

$S_3(\text{vendre})$	=	<i>acheteur</i>
$S_4(\text{vendre})$	=	<i>prix</i>
$S_1(\text{chimie})$	=	<i>chimiste</i>
$S_1(\text{muet}_{(\text{Adj})})$	=	<i>muet}_{(\text{N})}</i>

Russian

$S_1(\text{prodavat'})$	=	<i>prodavec</i>
$S_2(\text{prodavat'})$	=	<i>tovar</i>
$S_3(\text{prodavat'})$	=	<i>pokupatel'</i>
$S_4(\text{prodavat'})$	=	<i>cena</i>
$S_1(\text{ximija})$	=	<i>ximik</i>
$S_1(\text{nemoj}_{(\text{Adj})})$	=	<i>nemoj}_{(\text{N})}</i>

[14] Equip [French *équipage* 'crew']: team that allows L's referent to function

Equip(L) is a nominal lexical entity that designates an organized set of individuals (or, more generally, of living beings) involved in the functioning of L's referent; in other words, Equip(L) designates the set of L's DSyntAs **1**. Consequently, L is a quasi-predicative noun denoting a functioning entity.

Equip: nominal	
L	N
Equip(L)	N

English

Equip(<i>ship</i> _(N))	=	<i>crew</i>
Equip(<i>theater</i>)	=	<i>company</i>
Equip(<i>newspaper</i>)	=	「 <i>editorial staff</i> 」

French

Equip(<i>navire</i>)	=	<i>équipage</i>
Equip(<i>théâtre</i>)	=	<i>troupe</i>
Equip(<i>journal</i>)	=	<i>rédaction</i>

Russian

Equip(<i>sudno</i>)	=	<i>èkipaž</i>
Equip(<i>teatr</i>)	=	<i>truppa</i>
Equip(<i>gazeta</i>)	=	<i>redakcija</i>

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According to its nature, Equip applies only to lexical units whose DSyntA1 denotes a participant – $S_1(L)$ – that makes the entity L function. In other words, Equip means ‘set of S_1 ’; compare the above English illustrations of Equip with (21):

- (21) $S_1(ship_{(N)})$ = *sailor* [(*ship*’s) *crew* $\equiv$ *set of sailors*]
 $S_1(theater)$ = *actor* [(*theater*’s) *company* $\equiv$ *set of actors*]
 $S_1(newspaper)$ = *journalist* [*editorial staff* $\equiv$ *set of journalists*]

The meaning ‘set [of Xs]’ is represented by the LF Mult (Section 4.8, [26], p. 109). It is thus possible to propose a paraphrasing rule R_{Equip} that states the following semantic equivalence:

$R_{Equip}: \quad Equip(L) \equiv MultS_1(L)$ <i>crew</i> [<i>of a ship</i>] $\equiv$ <i>set of sailors</i> $_{S_1(ship)}$ [<i>of a ship</i>]
--

We can foresee a possible generalization of the LF Equip in order to cover cases corresponding to $MultS_2$, $MultS_3$, etc. With such a generalization, it would be necessary to distinguish $Equip_1$, $Equip_2$, $Equip_3$, etc.

[15] **Cap** [Latin *caput* ‘head’]: person in charge of L’s referent

Cap(L) is a nominal lexical entity that designates the individual in charge of a functioning entity L (in fact, in charge of the set of L’s DSyntAs1). As for Equip, L is a quasi-predicative noun.

Cap: nominal	
L	N
Cap(L)	N

Cap(L) denotes a specific element of the set of $S_1(L)$. Two cases have to be distinguished:

- if L denotes a group of individuals (e.g. a tribe, a pack of animals, etc.), Cap(L) denotes the main member of this group;
- if L denotes a functioning entity (e.g. a plane, a school, etc.) run by a group of individuals – that is, run by Equip(L), Cap(L) is the main member of Equip(L).

English

Cap(<i>kingdom</i>)	=	<i>king</i>
Cap(<i>mafia</i>)	=	<i>don, godfather</i>
Cap(<i>ship</i> _(N))	=	<i>captain</i> ₁
Cap(<i>plane</i> _(N))	=	<i>captain</i> ₂
Cap(<i>school</i> _(N))	=	<i>principal</i> _(N)
Cap(<i>tribe</i>)	=	<i>chief</i> (of ART ~)
Cap([<i>wolf</i>] <i>pack</i> _(N))	=	<i>leader</i> (of ART ~)

French

Cap(<i>royaume</i>)	=	<i>roi</i>
Cap(<i>mafia</i>)	=	<i>parrain</i>
Cap(<i>navire</i>)	=	<i>capitaine</i> ₁
Cap(<i>avion</i>)	=	<i>capitaine</i> ₂ , <i>commandant</i> (de bord)
Cap(<i>gare</i>)	=	「 <i>chef de gare</i> 」
Cap(<i>tribu</i>)	=	<i>chef</i> ₁ (de (ART) ~)
Cap(<i>meute</i>)	=	<i>chef</i> ₂ (de (ART) ~)

Russian

Cap(<i>korolevstvo</i>)	=	<i>korol'</i>
Cap(<i>mafija</i>)	=	<i>boss</i>
Cap(<i>sudno</i>)	=	<i>kapitan</i>
Cap(<i>samolët</i>)	=	「 <i>komandir korablja</i> 」
Cap(<i>plem ja</i>)	=	<i>vožd'</i> (~eni)
Cap(<i>sta ja</i>)	=	<i>vožak</i> (~i)

4.5.2 Nominal circumstantial derivatives

As seen above, S_i applies to a predicative or quasi-predicative lexical unit L and returns the general name for $DSyntA_i$ of L . In contrast, the next five LFs – S_{loc} , S_{instr} , S_{med} , S_{res} and S_{mod} – return for L the name of L 's prototypical *circumstantials* (a.k.a. adverbials, or adjuncts): locative, instrumental, means, result and manner circumstantials.

The domain of application of a circumstantial LF is determined by its semantic content, namely:

- S_{loc} applies to an L that denotes either an entity or a fact (action, event, process, state, ...), since both entities and facts can have localizations;
- S_{instr} and S_{med} apply to an L that denotes an action – i.e. a particular type of fact, since only actions involve instruments and means;

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- S_{res} and S_{mod} take an L that denotes a dynamic fact (action, event, process): only dynamic facts have results and happen in a certain way.

Because of their particular semantic content, the circumstantial LFs do not apply to adjectives and adverbs. All circumstantial LFs have the same PoS table:

$S_{loc/instr/med/res/mod}$: nominal				
L	V	N	Claus	
$S_{loc/instr/med/res/mod}(L)$	N	N	N	

The order of presentation of circumstantial LFs follows, roughly speaking, the hierarchy of circumstantials in natural languages: from more to less linguistically privileged circumstantials.



Historical remark

A different ordering of circumstantial LFs was used in previous publications, namely: S_{loc} , instead of heading the enumeration, was arbitrarily placed in the penultimate position.

[16] S_{loc} [Latin *locus* 'place']: name of the prototypical location

$S_{loc}(L)$ is a nominal lexical entity that is the name of the prototypical location of the referent of L.

English

$S_{loc}(\textit{surgery})$	=	「operation room」
$S_{loc}(\textit{smoke}_{(V)})$	=	「smoking area」; 「smoking room」
$S_{loc}(\textit{formal excrete})$	=	bathroom, restroom, toilet _(N) , washroom; privy; outhouse
$S_{loc}(\textit{orphan})$	=	orphanage
$S_{loc}(\textit{cattle})$	=	stable _(N)

French

$S_{loc}(\textit{opération [chirurgicale]})$	=	「salle d'opération」; 「bloc opératoire」
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$S_{loc}(fumer)$	=	<i>espace fumeur(s); fumer</i>
$S_{loc}(「faire ses besoins」)$	=	vulgar <i>chiottes, toilettes, W.-C.</i> ; formal <i>「lieu d'aisance(s)」; latrine</i>
$S_{loc}(orphelin)$	=	<i>orphelinat</i>
$S_{loc}(b��tail)$	=	<i>��table</i>

Russian

$S_{loc}(operacija)$	=	<i>operacionnaja</i>
$S_{loc}(kurit')$	=	<i>「mesto dlja kurenija」</i> , colloquial <i>kurilka</i>
$S_{loc}(\text{colloquial } 「spravljať nuždu」)$	=	<i>tualet, ubornaja</i> ; colloquial <i>sortir</i>
$S_{loc}(sirota)$	=	<i>「detskij dom」</i>
$S_{loc}(skot)$	=	<i>xlev</i>

[17] S_{instr} [Latin *instrumentum* 'tool']: name of the prototypical instrument

$S_{instr}(L)$ is a nominal lexical entity that is the name of the instrument, in the most general sense, which is prototypically used to do L.

English

$S_{instr}(drink_{(V)})$	=	<i>glass; cup</i>
$S_{instr}(shower_{(V)})$	=	<i>shower_{(N)}</i> [equipment]
$S_{instr}(wash_{(V)} [oneself])$	=	<i>washcloth</i>
$S_{instr}(fry_{(V)})$	=	<i>「frying pan」, skillet; fryer</i>
$S_{instr}(smell_{(V)})$	=	<i>nose_{(N)}</i>
$S_{instr}(murder_{(N)})$	=	<i>「murder weapon」</i>
$S_{instr}(Snip!)$	=	<i>scissors</i>

French

$S_{instr}(boire)$	=	<i>verre; tasse</i>
$S_{instr}(se doucher)$	=	<i>douche</i> [equipment]
$S_{instr}(se laver)$	=	<i>「gant de toilette」</i>
$S_{instr}(frire)$	=	<i>[une] po��le; friteuse</i>
$S_{instr}(sentir)$	=	<i>nez</i>
$S_{instr}(crime)$	=	<i>「arme du crime」</i>
$S_{instr}(Glouglou !)$	=	<i>verre</i>

Russian

$S_{instr}(pit')$	=	<i>stakan; ��a��ka; rjumka</i>
$S_{instr}([prinimat'] du��i)$	=	<i>du��</i> [equipment]
$S_{instr}(pri��esyvat'sja)$	=	<i>grebe��ok, ras���eska</i>
$S_{instr}(��arit')$	=	<i>skovoroda; fritjurnica</i>

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$S_{instr}(njuxat')$	=	<i>nos</i>
$S_{instr}(prestuplenie)$	=	「 <i>orudie prestuplenija</i> 」
$S_{instr}(\check{C}ik!)$	=	<i>nož; nožnicy</i>

In the above examples, English *Snip!*, French *Glouglou !* and Russian *Čik!* are clausative elements of the value of the LF $Claus_0SonS_0$ for the verbs English *cut*, French *boire* and Russian *rezat'* – cf. Son, Chapter 5, [63], p. 163.

[18] S_{med} [Latin *medius* ‘middle’; via such expressions as *by inter+med+iary of*]: name of the prototypical means

$S_{med}(L)$ is a nominal lexical entity that is the name of the means which is prototypically used to do L.

English

$S_{med}(shower_{(V)})$	=	<i>water</i> _(N) ; <i>soap</i> _(N)
$S_{med}(fry_{(V)})$	=	<i>oil</i> _(N)
$S_{med}(glue_{(V)})$	=	<i>glue</i> _(N)

French

$S_{med}(se\ doucher)$	=	<i>eau; savon</i>
$S_{med}(fire)$	=	<i>huile</i>
$S_{med}(coller)$	=	<i>colle</i>

Russian

$S_{med}([prinimat']\ duš\!i)$	=	<i>voda; mylo</i>
$S_{med}(\check{z}arit')$	=	<i>maslo</i>
$S_{med}(kleit')$	=	<i>klej</i>

S_{med} and S_{instr} are semantically distinguished by the fact that a means is “consummated” in the accomplishment of L, while an instrument remains with the user:

- (22) $S_{med}(nail_{(V)}) = nail_{(N)}$
 $S_{instr}(nail_{(V)}) = hammer_{(N)}$

[19] S_{res} [Latin *resultāre* ‘have as an outcome’]: name of the prototypical result

$S_{res}(L)$ is a nominal lexical entity that is the name of the prototypical expected result of L.

English

$S_{\text{res}}(\text{treat}_{(V)} [\text{medically}])$	=	(good) health
$S_{\text{res}}(\text{harm}_{(V)})$	=	harm _(N)
$S_{\text{res}}(\text{murder}_{(V)})$	=	corpse, (dead) body
$S_{\text{res}}(\text{reflection})$	=	fruits of ART ~

French

$S_{\text{res}}(\text{soigner})$	=	(bonne) santé
$S_{\text{res}}(\text{nuire})$	=	dommage
$S_{\text{res}}(\text{assassiner})$	=	mort _(N) ; corps
$S_{\text{res}}(\text{réflexion})$	=	fruit de ART ~

Russian

$S_{\text{res}}(\text{lečit'})$	=	(dobroe) zdorov'e
$S_{\text{res}}(\text{vredit'})$	=	vred
$S_{\text{res}}(\text{ubit'})$	=	ubityj; trup; (mertvoe) telo
$S_{\text{res}}(\text{razdum 'ja})$	=	plod ~ij

[20] S_{mod} [Latin *modus* 'manner']: generic name of manner

$S_{\text{mod}}(L)$ is a nominal lexical entity that is the name of the manner in which L takes place.

We do not use *prototypical*(ly) in the characterization of S_{mod} for the following reason. In contrast to the preceding four circumstantial LFs, $S_{\text{mod}}(L)$ is the name of the parameter "L's manner" and not the name of a particular manner in which L takes place. For instance, the noun *gait* – see English illustrations below – is the value of $S_{\text{mod}}(\text{walk}_{(V)})$ since it means just 'manner of walking', while *limp*_(N) denotes a particular manner of walking.

English

$S_{\text{mod}}(\text{live}_{(V)})$	=	<i>lifestyle</i> ; 'standard of living'
$S_{\text{mod}}(\text{walk}_{(V)})$	=	<i>gait</i>
$S_{\text{mod}}(\text{box}_{(V)})$	=	[boxing] style, technique; footwork

French

$S_{\text{mod}}(\text{vivre})$	=	'mode de vie'; 'train de vie'
$S_{\text{mod}}(\text{marcher})$	=	démarche
$S_{\text{mod}}(\text{boxer})$	=	style, technique [de boxe]; 'jeu de jambes'

Russian

$S_{\text{mod}}(\text{žit'})$	=	'obraz žizni'
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$S_{\text{mod}}(\textit{xodit'})$	=	<i>poxodka</i>
$S_{\text{mod}}(\textit{boksirovat'})$	=	<i>[boksërskij] stil', [boksërskaja] texnika;</i> <i>「rabota nog」</i>

4.6 Adjectival actantial derivatives

The three next LFs account for adjectival derivatives relative to L's DSyntAs. They all have the same PoS table:

A_i , Able_i , Qual_i : adjectival				
L	V	N	Adj	Claus
$A_i / \text{Able}_i / \text{Qual}_i(\text{L})$	Adj	Adj	Adj	Adj

[21] A_i : adjectival characterizer of L's DSyntA i

$A_i(\text{L})$ is an adjectival lexical entity that denotes the characteristic of being the DSyntA i , II , III , ... of L.

$A_i(\text{L})$ is a “pure” semantic derivative, contrary to two other adjectival actantial derivatives carrying additional chunks of meaning: Able_i and Qual_i – see [22] and [23].

For adjectival arguments, $A_1(\text{L}_{(\text{Adj})})$ returns $\text{L}_{(\text{Adj})}$ itself and therefore should not be used: an adjective is, by definition, its own A_1 . Of course, A_2 , A_3 , etc. can apply to adjectives, as illustrated below.

English

$A_1(\textit{harm}_{(\text{V})})$	=	<i>harmful</i>
$A_1(\textit{victory})$	=	<i>victorious</i>
$A_2(\textit{victory})$	=	<i>defeated</i>
$A_2(\textit{discussion})$	=	<i>under ~</i>
$A_2(\textit{effect}_{(\text{N})})$	=	<i>under the ~ [of N_X]</i>
$A_2(\textit{informed})$	=	<i>known</i>
$A_2(\textit{Yuck!})$	=	<i>disgusting, informal yucky</i>

French

$A_1(\textit{nuire})$	=	<i>nuisible</i>
$A_1(\textit{victoire})$	=	<i>victorieux</i>
$A_2(\textit{victoire})$	=	<i>vaincu_{(\text{Adj})}</i>
$A_2(\textit{discussion})$	=	<i>en ~</i>

$A_2(\textit{effet})$	=	<i>sous l'~ [de N_X]</i>
$A_2(\textit{jaloux}_{(Adj)})$	=	<i>jaloué</i>
$A_1(\textit{près})$	=	<i>proche_{(Adj)}</i>
$A_2(\textit{Beurk !})$	=	<i>dégoûtant</i>

Russian

$A_1(\textit{vredit'})$	=	<i>vrednyj</i>
$A_1(\textit{pobeda})$	=	<i>pobedonosnyj</i>
$A_2(\textit{pobeda})$	=	<i>pobežděnnij_{(Adj)}</i>
$A_2(\textit{vozdejstvi e})$	=	<i>pod ~em [N_{X-GEN}]</i>
$A_2(\textit{Fu!})$	=	<i>protivnyj</i>

[22] Able_i [Latin *habilis* ‘able, manageable’]: adjectival characterizer of L’s potential DSyntA i

Able_i(L) is an adjectival lexical entity that denotes the determining property of being the *potential* DSyntA i, II, III, ... of L (‘such that it easily can do L’ / ‘such that it easily can be L-ed’).

English

$\text{Able}_1(\textit{chat}_{(V)})$	=	<i>chatty</i>
$\text{Able}_2(\textit{read}_{(V)})$	=	<i>legible</i>
$\text{Able}_1(\textit{doubt}_{(V)})$	=	<i>doubtful a</i> [<i>Joey is doubtful of my success.</i>]
$\text{Able}_2(\textit{doubt}_{(V)})$	=	<i>doubtful b</i> [<i>For Joey, my success is doubtful.</i>]
$\text{Able}_1(\textit{emotion})$	=	<i>emotional</i>

French

$\text{Able}_1(\textit{bavarder})$	=	<i>bavard_{(Adj)}</i>
$\text{Able}_2(\textit{lire})$	=	<i>lisible</i>
$\text{Able}_1(\textit{peur})$	=	<i>peureux_{(Adj)}</i>
$\text{Able}_2(\textit{peur})$	=	Quebec <i>épeurant < effrayant</i>

Russian

$\text{Able}_1(\textit{boltat'})$	=	<i>boltlivyj</i>
$\text{Able}_2(\textit{čitat'})$	=	<i>razborčivyj</i> [počerk]
$\text{Able}_1(\textit{bojat'sja})$	=	<i>bojazlivyj</i>
$\text{Able}_1(\textit{èmocija})$	=	<i>èmocional'nyj</i>
$\text{Able}_2(\textit{somnevat'sja})$	=	<i>somnitel'nyj</i>

Able_i is an actantial derived adjective enriched, with respect to A_i (see [21] above), by the semantic content of potentiality, corresponding to such meaning

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as ‘be able to’, ‘be capable of’, ‘tend to’, ‘ought to’, etc. The exact meaning of each particular Able_i is specified by its lexicographic definition in its own lexical entry, just like for all similar cases.

[23] **Qual_i** [Latin *qualitas* ‘quality’]: adjectival characterizer of L’s probable DSyntA i

$\text{Qual}_i(\text{L})$ is an adjectival lexical entity that denotes the determining property of being the *probable* DSyntA I, II, III, ... of L (‘such that it is predisposed / strongly tends to do L’ / ‘such that it is predisposed / strongly tends to be L-ed’).

$\text{Qual}_i(\text{L})$ is a derived actantial adjective whose relation to L is of a more conceptual or cultural nature than a semantic one. Consequently, while $\text{Able}_i(\text{L})$ is very often a morphological derivative of L, a $\text{Qual}_i(\text{L})$ never is.

English

$\text{Qual}_1(\text{chat}_{(\text{V})})$	=	<i>voluble</i>
$\text{Qual}_1(\text{deceive})$	=	<i>dishonest</i>
$\text{Qual}_2(\text{deceive})$	=	<i>naïve < gullible</i>
$\text{Qual}_1(\text{fear}_{(\text{N})})$	=	<i>cowardly_{(\text{Adj})}</i>
$\text{Qual}_2(\text{fear}_{(\text{N})})$	=	<i>dangerous</i>

French

$\text{Qual}_1(\text{bavarder})$	=	<i>volubile</i>
$\text{Qual}_1(\text{tromper})$	=	<i>malhonnête</i>
$\text{Qual}_2(\text{tromper})$	=	<i>naïf</i>
$\text{Qual}_1(\text{peur})$	=	<i>lâche</i>
$\text{Qual}_2(\text{peur})$	=	<i>dangereux</i>

Russian

$\text{Qual}_1(\text{boltat}')$	=	<i>razgovorčivyj</i>
$\text{Qual}_1(\text{obmanyvat}')$	=	<i>nečestnyj</i>
$\text{Qual}_2(\text{obmanyvat}')$	=	<i>naivnyj</i>
$\text{Qual}_1(\text{ponimat}')$	=	<i>umnyj</i>
$\text{Qual}_2(\text{ponimat}')$	=	<i>lëgkij, prostoj</i>

4.7 Adverbial actantial derivatives

[24] **Adv_i**: adverb that means ‘while being L’s DSyntA i’

$\text{Adv}_i(\text{L})$ is an adverbial lexical entity with the following three properties:

- it is an adverbial that syntactically depends on a verb;
- it is subject-oriented, that is, it semantically bears on the DSyntA_I of this verb;
- it denotes the way L in which DSyntA_I of the verb in question does what it does being at the same time DSyntA_i of L (in L's government pattern).

In prose, the above means that Adv₁(L) is equivalent to *L-ing / being L* – see (23a), and Adv_{2/3/...}(L) is equivalent to *being L-ed / undergoing L* – see (23b).

- (23) a. Chris looked at Tony *suspiciously*_{Adv₁(suspect_(V))}. [≡ ... suspecting something.]
 b. Chris was arrested *under suspicion*_{Adv₂(suspect_(V))} of murder. [≡ ... being suspected of murder.]

Adv _i : adverbial				
L	V	N	Adj	Claus
Adv _i (L)	Adv	Adv	Adv	Adv[?]

English

Adv ₁ (<i>admire</i>)	=	<i>admiringly</i>
Adv ₁ (<i>deceive</i>)	=	<i>deceivingly</i>
Adv ₁ (<i>silence</i> _(N))	=	<i>in ~; silently</i>
Adv ₁ (<i>confrontation</i>)	=	<i>confrontationally; 'head to head'</i>
Adv ₁ (<i>alone</i>)	=	<i>single-handedly</i>
Adv ₁ (<i>many</i>)	=	<i>'in droves'</i>
Adv ₂ (<i>see</i> _(V) / <i>look</i> _(V))	=	<i>'under the eyes'</i> [of N _X]
Adv ₂ (<i>try</i> _(V) [in court])	=	<i>'in the dock'</i>

French

Adv ₁ (<i>admirer</i>)	=	<i>admirativement</i>
Adv ₁ (<i>tromper</i>)	=	<i>trompeusement</i>
Adv ₁ (<i>boiter</i>)	=	<i>clopin-clopant</i>
Adv ₂ (<i>voir</i>)	=	<i>'à la vue'</i> [de N _X]
Adv ₂ (<i>regarder</i>)	=	<i>'sous l'œil'</i> [de N _X]
Adv ₂ (<i>dire</i>)	=	<i>'aux dires'</i> [de N _X]
Adv ₁ (<i>majorité</i>)	=	<i>majoritairement</i>
Adv ₂ (<i>procès</i>)	=	<i>'sur le banc des accusés'</i>

Russian

$\text{Adv}_1(\text{vosxiščat'sja})$	=	<i>s vosxiščeniem; v vosxiščanii</i>
$\text{Adv}_1(\text{obmanyvat'})$	=	<i>obmančivo</i>
$\text{Adv}_1(\text{odin})$	=	「 <i>v odinočku</i> 」
$\text{Adv}_2(\text{govorit'})$	=	<i>po slovam</i> [$\text{N}_{\text{X-GEN}}$]
$\text{Adv}_2(\text{videt' / smotret'})$	=	「 <i>na glazax</i> 」 [$\text{N}_{\text{X-GEN}}$]
$\text{Adv}_2(\text{sudit'})$	=	「 <i>na skam'e podsudimyx</i> 」

Adv_1 does not express any semantic addition with respect to L, and therefore it should not be mistaken for the semantically full syntagmatic prepositional LFs Instr, which implies utilization, and Propt, which implies causation (Chapter 5, [43] and [44], p. 134–136).

Comment 1. For $\text{Adv}_1(\text{L})$, DSyntA I of L almost always⁵ coincides with DSyntA I of the governing verb; see (24), where CHRIS is DSyntA I of $\text{ANSWER}_{(\text{V})}$ and, at the same time, DSyntA I in the government pattern of HESITATE, and also of HESITATINGLY [= $\text{Adv}_1(\text{hesitate})$]:

(24) Chris answers *hesitatingly* _{$\text{Adv}_1(\text{hesitate})$} .

The semantics $\Leftrightarrow$ syntax correspondence for sentence (24) looks as in Figure 4.4.

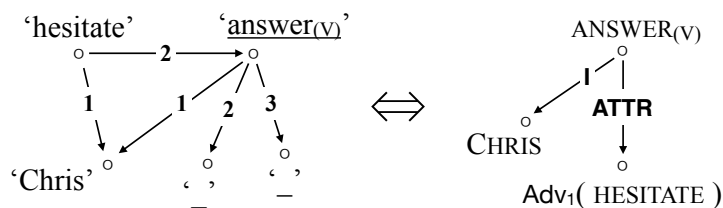


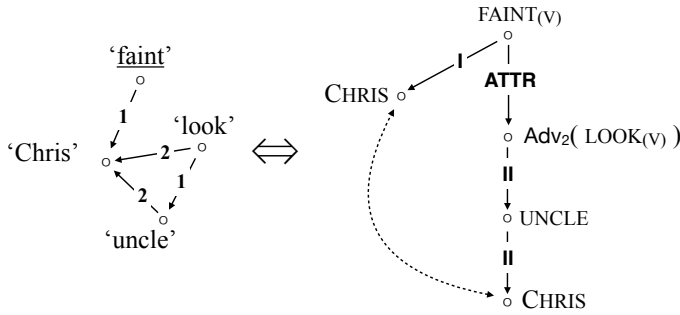
Figure 4.4: Sem $\Leftrightarrow$ DSynt correspondence for the sentence *Chris answers hesitatingly*.

For $\text{Adv}_{2/3...}(\text{L})$ things are more complicated since these adverbials involve a passive-like use of L; for instance, see (25), together with the corresponding Sem $\Leftrightarrow$ DSynt correspondence in Figure 4.5.

(25) Chris fainted 「*under the eyes*」 _{$\text{Adv}_2(\text{look}_{(\text{V})})$} of his uncle.

In the DSynt-tree of Figure 4.5, UNCLE appears as DSyntA II of $\text{Adv}_2(\text{look}_{(\text{V})})$ because this adverbial is equivalent to a passive form of $\text{LOOK}_{(\text{V})}$: 「*under the eyes of his uncle*」 $\equiv$ *being looked at by his uncle*.

⁵See Comment 2 below.



The dashed double-headed arrow in the DSynt-structure indicates a coreference relation.

Figure 4.5: Sem ↔ DSynt correspondence for the sentence *Chris fainted under the eyes of his uncle.*

Comment 2. A specific subset of lexical entities that are Adv₁(L) constitutes an exception to the statement in Comment 1: DSyntA I of these adverbials does not coincide with DSyntA I of the governing verb. Thus, in (26), DSyntA I of Adv₁ – i.e. CHRIS – does not coincide with DSyntA I of VOTE(V) – i.e. COMMITTEE:

(26) The committee voted *in the absence* of Chris.

Other Adv₁(L) of the same type are *in the presence* [of N_X], *with the departure* [of N_X], *with the illness* [of N_X], etc.

All such adverbials are encoded by putting the 1 actantial subscript between parentheses: Adv₍₁₎(L). This notation indicates that the subscript does not do its normal job, but marks simultaneously two things:

- the non-coincidence between L's DSyntA I and that of the governing verb;
- the obligatory character of the expression of L's DSyntA I with Adv₍₁₎(L).

Consequently, the adverbial expression *in the absence* in (26) is described by the LF formula (27).

(27) Adv₍₁₎(*absent*) = *in the absence* [of N_X]

The Adv₁ ~ Adv₍₁₎ contrast is parallel to the contrast between English regular V_{ING} phrases with DSyntA I coinciding with DSyntA I of the governing verb (28a) and special V_{ING} phrases whose DSyntA I does not (28b):

- (28) a. *Having left the house*, I heaved a sigh of relief.
 b. *My father having left the house*, I heaved a sigh of relief.

The V_{ING} construction in (28b) is known as the *absolute construction*. Since $\text{Adv}_{(1)}$ lexical entities are, so to speak, the lexical embodiment of the absolute construction, it is convenient to call them *absolute adverbials*.



Historical remark

In several previous publications, Adv_i was erroneously classified as a syntagmatic LF – see Mel’čuk (1982), Mel’čuk & Zholkovsky (1984), Mel’čuk et al. (1995) and Mel’čuk et al. (1999).

4.8 Singulative, collective and other inflection-like derivatives

The LFs introduced in this section correspond to semantic derivatives manifesting a strong analogy with the expression of such inflectional meanings as (noun) singular vs. plural, etc. These LFs come in two groups: (i) those that have to do with quantity as regards L – Sing and Mult – and (ii) those that characterize the accomplishment of event L – Imper, Perf, Imperf and Result_i.

[25] **Sing** [Latin *singulus* ‘singular, unique’]: **singulative**

Sing(L) is a lexical entity that denotes a unit / quantum of L’s referent.

Sing: verbal or nominal		
L	V	N
Sing(L)	V	N

English

$S_0\text{Sing}(\text{paint}_{(V)})$ = ‘brush stroke’

Note. This is not a “pure” Sing for a verb, but an $S_0\text{Sing}$, as it returns a nominal value. See the remark on verb singulatives after the illustrations.

$\text{Sing}(\text{rice})$ = grain of ~

Sing(<i>rain</i> _(N))	=	<i>raindrop</i>
Sing(<i>parquet</i>)	=	~ <i>slat</i>
Sing(<i>body</i>)	=	「 <i>body part</i> 」
Sing(<i>fur</i> _(N))	=	<i>hair 3</i> [cf. <i>hair 1</i> and <i>hair 2</i> under Mult, p. 110]
Sing(<i>vocabulary</i>)	=	<i>word</i>

French

S ₀ Sing(<i>peindre</i>)	=	「 <i>coup de pinceau</i> 」
Sing(<i>riz</i>)	=	<i>grain de</i> ~
Sing(<i>pluie</i>)	=	<i>goutte de</i> ~
Sing(<i>parquet</i>)	=	<i>lame de</i> ~
Sing(<i>corps</i>)	=	「 <i>partie du corps</i> 」
Sing(<i>pelage</i>)	=	<i>poil</i>
Sing(<i>vocabulaire</i>)	=	<i>mot</i>

Russian

S ₀ Sing(<i>pisat'</i> [<i>kartinu</i>])	=	<i>mazok</i>
Sing(<i>ris</i>)	=	「 <i>risovoe zërnyško</i> 」, <i>risinka</i>
Sing(<i>dožd'</i>)	=	<i>kaplja ~ja</i>
Sing(<i>parket</i>)	=	<i>parketina</i>
Sing(<i>telo</i>)	=	「 <i>čast' tela</i> 」
Sing(<i>mex</i>)	=	<i>volos, volosok</i>
Sing(<i>slovar'</i>)	=	<i>slovo</i>

In a *Standard Average European* [SAE] language, the singulative typically applies to nouns, less frequently to verbs. Here are examples of verbal singulatives:

- (29) English Sing(*beat*_(V)) = *hit*_(V)
 French Sing(*battre*) = *frapper*
 Russian Sing(*bit'*) = *udarit'*

Russian, with its inflectional category of aspect, has an important set of verbs whose perfective form expresses Sing:

- (30) Sing(*prygat'*) 'jump-IMPERF' = *prygnut'* 'jump-PERF = do one jump'
 Sing(*udarjat'*) 'hit-IMPERF' = *udarit'* 'hit-PERF = give one blow'
 Sing(*celovat'*) 'kiss-IMPERF' = *pocelovat'* 'kiss-PERF = give one kiss'

[26] **Mult** [Latin *multum* 'multitude']: collective (for nouns) / multiplicative (for verbs)

Mult(L) is a lexical entity that denotes a set of Ls forming a whole.

4 Paradigmatic lexical functions

Mult: verbal or nominal		
L	V	N
Mult(L)	V	N

English

Mult(<i>dog</i>)	=	<i>pack</i> _(N) .1 of ~s
Mult(<i>lie</i> _(N))	=	<i>pack</i> _(N) .4 of ~s
Mult(<i>hair</i> 1)	=	<i>hair</i> 2; 'head of hair'

French

Mult(<i>chien</i>)	=	<i>meute de</i> ~s
Mult(<i>mensonge</i>)	=	<i>tas de</i> ~s
Mult(<i>cheveu</i> 1)	=	<i>cheveux</i> 2, <i>chevelure</i>

Russian

Mult(<i>sobak</i> <i>a</i>)	=	<i>svora</i> ~
Mult(<i>l</i> <i>ož'</i>)	=	<i>kuča</i> ~ <i>ži</i>
Mult(<i>volos</i> 1)	=	<i>volosy</i> 2; <i>ševeljura</i>

In parallel with singulative verbs, SAE languages show multiplicative verbs:⁶

- (31) English AntiMagn+Mult(*jump*_(V)) = *hop*_(V)
 French AntiMagn+Mult(*taper*) = *tapoter*
 Russian AntiMagn+Mult(*xlopat'*) = *poxlopyvat'*

[27] Imper [Latin *imperāre* 'to] command, order']: imperative

Imper(L) is an utterance by which the Speaker incites the Addressee to do L.

Imper: verbal or clausative	
L	V
Imper(L)	V / Claus

English

Imper('go away')	=	colloquial 'Beat it!'
------------------	---	-----------------------

⁶On the AntiMagn LF used in (31), see Chapter 5, p. 122.

ImperPred(<i>quiet</i> _(Adj))	=	<i>Quiet!, Shush!, Shh!</i>
Imper(⌈ <i>give up</i> ⌋)	=	⌈ <i>Let it go!</i> ⌋
Imper(<i>Bang!</i>)	=	<i>Fire!</i>
ImperPrepar ₁ Real ₁ (<i>meal</i>)	=	<i>Breakfast <Lunch> <Supper> is ready!</i>

French

Imper(<i>partir</i>)	=	offensive <i>Dégage !</i> ; colloquial <i>Zou !</i>
Imper(<i>se taire</i>)	=	<i>Chut !</i> ; offensive ⌈ <i>Ta gueule !</i> ⌋
Imper(⌈ <i>tenir le coup</i> ⌋)	=	<i>Courage !</i>
Imper(<i>venir</i>)	=	<i>Ici !</i>
Imper _▷ (⌈ <i>laisser le passage</i> ⌋)	=	⌈ <i>Chaud devant !</i> ⌋
Imper(<i>Pan !</i>)	=	<i>Feu !</i>
Imper(<i>abandonner</i>)	=	⌈ <i>Laisse tomber !</i> ⌋
ImperPrepar ₁ Real ₁ (<i>repas</i>)	=	<i>À table !</i>

Russian

Imper(<i>ujti</i>)	=	offensive <i>Von (otsjuda)!</i> , colloquial offensive <i>Pošël von (otsjuda)!</i>
Imper(<i>ne šumet'</i>)	=	<i>Tixo!, Tss!, Tšš!</i>
Imper(<i>vzjat'</i>)	=	<i>Na(te)!</i>
Imper(child language <i>Pif-paf!</i>)	=	<i>Ogon'!</i>
ImperPrepar ₁ Real ₁ (<i>eda</i>)	=	<i>(Prošu vsex) za stol!, (Prošu vsex) k stolu!</i>

[28] Perf [Latin *perfectum*]: perfective

Perf(L) is a verbal or clausative lexical unit that denotes the fact of reaching the goal/limit inherent to the meaning of L – cf. the comment that follows the illustrations for the LF Result₁ below ([30], p. 114).

Perf and the following LF Imperf correspond to aspectual meanings grammaticalized in Slavic languages, where they are inflectional.

Perf: verbal or clausative	
L	V
Perf(L)	V / Claus

English

Perf(<i>end</i> _(V))	=	⌈ <i>be over</i> ⌋
Perf(<i>undress</i>)	=	<i>have undressed</i>

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Claus₀Perf(*dive*_(V)) = *Splash!*

French

Perf(*revenir*) = *「être de retour」*
 Perf(*se déshabiller*) = *s'être déshabillé*
 Claus₀Perf(*plonger*) = *Plouf!*

Russian

Perf(*razdevat'sja*) = *razdet'sja*
 Perf(*odevat'sja*) = *odet'sja*
 Claus₀Perf(*padat'* [in a liquid]) = *Bux!, Pljux!*

Values of Perf(L) and Imperf(L) are mostly trivial, in the sense that they are usually grammatical forms of a verb L. Perf is mainly used in complex LF formulas such as S_iPerf, A_iPerf or PerfOper_i.

- (32) S₁Perf(*「give birth」*) = *「new mother」*
 A₂Perf(*cook*_(V)) = *cooked*
 PerfOper₁(*divorce*_(N)) = *get* [ART ~]

[29] Imperf [Latin *imperfectum*]: imperfective

Imperf(L) is a verbal or clausative lexical unit that denotes fact L as ongoing process and, eventually (as in Slavic languages), as the iteration of fact L.

Imperf: verbal or clausative	
L	V
Imperf(L)	V / Claus

English

Imperf(*「be over」*) = *end*_(V), *finish*_(V)

French

Imperf(*se réveiller*) = *colloquial émerger*

Russian

Imperf(*vernut'sja*) = *vozvraščat'sja*
 Imperf(*odet'sja*) = *odevat'sja*

[30] Result_i [Latin **resultāre* ‘[to] result’]: resultative

Result_i(L) is a verbal or clausative lexical unit that denotes a situation *necessarily* resulting from reaching the goal / limit inherent to the meaning of L. DSyntA_i of Result_i(L) corresponds to DSyntA_i of L.

Let it be emphasized that the situation denoted by Result_i(L) is the necessary result of reaching L’s goal / limit. This is in contrast with some cases of Real_i(L) that are considered in Chapter 5 (Section 5.4.3, [49], “Realization verbs,” p. 149), where the situation referred to by Real_i(L) can be achieved in different ways.

Result _i : verbal or clausative	
L	V
Result _i (L)	V / Claus

English

Result₁(*reach*_(V) [N_Y]) = *be* [Loc_{in} N_Y]

Note. For the syntagmatic LF Loc_{in} (locative preposition meaning ‘being located in’), see Chapter 5, [40], p. 131.

Result₁(*colloquial* ‘*drift off*’) = *sleep*_(V); *Zzz!*
 Result₁(*agonize*) = *die*_(V)
 Result₁(*die*_(V)) = *colloquial* ‘*push up daisies*’
 Result₁(*acquire*) = *own*_(V)
 Result₂(*colloquial* ‘*shut up*’₂ [N_Y]) = *colloquial* [N_Y] ‘*shut up*’₁

French

Result₁(*atteindre* [N_Y]) = *être* [Loc_{in} N_Y]
 Result₁(*s’endormir*) = *dormir*; *Zzz!*
 Result₁(*agoniser*) = *mourir*
 Result₁(*mourir*) = *colloquial* ‘*manger les pissenlits par la racine*’
 Result₁(*acquérir*) = *posséder*
 Result₂(*colloquial* ‘*clouer le bec*’ [à N_Y]) = [N_Y] *se taire*

Russian

Result₁(*dostič’* [N_{Y-GEN}]) = *byt’* [Loc_{in} N_Y]
 Result₁(*zasnut’*) = *spat’*; *Xrrr...!*
 Result₁(*agonizirovat’*) = *umeret’*
 Result₁(*umeret’*) = *colloquial* ‘*byt’na tom svete*’

4 Paradigmatic lexical functions

$$\begin{aligned} \text{Result}_1(\text{priobresti}) &= \text{vladet'} \\ \text{Result}_2(\text{colloquial } \text{'}zatknut' rot'\text{ } [N_{Y-DAT}]) &= [N_{Y-NOM}] \text{zamolčat'}, \text{ colloquial} \\ &\quad \text{zatknut'sja; Ni_gugu}_{(Interj)} \end{aligned}$$

Result_i must be distinguished from Perf, for which it could be mistaken. For instance, (33a) has a different meaning from (33b):

- (33) a. *have reached*_{Perf(reach_{i(V)})} *Montreal*
 b. *be*_{Result_i(reach_{i(V)})} *in Montreal*

Perf(L) denotes the same fact as L but in a different state: 'have reached' is still 'reach', but 'completed'. On the contrary, Result_i(L) denotes a different fact: the situation which is necessarily entailed by reaching L's goal/limit and in which L's DSyntA *i* finds itself.

5 Syntagmatic lexical functions

Syntagmatic LFs are grouped into four major classes according to their part of speech [PoS] – i.e., the PoS of the collocate they encode. They are presented below in the same order as in the cooccurrence zone of a lexical entry (where they appear after paradigmatic LFs):

- Nominal collocates (Section 5.1)
- Adjectival / adverbial collocates (Section 5.2)
- Prepositional collocates (Section 5.3)
- Verbal collocates, distributed into seven subclasses corresponding to their increasing semantic load – i.e., the richness of their meaning ‘ σ^f ’ (Section 5.4)
 - Copula verb (Section 5.4.1)
 - Support [= light] verbs (Section 5.4.2)
 - Realization verbs (Section 5.4.3)
 - Phasic verbs (Section 5.4.4)
 - Verbs of causing (Section 5.4.5)
 - Verbs of mode of functioning (Section 5.4.6)
 - Other verbal collocates (Section 5.4.7)

We conclude this chapter with a summary of alternative approaches to the description of collocations, with a focus on lexicographic models – dictionaries and lexical databases (Section 5.5).

5.1 Nominal collocates

The two nominal collocates – Germ and Culm – are syntactic governors of L.

5 Syntagmatic lexical functions

[31] Germ [Latin *germen* ‘embryo’]: origin

Germ(L) is a nominal lexical unit that denotes L’s initial state, that is, something that will become L.

Germ: nominal	
L	N
Germ(L)	N

English

Germ(<i>flower</i> _(N))	=	[~] <i>bud</i>
Germ([<i>a</i>] <i>painting</i>)	=	<i>sketch</i> [of ART ~]
Germ(<i>idea</i>)	=	<i>germ</i> [of (ART) ~]

French

Germ(<i>fleur</i>)	=	<i>bouton</i> [de (ART) ~]
Germ([<i>une</i>] <i>peinture</i>)	=	<i>esquisse</i> [de ART ~]
Germ(<i>idée</i>)	=	<i>embryon, germe</i> [de (ART) ~]

Russian

Germ(<i>cvet</i> <i>ok</i>)	=	<i>buton</i> [~ <i>ka</i>]
Germ(<i>kartin</i> <i>a</i>)	=	<i>nabrosok</i> [~ <i>y</i>]
Germ(<i>ide</i> <i>ja</i>)	=	<i>zarodyš</i> [~ <i>i</i>]

[32] Culm [Latin *culmen* ‘highest point’]: culmination

Culm(L) is a nominal lexical unit that denotes L’s culminating / central point.

Culm: nominal	
L	N
Culm(L)	N

English

Culm(<i>anger</i> _(N))	=	<i>paroxysm</i> [of (ART) ~]
Culm(<i>crisis</i>)	=	<i>apogee, peak</i> [of ART ~]
Culm(<i>problem</i>)	=	<i>crux, heart</i> [of ART ~]
Culm(<i>night</i>)	=	<i>dead</i> [of (the) ~]

Culm(<i>earthquake</i>)	=	specialized focus, specialized hypocenter [of ART ~]; (specialized) epicenter [of ART ~]
Culm(<i>cyclone</i>)	=	<i>center, eye</i> [of ART ~]

Notes. The usage label **(specialized)** indicates that *epicenter* is a “runaway term,” i.e., a full-fledged term that nonetheless tends to be used in non-specialized discourse by speakers who do not necessarily master the corresponding terminological notion (Gotkova et al. 2024: 10). The phrases *center / eye* [of a cyclone] – cf. Culm(*cyclone*) – denote the central part of a cyclone where the intensity of the wind is minimal, but which is seen as the most dangerous place to be in.

French

Culm(<i>colère</i>)	=	<i>paroxysme</i> [de ART ~]
Culm(<i>joie</i>)	=	<i>comble</i> [de ART ~]
Culm(<i>problème</i>)	=	<i>cœur, noyau</i> [de ART ~]
Culm(<i>nuit</i>)	=	<i>cœur</i> [de ART ~]
Culm(「 <i>tremblement de terre</i> 」)	=	<i>foyer, specialized hypocentre</i> [de ART ~]; (specialized) épicerentre [de ART ~]
Culm(<i>cyclone</i>)	=	<i>œil</i> [de ART ~]

Russian

Culm(<i>ciklon</i>)	=	<i>glaz, oko</i> [~a]
Culm(<i>zemletrjaseni</i> e)	=	<i>épicerentr, očag</i> [~ja]; specialized gipocentr [~ja]
Culm(<i>problem</i> a)	=	<i>sut'</i> [~y]

Culm often appears in combination with the syntagmatic LF Loc_{in} (Section 5.3, [40], p. 131), forming a complex LF:

English

Loc _{in} Culm(<i>night</i>)	=	<i>in the dead</i> [of (the) ~]
Loc _{in} Culm(<i>storm</i>)	=	<i>at the heart</i> [of ART ~]

French

Loc _{in} Culm(<i>nuit</i>)	=	「 <i>en pleine</i> 」 [~], 「 <i>au plus noir</i> 」 [de la ~]
Loc _{in} Culm(<i>saison</i> [touristique])	=	「 <i>au plus gros</i> 」 [de la ~]

Russian

Loc _{in} Culm(<i>sezon</i>)	=	<i>v (samyj) razgar</i> [~a]
Loc _{in} Culm(<i>slav</i> a)	=	<i>na veršine</i> [~y]

Note that the expressions of Loc_{in}Culm for English and Russian in the above examples are themselves collocations – on collocates being collocations and so-called double-shot collocations, see Chapter 3, Section 3.3.2, footnote 9 (p. 48).



Historical remark

The LF Culm covers the former pair of LFs Culm and Centr (Latin *centrum* ‘center’) – the latter LF having been dropped. Note that the values of Culm(L) do not denote constitutive parts of L. The top of a tree is its extreme top part rather than its “culmination,” and the same can be said for the bow of a ship or the nose of a plane. The pairs *top* ~ *tree*, *bow* ~ *ship*, *nose* ~ *plane*, ... must be described as cases of *meronymy* (i.e., part / whole) relations. Such relations are encoded in the French Lexical Network [fr-LN] (Lux-Pogodalla & Polguère 2011) and other lexical network models by means of a formal tool called Mero – initially proposed in (Jousse 2010: 125–127).

Both syntagmatic LFs Germ and Culm are syntactically quite similar to the paradigmatic LF Figur; cf. *flame of passion* = Figur(*passion*) – see Chapter 4, [5], p. 83. However, Germ and Culm, on the one hand, and Figur, on the other, belong to two different LF families: Figur(L) is used by the Speaker as a substitute for L, whereas the syntagmatic Germ(L) and Culm(L) are used as collocates of L in order to designate something else than L.

5.2 Adjectival / adverbial collocates

The first five LFs in this section – Epit, Redun, Magn, Ver and Bon – are syntactic modifiers. Therefore, they are considered as deep adjectives when they modify a noun and as deep adverbs when they modify a verb, an adjective or another adverb; it allows us to speak of them as *adjectivals* / *adverbials*.

This treatment corresponds to the fact that all modifiers – be they modifiers of nouns or otherwise – are represented in deep-syntactic structure as dependents of the same DSynt-relation, namely **ATTR**.¹

[33] Epit [Latin *epitheton*]: clichéd modifier used for stylistic effects

Epit(L) is an adjectival / adverbial lexical unit used by the Speaker as a modifier of L in order to “fit in” by using a widely accepted cliché. For instance, in the collo-

¹On the **ATTR** DSynt-relation, see Chapter 2, Section 2.3, p. 28.

cation *chaotic jumble*, the adjective *chaotic* is semantically redundant – therefore, unnecessary (since *jumble* itself includes the semanteme ‘chaotic’); but it is used by the Speaker to abide by conventions of the corresponding genre, in this case, the journalese. Note that values of Epit(L) are not necessarily pleonastic in the strict sense and can encapsulate meanings such as that of (Anti)Magn, (Anti)Ver or (Anti)Bon. In such cases, they may be ideologically motivated in the context of a given genre – see, for instance, the Russian journalistic collocation of the Soviet era *svetloe buduščee* lit. ‘full.of.light future’ below.

Epit: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Epit(L)	Adv	Adj	Adv	Adv	Adj / Adv[?]

English

Epit(<i>chat</i> _(V))	=	<i>casually</i>
Epit(<i>mane</i> _(N))	=	<i>unruly</i>
Magn+Epit(<i>chasm</i>)	=	<i>deep < bottomless, yawning</i>
AntiBon+Epit(<i>crime</i>)	=	<i>heinous</i>
Epit(<i>winner</i> [lottery announcement])	=	<i>lucky</i>
Bon+Epit(<i>speaker</i> [academic conference])	=	<i>distinguished</i>

French

Epit(<i>insinuer</i>)	=	‘ <i>en douce</i> ’
Bon+Epit(<i>deviser</i>)	=	<i>gaiement</i>
Magn+Epit(<i>abîme</i>)	=	<i>profond < ‘sans fond’</i>
Bon+Epit(<i>chevalier</i> [romance of chivalry])	=	archaic <i>preux</i> anteposed
Bon+Epit(<i>princesse</i> [children’s tale])	=	<i>belle, jolie</i> anteposed
Epit(<i>gagnant</i> [lottery announcement])	=	<i>heureux</i> anteposed

Russian

Epit(<i>idti</i>)	=	<i>peškom</i>
Epit(<i>pocelovat’sja</i>)	=	‘ <i>drug s drugom</i> ’
Epit(<i>zabit’</i>)	=	<i>nasmert’</i>
Magn+Epit(<i>bezdna</i>)	=	<i>bezdonnaja, zizajuščaja</i>
Ver+Epit(‘ <i>na vysote</i> ’)	=	<i>dolžnaja</i> [<i>na dolžnoj vysote</i>]
Bon+Epit(<i>buduščee</i> [Soviet era journalism])	=	<i>svetloe</i>
Magn+Epit(<i>sveršenija</i> [Soviet era journalism])	=	<i>velikie</i>
Bon+Epit(<i>dévica</i> [folklore])	=	archaic <i>krasna</i>
Bon+Epit(archaic <i>mólodec</i> [folklore])	=	<i>dobryj</i>

[34] **Redun** [Latin *redundāre* ‘be in excess’]: pleonastic modifier used for word sense identification

Redun(L) is an adjectival / adverbial lexical unit used to help the Addressee identify the meaning of a lexical unit L used by the Speaker if L belongs to a polysemous vocable. Values of Redun(L) are lexical units that express explicitly a semantic component of L.

Redun is overwhelmingly used in terminologies (Polguère 2023), in the construction of an important subclass of so-called *complex terms* such as *chemical reaction*, *computer disk*, *money transfer*, ...

Redun: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Redun(L)	Adv	Adj	Adv	Adv	Adj / Adv[?]

English

Redun(<i>desire</i> _(V))	=	<i>sexually</i>
Redun(<i>desire</i> _(N))	=	<i>sexual</i>
Redun(<i>call</i> _(N))	=	<i>phone</i> _(N) [~]
Redun(<i>operation</i>)	=	<i>surgical</i>

French

Redun(<i>désirer</i>)	=	<i>sexuellement</i>
Redun(<i>appel</i>)	=	<i>téléphonique</i> postposed
Redun(<i>ébats</i>)	=	<i>amoureux</i> postposed
Redun(<i>opération</i>)	=	<i>chirurgicale</i> postposed

Russian

Redun(<i>želat'</i>)	=	formal <i>plotski</i>
Redun(<i>zvonok</i>)	=	<i>telefonnyj</i>
Redun(<i>ob"jat'ja</i>)	=	<i>ljubovnye</i>
Redun(<i>operacija</i>)	=	<i>xirurgičeskaja</i>



Historical remark

The distinction between Epit and Redun was introduced in the process of constructing the *DiCo* lexical database (Polguère 2000b) and that of compiling the *Lexique actif du français* [LAF] (Mel'čuk & Polguère 2007), where Redun was encoded with the “=” symbol. The LF name Redun was later systematically used in the French Lexical Network [fr-LN] project (Lux-Pogodalla & Polguère 2011). A different convention is used in Mel'čuk (2015b) in order to underscore the functional proximity of both LFs: Epit^{qual} for Epit and Epit^{restr} for Redun.

[35] Magn [Latin *magnus* ‘big, great’]: intensifier

Magn(L) is an adjectival / adverbial lexical unit used by the Speaker as a modifier of L to express a meaning of the type ‘very’, ‘a lot’, ‘intense(ly)’, etc.

Magn: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Magn(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

Magn(<i>sing</i> _(V))	=	‘at the top of N _X ’s lungs / voice’
Magn(<i>drink</i> _(V) [alcohol])	=	heavily, ‘like a fish’, ‘like a sponge’
Magn(<i>win</i> _(V))	=	‘with flying colors’
Magn(<i>love</i> _(N))	=	mad
Magn(<i>fear</i> _(N))	=	mortal
Magn(<i>dead</i> _(Adj))	=	‘as a doornail’, stone [-]
Magn(<i>pregnant</i>)	=	heavily, very
Magn(<i>Thanks!</i>)	=	[-] a lot

French

Magn(<i>chanter</i>)	=	‘à tue-tête’
Magn(<i>boire</i> [alcohol])	=	‘comme un trou’

5 Syntagmatic lexical functions

Magn(<i>gagner</i>)	=	「 <i>haut la main</i> 」
Magn(<i>amour</i>)	=	<i>fou</i> postposed
Magn(<i>peur</i>)	=	<i>bleue</i> postposed
Magn(<i>mort</i> _(Adj))	=	<i>raide</i> anteposed
Magn(<i>enceinte</i>)	=	informal 「 <i>jusqu'aux yeux</i> 」
Magn(<i>connu</i>)	=	「 <i>comme le loup blanc</i> 」
Magn(<i>Merci</i> !)	=	<i>bien</i> < <i>beaucoup</i> < <i>infiniment</i> postposed

Russian

Magn(<i>pet'</i>)	=	「 <i>vo ves' golos</i> 」 < 「 <i>vo vsë gorlo</i> 」
Magn(<i>pit'</i> [alcohol])	=	「 <i>kak sapožnik</i> 」
Magn(<i>pobeda</i>)	=	<i>blestjaščaja</i> < <i>polnaja</i>
Magn(<i>ljubov'</i>)	=	<i>bezumnaja, strasnaja</i>
Magn(<i>ranenyyj</i> _(N))	=	// <i>tjaželo</i> ~
Magn(<i>beremennaja</i>)	=	colloquial <i>sil'no</i> , // colloquial <i>na snosjax</i>
Magn(<i>izvestnyj</i>)	=	<i>široko</i> anteposed
Magn(<i>spasibo</i>)	=	<i>bol'soe</i> < <i>ogromnoe</i>

Five remarks concerning this LF follow. Note that remarks 1 to 4 also apply to the two other adjectival / adverbial LFs Ver and Bon, presented next.

1. The value of Magn(L) often contains fused elements, which are marked with the “//” symbol (on fusion, see Chapter 3, Section 3.3.3.1, p. 50); for instance:

- (1) Magn(*defeat*_(N)) = *crushing, heavy, major, massive, resounding, severe, shattering, stinging, stunning, thumping*
< *complete, thorough, total, // rout*_(N)

2. A fused Magn of L generally is at the same time its Syn_▷; for instance:²

- (2) Syn_▷(*defeat*_(N)) = *rout*_(N)

3. The LF Magn has an antonymic counterpart AntiMagn:

- (3) AntiMagn(*win*_(V)) = *narrowly*
AntiMagn(*rain*_(N)) = *light, // drizzle*_(N)
AntiMagn(*fever*) = *low-grade, slight*

²See also the comment on example (20) in Chapter 3, Section 3.3.3.1, p. 51.

4. The meaning of $\text{Magn}(\text{L})$ can bear on (i.e. have as target – see comment 5 below) the meaning of a DSyntA of L rather than on L itself, as is in the prototypical case. This is shown by a DSyntA number in subscript to Magn:

- (4) $\text{Magn}_1(\text{approval})$ = *general* [many Xs approve]
 $\text{Magn}_2(\text{check}_{(\text{N})})$ = *fat* [check's amount Y is significant]
 $\text{Magn}_3(\text{accusation})$ = *serious < grave* [Z of which X is accused is very bad]

5. The argument L of Magn, as well as that of Ver, must include as LF target in its lexicographic definition a semantic component capable of accepting intensification – for Magn – or evaluation of its “correctness” – for Ver. This remark does not apply to Bon, since any fact or entity is evaluated positively (or negatively) always from a subjective viewpoint. On LF targets, see Chapter 3, Section 3.3.2, p. 49.

[36] **Ver** [Latin *verus* ‘real, genuine’]: adequacy characterizer [$\approx$ ‘as it should be’]

$\text{Ver}(\text{L})$ is an adjectival / adverbial lexical unit that the Speaker uses as a modifier of L to express the meaning ‘adequate / as it should be ...’.

Ver: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Ver(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

$\text{Ver}(\text{sing})$	=	‘in key’, ‘in tune’
$\text{AntiVer}(\text{sing})$	=	<i>off-key</i>
$\text{Ver}(\text{talk}_{(\text{V})})$	=	informal <i>turkey</i>
$\text{AntiVer}(\text{talk}_{(\text{V})})$	=	‘in riddles’; informal // ‘beat around the bush’
$\text{Ver}(\text{aim}_{(\text{V})})$	=	<i>straight</i> postposed
$\text{Ver}(\text{truth}_{(\text{N})})$	=	<i>naked</i>
$\text{Ver}(\text{smile}_{(\text{N})})$	=	<i>compelling, frank</i>
$\text{AntiVer}(\text{smile}_{(\text{N})})$	=	<i>sour; wry</i>
$\text{Ver}(\text{success})$	=	<i>well-deserved</i>
$\text{AntiVer}(\text{success})$	=	<i>underserved</i>
$\text{Ver}(\text{choice})$	=	<i>correct, right</i>
$\text{AntiVer}(\text{choice})$	=	<i>wrong</i>
$\text{Ver}(\text{cooked})$	=	‘to perfection’

5 Syntagmatic lexical functions

French

Ver(<i>chanter</i>)	=	<i>juste</i> postposed
AntiVer(<i>chanter</i>)	=	<i>faux</i> postposed
Ver(<i>viser</i>)	=	<i>juste</i> postposed
AntiVer(<i>viser</i>)	=	«à côté»
Ver(<i>sourire</i> _(N))	=	<i>franc</i>
AntiVer(<i>sourire</i> _(N))	=	<i>forcé</i>
Ver(<i>vérité</i>)	=	«toute nue» postposed, <i>vraie</i> postposed
Ver(<i>succès</i>)	=	<i>mérité</i>
AntiVer(<i>succès</i>)	=	<i>immérité</i>
Ver(<i>choix</i>)	=	<i>approprié, judicieux</i>
AntiVer(<i>choix</i>)	=	<i>erroné</i>
AntiVer(<i>cerveau</i>)	=	<i>colloquial</i> «en compote»
Ver(<i>cuit</i>)	=	«à point»
Ver(<i>fait</i> _(Adj))	=	<i>bien</i> anteposed
AntiVer(<i>fait</i> _(Adj))	=	<i>mal</i> anteposed, «de travers» < «en dépit du bon sens», // «ni fait ni à faire»

Russian

Ver(<i>govorit'</i> [na ... jazyke])	=	<i>beglo, svobodno, xorošo < prekrasno</i>
AntiVer(<i>govorit'</i> [na ... jazyke])	=	<i>ploxo < edva</i>
Ver(<i>pet'</i>)	=	<i>čisto</i>
AntiVer(<i>pet'</i>)	=	<i>faľšivo</i>
Ver(<i>ulybka</i>)	=	<i>otkrytaja</i>
AntiVer(<i>ulybka</i>)	=	<i>kislaja</i>
Ver(<i>pravda</i>)	=	<i>čistaja</i>
Ver(<i>vybor</i>)	=	<i>pravil'nyj</i>
AntiVer(<i>vybor</i>)	=	<i>nepravil'nyj, ošibočnyj</i>
Ver(<i>uspex</i>)	=	<i>zaslužennyj</i>
AntiVer(<i>uspex</i>)	=	<i>nezaslužennyj</i>
Ver(<i>sšityj</i>)	=	<i>xorošo < prekrasno, «kak na zakaz»</i>
AntiVer(<i>sšityj</i>)	=	<i>ploxo < bezobrazno</i>
PredAntiVer(<i>mozg</i>)	=	// [U N _{X-GEN}] «šariki za roliki zaexali»

[37] **Bon** [Latin *bonus* 'good']: laudative characterizer [≈ 'good']

Bon(L) is an adjectival / adverbial lexical unit that the Speaker uses as modifier of L to express the meaning 'good', which constitutes the Speaker's positive evaluation of L's referent.

Bon: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Bon(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

Bon(<i>sing</i>)	=	<i>well</i> < <i>beautifully, delightfully</i> , 「like an angel」, 「like a bird / nightingale」
AntiBon(<i>sing</i>)	=	<i>poorly</i>
Bon(<i>choice</i>)	=	<i>fortunate</i>
AntiBon(<i>choice</i>)	=	<i>unfortunate</i>
Bon(<i>weather</i>)	=	<i>good, fine</i> < <i>beautiful</i>
AntiBon(<i>weather</i>)	=	<i>bad</i> < <i>filthy, foul</i>

French

Bon(<i>chanter</i>)	=	<i>bien</i> < 「comme un rossignol」, <i>divinement</i>
AntiBon(<i>chanter</i>)	=	<i>mal</i> < 「comme une casserole」
Magn+AntiBon(<i>chanter</i>)	=	// colloquial <i>gueuler</i>
Bon(<i>choix</i>)	=	<i>heureux</i>
AntiBon(<i>choix</i>)	=	<i>malheureux</i>
Bon(<i>temps</i> [weather])	=	<i>beau</i> anteposed < <i>magnifique</i>
AntiBon(<i>temps</i> [weather])	=	<i>mauvais</i> anteposed < 「à ne pas mettre un chat dehors」, colloquial 「de chien」

Russian

Bon(<i>pet'</i>)	=	<i>xorošo</i> < <i>prekrasno</i> , 「kak solovej」 < <i>božestvenno</i>
AntiBon(<i>pet'</i>)	=	<i>ploxox</i> < <i>otvratitel'no</i>
Magn+AntiBon(<i>pet'</i>)	=	// colloquial <i>orat'</i>
Bon(<i>vybor</i>)	=	<i>udačnyj</i>
AntiBon(<i>vybor</i>)	=	<i>neudačnyj</i>
Bon(<i>pogoda</i>)	=	<i>xorošaja</i> < <i>prekrasnaja</i> < <i>čudesnaja</i>
AntiBon(<i>pogoda</i>)	=	<i>ploxaja</i> < <i>merzkaia, užasnaja</i>
PredAntiBon(<i>pogoda</i>)	=	// 「V takuju pogodu xorošij xozjain sobaku na ulicu ne vygonit」



Historical remark

A particular case of Bon was previously encoded in the literature as Pos (now abandoned): Pos(L) is an adjectival / adverbial modifier of L that expresses the positive evaluation of L's Patient by L's Agent – that is, the evaluator. Since this LF is applicable only to lexical units denoting evaluations, it was decided to replace it with Bon₂ (for the use of actantial subscripts with Magn, Ver and Bon, see Remark 4, p. 123):

- (5) Bon₂(review_(N)) = *positive < glowing, rave*
 [The Speaker says that X, reviewer of Y, finds Y good.]
 AntiBon₂(review_(N)) = *lukewarm, mixed < negative < devastating*
 [The Speaker says that X, reviewer of Y, finds Y not very good or outright bad.]

In (5), with L denoting an evaluation, there is a slight semantic shift: contrary to “regular” Bon₂, the evaluation is done here by L's Agent, and not by the Speaker.

Magn, Ver and Bon: differences and similarities. The adjectival / adverbial LFs Magn, Ver and Bon form a triplet according to their semantic-pragmatic nature, in the sense that quite often two (or even three) of them can be expressed simultaneously by a single lexical unit; see (6)–(8) below.

- (6) Ver+Magn(*sleep*_(V)) = *soundly*
 Ver+Magn(*evidence*) = // *“smoking gun”*
 [L at a high degree (Magn) can fairly often be perceived as being as it should (Ver).]
 (7) Bon+Magn(*agree*) = *wholeheartedly*
 Magn+Bon(*smile*_(N)) = *infectious*
 [An inherently positive L at a high degree (Magn) is often perceived as being good (Bon).]

The order of constituent LFs in the second example of (7) above is inverted with respect to the first one because the positive nature of the smile (Bon) dominates its “size” (Magn). For the order of components in configurational LFs, see Chapter 3, Section 3.4.2, p. 55.

- (8) Bon+Magn+Ver(*health*) = *good*
 Bon+Magn+Ver(*sleep*_(V)) = *well*, 'like a baby'
 Bon+Magn+Ver(*lover* [sexual performance]) = *good*, *great* < *fantastic*

This notwithstanding, the three LFs Magn, Ver and Bon remain distinct and are often contrasted for the same L, as shown in our English illustrations with the verb SING – repeated below in (9).

- (9) Magn(*sing*) = 'at the top of N_X's voice'
 Ver(*sing*) = 'in key', 'in tune'
 Bon(*sing*) = *well* < *beautifully*, *delightfully*, 'like an angel', 'like a bird'

A formal shortcut: Degrad [Latin *degradāre* 'lower, degrade']. The verbal complex LF IncepPredAntiBon, which applies to nouns, is very much present in natural language lexicons: that is, it has many arguments and many possible values. Therefore, we use for this LF a shortcut notation: Degrad 'undergo degradation'.

Degrad: verbal	
L	N
Degrad(L)	V

English

- Degrad(*milk*_(N)) = 'turn / go sour'_I, *spoil*_(V)
 Degrad(*marriage*) = 'turn / go sour'_{II}
 Degrad(*flower*_(N)) = *wither*
 Degrad(*health*) = *decline*_(V), *deteriorate*
 Degrad(*discipline*) = *crumble*_(V)
 Degrad(*patience*) = 'wear thin'

French

- Degrad(*lait*) = *tourner*
 Degrad(*vin*) = 'tourner au vinaigre'_I
 Degrad(*situation*) = 'tourner au vinaigre'_{II}
 Degrad(*mariage*) = 'battre de l'aile'
 Degrad(*fleur*) = *se faner*
 Degrad(*santé*) = *se dégrader*, *s'étioler*
 Degrad(*discipline*) = 'foutre le camp'
 Degrad(*patience*) = *s'épuiser*

5 Syntagmatic lexical functions

Russian

Degrad(<i>moloko</i>)	=	<i>skisat'</i>
Degrad(<i>vino</i>)	=	<i>prokisat'</i>
Degrad(<i>situacija</i>)	=	<i>uxudšat'sja</i>
Degrad(<i>cvetok</i>)	=	<i>vjanut'</i>
Degrad(<i>zdorov'e</i>)	=	<i>pošatnut'sja</i>
Degrad(<i>disciplina</i>)	=	<i>rasšatyvat'sja</i>
Degrad(<i>terpenie</i>)	=	<i>istoščat'sja</i>

[38] Plus [Latin *plus* 'more']: positive comparative

The Plus LF means 'more'; it is used non-autonomously – generally in combination with the above three adjectival / adverbial LFs Magn, Ver and Bon – to build the three complex LFs PlusMagn, PlusVer and PlusBon. Plus being non-autonomous, its PoS is irrelevant (inherently, it is an adverbial – a modifier of adjectival LFs). Similar to paradigmatic LFs Sing, Mult, Imper, Perf, Imperf and Result_i (Chapter 4, Section 4.8), the LF Plus expresses a meaning of inflectional type: the positive comparative.

PlusMagn, PlusVer and PlusBon are typically found in combination with the complex LF IncepPred 'begin to be' = 'become': IncepPredPlusMagn, IncepPredPlusVer and IncepPredPlusBon. These three LFs take L as DSyntA₁, as seen in the illustrations below. As for the LF Plus, it functions as modifier of Magn / Ver / Bon and thus does not impact the syntactic structure of the whole LF formula. Among these LFs, it is IncepPredPlusMagn that is more present in lexicons.

English

IncepPredPlusMagn(<i>rain</i> _(N))	=	'pick up'
IncepPredPlusMagn(<i>battle</i> _(N))	=	'heat up'
IncepPredPlusMagn(<i>tension</i>)	=	<i>escalate</i>
IncepPredPlusMagn(<i>traffic</i> _(N))	=	'build up', <i>increase</i> _(V)
IncepPredPlusVer(<i>traffic</i> _(N))	=	'ease up'
IncepPredPlusVer(<i>sky</i> _(N))	=	'clear up'
IncepPredPlusBon(<i>situation</i>)	=	<i>improve</i>

French

IncepPredPlusMagn(<i>averse</i>)	=	<i>redoubler</i>
IncepPredPlusMagn(<i>bataille</i>)	=	<i>s'intensifier</i>
IncepPredPlusMagn(<i>tension</i>)	=	<i>monter</i>
IncepPredPlusMagn(<i>circulation</i> [<i>automobile</i>])	=	<i>s'intensifier</i>
IncepPredPlusVer(<i>circulation</i> [<i>automobile</i>])	=	<i>se fluidifier</i>

IncepPredPlusVer(<i>ciel</i>)	=	<i>se dégager</i>
IncepPredPlusVer(<i>vin</i>)	=	<i>se bonifier</i>
CausPredPlusVer(<i>vin</i>)	=	specialized <i>garder</i> [ART ~]
IncepPredPlusBon(<i>situation</i>)	=	<i>s'améliorer</i>

Russian

IncepPredPlusMagn(<i>liven'</i>)	=	<i>usilivat'sja</i>
IncepPredPlusMagn(<i>naprjažennost'</i>)	=	<i>rasti, vozrastat'</i>
IncepPredPlusMagn(<i>dviženie</i> [<i>na ulicax</i>])	=	<i>usilivat'sja</i>
IncepPredPlusVer(<i>nebo</i>)	=	<i>očičat'sja</i>
CausPredPlusVer(<i>vin o</i>)	=	<i>vyderživat'</i> [~o]
IncepPredPlusBon(<i>situacija</i>)	=	<i>uluččat'sja</i>

Plus can also be involved in configurational LFs:

English

PlusMagn+IncepOper ₁ (<i>weight</i> _(N))	=	<i>gain</i> _(V) , 'put on' [~]
PlusMagn+IncepOper ₁ (<i>speed</i> _(N))	=	'pick up' [~]
PlusMagn+IncepOper ₁ (<i>popularity</i>)	=	<i>grow</i> [in ~]

French

PlusMagn+IncepOper ₁ (<i>poids</i>)	=	<i>gagner, prendre</i> [du ~]
PlusMagn+IncepOper ₁ (<i>prudence</i>)	=	<i>redoubler</i> [de ~]
PlusBon+Caus ₁ Func ₀ (<i>technique</i>)	=	<i>affiner, améliorer</i> [ART ~]

Russian

PlusMagn+IncepOper ₁ (<i>skorost'</i>)	=	<i>nabirat'</i> [~]
PlusMagn+IncepOper ₁ (<i>usilija</i>)	=	<i>udvaivat', uveličivat'</i> [~]
PlusMagn+IncepOper ₁ (<i>potencial</i>)	=	<i>naraščivat'</i> [~]

Note that between complex and configurational LFs there exist, as a function of L, semantic equivalences of the following type:

- (10) IncepPredPlusMagn(*rain*_(N)) = 'pick up'
 ['rain begins to.be more.heavy']
 ≡
 PlusMagn+IncepFunc₀(*rain*_(N)) = 'pick up'
 ['more.heavy rain begins taking.place']

Equivalence (10) illustrates the following deep-syntactic paraphrasing rule:

R_{IncepPredPlusMagn}:

$L \leftarrow I \vdash \text{IncepPredPlusMagn}(L) \equiv \text{IncepFunc}_0(L) \rightarrow I \vdash L \text{--ATTR} \rightarrow \text{PlusMagn}(L)$

Remark on PlusMagn. The LF PlusMagn – in isolation or as a part of a complex LF formula – can be abbreviated as Plus for simplicity’s sake. This may create a formal ambiguity, which is effectively resolved by the PoS of the entity to which Plus thus applies. Two cases have to be considered.

1. When Plus takes as argument an adjective, it is to be interpreted as Plus, such as in (11) below:

$$\begin{aligned} (11) \quad \text{IncepPredPlus}(\text{weak}_{(\text{Adj})}) &= // \text{weaken} \\ \text{IncepPredPlus}(\text{solid}_{(\text{Adj})}) &= // \text{solidify} \end{aligned}$$

2. When Plus takes as argument anything other than an adjective, it is interpreted as PlusMagn, such as in (12) below:

$$\begin{aligned} (12) \quad \text{IncepPredPlus}(\text{confidence}) &= \text{grow} \\ \text{IncepPredPlus}(\text{cost}_{(\text{N})}) &= \text{rise}_{(\text{V})}, \text{‘go up’} < \text{skyrocket} \end{aligned}$$

This remark applies equally to MinusMagn, which can be abbreviated as Minus – see immediately below.

[39] Minus [Latin *minus* ‘less’]: negative comparative

Minus ‘less’ is the antonym of Plus: Minus = AntiPlus; as a result, Minus features the same properties as Plus.

English

$$\begin{aligned} \text{IncepPredMinusMagn}(\text{rain}_{(\text{N})}) &= \text{subside} \\ \text{IncepPredMinusMagn}(\text{muscle}_{(\text{N})}) &= \text{‘waste away’} \\ \text{IncepPredMinusMagn}(\text{fever}) &= \text{‘go down’} \\ \text{IncepPredMinusVer}(\text{sky}_{(\text{N})}) &= \text{‘cloud over’} \\ \text{MinusMagn+IncepOper}_1(\text{weight}_{(\text{N})}) &= \text{lose [-]} \end{aligned}$$

French

$$\begin{aligned} \text{IncepPredMinusMagn}(\text{muscle}) &= \text{s’atrophier, fondre} \\ \text{IncepPredMinusMagn}(\text{averse}) &= \text{se calmer, faiblir} \\ \text{IncepPredMinusMagn}(\text{fièvre}) &= \text{tomber} \end{aligned}$$

IncepPredMinusVer(<i>ciel</i>)	=	<i>se couvrir</i>
MinusMagn+IncepOper ₁ (<i>poids</i>)	=	<i>perdre</i> [<i>du ~</i>]
Russian		
IncepPredMinusMagn(<i>dožd'</i>)	=	<i>stixat'</i>
IncepPredMinusMagn(<i>muskul</i>)	=	<i>atrofirovat'sja</i>
IncepPredMinusMagn(<i>temperatura</i>)	=	<i>padat', snižat'sja</i>
IncepPredMinusVer(<i>nebo</i>)	=	<i>pokryvat'sja / zavolaktivat'sja</i> <i>oblakami</i>
MinusMagn+IncepOper ₁ (<i>ves</i>)	=	<i>terjat'</i> [<i>~</i>]

5.3 Prepositional collocates

[40] **Loc_{in}** [Latin *locus* 'place']: position locative preposition

[41] **Loc_{ad}**: destination locative preposition

[42] **Loc_{ab}**: origin locative preposition

Loc_{in}(L), Loc_{ad}(L) and Loc_{ab}(L) are prepositions that take the noun L as complement and mean:

- Loc_{in}(L) 'in L': 'having L as location';
- Loc_{ad}(L) 'into L': 'having L as destination';
- Loc_{ab}(L) 'out_of L': 'having L as origin'.

In these glosses the notion of location is to be understood in the vaguest, often metaphorical way. Hence, the LFs Loc_{in/ad/ab} describe both spatial and metaphorical (temporal and abstract) localizations.

The three LFs semantically correspond to three grammatical cases, which are found, for instance, in Hungarian and Finnish: locative for Loc_{in}, allative for Loc_{ad} and ablative for Loc_{ab}.

Loc _{in/ad/ab} : adverbial	
L	N
Loc _{in/ad/ab} (L)	Adv

The value of Loc_{in/ad/ab}(L), which is a preposition, is described at the deep-syntactic level as a particular type of Adverb [Adv] (cf. Chapter 3, Section 3.5.1).

5 Syntagmatic lexical functions

English

Loc _{in} (<i>France</i>)	=	<i>in</i> [~]
Loc _{ad} (<i>France</i>)	=	<i>to</i> [~]
Loc _{in} (<i>London</i>)	=	<i>in</i> [~]
Loc _{ad} (<i>London</i>)	=	<i>to</i> [~]
Loc _{in} (<i>face</i> _(N))	=	<i>on</i> [ART ~]
Loc _{ad} (<i>face</i> _(N))	=	<i>in</i> [ART ~]
Loc _{in/ad} (<i>imagination</i>)	=	<i>in</i> [ART ~]

French

Loc _{in/ad} (<i>France</i>)	=	<i>en</i> [~]
Loc _{in/ad} (<i>Canada</i>)	=	<i>à</i> [<i>le</i> ~] [⇒ <i>au Canada</i>]
Loc _{in/ad} (<i>Avignon</i>)	=	<i>à</i> , <i>en</i> [~]
Loc _{in} (<i>visage</i>)	=	<i>sur</i> [ART ~]
Loc _{ad} (<i>visage</i>)	=	<i>dans</i> [ART ~] < « <i>en plein</i> » [~]
Loc _{in/ad} (<i>travail</i>)	=	<i>à</i> [<i>le</i> ~] [⇒ <i>au travail</i>] < « <i>en plein</i> » [~]
Loc _{in/ad} (<i>imagination</i>)	=	<i>dans</i> [ART ~], <i>en</i> [~]
Loc _{in/ad} (<i>rue</i>)	=	Ø _(Prep, loc) 1 [~ → N _{'street name'}] [Ex.: <i>se trouver rue Balzac / rue du Bac</i>]; <i>dans</i> , <i>sur</i> [ART ~]

Note. Ø_(Prep, loc) **1** is a locative zero preposition – see Mel'čuk (2018).

Russian

Loc _{in} (<i>Franci ja</i>)	=	<i>vo 1</i> [~i]
Loc _{ad} (<i>Franci ja</i>)	=	<i>vo 2</i> [~ju]
Loc _{in} (<i>London</i>)	=	<i>v 1</i> [~e]
Loc _{ad} (<i>London</i>)	=	<i>v 2</i> [~]
Loc _{in} (<i>lic o</i>)	=	<i>na 1</i> [~e]
Loc _{ad} (<i>lic o</i>)	=	<i>v 2</i> [~o]
Loc _{in} (<i>ulic a</i>)	=	<i>na 1</i> [~e]
Loc _{ad} (<i>ulic a</i>)	=	<i>na 2</i> [~u]
Loc _{in} (<i>vokzal</i>)	=	<i>na 1</i> [~e]
Loc _{ad} (<i>vokzal</i>)	=	<i>na 2</i> [~]

Note. The prepositions *na 1* / *na 2* and *v(o) 1* / *v(o) 2* are different lexemes of the vocables *na* and *v(o)*. *na 1* and *v(o) 1* govern the locative case and indicate a location; *na 2* and *v(o) 2* govern the accusative case and indicate a destination.

As one can see from the above illustrations, French systematically does not distinguish values for Loc_{in} and Loc_{ad}. (This is a common feature of Romance languages.) The distinction is, however, quite relevant in other languages – in the first place, in those that have grammatical cases.

Temporal localization, as mentioned earlier, is covered by Loc_{in} , just like physical localization:

English

$\text{Loc}_{\text{in}}(\textit{weather}_{(\text{N})})$	=	<i>in</i> [$\sim$ with a modifier]
$\text{Loc}_{\text{in}}(\textit{reign}_{(\text{N})})$	=	<i>under</i> [ART $\sim$]
$\text{Loc}_{\text{in}}(\textit{antiquity})$	=	<i>in</i> [<i>the</i> $\sim$]
$\text{Loc}_{\text{in}}(\textit{Middle Ages})$	=	<i>in</i> [<i>the</i> $\sim$]
$\text{Loc}_{\text{in}}(\textit{epoch})$	=	<i>in</i> [ART $\sim$]
$\text{Loc}_{\text{in}}(\textit{September})$	=	<i>in</i> [$\sim$]
$\text{Loc}_{\text{in}}(\textit{Monday})$	=	$\emptyset_{(\text{Prep, temp})}$, <i>on</i> [$\sim$]

French

$\text{Loc}_{\text{in}}(\textit{temps})$	=	<i>par</i> [ART $\sim$ with a modifier]
$\text{Loc}_{\text{in}}(\textit{règne})$	=	<i>pendant, sous</i> [ART $\sim$]
$\text{Loc}_{\text{in}}(\textit{Antiquité})$	=	<i>dans</i> [<i>l'</i> $\sim$]
$\text{Loc}_{\text{in}}(\textit{Moyen-Âge})$	=	<i>à</i> [<i>le</i> $\sim$] [$\Rightarrow$ <i>au Moyen-Âge</i>]
$\text{Loc}_{\text{in}}(\textit{époque})$	=	<i>à</i> [ART $\sim$]
$\text{Loc}_{\text{in}}(\textit{septembre})$	=	<i>en</i> [$\sim$]
$\text{Loc}_{\text{in}}(\textit{lundi})$	=	$\emptyset_{(\text{Prep, temp})} \mathbf{2}$ [(ART) $\sim$] [Ex.: <i>arriver</i> (ART) <i>lundi</i>]

Note. $\emptyset_{(\text{Prep, temp})} \mathbf{2}$ is the temporal counterpart of the locative zero preposition $\emptyset_{(\text{Prep, loc})} \mathbf{1}$, which was commented upon in the preceding illustration table.

Russian

$\text{Loc}_{\text{in}}(\textit{pogoda} a)$	=	$\mathbf{v2}$ [$\sim u$ with a modifier]
$\text{Loc}_{\text{in}}(\textit{carstvovani} e)$	=	$\mathbf{v2}$ [$\sim e$]
$\text{Loc}_{\text{in}}(\textit{antičnost} ')$	=	$\mathbf{v1}$ [$\sim i$]
$\text{Loc}_{\text{in}}(\textit{Sredni} e \textit{Vek} a')$	=	$\mathbf{v2}$ [$\sim e \sim a'$]
$\text{Loc}_{\text{in}}(\textit{èpox} a)$	=	$\mathbf{v2}$ [$\sim u$]
$\text{Loc}_{\text{in}}(\textit{sentjabr} ')$	=	$\mathbf{v1}$ [$\sim e$]
$\text{Loc}_{\text{in}}(\textit{ponedel'nik})$	=	$\mathbf{v2}$ [$\sim$]



Historical remark

Loc_{in} expressing temporal localization was previously encoded as $\text{Loc}_{\text{in}}^{\text{temp}}$.

[43] **Instr** [Latin *instrūmentum* ‘instrument’]: instrumental preposition

Instr(L) is a preposition that means ‘[do something] by means of L’ and takes the noun L as DSyntA II.

Instr: adverbial	
L	N
Instr(L)	Adv

English

Instr(<i>knife</i> _(N))	=	<i>with</i> [ART ~]
Instr(<i>car</i>)	=	<i>by</i> [~], // <i>driving</i>
Instr(<i>plane</i> _(N))	=	<i>aboard</i> [ART ~], <i>by</i> [~], // <i>flying</i>
Instr(<i>horse</i> _(N))	=	// ‘ <i>on horseback</i> ’, // <i>riding</i>
Instr(<i>mule</i>)	=	<i>on</i> [ART ~]
Instr(<i>mail</i> _(N))	=	<i>by</i> [~]

French

Instr(<i>couteau</i>)	=	<i>à</i> [le ~] [⇒ <i>au couteau</i>], <i>avec</i> [ART ~]
Instr(<i>voiture</i>)	=	<i>en</i> [~]
Instr(<i>avion</i>)	=	<i>en</i> , <i>par</i> [~]
Instr(<i>cheval</i>)	=	‘ <i>à dos</i> ’ [de ~] syntactic governor means ‘transport [something]’; <i>à</i> [~] syntactic governor means ‘go [somewhere]’
Instr(<i>mulet</i>)	=	‘ <i>à dos</i> ’ [de ~]
Instr(<i>courrier</i>)	=	<i>par</i> [~]

Russian

Instr(<i>nož</i>)	=	// ~om
Instr(<i>mašin</i> a)	=	<i>na</i> 1 [~e], // ~oj
Instr(<i>samolët</i>)	=	<i>na</i> 1 [~e], // ~om
Instr(<i>lošad</i> ’)	=	(<i>verxom</i>) <i>na</i> 1 [~i], // <i>verxom</i>
Instr(<i>mul</i>)	=	(<i>verxom</i>) <i>na</i> 1 [~e]
Instr(<i>počt</i> a)	=	<i>po</i> [~e]

The syntagmatic LF Instr is semantically equivalent to the complex paradigmatic LF Adv₁Real_i: Instr(L) = Adv₁Real_i(L).³ This equality is exemplified in (13), using one of the English illustrations for Instr above: the value of Instr(*car*) combined with *car* is semantically equivalent to Adv₁Real₁(*car*) semantic derivative.

³For Adv₁, see Chapter 4, [24], p. 104; for Real_i, see [49] below, p. 147.

- (13) Instr(*car*) = *by* [~], // *driving*
 Adv₁Real₁(*car*) = *by* ~, *driving*

[44] **Propt_i** [Latin *propter* ‘because of’]: causal preposition

Propt_i(L) is a preposition that means ‘[P takes place] being caused by L’. Its position in the syntactic structure of the sentence is as follows:

- Propt_i(L) takes the noun L as DSyntA II;
- it is itself an **ATTR** syntactic dependent of a predicative lexical unit P;
- P takes as DSyntA I L’s DSyntA i.

Figure 5.1 illustrates the participation of Propt_i in the Sem-structure $\Leftrightarrow$ DSynt-structure correspondence with the case of Propt₁(*boredom*) used in sentence (14):

- (14) *Rachel dated Joey out_of_{Propt₁(boredom)} boredom.*

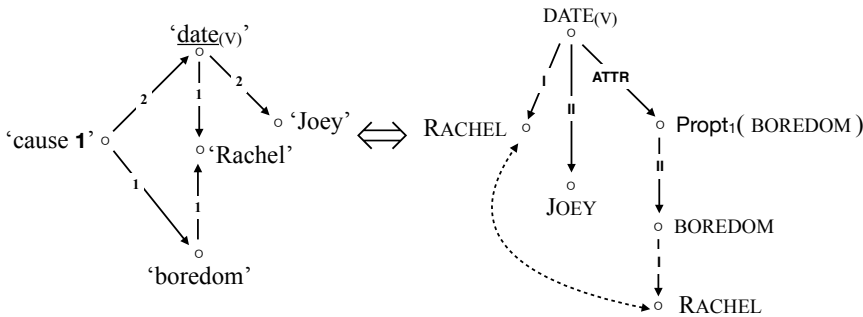


Figure 5.1: Illustration of Propt₁ in the Sem $\Leftrightarrow$ DSynt correspondence⁴

Because of its meaning, Propt_i is naturally related to the LF Caus (Section 5.4.5, [57], p.157), such that it is possible to write the following deep-syntactic paraphrasing rule for Propt₁:

R_{Propt₁}: $L_{(V)} - \text{ATTR} \rightarrow \text{Propt}_i(L) - \text{II} \rightarrow L \equiv L \leftarrow \text{I} - \text{CausFunc}_0(S_0(L_{(V)})) - \text{II} \rightarrow S_0(L_{(V)})$ <i>The bar closed for_{Propt₁(lack_(N))} lack of customers.</i> $\equiv$ <i>The lack of customers lead_{CausFunc₀(S₀(close_(V)))} to the closure_{S₀(close_(V))} of the bar.</i>

⁴On the non-agentive ‘cause 1’ vs. the agentive ‘cause 2’, see “Verbs of causing,” p.159.

5 Syntagmatic lexical functions

Propt _i : adverbial	
L	N
Propt _i (L)	Adv

English

Propt ₁ (<i>interest</i> _(N))	=	<i>out of</i> [ART ~]
Propt ₁ (<i>alcohol</i>)	=	<i>under the effect</i> [of ~]; // <i>under the influence</i> ⁷
Propt ₁ (<i>combat</i> _(N))	=	<i>in</i> [~]
Propt ₁ (<i>heart attack</i> ⁷)	=	<i>of</i> [(ART) ~]
Propt ₁ (<i>lack</i> _(N))	=	<i>for</i> [~]
[N _X 's <i>lack of</i> N _Y]		
Propt ₂ (<i>attack</i>)	=	<i>in</i> [ART ~]

French

Propt ₁ (<i>intérêt</i>)	=	<i>par</i> [~]
Propt ₁ (<i>alcool</i>)	=	<i>sous l'emprise</i> [de ART ~]
Propt ₁ (<i>combat</i>)	=	<i>à</i> [le ~] [⇒ <i>au combat</i>]
Propt ₁ (<i>crise cardiaque</i> ⁷)	=	<i>de</i> [(ART) ~]
Propt ₂ (<i>manque</i>)	=	<i>par</i> [~]
[<i>manque de</i> N _X <i>pour</i> N _Y]		
Propt ₂ (<i>attentat</i>)	=	<i>dans</i> [ART ~]

Russian

Propt ₁ (<i>interes</i>)	=	<i>iz</i> [~a]
Propt ₁ (<i>alkogol</i> ')	=	<i>pod vozdejstviem</i> [~ja]
Propt ₁ (<i>boj</i>)	=	<i>v</i> 1 [~u]
Propt ₁ (<i>infarkt</i>)	=	<i>ot</i> [~a]
Propt ₂ (<i>otsutstvi</i> e)	=	<i>iz-za</i> [~ja]
[<i>otsutstvie</i> N _{X-GEN} <i>u</i> N _{Y-GEN}]		
Propt ₂ (<i>pokušeni</i> e)	=	<i>v rezul'tate</i> [~ja]

The syntagmatic LF Propt₁, which includes causation, should not be mistaken for the non-causative paradigmatic LF Adv₁; cf. the following contrast:

- (15) Adv₁(*affection*) = *affectionately, with* ~
 vs.
 Propt₁(*affection*) = *out_of* [~]

See also, on Adv_i, Chapter 4, Section 4.7, [24], p.104. For a detailed study of Propt_i, see Iordanskaja & Mel'čuk (2007: 503–617).

5.4 Verbal collocates

5.4.1 Copula verb

[45] **Copul** [Latin *copula*]: verb that means ‘be’

Copul(L) is a verb that means ‘be_(copula)’ and applies to a nominal or adjectival / adverbial lexical unit L. It takes L as DSyntA II; its DSyntA I is the DSyntA I of L.

Copul: verbal			
L	N	Adj	Adv
Copul(L)	V	V	V

English

Copul(<i>example</i>)	=	<i>be</i> [ART ~], <i>serve</i> [as an ~]
Copul(<i>challenge</i> _(N))	=	<i>be</i> , <i>constitute</i> , <i>represent</i> [ART ~]
Copul(<i>true</i>)	=	<i>be</i> , <i>hold</i> [~]
IncepCopul(<i>dead</i> _(Adj))	=	<i>drop</i> [~]

French

Copul(<i>exemple</i>)	=	<i>être</i> [ART ~], <i>servir</i> [d’~]
Copul(<i>défi</i>)	=	<i>être</i> , <i>constituer</i> , <i>représenter</i> [ART ~]
IncepCopul(<i>amoureux</i> _(Adj))	=	<i>tomber</i> [~]

Russian

Copul(<i>primer</i>)	=	<i>byt’</i> , <i>služit’</i> [~om]
IncepCopul(<i>zamertvo</i>)	=	<i>upast’</i> [~]
IncepCopul(<i>vljublënnij</i>)	=	// <i>vljubit’sja</i>

Copul is an analytic variant of the paradigmatic LF Pred (Chapter 4, [12], p. 92), so that the following equivalence holds: Copul(L) $\rightarrow$ L $\equiv$ Pred(L). For instance:

- (16) Her declaration was_{Copul(*unexpected*)} unexpected.
 $\equiv$
 Her declaration ‘came as a bolt from the blue’_{Pred(*unexpected*)}.

Copul does not have many values in any given natural language. It is mainly used within complex LFs, as in IncepCopul, etc.

5.4.2 Support [= light] verbs

All LFs f introduced in this section are “pure” verbalizers. They are such that:

- f applies to a non-verbal lexical unit L , and $f(L)$ takes L as DSyntA I (i.e., subject) or DSyntA II / III / ... (i.e., object);
- $f(L)$ is not selected by the Speaker to express a given meaning, but in order to construct a well-formed verbal syntactic structure with the same meaning as L .

Each $f(L)$ of this type, so to speak, “supports” a non-verbal L , by enabling the collocation $f(L) \rightarrow L$ to function as the syntactic head of the clause; hence the term *support verb* (Gross 1981). Because support verbs are semantically “light,” i.e. they do not semantically contribute to the sentence, they are also referred to as *light verbs*, a term generally attributed to Jespersen (1946: Ch. VII, § 7.21–5).



A support verb is semantically empty in the collocation it forms with L in actual utterances. But this does not prevent it from having an inherent meaning in the lexicon, this meaning becoming redundant with respect to that of L . Such is the case, for instance, of the verb $\text{FEEL}_{(V)}$ in the collocation *feel pain*, the verb $\text{FACE}_{(V)}$ in *face an ultimatum*, etc.

There are three types of support verbs according to the position of the collocation base L in their DSynt-actantial structure: Oper_i , Func_i and Labor_{ij} . These three LFs have the same PoS table:

$\text{Oper}_i, \text{Func}_i, \text{Labor}_{ij}$: verbal			
L	N	Adj	Adv
$\text{Oper}_i / \text{Func}_i / \text{Labor}_{ij}(L)$	V	V	V

A typical support verb collocation is controlled by a nominal base; almost all examples that follow present this type of collocation. However, adjectival and adverbial bases for such collocations are limitedly possible; a few illustrations are provided at the end of this section.

A support verb LF, being empty by definition, cannot be semantically characterized. We will, thus, supply for each LF described below – Oper_i , Func_i and Labor_{ij} – a prototypical lexical equivalent, so that the reader can have an intuitive feeling of its usage (cf. $\text{DO}_{(V)}$ for Oper_i).

[46] Oper_i [Latin *operāri* ‘do, carry out’]: support verb analogous to $\text{DO}_{(V)}$

$\text{Oper}_i(L)$ is a support verb that takes L as DSyntA II and the DSyntA I of L as DSyntA I .

Oper_i participates in the $\text{Sem} \Leftrightarrow \text{DSynt}$ correspondence shown in Figure 5.2.

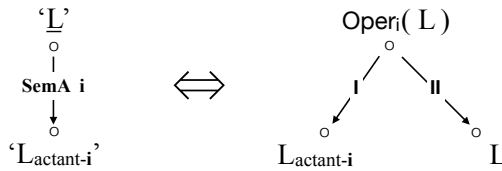


Figure 5.2: Oper_i in the $\text{Sem} \Leftrightarrow \text{DSynt}$ correspondence

Figure 5.3 illustrates the correspondence template in Figure 5.2 for Oper_1 with $L = \text{FEVER}$.

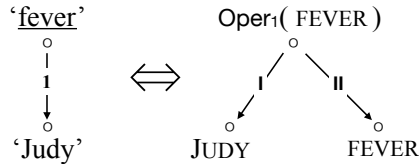


Figure 5.3: Oper_1 in the $\text{Sem} \Leftrightarrow \text{DSynt}$ correspondence with $L = \text{FEVER}$

Sentences (17a–b) are possible implementations of the DSynt -structure given in Figure 5.3:

- (17) a. Judy *has* a fever.
b. Judy *runs* a fever.

$\text{Oper}_i(L)$ can be an impersonal support verb, which has no DSyntA I . In such a case, the notation $\text{Oper}_0(L)$ is used (see French and Russian illustrations below).

English

- $\text{Oper}_1(\text{crime})$ = *commit, perpetrate* [$\text{ART} \sim$]
 $\text{Oper}_1(\text{supremacy})$ = *detain* [$\text{ART} \sim$]

5 Syntagmatic lexical functions

Oper ₂ (<i>bullying</i>)	=	<i>undergo</i> [~]
Oper ₂ (<i>danger</i>)	=	<i>run</i> [ART ~]
Oper ₂ (<i>applause</i>)	=	<i>draw, get, win</i> [~ with a modifier]
Oper ₃ (<i>order</i> _(N))	=	<i>get, receive</i> [ART ~]
[N _X 's order to V _Y to N _Z]		
Oper ₃ (<i>refusal</i>)	=	<i>meet</i> [with (ART) ~]
[N _X 's refusal to V _Y to N _Z]		

French

Oper ₀ (<i>nuit</i>)	=	[<i>il</i> _(impers)] <i>faire</i> [~]
Oper ₀ (<i>soif</i>)	=	informal [<i>il</i> _(impers)] <i>faire</i> [~]
Oper ₁ (<i>méfait</i>)	=	<i>perpétrer</i> [ART ~]
Oper ₁ (<i>suprémie</i>)	=	<i>détenir</i> [ART ~]
Oper ₂ (<i>calomnie</i>)	=	<i>subir</i> [ART ~]
Oper ₂ (<i>danger</i>)	=	<i>courir</i> [ART ~]
Oper ₂ (<i>applaudissements</i>)	=	<i>recueillir</i> [ART ~]
Oper ₃ (<i>consensus</i>)	=	<i>faire</i> [~]
[consensus entre N _X et N _Y à propos de N _Z]		
Oper ₃ (<i>refus</i>)	=	<i>essuyer</i> [ART ~], <i>se heurter</i> [à ART ~]
[refus de N _X de V _{Y-INF} à N _Z]		

Russian

Oper ₀ (<i>zapax</i>)	=	<i>tjanut'</i> [~om]
Oper ₀ (<i>proxlad a</i>)	=	<i>vejat'</i> [~oj]
Oper ₁ (<i>prestupleni e</i>)	=	<i>soveršit'</i> [~e]
Oper ₁ (<i>prevosxodstv o</i>)	=	<i>imet'</i> [~o]
Oper ₂ (<i>izdevatel'stv a</i>)	=	<i>podvergat'sja</i> [~am]
Oper ₂ (<i>opasnost ')</i>	=	<i>podvergat'sja</i> [~i]
Oper ₂ (<i>aplodisment y</i>)	=	<i>byt' vstrečen</i> [~ami]
Oper ₃ (<i>prikaz</i>)	=	<i>polučit'</i> [~]
[prikaz N _{X-GEN} N _{Z-DAT} V _{Y-INF}]		
Oper ₃ (<i>otkaz</i>)	=	<i>natolknut'sja</i> [naz ~], <i>polučit'</i> [~]
[otkaz N _{X-GEN} N _{Z-DAT} V _{Y-INF}]		

Oper_i is the most frequent among support verbs. This is explained by the following fact: the base of an Oper_i collocation is in most cases the direct object of the verb, and semantic constraints are the strongest between a transitive verb and its direct object – stronger than between a verb and its other actants. It is for this reason that natural languages phraseologize in the first place phrases of the form V –DirO→ N.

Understandably, Oper_i has the pride of place in paraphrasing systems of natural languages, as illustrated by the following two cases.

1. $\text{Oper}_1(\text{S}_0(\text{L}_{(V)})) \dashv\!\!\dashv\!\!\rightarrow \text{S}_0(\text{L}_{(V)})$ is a paraphrase of $\text{L}_{(V)}$:

(18) Joey joked.
 $\equiv$
 Joey cracked a joke.

On such equivalence, see syntactic paraphrasing rule $\mathbf{R}_{\text{Oper}_1}$, introduced in Chapter 3, Section 3.2.3.3 (p. 44).

2. Oper_2 is in the same paraphrastic relation – i.e., syntactic conversion – with Oper_1 as found between the passive and the active voice of a verb, cf. (19a–b):

(19) a. Rachel gave _{Oper_1} a kiss to Joey.
 $\equiv$
 Rachel kissed_{active voice} Joey.
 b. Joey got _{Oper_2} a kiss from Rachel.
 $\equiv$
 Joey was kissed_{passive voice} by Rachel.

Transfer of L's actants to support verbs. In collocations $\text{Oper}_i(\text{L}) \dashv\!\!\dashv\!\!\rightarrow \text{L}$, the DSyntAs of L can migrate – i.e., be transferred under paraphrasing – to $\text{Oper}_i(\text{L})$. In other words, $\text{Oper}_i(\text{L})$ can host, besides its own two obligatory DSyntAs, additional DSyntAs coming from L. This transfer of actants is known as (*syntactic*) *raising* – see Chapter 2, Section 2.3, p. 30. It is illustrated in (20) with $\text{Oper}_1(\text{crime})$.⁵

(20) Joey committed _{Oper_1} a terrible crime $\dashv\!\!\dashv\!\!\rightarrow$ *against decency*.
 $\cong$
 a terrible crime that Joey committed $\dashv\!\!\dashv\!\!\rightarrow$ *against decency*

The transfer of actants from the collocation base to the verbal collocate may be performed for all types of support verbs LF_s – Oper_i , Func_i and Labor_{ij} , as well as for corresponding complex LF_s (for instance, CausFunc_0).

⁵The two utterances in (20) are semantically quasi-equivalent: they express the same propositional meaning, but have different communicative organizations.



Further reading

On support / light verbs, see Gross (1981), Mel'čuk (2022) and Fleischhauer & Riccio (2025); on transfer of actants from a collocation base to the collocative verb, see Alonso Ramos (2004: 253–272, 2007).

[47] **Func_i** [Latin **functionāre* '[to] function']: support verb analogous to 'TAKE PLACE'

Func_i(L) is a support verb that takes L as DSyntA_I and the DSyntA_i of L as DSyntA_{II}. However, it can also have only one actantial dependent: L as DSynt_I. In such a case – an absolutely intransitive construction – the LF is denoted as Func₀.

Func_i participates in the Sem ⇔ DSynt correspondence shown in Figure 5.4; it is to be compared to the Oper_i(L) correspondence in Figure 5.2 (p. 139).

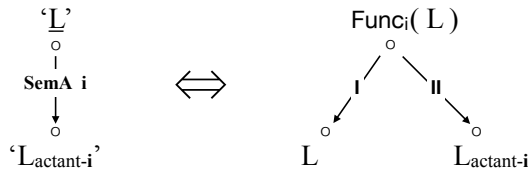


Figure 5.4: Func_i in the Sem ⇔ DSynt correspondence

Figure 5.5 illustrates the correspondence template in Figure 5.4 for Func₁ with L = RESPONSIBILITY.

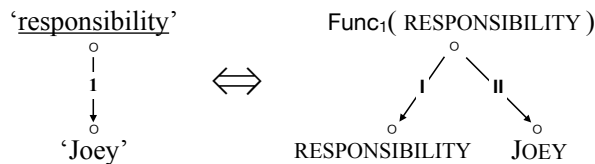


Figure 5.5: Func₁ in the Sem ⇔ DSynt correspondence with L = RESPONSIBILITY

Sentences (21a–b) are possible implementations of the DSynt-structure given in Figure 5.5:

- (21) a. The responsibility *lies* with Joey.
 b. The responsibility *rests* with Joey.

English

Func ₀ (<i>thunder</i> _(N))	=	<i>clap</i> _(V) , <i>crash</i> _(V) , <i>roll</i> _(V) , <i>rumble</i> _(V)
Magn+Func ₀ (<i>fire</i> _(N) [disaster])	=	<i>rage</i> _(V)
Magn+Func ₁ (<i>fire</i> _(N) [disaster])	=	<i>rip</i> _(V) , <i>sweep</i> _(V) [through N _X]
Func ₁ (<i>responsibility</i> [of N _X])	=	<i>lie</i> _(V) , <i>rest</i> _(V) [with N _X]
Func ₁ (<i>smell</i> _(N) [of N _X])	=	<i>come</i> _(V) [from N _X]
Func ₂ (<i>danger</i> [for N _Y])	=	<i>threaten</i> [N _Y]
Func ₂ (<i>accusation</i> [against N _Y])	=	<i>hang</i> _(V) [over N _Y]
Func ₃ (<i>conversation</i>)	=	<i>be</i> [about N _Z]
[<i>conversation between N_X and N_Y about N_Z</i>]		

French

Func ₀ (<i>tonnerre</i>)	=	<i>gronder</i> , <i>retentir</i>
Magn+Func ₀ (<i>incendie</i>)	=	「 <i>faire rage</i> 」
Func ₁ (<i>responsabilité</i> [de N _X])	=	<i>incomber</i> [à N _X]
Func ₁ (<i>odeur</i> [de N _X])	=	<i>se dégager</i> , <i>émaner</i> [de N _X]
Func ₂ (<i>danger</i> [pour N _Y])	=	<i>menacer</i> [N _Y]
Func ₂ (<i>accusation</i> [contre N _Y])	=	<i>frapper</i> [N _Y]
Func ₃ (<i>conversation</i>)	=	<i>porter</i> [sur N _Z]
[<i>conversation entre N_X et N_Y sur N_Z</i>]		

Russian

Func ₀ (<i>grom</i>)	=	<i>gremet'</i> , <i>gromyxať</i>
Magn+Func ₀ (<i>požar</i>)	=	<i>buševat'</i>
Func ₁ (<i>otvetstvennost'</i> [N _{X-GEN}])	=	<i>ležat'</i> [na 1 N _{X-PREPOS}]
Func ₁ (<i>zapax</i> [N _{X-GEN}])	=	<i>idti</i> , <i>isxodit'</i> [ot N _{X-GEN}]
Func ₂ (<i>opasnost'</i> [dlja N _{Y-GEN}])	=	<i>grozit'</i> , <i>ugrožat'</i> [N _{Y-DAT}]
Func ₂ (<i>obvinenie</i> [protiv N _{Y-GEN}])	=	<i>navisat'</i> [nad N _{Y-INSTR}]
Func ₃ (<i>razgovor</i>)	=	<i>idti</i> [o N _{Z-PREPOS}], <i>vertet'sja</i> [vokrug N _{Z-GEN}]
[<i>razgovor meždu N_{X-INSTR} i N_{Y-INSTR} o N_{Z-PREPOS}</i>]		

[48] **Labor_{ij}** [Latin *laborāre* 'to work']: support verb analogous to SUBMIT
 [N₁ to N₂]

Labor_{ij}(L) is a support verb that takes L as its most oblique [= most peripheral] DSyntA. Its DSyntA I corresponds to L's DSyntA i and its DSyntA II corresponds to L's DSyntA j.

5 Syntagmatic lexical functions

Labor_{ij} participates in the Sem $\Leftrightarrow$ DSynt correspondence shown in Figure 5.6; Figure 5.7 illustrates the correspondence template in Figure 5.6 for Labor₁₂ with L = APPLAUSE.

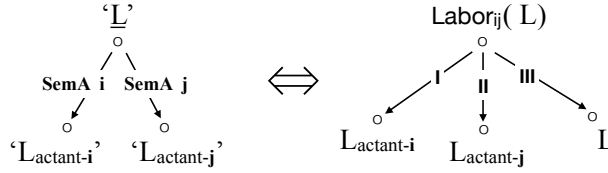


Figure 5.6: Labor_{ij} in the Sem $\Leftrightarrow$ DSynt correspondence

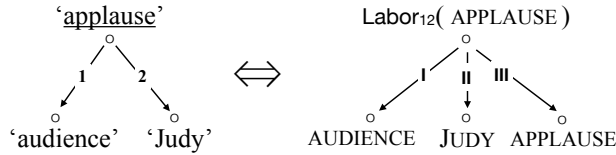


Figure 5.7: Labor₁₂ in the Sem $\Leftrightarrow$ DSynt correspondence with L = APPLAUSE

The DSynt-structure given in Figure 5.7 is expressed by sentence (22):

(22) The audience *meets* Judy with applause.

English

Labor ₁₂ (<i>esteem</i> _(N) [of N _X for N _Y])	=	<i>hold</i> [N _Y in ~]
Labor ₁₂ (<i>use</i> _(N) [by N _X of N _Y])	=	<i>put</i> [N _Y to ~]
Labor ₂₁ ([N _X ,] <i>result</i> _(N) [of N _Y])	=	<i>have</i> [N _X as a ~]

French

Labor ₁₂ (<i>estime</i> [de N _X pour N _Y])	=	<i>avoir, tenir</i> [N _Y en ~]
Labor ₁₂ (<i>soins</i> [de N _X destinés à N _Y])	=	<i>entourer</i> [N _Y de ~]
Labor ₂₁ ([N _X ,] <i>résultat</i> [de N _Y])	=	<i>avoir</i> [N _X comme ~]

Russian

Labor ₁₂ (<i>vlast</i> [N _{X-GEN} nad N _{Y-INSTR}])	=	<i>deržat'</i> [N _{ACC} pod (svoej) ~'ju]
Labor ₁₂ (<i>zabot</i> a [N _{X-GEN} o N _{Y-PREPOS}])	=	<i>okružít'</i> [N _{Y-ACC} ~oj]
Labor ₂₁ ([N _{X-NOM} ,] <i>rezul'tat</i> [N _{Y-GEN}])	=	<i>imet'</i> [N _{X-ACC} ~om < v kačestve ~a>]

Figure 5.6 and the above illustrations describe the prototypical case, where the argument L of Labor_{ij} is bi-actantial. In such a case, Labor_{ij}(L) governs three

DSyntAs, L filling the most oblique DSynt-slot. However, there are Ls with more than two DSyntAs. As a result of the transfer⁶ of L's DSyntAs to the support verb, one obtains Labor verbs with more than three DSyntAs. Take, for instance, the noun INHERITANCE 'belongings Y that X inherits from Z', and its French and Russian counterparts: they control three four-actantial Labor collocates, as illustrated below.

English

Labor ₁₂₃ (<i>inheritance</i>)	=	<i>receive</i> [$N_{Y \leftrightarrow II}$ <i>from</i> $N_{Z \leftrightarrow III}$ <i>as</i> $\sim_{L \leftrightarrow IV}$]
Labor ₃₂₁ (<i>inheritance</i>)	=	<i>leave</i> _(V) [$N_{Y \leftrightarrow II}$ <i>to</i> $N_{X \leftrightarrow III}$ <i>as</i> $\sim_{L \leftrightarrow IV}$]
Labor ₂₁₃ (<i>inheritance</i>)	=	<i>come</i> _(V) [<i>to</i> $N_{X \leftrightarrow II}$ <i>from</i> $N_{Z \leftrightarrow III}$ <i>as</i> $\sim_{L \leftrightarrow IV}$]

French

Labor ₁₂₃ (<i>héritage</i>)	=	<i>obtenir, recevoir</i> [$N_{Y \leftrightarrow II}$ <i>de</i> $N_{Z \leftrightarrow III}$ <i>en</i> $\sim_{L \leftrightarrow IV}$]
Labor ₃₂₁ (<i>héritage</i>)	=	<i>laisser, léguer</i> [$N_{Y \leftrightarrow II}$ <i>à</i> $N_{X \leftrightarrow III}$ <i>en</i> $\sim_{L \leftrightarrow IV}$]
Labor ₂₁₃ (<i>héritage</i>)	=	specialized <i>échoir, venir</i> [<i>à</i> $N_{X \leftrightarrow II}$ <i>de</i> $N_{Z \leftrightarrow III}$ <i>en</i> $\sim_{L \leftrightarrow IV}$]

Russian

Labor ₁₂₃ (<i>nasledstv o</i>)	=	<i>polučit'</i> [$N_{-ACC} Y \leftrightarrow II$ <i>ot</i> $N_{-GEN} Z \leftrightarrow III$ $v2 \sim o_{L \leftrightarrow IV}$]
Labor ₃₂₁ (<i>nasledstv o</i>)	=	<i>ostavit'</i> [$N_{-ACC} Y \leftrightarrow II$ $N_{-DAT} X \leftrightarrow III$ $v2 \sim o_{L \leftrightarrow IV}$]
Labor ₂₁₃ (<i>nasledstv o</i>)	=	<i>dostat'sja</i> [$N_{-DAT} X \leftrightarrow II$ <i>ot</i> $N_{-GEN} Z \leftrightarrow III$ $v2 \sim o_{L \leftrightarrow IV}$]

As we have seen, a collocational support verb Labor_{ij} relegates the collocation's base L to the most oblique syntactic role: thus, L occupies the least privileged syntactic position, and this is anti-iconic. As a consequence, cases of Labor_{ij} are the least frequent ones among support verbs.

It is important to emphasize that the three support verb LFs are syntactically converse to each other, so that $Func_1(L) = Conv_{21}Oper_1(L)$, etc.:

- (23) a. $Oper_1(analysis)$ = 'carry out', *conduct*_(V), *make*_(V) [$ART \sim$]
 $Oper_2(analysis)$ = *undergo* [$ART \sim$]
 b. $Func_1(analysis)$ = *come*_(V) [*from* N_X]
 $Func_2(analysis)$ = *concern*_(V) [N_Y]
 c. $Labor_{12}(analysis)$ = *subject*_(V) [N_Y *to* $ART \sim$]

To round off the present section, we will consider three cases of support verb collocations having a non-nominal base.

1. In SAE languages, all adjectives used as attributive complement of a copula verb have the same "joker" $Oper_1$ support verb; for English, French and Russian: BE_(copula), ÊTRE_(copula) and BYT'_(copula).

⁶That is, syntactic raising – see above, "Support [= light] verbs," p. 141.

5 Syntagmatic lexical functions

$\text{Oper}_1(\text{L}_{(\text{Adj})})$ is equivalent to $\text{Copol}(\text{L}_{(\text{Adj})})$ – Section 5.4.1, [45], p. 137. Since the corresponding phrases are free (non-phraseologized), they are not to be considered here.

2. However, adjectives and adverbs denoting meteorological phenomena behave differently. They have a specific Oper_i : an impersonal Oper_0 .

English

$\text{Oper}_0(\text{hot} / \text{misty}) = [\text{it}_{(\text{impers})}] \text{be}_{(\text{meteo})} [\sim]$
It was hot / misty in Montreal.

French

$\text{Oper}_0(\text{chaud}) = [\text{il}_{(\text{impers})}] \text{faire}_{(\text{meteo})} [\sim]$
Il faisait chaud à Montréal.

$\text{Oper}_0(\text{brumeux}) = [\text{il}_{(\text{impers})}] \text{faire}_{(\text{meteo})}, [\text{ce}_{(\text{impers})}] \text{être}_{(\text{meteo})} [\sim]$
Il faisait <C'était> brumeux à Montréal.

Russian

$\text{Oper}_0(\text{žarko} / \text{tumanno}) = [\emptyset_{(\text{impers})}] \text{byt}'_{(\text{meteo})} [\sim]$
V Monreale bylo žarko / tumanno.

3. 'Prep → N' adverbial idioms having several Oper_1 collocates are not uncommon, as illustrated by the English idiom 'ON CLOUD NINE' 'extremely happy':⁷

(24) $\text{Oper}_1(\text{'on cloud nine'}) = \text{be}, \text{float}_{(\text{V})}, \text{fly}_{(\text{V})} [\sim]$

5.4.3 Realization verbs

This section introduces three types of verbal collocates that are similar in their syntactic behavior to the three support verbs described in the previous section. Semantically, however, *realization verbs* are very different. They are not empty, but express a specific meaning: 'fulfill an inherent requirement of L' – in the most general sense (more specifically, 'realize', 'use', 'accomplish', etc.). This semantic load justifies the name *realization verbs*. Just as the support verb LFs, the three realization verb LFs are mutual conversives.

⁷See also Mel'čuk & Polguère (2021: 136) for a detailed analysis of the French idiom 'DANS LA SEMOULE' (lit. 'in the semolina') 'not knowing what to do in a difficult situation' and its Oper_1 collocates: *être* 'be', *nager* 'swim', *patauger* 'wade' and *pédaler* 'pedal'.

A fourth LF that is semantically related to realization verb LFs is also introduced at the end of this section: *Prepar*. It is, so to speak, a realization verb by extension.

All four verbal LFs presented in this section have the same PoS table:

Real _i , Fact _i , Labreal _{ij} , Prepar: verbal		
L	V	N
Real _i / Fact _i / Labreal _{ij} / Prepar(L)	V	V

This table shows that the realization verbs prototypically have a nominal base. The shading of the column for verbal bases indicates that these cases present some complications – see “Realization verbs” below, p. 149.

[49] Real_i [Latin *realis* ‘real’] : realization verb that means ‘fulfill [L]’

Real_i(L) is a realization verb that takes L as DSyntA II and the DSyntA i of L as DSyntA I. It controls the same syntactic structure as support verbs of the type Oper_i.

English

Real ₁ (<i>promise</i> _(N))	=	<i>keep</i> [ART ~]
Real ₂ (<i>expectations</i>)	=	‘ <i>come up</i> ’ [to ART ~], <i>fulfill</i> [ART ~]; ‘ <i>live up</i> ’ [to ART ~]
Real ₃ (<i>order</i> _(N))	=	‘ <i>carry out</i> ’, <i>execute</i> , <i>fulfill</i> [ART ~]
[N _X ’s order to V _Y to N _Z]		
Real ₁ (<i>plane</i> _(N))	=	<i>pilot</i> _(V) [ART ~]

French

Real ₁ (<i>promesse</i>)	=	<i>tenir</i> [ART ~]
Real ₂ (<i>attentes</i>)	=	<i>répondre</i> [à ART ~]
Real ₃ (<i>ordre</i>)	=	<i>exécuter</i> , <i>suivre</i> [ART ~]
[ordre de N _X de V _{Y-INF} à N _Z]		
Real ₁ (<i>avion</i>)	=	<i>piloter</i> [ART ~]
Real ₁ (‘ <i>chasse d’eau</i> ’)	=	<i>actionner</i> , <i>activer</i> , <i>tirer</i> [ART ~]

Russian

Real ₁ (<i>obeščani</i> _e)	=	<i>sderžat’</i> , <i>vypolnit’</i> [~e]
Real ₂ (<i>ožidani</i> _{ja})	=	<i>opravdat’</i> [~ja]
Real ₃ (<i>prikaz</i>)	=	<i>vypolnit’</i> [-]
[prikaz N _{X-GEN} V _{Y-INF} N _{Z-DAT}]		
Real ₁ (<i>samolët</i>)	=	<i>pilotirovat’</i> , <i>vesti</i> [~]

The argument L of *Real_i* – just as of the two other realization verbs – must contain in its lexicographic definition at least one particular semantic component which expresses a “requirement” of L and thus is compatible with the meaning of realization. It is this component that is the target for *Real_i*.⁸ Consider, for instance, the definition of the noun *DEBT*:

- (25) *X's debt of Y to Z*: sum of money Y that X got from Z
 • that X is expected to 'pay back' to Z

The emphasized component ('is expected to 'pay back') corresponds to the inherent requirement of DEBT, which is targeted in the Real₁ collocations: *clear*, 'pay back', 'pay off', *repay a debt*.

Degree of realization. The elements of the value of $\text{Real}_i(L)$ may differ by the degree of realization. Three degrees are distinguished, using the superscripts ^I, ^{II} and ^{III}:

- (26) a. $\text{Real}_1^I(\textit{suitcase}) = \textit{fill}_{(V)}, \textit{pack}_{(V)} [\text{ART} \sim]$
 $\text{Real}_1^{II}(\textit{suitcase}) = \textit{carry}_{(V)} [\text{ART} \sim]; \textit{drag}_{(V)} [\text{ART} \sim]; \textit{wheel}_{(V)} [\text{ART} \sim]$
 $\text{Real}_1^{III}(\textit{suitcase}) = \textit{empty}_{(V)}, \textit{unpack} [\text{ART} \sim]$
 b. $\text{Real}_2^I(\textit{task}_{(N)}) = \textit{tackle}_{(V)} [\text{ART} \sim]$
 $\text{Real}_2^{II}(\textit{task}_{(N)}) = \textit{fulfill} [\text{ART} \sim]$

Some details on the distinction between Real₁ and the LF Prepar are given below ([52], p. 152).

External participant as realizer. The “realizer” of $\text{Real}_i(\text{L})$ can be an external participant of the situation denoted by L (cf. Chapter 2, Section 2.2, p. 24). In such cases, a special actantial subscript Ω is used: Real_Ω . The symbol Ω indicates that the DSyntA of $\text{Real}_\Omega(\text{L})$ does not correspond to one of L’s DSyntAs . Here are a few illustrations:

- (27) $\text{Real}_{\Omega}(\text{mountain}) = \text{climb}_{(V)}, \text{scale}_{(V)} [\text{ART } \sim]$
 $\text{Real}_{\Omega}(\text{costs}_{(N)}) = \text{cover}_{(V)} [\text{ART } \sim]$
 $\text{Real}_{\Omega}(\text{baby}_{(N)}) = \text{「take care」} [\text{of ART } \sim], // \text{babysit}$
 $\text{Real}_{\Omega}(\text{pulse}_{(N)}) = \text{feel}_{(V)}, \text{take}_{(V)} [\text{ART } \sim]$

⁸See the presentation of targets of LFs in Chapter 3, Section 3.3.2, p. 49.



Historical remark

In previous lexicographic projects – in particular, in the development of the *DiCo* lexical database (Polguère 2000b) –, subscripts $\diamond$ or @ have been used in place of Ω .

Verbal argument of Real_i . Real_i can also apply to a verb. In this case, $\text{Real}_i(\text{L})$ normally gives fused values:

- (28) $\text{Real}_1(\text{approach}_{(V)} [\text{N}_Y]) = // \text{reach}_{(V)} [\text{N}_Y]$
 $\text{Real}_1(\text{look}_{(V)} [\text{for } \text{N}_Y]) = // \text{find}_{(V)} [\text{N}_Y]$
 $\text{Real}_1(\text{pursue} [\text{N}_Y]) = // \text{catch}_{(V)} [\text{N}_Y]$
 $\text{Real}_1(\text{stagger}_{(V)}) = // \text{fall}_{(V)}$
 $\text{Real}_2(\text{rock}_{(V)} [\text{N}_Y]) = // \text{fall asleep}$

These fused $\text{Real}_i(\text{L})$ should not be mistaken for cases of $\text{Result}_i(\text{L})$, which are in a *paradigmatic* relation with L (Chapter 4, [30], p. 113). The situation denoted by $\text{Real}_i(\text{L})$, contrary to $\text{Result}_i(\text{L})$, is not necessarily a result of the situation denoted by L. For instance, one can fall asleep without being rocked – cf. last illustration in (28).

Frequent combination with *Anti*. Because it is not semantically empty, Real_i – and the two other LFs of realization as well – frequently combines with *Anti*. As a result, $\text{AntiReal}_i(\text{L})$ expresses the meaning ‘do the opposite of fulfilling an inherent requirement of L’.

- (29) a. $\text{Real}_1(\text{impulse}_{(N)}) = \text{yield}_{(V)} [\text{to ART } \sim]$
 $\text{AntiReal}_1(\text{impulse}_{(N)}) = \text{control}_{(V)}, \text{resist}_{(V)} [\text{ART } \sim]$
 b. $\text{Real}_1(\ulcorner \text{red light} \urcorner) = \text{stop}_{(V)} [\text{at ART } \sim]$
 $\text{AntiReal}_1(\ulcorner \text{red light} \urcorner) = \text{jump}_{(V)}, \text{run}_{(V)} [\text{ART } \sim]$
 c. $\text{Real}_2(\text{exam}) = \text{pass}_{(V)} [\text{ART } \sim]$
 $\text{AntiReal}_2(\text{exam}) = \text{fail}_{(V)} [\text{ART } \sim]$

[50] **Fact_i** [Latin *factum* ‘action, fact’]: realization verb that means ‘[L] be fulfilled’

$\text{Fact}_i(\text{L})$ is a realization verb that takes L as $\text{DSyntA } \text{I}$ and the expression of the $\text{DSyntA } \text{I}$ of L as $\text{DSyntA } \text{II}$.

5 Syntagmatic lexical functions

This verb of realization is a converse of Real_i ; it controls the same syntactic configuration as the support verb Func_i .

Fact_0 (i.e., with $i = 0$) means that the verbal collocate does not govern an object expressing a DSyntA of L.

English

$\text{Fact}_0(\text{dream}_{(N)})$	=	'come true'
$\text{Fact}_1(\text{fright}_{(N)})$	=	paralyze $[\text{N}_X]$
$\text{Fact}_2(\text{insolence})$	=	offend, shock $_{(V)}$ $[\text{N}_Y]$

French

$\text{Fact}_0(\text{r\^e} \text{ve})$	=	se r\^ealiser
$\text{Fact}_1(\text{effroi})$	=	paralyser $[\text{N}_X]$
$\text{Fact}_2(\text{insolence})$	=	choquer, offenser, offusquer $[\text{N}_Y]$

Russian

$\text{Fact}_0(\text{me}\check{\text{c}}\text{ta})$	=	sbyt'sja
$\text{Fact}_1(\text{u}\check{\text{z}}\text{as})$	=	paralizovat' $[\text{N}_{X-\text{ACC}}]$
$\text{Fact}_2(\text{naglost'})$	=	potrjasti, vozmutit' $[\text{N}_{Y-\text{ACC}}]$

[51] **Labreal_{ij}** [hybrid of **Labor** and **Real**]: realization verb that means 'submit to the action [of L], thereby fulfilling [L]'

Labreal_{ij}(L) is a realization verb that takes L as its most oblique [= most peripheral] DSyntA. Its DSyntA_i corresponds to the DSyntA_i of L and its DSyntA_{II} corresponds to the DSyntA_j of L. Labreal_{ij} controls the same syntactic configuration as support verbs of the Labor_{ij} type.

The communicative effect of Labreal_{ij} is to give a peripheral position to the expression of L in the sentence, while communicatively favoring the expression of the actants i and j .

English

Labreal ₁₂ (microwave $_{(N)}$)	=	heat $_{(V)}$ $[\text{N}_Y \text{ in ART } \sim]$, // microwave $_{(V)}$ $[\text{N}_Y]$
Labreal ₁₂ (trap $_{(N)}$)	=	catch $_{(V)}$ $[\text{N}_Y \text{ in ART } \sim]$, // trap $_{(V)}$ $[\text{N}_Y]$
Labreal ₁₂ (imagination)	=	see $_{(V)}$ $[\text{N}_Y \text{ in (ART) } \sim]$, // imagine $[\text{N}_Y]$
Labreal ₂₃ (mailbox)	=	drop $_{(V)}$, put $[\text{N}_Z \text{ in/ into ART } \sim]$
[mailbox of N_X into which N_Y puts N_Z]		

French

Labreal ₁₂ (micro-ondes)	=	passer $[\text{N}_Y \text{ au } \sim]$
Labreal ₁₂ (pi\^e\grave{g}e)	=	prendre $[\text{N}_Y \text{ dans ART } \sim]$, // pi\^e\grave{g}er $[\text{N}_Y]$

Labreal ₁₂ (<i>imagination</i>)	=	<i>voir</i> [N _Y <i>en</i> ~], // <i>imaginer</i> [N _Y]
Labreal ₂₃ (<i>boîte aux lettres</i>)	=	<i>déposer, glisser, mettre</i> [N _Z <i>dans</i> ART ~]
[<i>boîte aux lettres de</i> N _X <i>dans laquelle</i> N _Y <i>met</i> N _Z]		

Russian

Labreal ₁₂ (<i>mikrovolnovk a</i>)	=	<i>razogret'</i> [N _{Y-ACC} <i>v1</i> ~ <i>e</i>]
Labreal ₁₂ (<i>lovušk a</i>)	=	<i>pojmat'</i> [N _{Y-ACC} <i>v2</i> ~ <i>u</i>]
Labreal ₁₂ (<i>voobraženi e</i>)	=	<i>videt'</i> [N _{Y-ACC} <i>v1 svoëm</i> ~ <i>i</i>], // <i>voobrazit'</i> [N _{Y-ACC}]
Labreal ₂₃ (<i>počtovyj jaščik</i>)	=	<i>brosit', opustit'</i> [N _{Z-ACC} <i>v2</i> ~]
[<i>počtovyj jaščik</i> N _{X-GEN} , <i>kuda</i> N _{Y-NOM} <i>pomeščæt</i> N _{Z-ACC}]		

[52] **Prepar** [Latin *praeparāre* 'prepare']: verb that means 'prepare [L] for functioning'

Prepar takes as argument L only nouns that are semantic quasi-predicates.⁹ It means 'cause that L is ready to function / to be used'.

Prepar: verbal	
L	N
Prepar(L)	V

Prepar is equivalent to CausPredAble₁Fact₀: it denotes an action that makes it possible for the realization of L to take place. For this reason, it is introduced in the section dedicated to realization verbs.

English

Prepar(<i>knife</i> _(N))	=	<i>sharpen</i> [ART ~]
Prepar(<i>shoes</i>)	=	<i>shine</i> _(V) [ART ~]
Prepar(<i>needle</i> _(N))	=	<i>thread</i> _(V) [ART ~]
Prepar(<i>oven</i> _(N))	=	<i>preheat</i> [ART ~]
Prepar(<i>vocal cords</i>)	=	// <i>clear</i> [N _X 's] <i>throat</i> ⁷

French

Prepar(<i>couteau</i>)	=	<i>aiguiser</i> [ART ~]
Prepar(<i>chaussures</i>)	=	<i>cirer</i> [ART ~]
Prepar(<i>aiguille</i>)	=	<i>enfiler</i> [ART ~]

⁹On quasi-predicates, see Chapter 2, Section 2.2, p. 24.

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Prepar(*four* [pour la cuisine]) = *préchauffer* [ART ~]
 Prepar(⌈*cordes vocales*⌋) = // ⌈*s'éclaircir la gorge*⌋; // *vocaliser*

Russian

Prepar(*nož*) = *točit'* [~]
 Prepar(*botink*|*i*) = *čistit'*, *načistit'* [~i]
 Prepar(*igolk*|*a*) = *vdet' nitku* [v2 ~u]
 Prepar(*duxovk*|*a*) = *razogret'* [~u]
 Prepar(⌈*golosovye svjazki*⌋) = // *kašljanut'*, // *otkašljat'sja*,
 // ⌈*pročistit' glotku*⌋,
 // ⌈*pročistit' gorlo*⌋, // *prokašljat'sja*

Prepar is semantically related to two other syntagmatic LFs, for which it should not be mistaken: Real_1^1 and CausPredAble_1 .

- Examples in (30a–b) illustrate the difference between Prepar and Real_1^1 (i.e., the first degree of Real_1 – see [49], “Realization verbs,” p. 148):

(30) a. Prepar(*rifle*_(N)) = *load*_(V) [ART ~]
 vs.
 Real_1^1 (*rifle*_(N)) = *cock*_(V) [ART ~]; *shoulder*_(V) [ART ~]
 b. Prepar(*horse*_(N)) = *saddle*_(V) [ART ~]
 vs.
 Real_1^1 (*horse*_(N)) = *mount*_(V) [ART ~], // ⌈*mount up*⌋

The examples in (30) show that the user of L's referent does not necessarily correspond to the DSyntA_1 of Prepar, while it is, by definition, the DSyntA_1 of Real_1^1 : the person who loads the rifle is not necessarily the shooter, whereas it is necessarily the shooter who shoulders it; the person who saddles the horse is not necessarily the rider, whereas it is necessarily the rider who gets into the saddle.

- As stated earlier, Prepar is semantically equivalent to $\text{CausPredAble}_1\text{Fact}_0$, and therefore it can be easily mistaken for the complex LF CausPredAble_1 , illustrated in (31).

(31) $\text{Caus}_1\text{PredAble}_1$ (*sleep*_(V)) = // ⌈*lie down*⌋
 $\text{Caus}_1\text{PredAble}_1$ (*exercise*_(V)) = // ⌈*warm up*⌋

The difference between Prepar and CausPredAble_1 is that the latter applies to semantic predicates, while Prepar applies only to quasi-predicates.



Historical remark

In previous publications, Prepar was not presented as intrinsically related to realization verbs and was occupying a different position in lists of LFs. Prepar was also conceptualized as being applicable to verbs, with the use of an actantial subscript indicating which DSyntA of L is the DSyntA_I of the verbal collocate – see, e.g., Mel'čuk (2015b: 231).

5.4.4 Phasic verbs

LFs presented in this section – Incep, Fin and Cont – are *phasic verbs*, which apply only to verbal lexical units L: they take L as DSyntA_{II} and the expression of the DSyntA_I of L as DSyntA_I. As a result, a phasic LF hooks up to L without altering L's actantial structure, which implies that it does not carry an actantial subscript. The three phasic LFs Incep, Fin and Cont respectively mean 'begin', 'cease' and 'continue', i.e., they denote the three phases of the dynamic unfolding of L: beginning, cessation and continuation. In the extralinguistic reality, these three phases temporally follow each other in a different order: beginning, continuation and cessation. From a semantic viewpoint, however, the meaning of continuation is the most complex one: 'X continues' = 'X does not cease'; it has, therefore, to be considered last.

The following two equations express semantic relations holding between phasic LFs:

$$\begin{aligned}\text{Fin}(L) &= \text{IncepNon}(L) \\ \text{Cont}(L) &= \text{NonIncepNon}(L)\end{aligned}$$

A phasic LF, because of its meaning, applies only to Ls denoting non-punctual facts. Consequently, by saying *begin to explode* one considers an explosion from a special perspective (e.g., an explosion considered as a physical process that is technologically analyzable or describing explosions of multiple entities).

All three phasic LFs have the following PoS table:

Incep, Fin, Cont: verbal	
L	V
Incep / Fin / Cont(L)	V

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As Incep, Fin and Cont apply to verbs only, they require the introduction of a support verb – or of Copul – in case of a non-verbal argument. They frequently appear, therefore, in complex LFs. Note that in English, and in many other languages as well, cases of application of “pure” phasic LFs (simple LFs with a verbal L argument) systematically return fused values (see illustrations below).

[53] Incep [Latin *incipĕre*]: verb that means ‘begin’

English

Incep(<i>sleep</i> _(V))	=	// <i>fall asleep</i>
Incep(<i>drive</i> _(V))	=	// [⌈] <i>take the wheel</i> [⌋]
IncepOper ₁ ([⌈] <i>in love</i> [⌋])	=	<i>fall</i> [~]
IncepOper ₁ (<i>cold</i> _(N))	=	<i>catch</i> [ART ~]
IncepFunc ₀ (<i>day</i>)	=	<i>break</i> _(V) , <i>dawn</i> _(V)
IncepFunc ₀ (<i>night</i>)	=	literary <i>descend</i> , <i>fall</i> _(V)

French

Incep(<i>dormir</i>)	=	// <i>s’endormir</i>
Incep(<i>conduire</i>)	=	// [⌈] <i>prendre le volant</i> [⌋]
IncepOper ₁ (<i>amoureux</i> _(Adj))	=	<i>tomber</i> [~]
IncepOper ₁ (<i>rhume</i>)	=	<i>attraper</i> , colloquial <i>choper</i> [ART ~], // <i>s’enrhumer</i>
IncepFunc ₀ (<i>jour</i>)	=	<i>poindre</i> < <i>se lever</i>
IncepFunc ₀ (<i>nuit</i>)	=	<i>tomber</i>

Russian

Incep(<i>spat’</i>)	=	// <i>zasnut’</i>
Incep(<i>vesti’</i> [mašinu])	=	// [⌈] <i>sest’ za rul’</i> [⌋]
IncepOper ₁ (<i>vljublĕnnij</i> _(Adj))	=	// <i>vljubit’sja</i>
IncepOper ₁ (<i>nasmork</i>)	=	<i>podxvatit’</i> , <i>pojmat’</i> [~]
IncepFunc ₀ (<i>den’</i>)	=	<i>zabrezžit’</i> < <i>zanjat’sja</i> ; // <i>nastat’ utro</i>
IncepFunc ₀ (<i>noč’</i>)	=	<i>nastat’</i> , literary <i>opustit’sja</i> , <i>prijti</i>

[54] Fin [Latin *finĭre*]: verb that means ‘cease’

English

Fin(<i>sleep</i> _(V))	=	// <i>awake</i> _(V) , // [⌈] <i>wake up</i> [⌋]
FinOper ₁ (<i>sleep</i> _(N))	=	<i>awake</i> _(V) [from ART ~]
FinOper ₁ (<i>illness</i>)	=	<i>recover</i> [from ART ~]

FinOper ₁ (<i>efforts</i>)	=	<i>abandon</i> _(V) [~]
FinOper ₁ (<i>in love</i>)	=	// <i>fall out of love</i>
FinFunc ₀ (<i>fog</i> _(N))	=	<i>lift</i> _(V)

French

Fin(<i>dormir</i>)	=	// <i>s'éveiller, se réveiller</i>
FinOper ₁ (<i>sommeil</i>)	=	<i>émerger, sortir</i> [de ART ~]
FinOper ₁ (<i>maladie</i>)	=	<i>guérir</i> [de ART ~]
FinFunc ₀ (<i>brouillard</i>)	=	<i>se dissiper</i>

Russian

Fin(<i>spat'</i>)	=	// <i>prosnuť'sja</i>
FinOper ₁ (<i>bolezn '</i>)	=	<i>izlečit'sja < opravit'sja</i> [ot ~i], // <i>vyzdorovet'</i>
FinOper ₁ (<i>usili ja</i> _(N))	=	<i>prekratit'</i> [~ja]
FinOper ₁ (<i>vljublěnnjy</i> _(Adj))	=	// <i>razljubit'</i>
FinFunc ₀ (<i>tuman</i>)	=	<i>rassejat'sja</i>

[55] **Cont** [Latin *continuāre*]: verb that means 'continue'

English

ContOper ₁ (<i>impassive</i>)	=	<i>remain</i> [~]
ContOper ₁ (<i>silence</i> _(N))	=	<i>maintain</i> [~]
ContFunc ₀ (<i>doubt</i> _(N))	=	<i>persist</i>

French

ContOper ₁ (<i>impassible</i>)	=	// <i>rester de marbre</i>
ContOper ₁ (<i>silence</i>)	=	<i>garder</i> [ART ~]
ContFunc ₀ (<i>doute</i>)	=	<i>persister</i>

Russian

ContOper ₁ (<i>besstrastn yj</i>)	=	<i>ostavat'sja</i> [~ym]
ContOper ₁ (<i>molčani e</i>)	=	<i>xranit'</i> [~e]
ContFunc ₁ (<i>somnenie</i>)	=	<i>ne pokidat'</i> [N _{X-ACC}]

For semantic reasons, we introduce in the present section also the non-phasic verbal LF Prox, as it is directly linked to the expression of the beginning or cessation phase of a fact L.

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[56] **Prox** [Latin *proximāre* ‘[to] approach’]: verb that means ‘be ‘on the verge’ [of L]’

Prox(L) is a verbal lexical unit that denotes the proximity of the initial or final phase of L. It takes the verbal lexical unit L as DSyntA II and the expression of the DSyntA I of L as DSyntA I.

Prox: verbal	
L	V
Prox(L)	V

Prox often combines with two other phasic LFs: Incep and Fin. It is clearly not compatible with Cont, as it would be odd to say that someone is on the verge of continuing to do something. ProxIncep is abridged as Prox.

English

Prox(<i>die</i> _(V))	= // ‘be fighting for life’, // ‘be hovering between life and death’
A ₁ Prox(<i>die</i> _(V))	= // ‘at the point of death’, // ‘at death’s door’
ProxOper ₁ (<i>collapse</i> _(N))	= // ‘be on the brink’ [of ~]
ProxFunc ₀ (<i>revolt</i> _(N))	= <i>brew</i> _(V) , <i>rumble</i> _(V)
ProxFin(‘take place’)	= ‘near’ _(V) [A _(poss) X] <i>end</i> ’
ProxFinFact ₀ (<i>train</i> _(N))	= <i>pull into the station</i>

French

Prox(<i>mourir</i>)	= // <i>agoniser</i>
A ₁ Prox(<i>mourir</i>)	= ‘à l’article de la mort’
ProxOper ₁ (<i>victoire</i>)	= <i>courir</i> [à ART ~]
ProxFunc ₀ (<i>révolte</i>)	= <i>gronder</i>
ProxFin(‘avoir lieu’)	= ‘toucher à sa fin’
ProxFinFact ₀ (<i>train</i>)	= <i>entrer en gare</i>

Russian

Prox(<i>umeret’</i>)	= // ‘borot’sja za svoju žizn’
A ₁ Prox(<i>umeret’</i>)	= ‘na poroge smerti’; ‘odnoj ногоj v mogile’
A ₁ Prox(<i>krax</i>)	= ‘na grani’ [~a]
ProxFunc ₀ (<i>mjatež</i>)	= <i>gotovit’sja</i> , <i>podgotovljat’sja</i>
ProxFin(‘imet’ mesto’)	= ‘blizit’sja k (svoemu) koncu’
ProxFinFact ₀ (<i>poezd</i>)	= <i>podxodit’ k vokzalu</i> (stancii) < <i>podxodit’ k peronu</i>

5.4.5 Verbs of causing

The three LFs presented in this section – Caus ‘cause’, Liqu ‘cause to cease’ and Perm ‘not cause to cease’ [= ‘let continue’] – are *verbs of causing*, which take as DSyntA II a verbal lexical unit L and as DSyntA I the Cause / Causer.



Most of the time, the argument L of a verb of causing is another verbal LF (CausOper₁, CausReal₁, CausCont, etc.). L being a genuine verb of the language is, of course, possible, but much rarer in such languages as English and French, which have a regular means of expressing causation (see the remark on joker values of Caus below, p. 159).

By *verb* or *LF of causing*, we mean a verb or a verbal LF that denotes causation as such. A verb that denotes a fact that is not a causation proper, but includes a causation, is called *causative verb*. For instance, *make* [*something happen*] is a verb of causing, whereas *kill* ‘do something that causes someone to die’ is a causative verb.

The Cause / Causer is, generally speaking, an additional semantic and deep-syntactic actant – additional with respect to the situation denoted by L. For this reason the LFs of causing do not carry actantial subscripts. However, it is possible that the Cause / Causer actant coincides with one of L’s actants; then, the LF of causing receives an actantial subscript corresponding to this actant of L (Caus₁, Caus₂, ..., Liqu₁, Liqu₂, ..., Perm₁, Perm₂, ...) – see also Chapter 3, Section 3.2.3.4.

The three LFs of causing are intimately related to the three phasic LFs (see Section 5.4.4, p. 153) and both sets of LFs have the same PoS table.

Caus, Liqu, Perm: verbal	
L	V
Caus / Liqu / Perm(L)	V

[57] **Caus** [Latin *causāre*]: verb that means ‘cause [to begin]’

Cause that L carries one of the following two meanings:

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- ‘cause that L begins’, if L is non-punctual – with the indication of the beginning phase;
- ‘cause that L takes place’, if L is punctual – without the indication of a phase.

In theory, one should use the complex LF CausIncep in the first case and the simple LF Caus in the second case. To simplify, however, CausIncep is abridged as Caus, and it is the punctual or non-punctual semantic nature of L that allows for the disambiguation of the formula.

English

Caus(<i>sleep</i> _(V)) [= CausIncep(<i>sleep</i> _(V))]	=	<i>put</i> [N _X to ~]
Caus(<i>colloquial</i> ‘ <i>shut up</i> ’)	=	// <i>hush</i> _(V) [N _X]; // <i>mute</i> _(V) [N _X]
Caus ₁ Oper ₁ (<i>prisoner</i>)	=	// ‘ <i>give oneself up</i> ’
CausFunc ₀ (<i>wound</i> _(N))	=	<i>inflict</i> [ART ~]
Caus ₁ Func ₀ ([<i>sports</i>] <i>record</i> _(N))	=	<i>set</i> [ART ~]
Caus ₁ Fact ₀ ([<i>electric</i>] <i>light</i> _(N))	=	‘ <i>switch on</i> ’, ‘ <i>turn on</i> ’ [ART ~]

French

Caus(<i>dormir</i>) [= CausIncep(<i>dormir</i>)]	=	// <i>endormir</i> [N _X]
Caus(<i>se taire</i>)	=	// <i>colloquial</i> ‘ <i>clouer le bec</i> ’ [à N _X]
Caus ₁ Oper ₁ (<i>prisonnier</i>)	=	<i>se constituer</i> [~]
CausFunc ₀ (<i>lésion</i>)	=	<i>entraîner, provoquer</i> [ART ~]
Caus ₁ Func ₀ (<i>record</i>)	=	<i>établir</i> [ART ~]
Caus ₁ Fact ₀ (<i>lumière</i> [<i>électrique</i>])	=	<i>allumer</i> [ART ~]

Russian

Caus(<i>spat</i> ′) [= CausIncep(<i>spat</i> ′)]	=	// <i>usypit</i> ′ [N _{X-ACC}]; // <i>ubajukat</i> ′ [N _{X-ACC}] [by rocking]
Caus(<i>zamolč</i> <i>at</i> ′)	=	<i>zastavit</i> ′ [N _{X-ACC} ~ <i>at</i> ′], // <i>offensive</i> ‘ <i>zatknut</i> ′ [N _{X-DAT}] <i>rot</i> ′
Caus ₁ Oper ₁ (<i>plennyj</i> _(N))	=	// ‘ <i>sdat’sja</i> (v <i>plen</i>)’
CausFunc ₀ (<i>povreždeni</i> <i>e</i>)	=	<i>pričinit</i> ′ [~ <i>e</i>]
Caus ₁ Func ₀ (<i>rekord</i>)	=	<i>postavit</i> ′, <i>ustanovit</i> ′ [~]
Caus ₁ Fact ₀ (<i>svet</i> [<i>električestv</i> <i>o</i>])	=	<i>vključit</i> ′, <i>zažeč</i> ′ [~ <i>o</i>]

As shown in the above illustrations, for nominal arguments, Caus is used in combination with a support verb LF or a realization verb LF. This behavior is similar to that of phasic verbs and concerns all LF of causing.

Joker values of Caus. In English, the joker Caus for verbs is MAKE_(V) (*make* [someone] *fall / read ...*); in French, it is FAIRE (*faire tomber / lire ...*); Russian does not have such a joker Caus.

Non-agentive vs. agentive causation. The LF Caus and the two other LFs of causation encompass both types of causation (Kahane & Mel'čuk 2006, Mel'čuk 2012: Ch. 5), that is, the non-agentive 'cause 1' [$\approx$ 'X is the Cause of Y'] and the agentive 'cause 2' [$\approx$ 'X is the Causer of Y'].

[58] **Liqu** [Latin **liquidāre*]: verb that means 'cause to cease'

The meaning of the LF Liqu is built on that of Fin ([54], p. 154):

$$\text{Liqu(L)} = \text{CausFin(L)}.$$

English

Liqu(<i>sleep</i> _(V))	=	// <i>awake</i> [N _X], // <i>wake</i> [N _X]
Liqu ₍₁₎ Func ₁ (<i>smile</i> _(N))	=	<i>wipe</i> [ART ~ off N _X 's face]
LiquFunc ₀ (<i>ambiguity</i>)	=	<i>resolve</i> [ART ~]
Liqu ₁ Fact ₀ (<i>engine</i>)	=	「 <i>shut off</i> 」 [ART ~]

French

Liqu(<i>dormir</i>)	=	// <i>éveiller</i> [N _X]; // <i>réveiller</i> [N _X]
Liqu ₁ Oper ₁ (<i>sourire</i>)	=	<i>se départir</i> [de ART ~]
LiquFunc ₀ (<i>ambiguïté</i>)	=	<i>lever</i> [ART ~]
Liqu ₁ Fact ₀ (<i>moteur</i>)	=	<i>couper</i> [ART ~]

Russian

Liqu(<i>spat'</i>)	=	// <i>razbudit'</i> [N _{X-ACC}]
Liqu ₁ Oper ₁ (<i>ulybk a</i>)	=	「 <i>steret's lica</i> 」 [~u]
LiquFunc ₀ (<i>neodnoznačnost '</i>)	=	<i>razrešit'</i> [~]; <i>snjat'</i> [~]
Liqu ₁ Fact ₀ (<i>motor</i>)	=	<i>vyključit'</i> [~]

[59] **Perm** [Latin *permittere*]: verb that means 'not cause to cease' [$\approx$ 'allow']

English

Perm ₂ Oper ₁ (<i>member</i>)	=	<i>accept</i> [N _X as ~]
Perm ₁ IncepPredMinusMagn(<i>vigilance</i>)	=	<i>decrease, diminish</i> [ART ~]
NonPerm ₁ Func ₀ (<i>smile</i> _(N))	=	<i>suppress</i> [ART ~]

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French

PermOper ₁ (<i>abandon</i>)	=	<i>laisser</i> [N _Y à l'~]
Perm ₁ IncepPredMinusMagn(<i>vigilance</i>)	=	<i>relâcher</i> [ART ~]
NonPerm ₁ Func ₀ (<i>sourire</i> _(N))	=	<i>retenir</i> [ART ~]

Russian

Perm ₂ Oper ₁ (<i>člen</i>)	=	<i>prinjat'</i> [N _{X-ACC} v 2 ~y]
Perm ₁ IncepPredMinusMagn(<i>bditel'nost'</i>)	=	<i>oslabit'</i> [~]
NonPerm ₁ Func ₀ (<i>ulybk a</i>)	=	<i>prjatat', starat'sja skryt'</i> [~u]

The three LFs of causing are connected by the following equations – to be compared to the equations connecting phasic LFs (cf. Section 5.4.4, p. 153):

$$\text{Perm}(L) = \text{NonLiqu}(L) = \text{NonCausFin}(L).$$

Joker values for Perm. In English, the joker Perm for verbs is LET_(V) (as in *Let me go*); in French, it is LAISSER; just as for Caus (see above, p. 159), there is no equivalent Perm in Russian.

5.4.6 Verbs of mode of functioning

The three LFs presented in this section – Obstr, Stop and Excess – are verbs that take a nominal lexical unit L as DSyntA1 and denote three abnormal (approximately, AntiVer) modes of functioning of L's referent: hindered functioning, interrupted functioning and excessive functioning.

By virtue of their semantic content (and syntactic behavior), these LFs are linked to two previously introduced LFs:

- the realization verb LF Fact₀ ([50], p. 149), which denotes L's functioning;
- the support verb LF Func₀ ([47], p. 142), in case L itself denotes a functioning.

The individual presentations of Obstr, Stop and Excess below explain these relations.

LFs of mode of functioning are mainly used to form a configurational LF with Sympt_{ijk} ([66], p. 167). They all have the following PoS table:

Obstr, Stop, Excess: verbal	
L	N
Obstr/Stop/Excess(L)	V

[60] **Obstr** [Latin *obstruere* ‘obstruct’]: verb that means ‘not function as it should’

Obstr is equivalent to either the complex LF AntiVerFact₀ or the configurational LF AntiVer+Func₀, depending on L’s meaning.

English

Obstr(<i>memory</i>) [= AntiVerFact ₁ (<i>memory</i>)]	= ‘play tricks’ [on N _X]
Obstr(<i>negociations</i>) [= AntiVerFact ₀ (<i>negociations</i>)]	= stall _(V) ; ‘drag on’
Conv ₂₁ Obstr(<i>breath</i>) [= AntiVer+Oper ₁ (<i>breath</i>)]	= be short [of ~]; cannot catch [A _(poss) X ~]

French

Obstr(<i>mémoire</i>) [= AntiVerFact ₀ (<i>mémoire</i>)]	= <i>flancher</i>
Obstr(<i>négociations</i>) [= AntiVerFact ₀ (<i>négociations</i>)]	= <i>patiner</i> < ‘être au point mort’
Obstr(<i>souffle</i>) [= AntiVer+Func ₀ (<i>souffle</i>)]	= <i>manquer</i> [à N _X]

Russian

Obstr(<i>pamjat’</i>) [= AntiVerFact ₁ (<i>pamjat’</i>)]	= <i>podvodit’</i> [N _{X-ACC}]
Obstr(<i>peregovory</i>) [= AntiVerFact ₀ (<i>peregovory</i>)]	= <i>buksovat’</i> , <i>probuksovyvat’</i> ; <i>zastrajat’</i> , <i>zastoporit’sja</i>
Obstr(<i>dyxani e</i>) [= AntiVer+Func ₁ (<i>dyxani e</i>)]	= <i>ne xvatat’</i> [N _{X-DAT} ~ja]
Conv ₂₁ Obstr(<i>dyxani e</i>) [= AntiVer+Oper ₁ (<i>dyxani e</i>)]	= // <i>zadyxat’sja</i>

[61] **Stop** [Latin **stuppāre* ‘stop up, plug’]: verb that means ‘be interrupted’ [≈ ‘stop abnormally’]

Stop is equivalent to either AntiVerFinFunc₀ or AntiVerFinFact₀, depending on L’s meaning.

English

Stop(<i>match</i> _(N)) [= AntiVerFinFunc ₀ (<i>match</i> _(N))]	= <i>be interrupted</i>
Stop(<i>heart</i> _(N)) [= AntiVerFinFact ₀ (<i>heart</i> _(N))]	= <i>fail</i> _(V) , ‘give out’
Stop(<i>engine</i>) [= AntiVerFinFact ₀ (<i>engine</i>)]	= <i>stall</i> _(V)

French

Stop(<i>match</i>) [= AntiVerFinFunc ₀ (<i>match</i>)]	= <i>s’interrompre</i>
Stop(<i>cœur</i>) [= AntiVerFinFact ₀ (<i>cœur</i>)]	= <i>s’arrêter</i> , <i>flancher</i>
Stop(<i>moteur</i>) [= AntiVerFinFact ₀ (<i>moteur</i>)]	= <i>caler</i>

Russian

Stop(<i>matč</i>) [= AntiVerFinFunc ₀ (<i>matč</i>)]	= <i>prervat’sja</i>
---	----------------------

- c. The wind *caresses*_{Involv(wind_(N))} his face.
 d. His lips *are quivering*_{Excess^{trem}(lips)—Sympt₂₃(desire_(N))} with desire.

In the case of *Manif*, *Involv* and *Sympt_{ijk}*, the verbal collocate puts L into relation with an additional participant, not foreseen by L's meaning. This participant is denoted in the above examples by *eyes* (32b), *face* (32c) and *lips* (32d).

All four verbal LFs presented in this section have the same PoS table, namely:

Son, Manif, Involv, Sympt _{ijk} : verbal	
L	N
Son / Manif / Involv / Sympt _{ijk} (L)	V

[63] **Son** [Latin *sonāre* 'to sound']: verb that means 'produce a typical sound'

Son is a verb that takes the nominal lexical unit L as DSyntA I.

English

- Son(*wind*_(N)) = *murmur*_(V), *whisper*_(V) < *howl*_(V), *roar*_(V)
 Son(*stream*_(N)) = *murmur*_(V) < *bubble*_(V)
 Son(*fire*_(N)) = *crackle*_(V)
 Son(*dog*_(N)) = *yap*_(V) < *bark*_(V) < *howl*_(V); *growl*_(V)
 Son(*fly*_(N)) = *buzz*_(V)
 Son(*steps*_(N)) = *sound*_(V)
 Son(*whip*_(N)) = *crack*_(V)
 Son(S₁(*sleep*_(V))) = *snore*_(V)

Note. The last example is a case of composition of LFs (Chapter 3, Section 3.4.1, p. 53). *X snores* means 'X produces the typical sound_{Son} of a sleeper_{S₁(sleep_(V))}'. From the viewpoint of LF combination, S₁ applies to *sleep*_(V) and returns *sleeper*, after which Son applies to *sleeper* [= S₁(*sleep*_(V))] to produce *snore*_(V). Such a composition of LFs is characteristic of Son, since it describes a sound produced by an entity S_i involved in a particular situation L – e.g., Son(S₁([physical] *pain*_(N))) = *scream*_(V).

French

- Son(*vent*) = *gémir*, *murmurer* < *siffler* < *hurler*, *mugir*, *rugir*
 Son(*ruisseau*) = *chanter*; *gazouiller*; *murmurer*
 Son(*flamme*) = *crépiter*
 Son(*chien*) = *japper* < *aboyer* < *hurler*
 Son(*mouche*) = *bourdonner*

5 Syntagmatic lexical functions

Son(<i>pas</i> _(N))	=	<i>résonner</i>
Son(<i>fouet</i>)	=	<i>claquer</i>
Son(<i>S</i> ₁ (<i>dormir</i>))	=	<i>ronfler</i>

Russian

Son(<i>veter</i>)	=	<i>svistet' < revet'; vyt', zavyvat'</i>
Son(<i>ručej</i>)	=	<i>žurčat'</i>
Son(<i>ogon'</i>)	=	<i>potreskivat' < treščat'</i>
Son(<i>sobaka</i>)	=	<i>gavkat', tjavkat' < lajat' < vyt'; ryčat'</i>
Son(<i>muxa</i>)	=	<i>žužžat'</i>
Son(<i>šagi</i>)	=	<i>poslyšat'sja < razdat'sja</i>
Son(<i>xlyst</i>)	=	<i>ščelknut'</i>
Son(<i>S</i> ₁ (<i>spat'</i>))	=	<i>xrapet'</i>

The LF Son may be semantically related to two other syntagmatic LFs, depending on the meaning of their argument L.

- L denotes a sound. In this case, Son(L) = Func₀(L).

$$(33) \quad \begin{aligned} \text{Son}(\textit{thunder}_{(N)}) &= \text{Func}_0(\textit{thunder}_{(N)}) = \textit{rumble}_{(V)} \\ \text{Son}(\textit{gunshot}) &= \text{Func}_0(\textit{gunshot}) = \textit{'ring out'} \end{aligned}$$

- L denotes an artefact designed to produce a sound. In this case, Son(L) = Fact₀(L).

$$(34) \quad \begin{aligned} \text{Son}(\textit{bell}_{(N)}) &= \text{Fact}_0(\textit{bell}_{(N)}) = \textit{ring}_{(V)} \\ \text{Son}(\textit{gun}_{(N)}) &= \text{Fact}_0(\textit{gun}_{(N)}) = \textit{'bang out'} \end{aligned}$$

The meaning of Son is often expressed in natural languages by onomatopoeic interjections, i.e., clausative lexical units. This implies that the LF Son naturally combines with the structural derivative LF Claus₀ (see Chapter 4, [11], p. 90), as illustrated below.

English

Claus ₀ Son(<i>dog</i> _(N))	=	<i>Bow-wow!, Woof-woof!</i>
Claus ₀ Son(<i>cat</i> _(N))	=	<i>Meow!; Purr-purr!</i>
Claus ₀ Son(<i>doorbell</i>)	=	<i>Ding-dong!, Ring-ring!</i>
Claus ₀ Son(<i>collision</i>)	=	<i>Bam!</i>

French

Claus ₀ Son(<i>chien</i>)	=	<i>Ouah, ouah !</i>
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Claus ₀ Son(<i>chat</i>)	=	<i>Miaou !; Ronron !</i>
Claus ₀ Son(<i>sonnette</i>)	=	<i>Ding-dong !, Drelin !, Dring !</i>
Claus ₀ Son(<i>goutte</i> [<i>d'eau</i>])	=	<i>Plic !, Ploc !</i>

Russian

Claus ₀ Son(<i>sobaka</i>)	=	<i>Gav, gav!</i>
Claus ₀ Son(<i>koška</i>)	=	<i>Mjau!</i>
Claus ₀ Son(<i>zvonok</i>)	=	<i>Dzin', dzin'!</i>
Claus ₀ Son(<i>kapli</i>)	=	<i>Kap, kap!</i>

[64] Manif [Latin *manifestāre* ‘manifest’]: verb that means ‘manifest itself in something’

Manif is a verb that takes as DSyntA_I the nominal lexical unit L and as DSyntA_{II} the expression of an additional participant – i.e., the “something” in which L is manifested.

In the value’s government pattern appearing in the illustrations below, “N” stands for the nominal expression of the additional participant.

English

Manif(<i>tiredness</i>)	=	<i>show</i> _(V) [Loc _{in} N] [<i>Tiredness was showing in his eyes</i> ⟨ <i>on his face</i> ⟩.]
Manif(<i>joy</i> _(N))	=	<i>beam</i> _(V) [Loc _{in} N] < <i>explode</i> [Loc _{in} N]
A ₂ Manif(<i>joy</i> _(N))	=	<i>joyful</i> [<i>joyful look</i>]
A ₂ Manif(<i>admiration</i>)	=	<i>admiring</i> < <i>full of</i> ~

French

Manif(<i>fatigue</i>)	=	<i>se lire</i> [Loc _{in} N] [<i>La fatigue se lisait dans ses yeux</i> ⟨ <i>sur son visage</i> ⟩.]
Manif(<i>joie</i>)	=	<i>briller, luire, rayonner</i> [Loc _{in} N]
A ₂ Manif(<i>joie</i>)	=	<i>joyeux</i> [<i>regard joyeux</i>]
A ₂ Manif(<i>admiration</i>)	=	<i>admiratif</i> < <i>plein de</i> (ART) ~, <i>rempli de</i> (ART) ~

Russian

Manif(<i>ustalost'</i>)	=	<i>čitat'sja</i> [Loc _{in} N] [<i>Ustalost' čitalas' v ego glazax</i> ⟨ <i>na ego lice</i> ⟩.]
Manif(<i>radost'</i>)	=	<i>svetit'sja</i> [Loc _{in} N] < <i>sverkat</i> [Loc _{in} N]
A ₂ Manif(<i>radost'</i>)	=	<i>radostnyj</i> [<i>radostnyj vzgljad</i>]
A ₂ Manif(<i>vosxiščeni e</i>)	=	<i>vosxiščennyj</i> < <i>polnyj ~ja</i> < <i>ispolnennyyj ~ja</i>

[65] **Involv** [Latin *involvĕre* ‘drag along’]: verb that means ‘impact something’

Involv controls the same deep-syntactic configuration as Manif: it is a verb that takes as DSyntA I the nominal lexical unit L and as DSyntA II the designation of an additional participant, that is, of the entity / fact impacted by L.

L’s impact on something can be good or bad for the referent of L – DSyntA I of Involv(L) – or for the thing impacted – DSyntA II of Involv(L). Therefore, Involv is often combined with Bon_{1/2} and AntiBon_{1/2}, as shown in illustrations below.

English

Involv(<i>crowd</i> _(N))	=	<i>fill</i> _(V) [N]; <i>rush</i> _(V) [Loc _{ad} N]
Involv(<i>voice</i> _(N))	=	<i>fill</i> _(V) [N]
Bon ₂ Involv(<i>river</i>)	=	<i>irrigate</i> [N]
AntiBon ₂ Involv(<i>river</i>)	=	<i>flood</i> _(V) [N]
AntiBon ₁ Involv(<i>car</i>)	=	<i>crash</i> _(V) , <i>smash</i> _(V) [into N]
AntiBon ₂ Involv(<i>car</i>)	=	<i>hit</i> _(V) , <i>mow</i> _(V) , ‘ <i>run over</i> ’ [N N is a living being]

Note. The LF (Anti)Bon_i in the last four complex LFs characterizes the effect produced by the fact denoted by Involv on the DSyntA i of Involv. Thus, *The car crashed*_{AntiBon₁Involv(car)} *into N* implies a negative effect_{AntiBon} for the car [= DSyntA I of Involv], while *The car ran over*_{AntiBon₂Involv(car)} *N* implies a negative effect_{AntiBon} for N [= DSyntA II of Involv].

French

Involv(<i>foule</i>)	=	<i>emplir</i> [N]; <i>déferler</i> [Loc _{ad} N]
Involv(<i>voix</i>)	=	<i>emplir</i> [N]
Bon ₂ Involv(<i>fleuve</i>)	=	<i>irriguer</i> [N]
AntiBon ₂ Involv(<i>fleuve</i>)	=	<i>inonder</i> [N]
AntiBon ₁ Involv(<i>voiture</i>)	=	<i>s’écraiser</i> [contre N]
AntiBon ₂ Involv(<i>voiture</i>)	=	<i>faucher</i> , <i>renverser</i> [N N is a living being]

Russian

Involv(<i>tolpa</i>)	=	<i>zapolnit’</i> [N _{-ACC}]; <i>xlynut’</i> [Loc _{ad} N]
Involv(<i>golos</i>)	=	<i>napolnit’</i> [N _{-ACC}] < <i>zapolnit’</i> [N _{-ACC}]
Bon ₂ Involv(<i>reka</i>)	=	<i>orošat’</i> [N _{-ACC}]
AntiBon ₂ Involv(<i>reka</i>)	=	<i>navodnit’</i> < <i>zalit’</i> , <i>zatopit’</i> [N _{-ACC}]
AntiBon ₁ Involv(<i>avtomobil’</i>)	=	<i>naletet’</i> [na2 N _{-ACC}], <i>vrezat’sja</i> [v2 N _{-ACC}]
AntiBon ₂ Involv(<i>avtomobil’</i>)	=	<i>pereexat’</i> , <i>sbit’</i> [N _{-ACC}] < <i>zadavit’</i> [N _{-ACC} N is a living being]

[66] **Sympt_{ijk}** [Latin **symp_{to}ma*]: verb that means ‘show symptoms [of L] in a part of one’s body [literally or metaphorically]’

The LF Sympt_{ijk} is meant to describe *somatic phrasemes* denoting psychological states – e.g., *shake with fear* – or physiological states – e.g., *choke with thirst*. It is transgressive as regards the very notion of LF, for at least two reasons.

1. From the viewpoint of its domain of application – i.e., the range of Ls that Sympt_{ijk} can take as argument, it is very constrained. Sympt_{ijk} can take as argument only nouns of feeling – more precisely, nominal lexical units that denote psychological or physiological states of human beings (marginally, of animals). Its inclusion within the system of standard LFs, however, is justified by the fact that Sympt_{ijk} reflects the anthropocentric organization of natural languages.
2. From the viewpoint of its structure, Sympt_{ijk} always functions in combination with one of the three LFs of mode of functioning (Section 5.4.6, p. 160): Obstr, Stop or Excess ([60], [61], [62]), with these LFs applying to their own argument L', different from L. More precisely, Sympt_{ijk}(L) is encoded using a very peculiar template shown in (35), where “X” refers to the DSyntA I of L, that is, to the human being:

(35) Obstr / Stop / Excess(L'('body element of X'))—Sympt_{ijk}(L).

For the LFs Obstr and Excess, the following superscripts may be used to specify the aspect of obstruction or excessive functioning:

color	: ‘with respect to color’	motor	: ‘with respect to movements’
dim	: ‘with respect to dimension / size’	stat	: ‘with respect to vertical position’
express	: ‘with respect to expression’	trem	: ‘with respect to trembling’
fulg	: ‘with respect to brightness’	t'	: ‘with respect to temperature’

Syntactically, Sympt_{ijk} is a verb that controls three DSyntAs:

- One corresponds to X, i.e., the DSyntA I of L; it is designated in Sympt_{ijk} formulas by the subscript “₁”.
- One corresponds to X’s body element that reacts to L; it is designated by the subscript “₂”.
- One corresponds to L; it is designated by the subscript “₃”.

5 Syntagmatic lexical functions

Finally, in the formula Sympt_{ijk} , the *linear order* of actantial subscripts is to be interpreted as follows:

- i (first position) is the DSyntA I of Sympt_{ijk} ;
- j (second position) is the DSyntA II of Sympt_{ijk} ;
- k (third position) is the DSyntA III of Sympt_{ijk} .

English

Obstr ^{color} (<i>face</i> _(N))—Sympt ₁₃ (<i>fear</i> _(N))	= [N _X] <i>turned pale</i> ⟨ <i>livid</i> ⟩ [with ~]
Stop(<i>body</i>)—Sympt ₁₃ (<i>fear</i> _(N))	= [N _X] <i>be paralyzed, freeze</i> _(V) [with ~]
Excess ^{trem} (<i>body</i>)—Sympt ₁₃ (<i>fear</i> _(N))	= [N _X] <i>shake</i> _(V) , <i>tremble</i> _(V) [with ~]
Excess ^{trem} (<i>lips</i>)—Sympt ₂₃ (<i>fear</i> _(N))	= [N _X 's] <i>lips quiver</i> _(V) [with ~]
Excess ^{motor} (<i>excrement</i>)—Sympt ₁₃ (<i>fear</i> _(N))	= vulgar [N _X] <i>shit one's pants</i> [with ~] literal; vulgar [N _X] 「 <i>shit one's pants</i> 」 [with ~] metaph.

Note. This case is chosen for its clear-cut character to illustrate the fact that $\text{Sympt}_{ijk}(L)$ can be literal or metaphorical. We do not indicate this difference in other illustrations.

Excess ^{dim} (<i>eyes</i>)—Sympt ₂₃ (<i>fright</i> _(N))	= [N _X 's eyes] 「 <i>pop out</i> 」 of their sockets [from ~]
Excess ^{express} (<i>eyes</i>)—Sympt ₂₃ (<i>passion</i>)	= [N _X 's eyes] <i>be ablaze</i> [with ~]
Excess ^{fulg} (<i>eyes</i>)—Sympt ₂₃ (<i>joy</i> _(N))	= [N _X 's eyes] <i>be shining</i> ⟨ <i>radiant</i> ⟩ [with ~]
Excess ^{fulg} (<i>eyes</i>)—Sympt ₃₂ (<i>joy</i> _(N))	= [~] <i>be shining</i> [in N _X 's eyes]
Obstr ^{stat} (<i>legs</i>)—Sympt ₂₃ (<i>fear</i> _(N))	= [N _X 's legs] 「 <i>give way</i> 」 [from ~]
Excess ^{t*} (<i>cheeks</i>)—Sympt ₂₃ (<i>shame</i> _(N))	= [N _X 's cheeks] <i>burn</i> [from ~]

French

Obstr ^{color} (<i>visage</i>)—Sympt ₁₃ (<i>peur</i>)	= [N _X] <i>blêmir</i> [de ~]
Stop(<i>corps</i>)—Sympt ₁₃ (<i>peur</i>)	= [N _X] <i>être paralysé</i> [de ~]
Excess ^{trem} (<i>corps</i>)—Sympt ₁₃ (<i>peur</i>)	= [N _X] <i>trembler</i> [de ~]
Excess ^{trem} (<i>lèvres</i>)—Sympt ₂₃ (<i>peur</i>)	= [les lèvres de N _X] <i>trembler</i> [de ~]
Excess ^{motor} (<i>excréments</i>)—Sympt ₁₃ (<i>peur</i>)	= vulgar [N _X] <i>chier dans son froc</i> [de ~]

Excess ^{dim} (<i>yeux</i>)—Sympt ₁₂₃ (<i>effroi</i>)	= [N _X] <i>écarquiller</i> [<i>les yeux d'~</i>]
Excess ^{express} (<i>visage</i>)—Sympt ₂₃ (<i>joie</i>)	= [<i>le visage de N_X</i>] <i>rayonner</i> [<i>de ~</i>]
Excess ^{fulg} (<i>yeux</i>)—Sympt ₂₃ (<i>désir</i>)	= [<i>les yeux de N_X</i>] <i>briller</i> [<i>de ~</i>]
Obstr ^{stat} (<i>jambes</i>)—Sympt ₂₁₃ (<i>effroi</i>)	= // [<i>les jambes de N_X</i>] <i>se dérober</i> [<i>sous N_X</i>]
Excess ^t (<i>joues</i>)—Sympt ₂₃ (<i>honte</i>)	= [<i>les joues de N_X</i>] <i>brûler</i> [<i>de ~</i>]
Russian	
Obstr ^{color} (<i>lico</i>)—Sympt ₁₃ (<i>strax</i>)	= [N _{X-NOM}] <i>poblednet' < stat' ves' belym</i> [<i>ot ~a</i>]
Stop(<i>telo</i>)—Sympt ₁₃ (<i>strax</i>)	= [N _{X-NOM}] <i>ocepenet', okamenet'</i> [<i>ot ~a</i>]
Excess ^{trem} (<i>telo</i>)—Sympt ₁₃ (<i>strax</i>)	= [N _{X-NOM}] <i>drožat', trjastis' (vsem telom)</i> [<i>ot ~a</i>]
Excess ^{trem} (<i>guby</i>)—Sympt ₂₃ (<i>strax</i>)	= [<i>guby u N_{X-GEN}</i>] <i>drožat', trjastis'</i> [<i>ot ~a</i>]
Excess ^{motor} (<i>èkskrementy</i>)—Sympt ₁₃ (<i>strax</i>)	= colloquial [N _{X-NOM}] <i>naklast' < naložit' v štany</i> [<i>ot ~a</i>], colloquial <i>naklast' < naložit' polnye štany</i> [<i>ot ~a</i>], colloquial <i>obdelat'sja</i> [<i>ot ~a</i>]
Excess ^{dim} (<i>glaza</i>)—Sympt ₂₃ (<i>užas</i>)	= [<i>glaza u N_{X-GEN}</i>] <i>vylezti iz orbit</i> [<i>ot ~a</i>]
Excess ^{express} (<i>glaza</i>)—Sympt ₂₃ (<i>strast' '</i>)	= [<i>glaza N_{X-GEN}</i>] <i>goret' [ognëm ~i]</i>
Excess ^{fulg} (<i>glaza</i>)—Sympt ₂₃ (<i>radost' '</i>)	= [<i>glaza N_{X-GEN}</i>] <i>lučit'sja [~'ju], sverkat'</i> [<i>ot ~i</i>]
Excess ^{fulg} (<i>glaza</i>)—Sympt ₃₂ (<i>radost' '</i>)	= [<i>v glazax N_{X-GEN}</i>] <i>sverkat' [~]</i>
Excess ^{fulg} (<i>glaza</i>)—Sympt ₃₂ (<i>otčajani e</i>)	= [<i>v glazax N_{X-GEN}</i>] <i>čitat'sja [~e]</i>
Obstr ^{stat} (<i>nogi</i>)—Sympt ₂₃ (<i>strax</i>)	= [<i>nogi N_{X-GEN}</i>] <i>sdelat'sja < stat' kak vatnye</i> [<i>ot ~a</i>]
Excess ^t (<i>ščěki</i>)—Sympt ₂₃ (<i>styd</i>)	= [<i>ščěki u N_{X-GEN}</i>] <i>goret' [ot ~a]</i>

More illustrations for Sympt_{ijk} can be found in the lexical entry of ANGER_(N), given in the Appendix (p. 209).

Sympt_{ijk} and double-shot collocations. As noted in Ha (2026), phrasemes encoded by Sympt_{ijk} are often more complex than basic base-collocate structures. For instance, the LF formula (36) that appears in the English illustrations above:

$$(36) \text{ Excess}^{\text{trem}}(\textit{lips})\text{---Sympt}_{23}(\textit{fear}_{(N)})$$

encodes a complex phraseological structure, as shown in (37).

$$(37) \text{ [[} \textit{lips quiver} \text{]}_{\text{Excess}^{\text{trem}}(\textit{lips})} \text{ with } \textit{fear} \text{]}_{\text{Sympt}_{23}(\textit{fear}_{(N)})}$$

Such embedding of collocations has been termed *double-shot collocation* in Chapter 3 (Section 3.3.2, footnote 9, p. 48). As a rule, any $\text{Sympt}_{ijk}(\text{L})$ formula that contains at least the two subscripts “₂” [= X’s body element] and “₃” [= L] encodes a double-shot collocation, where the noun for X’s body element and the noun L are DSyntAs of the verb denoting a physical phenomenon; L is the base of the collocation and another collocation (of the lexeme denoting X’s body element) is its collocate. By contrast, an LF such as Sympt_{13} corresponds to a syntactic structure in which the expression of X’s body element (subscript “₂”) is absent and encodes, by default, simple collocations, as in (38) below.

(38) $\text{Exess}^{\text{motor}}(\text{body}) - \text{Sympt}_{13}(\text{pain}_{(\text{N})}) = [\text{N}_x] \text{ writhe} [\text{with } \sim]$

Alternative approach to Sympt_{ijk} . The above presentation of the LF Sympt_{ijk} corresponds to the standard approach, where Sympt_{ijk} is described as a two-headed LF with two arguments (Iordanskaja 1972, 1986).

An alternative approach – introduced in Mel’čuk & Polguère (2021) – considers that the Speaker does not *necessarily* have in mind a given lexical unit denoting a body element when expressing $\text{Sympt}_{ijk}(\text{L})$. It is possible that only the actual body element matters, regardless of which lexical unit expresses it. As a matter of fact, all above illustrations where the subscript “₂” does not appear next to Sympt correspond to phrasemes where the body element is not even lexicalized. For instance, when saying *Joey_X turns pale with fear* [$\text{Sympt}_{13}(\text{fear}_{(\text{N})})$], the Speaker may think of Joey’s face without having any expressive goal regarding specific lexical units such as $\text{FACE}_{(\text{N})}$ or colloquial $\text{MUG}_{(\text{N})}$ or literary VISAGE , etc. By contrast, $\text{Sympt}_{13}(\text{fear}_{(\text{N})})$ is indeed a collocate of the lexical unit $\text{FEAR}_{(\text{N})}$, the base of the collocation.

For this, in the alternative approach to Sympt_{ijk} , the Obstr, Stop and Excess components of Sympt_{ijk} formulas are not considered as actual *lexical* functions, their arguments being *meanings* rather than lexical units. Thus, template (35) given p. 167 for Sympt_{ijk} can be rephrased as (39):

(39) $\text{Obstr/Stop/Excess}(\text{‘body element of X’}) - \text{Sympt}_{ijk}(\text{L})$

and the above illustrations should follow template (39) with arguments of Obstr, Stop and Excess written in the form of meanings (i.e., between ‘...’) rather than lexical units (in italics) – for instance:

(40) $\text{Obstr}^{\text{color}}(\text{‘face’}) - \text{Sympt}_{13}(\text{fear}_{(\text{N})}) = [\text{N}_x] \text{ turned pale} \langle \text{livid} \rangle [\text{with } \sim]$
 $\text{Stop}(\text{‘body’}) - \text{Sympt}_{13}(\text{fear}_{(\text{N})}) = [\text{N}_x] \text{ be paralyzed, freeze}_{(\text{V})} [\text{with } \sim]$

5.5 Non-LF approaches to the description of collocations

Though paradigmatic LFs can be considered primary from a theoretical point of view,¹⁰ the originality and relevance of LFs are better understood from the perspective of the description of collocations – i.e., from the perspective of syntagmatic LFs. For instance, the first glimpse at the notion of LF that we offer in the Introduction of the book starts with the analysis of the two English collocations *pay tribute* and *give recognition* (Chapter 1, Section 1.1). As a matter of fact, LFs were invented in the early 1960s as a solution to modeling the problem of phraseologically constrained lexical choice in Machine Translation, with the conceptualization of the two iconic syntagmatic LFs Magn and Oper₁ (Mel'čuk 2015b: Ch. 14, Sec. 8, p. 269). It is thus reasonable to conclude this chapter on syntagmatic LFs – often perceived as LFs par excellence – with an overview of alternative means of description of collocations found in current lexical resources: dictionaries (Section 5.5.1) and lexical databases (Section 5.5.2).

5.5.1 Collocations in dictionaries

General-purpose dictionaries, as a rule, do not systematically describe collocations. One common strategy used by lexicographers is to simply mention some collocations in the entries for lexical units that are collocation bases, in particular, in lexicographic examples. For instance, the entry for the lexeme **DISTINCTION 1a** of the online *American Heritage Dictionary of the English Language* (American Heritage 2022) smuggles an Oper₁ for this noun into a lexicographic example: “*economists making[= Oper₁] an ongoing distinction between domestic and foreign markets.*”

The *Longman Dictionary* (LDOCE 2025) provides more collocations in its entry for the corresponding lexeme – numbered **DISTINCTION 1**, but with minimal information. It simply lists phrases (41a–c) together with illustrative examples, under the lexicographic definition:

- (41) a. *distinction between*
 [i.e., part of **DISTINCTION 1**'s government pattern – see below in footnote 11,
 the so-called “grammatical collocations” of the *BBI Dictionary*]
 b. *clear / sharp distinction*
 [i.e., Magn(*distinction 1*)]
 c. *make / draw a distinction*
 [i.e., Oper₁(*distinction 1*)]

¹⁰Cf. Chapter 4, where the paradigmatic LF Syn is presented as central in the system of LFs (Section 4.1, p. 71).

In a dictionary where collocations are classified, this classification is mainly based on the PoS of the collocate or on the syntactic structure of the collocations. Such is prototypically the case in *dictionaries of collocations* (also called *collocation(s) dictionaries*). For instance, the *Oxford collocations dictionary for students of English* (OCD 2022) lists four groups of collocations in its entry for $ANGER_{(N)}$, presented and commented in Table 5.1.

Table 5.1: Classified collocations of $ANGER_{(N)}$ in the OCD (2022)

OCD's description	Comments
<u>adj.</u> bitter, black [...] terrible genuine, real [...]	A significant proportion of collocates listed in this section of the OCD entry correspond to the two modifying syntagmatic LFs Magn and Ver. Note that the symbol “ ” is used in the OCD to separate different subclasses of collocates that correspond to different (families of) LFs.
<u>quant.</u> burst, fit, outburst <i>He slammed the door in a fit of anger.</i>	The collocations these lexical units form with their base $ANGER_{(N)}$ correspond to values of the <i>paradigmatic</i> LF Sing – cf. the lexicographic entry for $ANGER_{(N)}$ in the Appendix (p. 210). On analytic values of paradigmatic LFs, see Chapter 3, Section 3.3.3.1 (p. 50).
<u>verb + anger</u> be filled with, feel, shake with, tremble with [...] express, give vent to, show [...]	This section of the OCD entry lists a heterogeneous set of collocates, from the viewpoint of syntagmatic LFs: in the case of collocates listed here, $Oper_1$, $Sympt_{13}$ and $Perm_1Fact_0$.
<u>anger + verb</u> boil over/up, bubble up [...] abate, drain [...]	This section contains two families of collocations corresponding to IncepPredPlusMagn vs. IncepPredMinusMagn.

Note that the online *Longman Dictionary* (LDOCE 2025) – mentioned above to illustrate the strategies used in general-purpose dictionaries – encapsulates a full-fledged dictionary of collocations. In the case of the collocations (41b–c), which are embedded in the entry for $DISTINCTION_{1a}$, they are further described among other collocations in a table positioned at the end of the vocable's entry (instead of being inserted in the entry for $DISTINCTION_{1a}$, where they belong). As for most dictionaries of collocations, a syntactic classification is used in this table: two sections “verbs” and “adjectives” under which collocations are grouped according to the PoS of collocates.

The *BBi Combinatory Dictionary of English* (Benson et al. 2010) deserves special attention: while not explicitly using LFs to describe collocations, it has borrowed

from the LF approach the idea of classifying lexical collocations¹¹ both syntactically and semantically (see the dictionary's preface, p. ix). The classification used by this dictionary is based on explicitly defined *types*. For instance, the L4 type is defined as follows (Benson et al. 2010: xxxiii):

L4 collocations consist of a noun and verb; the verb names an action characteristic of the person or thing designated by the noun: *adjectives modify*, *alarms go off* (ring, sound), *bees buzz* (sting, swarm), *blizzards rage*, *blood circulates* (clots, congeals, flows, runs), *bombs explode* (go off), etc. The *Dictionary* does not include predictable combinations such as *bakers bake*, *boxers box*, *cooks cook*, *dancers dance*, *fencers fence*, etc.

The L4 type encompasses at least the two LFs Fact_i and Son , together with their phasic and causal variants – cf. $\text{IncepSon}(\text{alarm}_{(N)}) = \text{'go off'}$. As a rule, the BBI classification system is significantly coarser than that of LFs, which is justified by the fact that this dictionary targets the general public. For the same reason, type names (L1, L2, ...) are not mentioned in the entries. Types are used to group and list collocations in a canonical order throughout the dictionary,¹² and illustrative phrases are used to help the users grasp the semantic-syntactic value of collocations.

Presenting an extensive review of the description of collocations in dictionaries is beyond the scope of the present section and we have to limit ourselves to a selection of representative and well-known lexicographic approaches. This also applies to our next topic: the description of collocations in lexical databases that are not making use of LFs.

¹¹The BBI dictionary distinguishes between *lexical* collocations – i.e., constrained combinations of lexical units that we model by means of syntagmatic LFs – and *grammatical* collocations – i.e., prepositional / conjunctive constructions controlled by verbs, nouns, etc., to express their actants (*cope with*, *rely on / upon*, ...). These phrases are actually not collocations as they pertain to the active syntactic valence of predicative lexical units, modeled by government patterns in Explanatory Combinatorial Dictionaries (see Chapter 2, Section 2.3, p. 31).

¹²In the same vein, LF relations are always listed in a canonical order in Explanatory Combinatorial Dictionaries – cf. “System of standard simple lexical functions” in Chapter 3 (Section 3.8, p. 67).



Further reading

On techniques used by lexicographers for embedding collocational information in general-purpose dictionaries, see Gouws (2015); on dictionaries of collocations, see Hausmann (1989) and Corpas Pastor (2016); on the use of popularization formulas to paraphrase LF relations, see Polguère (2000a), Mel'čuk & Polguère (2007) and L'Homme (2008); on the automatic extraction and syntactic classification of collocations for the *Antidote* software suite, see Charest et al. (2010).¹³

5.5.2 Collocations in lexical databases

By *lexical databases*, we mean descriptions of lexical entities that have the following properties:

- the information they contain is structured, stored and made available as computational databases (simple spreadsheets, relational SQL tables, etc.);
- even though they can be accessed directly by human users, they are meant to be compatible with computational exploitations, such as in Natural Language Processing;
- for this reason, they abide by well-defined formalization principles.

As is the case with dictionaries, we have to make a drastic choice from among the existing lexical resources. We focus on one of them, the set of *lexicon-grammar tables*, which offers a broad-scale description of the French lexicon and has given rise to significant exploitation in linguistic research and Natural Language Processing.

The *lexicon-grammar* (Gross 1988b, 1994) originated from the *distributional analysis* approach put forward by Zellig Harris (Harris 1954, Sahlgren 2008). It was

¹³The current bilingual distribution of *Antidote* (Antidote 2025) contains a range of linguistic resources: dictionaries, grammars and text correctors. Its dictionaries of collocations – named *Dictionary of combinations*, for English, and *Dictionnaire des cooccurrences*, for French – list collocations according to the syntactic dependencies that connect bases and their collocates.

oriented toward the lexicographic construction of lexical resources deemed to be exploitable in Natural Language Processing, with a focus on the French language.¹⁴ The study of phraseology is at the center of the lexicon-grammar, which is well-known for its description of support verbs (Section 5.4.2, p. 138), seen as tools for verbalizing nominal predicates and, therefore, as sentence-builders.¹⁵ As for formalization, support verb collocations are described in tables where rows correspond to predicative nouns and columns correspond to potential verbal collocates. The acceptability vs. non-acceptability of a given collocation is indicated in the corresponding table cell by “+” vs. “–” signs, as illustrated in Table 5.2.¹⁶

Table 5.2: Extract from the *N_aa.lgt* lexicon-grammar table¹⁷

<ID>	<ENT>N	avoir = concevoir	avoir = éprouver	avoir = ressentir	Dét =: un	Dét =: un-Modif	Dét =: du	Prép1 =: à l'égard de
1	admiration	+	+	+	–	+	+	–
2	affection	+	+	+	–	+	+	–
3	amitié	+	+	+	–	+	+	–
4	amour	+	+	+	–	+	+	–
5	animosité	+	+	+	–	+	+	–
6	antipathie	+	+	+	–	+	+	+
7	appréhension	–	–	+	–	+	+	–
8	attachement	–	+	+	–	+	+	+

Explanations. The second column lists French nouns of feelings: *admiration*, *affection*, *amitié* ‘friendship’, *amour* ‘love_(N)’, *animosité* ‘animosity’, *antipathie* ‘antipathy’, *appréhension* ‘apprehension’ and *attachement* ‘attachment’. Columns 3 to 5 indicate the (in)compatibility with French support verbs *concevoir* lit. ‘conceive’, *éprouver* ‘experience_(V)’ and *ressentir* ‘feel’. The specific class of support verbs treated here is identified with the formula “avoir =” in reference to the joker *Oper*₁ for French nouns of feeling: *avoir* ‘have’. The remaining columns are used to specify the other syntactic elements of the collocation (determiners and noun complements) – cf. government patterns of value elements in LF applications (Chapter 3, Section 3.1.2, p. 38).

¹⁴For an application to English, see Garcia-Vega & Machonis (2014).

¹⁵Cf. Maurice Gross’s semantic-syntactic credo: “The unit of meaning is the elementary sentence, and not the word” (Gross 1988b: 182).

¹⁶This is a simplification; in fact, lexicon-grammar tables make use of a richer set of cell values.

¹⁷Lexicon-grammar tables are archived on the website of the LIGM laboratory: <https://infolingua.univ-eiffel.fr> (last consulted 15/12/2025).

Lexicon-grammar tables describe all three families of support verbs identified in terms of LFs as $Oper_i$, $Func_i$ and $Labor_{ij}$; they also encompass verbal collocates that incorporate phasic meanings. Interestingly, working on support verbs for nouns of feelings lead lexicon-grammar research to take into consideration phraseological expressions corresponding to the $Sympt_{ijk}$ LF – cf. the construction of a *local grammar*¹⁸ of French nouns of feelings in Gross (1988a: Sec. 4.2) and Gross (1995: Sec. 5).

To conclude, let us mention that there exist numerous lexical databases that use LFs to encode collocations (and paradigmatic lexical relations); to name a few: for French, the French Lexical Network [fr-LN] (Lux-Pogodalla & Polguère 2011),¹⁹ for Spanish, DiCE (Alonso Ramos et al. 2010) and Diretes (Barrios Rodríguez & Boguslavsky 2019); for English and Russian, the multilingual dictionaries of the ETAP-3 linguistic processor (Boguslavsky et al. 2004).



Further reading

On the automatic enrichment of the Princeton WordNet with collocational information, see Espinosa-Anke et al. (2016); on collocational information in the FrameNet lexical database, see Ruppenhofer et al. (2002); for connections between syntagmatic LFs and collocational data in FrameNet, see Langer (2005: 3–5) and Bouveret & Fillmore (2008).

¹⁸Local grammars are formal graph-based representations of the structure of utterances used in Natural Language Processing (in particular, for sentence and phrase extraction from corpora).

¹⁹The fr-LN is presented in detail in Chapter 6.

6 Lexical networks

This concluding chapter examines the role played by lexical functions [LFs] in the structure of natural language lexicons and, consequently, their relevance to the structuring of lexicon models. Because lexicons are large informational systems involved in complex mental processes (speech production / understanding, language acquisition, reasoning with elaborate concepts, etc.), they can be expected to have a structure that supports their involvement in such processes, and it is there that LFs come into play.

The preceding chapters deal with the relevance of LFs from a dual perspective:

- writing lexical entries – cf. the cooccurrence zone of corresponding lexical entries, as illustrated with the entry for $ANGER_{(N)}$ (Appendix, p. 209);¹
- modeling lexical selection in speech production (Chapter 3, Section 3.2.3.2) – cf. the role of LFs in deep-syntactic structures and in deep-syntactic paraphrasing rules described throughout the book.

We now take a broader view of the system of standard LFs and explain the key role they play in organizing the Speaker's *lexical knowledge*. The implication of what will be presented now is twofold.

First, the system of lexical relations induced by LF applications can be used to determine the shape of new *models of the lexicon* – for short, *lexicon models* – and, therefore, can be exploited to develop a new type of lexicography that goes beyond traditional dictionary writing (as demonstrated in Section 6.3, p. 204).

Second, in terms of applicability, lexicon models primarily structured by LF relations are expected to be better *data structures* than ordinary – textual – dictionaries, in the sense that they are better tools for the study and description of essential lexicon-related phenomena, such as lexical acquisition and lexical access / selection by the Speaker.

The present chapter contains the following sections:

¹LF applications that systematically illustrate the presentation of each individual LFs in Chapters 4 and 5 are in reality part of lexicographic information: an expression of the form “ $f(L) = L'$ ” belongs to the cooccurrence zone of L 's lexical entry.

- What is the “shape” of natural language lexicons? (Section 6.1)
- Lexical Systems as network lexicon models (Section 6.2)
- Proof of concept: the French Lexical Network [fr-LN] (Section 6.3)

6.1 What is the “shape” of natural language lexicons?

The question of the structure of lexical knowledge – and, consequently, the formal structure of models that describe it – has been largely disregarded by lexicologists and lexicographers in favor of a dictionary-based approach. Under this approach, the *macrostructure* (Wiegand 1996) of a lexicon model is an alphabetically sorted list of lexical entries; this is the common way to structure most of the known dictionaries. However, there have been notable exceptions, for instance:

- For dictionaries: *production dictionaries*, inaugurated for English by the *Longman Language Activator* (Summers 1993). Its thesaurus-like macrostructure gives the users access to lexical entries via a set of conceptual keywords, thus breaking the linear alphabetical barrier and allowing for a semantic navigation through lexical knowledge.²
- For lexical databases: *-Net models* (Fontenelle 2012), such as the family of *WordNets* (Princeton WordNet, EuroWordNet, etc.), which adopt a *taxonomic* perspective on the structuring of lexical knowledge (Fellbaum 1998), or *FrameNet*, which is based on lexical relations induced by Frame Semantics (Fillmore et al. 2003).

Given this state of affairs, we have to answer the following question: What is the “shape” of a natural language lexicon? Of course, a real lexicon of natural language is a purely informational (non-physical) entity and does not have actual physical shape. We use the figurative expression *shape of the lexicon* in order to stress the fact that, in advanced formal lexicology and lexicography, we need adequate structuring and visualizations of lexicon models (see Figure 6.1, p. 181) that could support reasoning on such knowledge representations.

To develop the notion of lexicon model, we will introduce two postulates: one immediately below and the second in Section 6.2 (p. 185).

²Cf. the illustration given in Rundell & Ham (1994: 176): the ANGRY concept, which gives access to (dictionary entries for) lexical realizations such as: *angry, cross, enrage, fly off the handle, furious, incensed, mad, maddening, mollify, rage, stormy, wind someone up*.



Postulate 1: structure of lexicons

Natural language lexicons are, first and foremost, sets of lexical entities connected by lexical relations. Formally, they are *lexical networks*; therefore, lexicon models should be structured as lexical networks.

This postulate is taken for granted – by definition, a postulate does not call for a proof. However, its *adequacy* can be evaluated in the context of descriptive lexicology – i.e., lexicography – and in that of its applications – language teaching, Natural Language Processing, etc. Postulate 1 is also in line with a whole array of observations made in disciplines as varied as psycholinguistics, speech disorder research or neurolinguistics.³

Lexical networks are often associated in contemporary research with the notion of *Mental Lexicon* – roughly, lexical knowledge as stored in the brain (Aitchison 2012, De Deyne et al. 2017). We believe that network models of the lexicon are indeed plausible models of the Mental Lexicon or, to put it more broadly, of the lexicon in its “cognitive reality” (Goddard 2026: Sec. 6).

Mere adherence to the principle of representing lexical knowledge as lexical network is insufficient to properly determine the formal nature of resulting lexicon models. The lexicon is a set of lexical entities, but which ones? They are connected by relations, but which ones? The nature of these entities and these relations determines the actual formal structure of lexicon models. To illustrate this, let us contrast two approaches to structuring lexical networks: taxonomic networks, exemplified with the Princeton WordNet, vs. small-world networks, exemplified with the French Lexical Network [fr-LN], presented in Section 6.3 (p. 204).

The Princeton WordNet for English⁴ is a computational lexicon model – a lexical database – that follows taxonomic principles postulated in the context of psycholinguistic research (Miller et al. 1990).⁵ It is probably the most well-

³This is no place for a detailed review of the huge literature on the application of lexical networks in these areas. We content ourselves with the mention of particularly significant texts.

⁴We focus here on the Princeton WordNet for the English language. The WordNet approach has been applied to a broad range of natural languages and has also been adapted to the construction of multilingual lexical resources (Fellbaum & Vossen 2012).

⁵A taxonomy – in the most general sense – is a model of a vast natural domain organized as a hierarchy of classes and subclasses of entities. Linnaeus’s classification of species is generally cited as prototypical taxonomic model in science. For a presentation and critique of this taxonomy, see Ereshefsky (1994).

known and most widely used lexical database in Natural Language Processing. In the Princeton WordNet, lexical units, called *senses*, are grouped into small sets of (quasi-)synonyms, called *synsets*, and it is those synsets that are linked within the original WordNet’s lexical network. They constitute lexical classes primarily connected by the relation of hyperonymy / hyponymy. This entails a global taxonomic organization of the lexicon model, characterized by its creators as being an *inheritance system* (Miller et al. 1990: 242).

The fr-LN (Lux-Pogodalla & Polguère 2011) is a computational lexicon model built according to the principles of Explanatory Combinatorial Lexicology. Its structure has the characteristic properties of small-world networks (see details in Sections 6.2 and 6.3); it can be loosely characterized as being a “social network” of lexical entities (mainly, lexical units) connected by paradigmatic and syntagmatic lexical relations, that are, in most part, LF relations. Though not taxonomic, its small-world organization puts order in the apparent mess of the lexicon: this organization allows for the computation of meaningful lexical clusters and, more generally, of topological proximity.⁶

Figure 6.1 visualizes the formal gap between the two approaches; it features:

- a sample of WordNet’s taxonomy centered around the lexical unit WATERCOURSE 2 – a member of the “{STREAM 1, WATERCOURSE 2}” *synset*;
- a sample of the fr-LN centered around the corresponding French lexical unit ‘COURS D’EAU’, where the geometric distribution of lexical nodes reflects their topological proximity in the lexical network.

Though both the Princeton WordNet and the fr-LN have the ambition to reflect the organization of the Mental Lexicon, Figure 6.1 clearly shows how different structuring principles adopted for lexical network models induce different characteristic “shapes” of these models.⁷

We present in Section 6.2 (p. 183) lexicon models called Lexical Systems, of which the fr-LN is a particular instance, and highlight the fundamental role played by LF relations in such network lexicon models. Before we do that, let us briefly examine paradigmatic lexical relations encoded in WordNet from the perspective of LFs.

⁶On the topological proximity of lexical nodes in networks / graphs, see Section 6.2.2.2, p. 201.

⁷For arguments in favor of choosing small-world lexical networks over taxonomic ones, see Polguère (2014) and Gotkova et al. (2024). The computable nature of small-world lexical networks as data structures is presented in Section 6.2.2 (p. 194).

6.1 What is the “shape” of natural language lexicons?

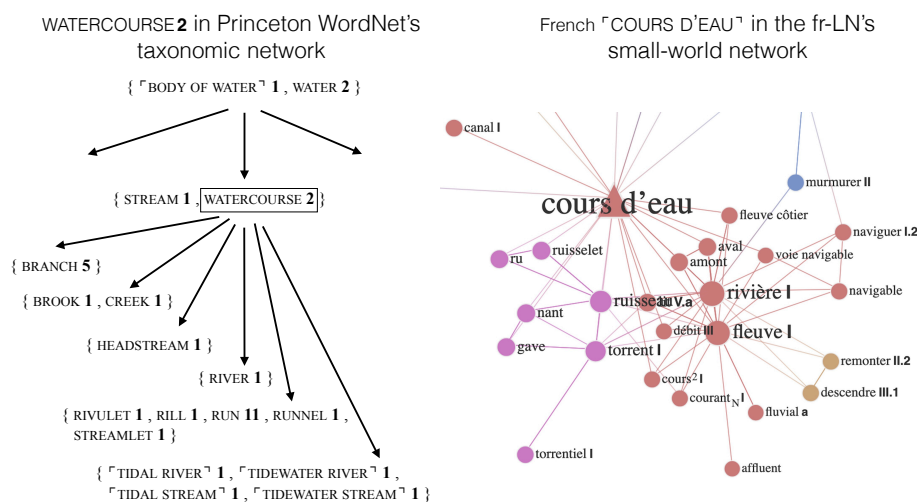


Figure 6.1: Taxonomic vs. small-world lexical networks

WordNet’s paradigmatic relations from the LF perspective. The taxonomic organization of WordNet – illustrated in Figure 6.1 – operates only in the nominal component of this lexical database (Miller 1990), which is subdivided according to part of speech classification (nouns, verbs, adjectives and adverbs) and follows different structuring principles for each part of speech.

Furthermore, WordNet has a hybrid network structure where *senses* themselves (and not only *synsets*) can be connected. Such is the case for WordNet’s so-called *morphosemantic links*, which were grafted onto the original version of the database (Miller & Fellbaum 2003): they indicate lexical relations that would be encoded by us with paradigmatic LFs. In WordNet, these links are not typed, but identified under the same label: *derivation*, described as being a “derivational morphology pointer.” This is illustrated in Figure 6.2, which shows the WordNet subnetwork of derivation links extracted by taking the sense MARRY 1 as point of origin and recursively following these links until all available derivation links are exhausted.⁸

In order to clarify the nature of WordNet senses appearing in Figure 6.2, we list in Table 6.1 their definitions in WordNet – called *glosses* (Miller et al. 1990: 240) – together with the relevant neighboring polysemy.

Let us make three comments about Figure 6.2, taking the perspective of the encoding of paradigmatic relations by means of LFs.

⁸The present discussion is based on data extracted from version 3.1 of the Princeton WordNet.

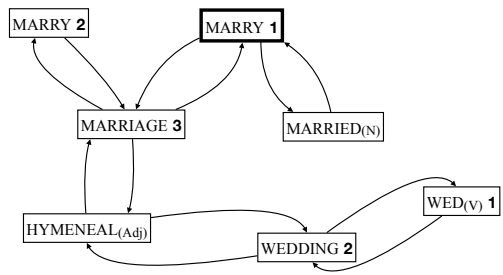


Figure 6.2: Subnetwork of derivation links in WordNet originating from MARRY 1

Table 6.1: Extract of WordNet’s polysemy around the vocable MARRY

Senses	Glosses
MARRY 1	<i>take in marriage</i>
MARRY 2	<i>perform a marriage ceremony</i>
MARRIAGE 1	<i>the state of being a married couple voluntarily joined for life (or until divorce)</i>
MARRIAGE 2	<i>two people who are married to each other</i>
MARRIAGE 3	<i>the act of marrying; the nuptial ceremony</i>
MARRIED _(N)	<i>a person who is married</i>
HYMENEAL _(Adj)	<i>of or relating to a wedding or marriage</i>
WEDDING 1	<i>the social event at which the ceremony of marriage is performed</i>
WEDDING 2	same gloss as its synonym MARRIAGE 3: it belongs to the same synset (WordNet’s glosses are associated to synsets, not to individual senses)
WED _(V) 1	same gloss as its synonym MARRY 1
WED _(V) 2	same gloss as its synonym MARRY 2

1. WordNet’s derivation links encode a very heterogeneous set of paradigmatic relations that can be discriminated with LFs; e.g., in Figure 6.2:

- (1)
- MARRY 1

MARRIAGE 3

MARRY 1
- ⊢_{S₀}→

⊢_{V₀}→

⊢_{S_{1/2}Perf}→
- MARRIAGE 3

MARRY 1

MARRIED_(N)

This is a known issue with derivation links in WordNet. For a proposal to improve the description, see Fellbaum et al. (2009).

2. A derivation link does not necessarily encode a morphological derivation, in spite of being termed *morphosemantic* in the WordNet literature. Such is

the case of the link between MARRIAGE 3 and HYMENEAL_(Adj) in Figure 6.2: a case of A_0 , in terms of LFs. Because morphology is not necessarily involved, these links can be considered as encoding *semantic* derivations, just like paradigmatic LFs.

3. The encoding of semantic derivations in WordNet can be implemented in surprising ways. For instance, the absence of SPOUSE in Figure 6.2 is somehow strange, given the presence of the very marginal term MARRIED_(N). The noun SPOUSE happens to be connected to MARRIAGE 2 – not MARRIAGE 3 – by the WordNet relation of meronymy. Because MARRIAGE 2 is glossed as denoting a set of two persons rather than a social relation between two individuals (cf. Table 6.1),⁹ SPOUSE is modeled as denoting a unit of a marriage 2 and therefore does not surface in Figure 6.2.

This last observation illustrates the importance of organizing the modeling of paradigmatic – and syntagmatic! – relations by means of a coherent, well-defined *system* of such relations that guides the process of exploring – not just encoding – lexical relations.

6.2 Lexical Systems as network lexicon models

A *Lexical System* (Polguère 2009, 2014) – with capitalized initials – is a lexicon model structured as a network of lexical entities connected, primarily, by paradigmatic and syntagmatic LF relations. Though built according to the principles of Explanatory Combinatorial Lexicology, Lexical Systems have a formal structure radically different from that of Explanatory Combinatorial Dictionaries (Mel’čuk & Zholkovsky 1984, 2016, Mel’čuk et al. 1984, 1988, 1992, 1999), these latter being textual (therefore, linear) representations of lexical knowledge.

This section contains two parts: main characteristics of Lexical Systems (Section 6.2.1) and formal properties of Lexical Systems (Section 6.2.2).

The Lexical System approach has been applied to the construction of several lexical networks (LNs), in particular, for French (fr-LN), English (en-LN) and Russian (ru-LN). Because at the present time only the fr-LN (Section 6.3) can be considered as being a full-fledged Lexical System, our illustrations are mostly borrowed from this particular lexical database.¹⁰

⁹WordNet illustrates this sense with the following example: *his second marriage was happier than the first*.

¹⁰For information on exploratory lexicographic work conducted on the en-LN, see Gader et al. (2014); on the ru-LN, see Krylosova (2017) and Mikhel (2022).

6.2.1 Main characteristics of Lexical Systems

6.2.1.1 LF relations: the backbone of Lexical Systems

Going back to Postulate 1 (the network structure of lexicons: Section 6.1, p. 178), we can characterize Lexical Systems as information structures made up of two types of elements:

1. lexical entities, as defined in Chapter 2 (Section 2.1, p. 22) – primarily, lexical units, but also, compositional phrasemes¹¹ and word-formation means analogous to lexemes;
2. lexical relations that connect these entities, as defined in Chapter 3 (Section 3.2.1, p. 39) – primarily, LF relations (other relations that play an important structuring role in Lexical Systems are presented in Section 6.2.1.2, p. 189).

Elements of the first type correspond to the *nodes* of the lexical network and elements of the second type correspond to the *arcs* of the network. Figure 6.3 illustrates these two notions. It is a blow-up of a segment of the fr-LN's subnetwork given in Figure 6.1 (p. 181) that features two lexical nodes – French CANAL1 '[water] canal' and 'COURS D'EAU' 'watercourse', which are connected by an arc corresponding to the LF relation *Gener*; i.e., in terms of LF application, it encodes $\text{Gener}(\text{canal}_1) = \text{'cours d'eau'}$. In the pop-up of Figure 6.3, "FL" stands for French *fonction lexicale*.

As a rule, an LF application $f(L)$ translates within a Lexical System as a set of arcs labeled with the LF f that originate from the lexical node L and lead to each individual value elements of the LF application. They are called *outgoing arcs*, relative to L , and *incoming arcs*, relative to these elements.

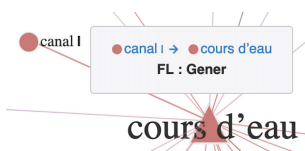


Figure 6.3: LF application $\text{Gener}(\text{canal}_1 \text{ '[water] canal' }) = \text{'cours d'eau'}$ 'watercourse' as represented in the fr-LN

¹¹I.e., collocations and (linguistic) clichés. An illustration of a lexical entity of the fr-LN that is a linguistic cliché can be found in Figure 6.11 (Section 6.2.2.2, p. 203): French *Fumer tue* 'Smoking kills' (bottom left corner of the figure). This cliché is a pragmateme (Mel'čuk 2020) used as a warning to smokers, written on packs of cigarettes.



Visualization of Lexical Systems

In the present chapter, we make use of a dedicated *network browser*, called *Spiderlex*, for the visualization of the content of Lexical Systems (Ollinger et al. 2020, Ollinger & Polguère 2020). *Spiderlex* exploits the mathematical properties of the small-world structure of Lexical Systems (Section 6.2.2, p. 194) and plays an important role for the utilization of lexical data both by lexicographers and users of the fr-LN – see Section 6.3.2, p. 206. Note that the orientation of arcs and their labels are not displayed by default, as in Figure 6.1, in order to keep the visualization legible; this information can be shown by pointing at individual arcs, as in Figure 6.3.¹²

The notion of Lexical System, first introduced in Polguère (2009), gradually took shape out of the practice of computational Explanatory Combinatorial Lexicography (Steinlin et al. 2005, Polguère & Mel’čuk 2006). It was based on the observation that the system of simple standard LFs – in particular LFs identified in this book as being central (Chapter 3, Section 3.8.1, p. 65) – was a good candidate as main contributor to the structuring of lexical knowledge, better by all means than mainstream taxonomic principles. This has lead to a second postulate, this time concerning the structure of Lexical Systems.



Postulate 2: structure of Lexical Systems

Lexical Systems, as network lexicon models, are primarily structured by standard LF relations.

Postulate 2 implies that LF relations form the backbone of Lexical Systems. From a methodological viewpoint, the lexicographic process of building a Lexical System for a given natural language is driven by the weaving of standard LF

¹²Data from the fr-LN used in this chapter and the screenshots from *Spiderlex* reflect the state of this constantly evolving model at time of writing.

relations within the lexical network (Lux-Pogodalla & Polguère 2011, Polguère 2012b).

Two properties of standard LFs can be put forward in support of the adoption of Postulate 2:

1. Standard LFs express constrained meanings (Chapter 3, 33, p. 35), present in all natural languages; these meanings determine fundamental paradigmatic and syntagmatic relations between lexical units.
2. A standard LF is essentially related to the meaning of each individual lexical unit *L* it takes as argument, in the sense that it is anchored in *L*'s definition – cf. the notion of LF target (Chapter 3, Section 3.3.2, p. 49). Conversely, a standard LF can serve as a revealer of the presence of a particular semantic chunk in *L*'s meaning (Iordanskaja & Polguère 2005).

Relational profile of a node / lexical unit. Because of the above properties, it is justified to consider that each node *n* of a Lexical System – and the corresponding lexical unit *L* – has a *relational profile* consisting of two sets: the set of *n*'s outgoing arcs and the set of its incoming arcs.

- The set of *n*'s outgoing arcs – i.e., *L*'s *outgoing profile* – is the main constituent of *L*'s relational profile as it is a reflection of the essential semantic and combinatorial properties of *L*. It corresponds to the information listed in the cooccurrence zone in an Explanatory Combinatorial Dictionary – see Appendix, p. 210.
- The set of *n*'s incoming arcs – i.e., *L*'s *incoming profile* – is a relational information that is *not* present in the cooccurrence zone of *L*'s lexical entry in an Explanatory Combinatorial Dictionary and, more importantly, that is simply not encoded in this type of model. In a Lexical System, by contrast, incoming profiles are embedded in the model's network structure and can be accessed directly.

This is illustrated in Figure 6.4, with an adaptation of the Spiderlex visualization of both the current outgoing and incoming profiles of French SENTIMENT 1a 'feeling' in the fr-LN. The complementarity of the two profiles is well illustrated by the contrast between the unique outgoing Gener relation (towards French *état psychique* 'psychological state') and the large set of incoming Gener relations, which characterizes French SENTIMENT 1a as being a generic term (i.e., classifying lexical unit) in the French lexicon.

6.2 Lexical Systems as network lexicon models

Outgoing LF relations		Incoming LF relations	
Syn _⊃	: <i>émotion 2</i>	Syn _⊃	for <i>émotion 2</i>
Syn _⊂	: <i>sensation 1, spec affect a; spec état émotionnel</i>	Syn _⊂	for <i>affect a – état émotionnel – sensation 1</i>
Gener	: <i>(spec) état psychique</i>	Gener	for <i>admiration – allégresse – amour 1.1a – amour 1.1b – animosité 1 – appitoiement a – appitoiement b – aversion – bonheur 1 – cafard 1 – chagrin_(N) 1 – colère 1 – compassion – contentement – curiosité 1.2 – déception – dégoût 1 – déprime – désespoir 1 – désir 1 – désir 2 – dévastation 1.2 – doute – effroi – émoi – émotion 1 – empathie – étonnement – fierté 1.1 – frayer – ‘froid dans le dos’ – frousse – frustration – fureur 1 – gratitude – haine 1 – haine 2 – honte 1.1 – horreur 1 – horreur 1.1 – humiliation 2 – jalousie 1 – ‘mal au cœur’ 1 – ‘mal du pays’ – malaise 3 – mécontentement – mélancolie 1.1 – mépris – nausée 1 – nostalgie 1 – orgueil 2a – peur 1 – plaisir 1.1 – rage 1 – ravissement – reconnaissance 1 – répugnance 1 – répulsion 1 – ressentiment – satisfaction 1 – sentiment 1b – souci 1 – spleen – tristesse 1 – truille</i>
S _{loc} ^{prototyp}	: <i>cœur 1.1; âme 1, esprit 1.1</i>	S ₂	for <i>éprouver a</i>
Magn	: <i>vif 1.2, profond_(Adj) 1, fort_(Adj) 1.2</i>	S ₂ ^{prototyp}	for <i>emoji</i>
Oper ₁	: <i>éprouver a</i> [ART ~], <i>ressentir</i> [ART ~]; <i>nourrir 1.1</i> [ART ~], <i>avoir 1.2</i> [ART ~]	S ₃ ^{prototyp}	for <i>chantage 1</i>
Func ₁	: <i>habiter 1</i> [N _X] < <i>êtreindre 1</i> [N _X] < <i>submerger 1</i> [N _X]		
IncepFunc ₁	: <i>naître 1</i> [en N _X], <i>envahir 1</i> [N _X], <i>saisir 1</i> [N _X], <i>prendre 1.1</i> [N _X] < <i>s’emparer 1.2</i> [de N _X]		
Caus ₁ Func ₂	: <i>transférer 1</i> [ART ~ sur N _Y]		
Fact ₁	: <i>guider 1</i> [N _X], <i>animer 1</i> [N _X]		
IncepManif	: <i>se peindre</i> [sur N]		

Figure 6.4: Outgoing vs. incoming profiles of French SENTIMENT 1a ‘feeling’ in the fr-LN

Figure 6.4 shows that Lexical Systems’ data need not be visualized graphically. Textual, dictionary-like visualizations similar to lexical entries of a traditional dictionary can also be generated from the same data. For this reason, the construction of Lexical Systems is presented in Polguère (2012a) as being a *lexicography of virtual dictionaries*.

Incoming profiles can play an important role in applications of lexical network models, for instance, in language teaching. Thus, the meaning of the generic term PRECIPITATION 1 ‘type of weather condition’¹³ is better taught to language learners by relying on this term’s incoming profile, i.e., by listing some of the most typical lexical units for which PRECIPITATION 1 is a Gener – RAIN_(N), SNOW_(N), HAIL_(N), etc. – as most dictionaries do.¹⁴ This is probably a better pedagogical

¹³The lexicographic number of PRECIPITATION 1 comes from the LDOCE.

¹⁴See, for instance, the largely enumerative definition of LDOCE: ‘any form of water, such as rain, snow, sleet, or hail, that falls to the Earth’s surface’. On defining so-called *functional collective superordinates* within the Natural Semantic Metalanguage approach, see Goddard (2017).

strategy than trying to first provide learners with a lexicographic definition that can only be non-specific and will be felt as more confusing than explanatory.

This observation does not necessarily apply to any generic terms, in particular to generic terms with more specific meanings, for which a lexicographic definition can be a good starting point in vocabulary teaching – e.g., generic terms such as BIRD, TREE_(N), etc. Collocative lexical units – lexical units that are essentially selected by the Speaker as collocates in collocations – are another illustration of the usefulness of incoming profiles. Thus, the intensifier HEAVY_(Adj) 2 [*heavy rain*]¹⁵ controls very few outgoing LF relations and its relational profile resides essentially in all its incoming Magn relations – the set of all lexical units for which it is a Magn (ACCENT_(N), ACCUSATION, APPLAUSE, BATTLE_(N), BLOW_(N), EMPHASIS, FOG_(N), etc.). The meaning of HEAVY_(Adj) 2 is simply that of Magn – the same constrained meaning as for dozens of adjectival intensifiers in English; consequently, its incoming profile is as important as its definition in the acquisition of this lexical unit by language learners.¹⁶

Semantic weight of LF relations. All LF relations found in the relational profile of a given lexical unit L do not carry the same *semantic weight*.¹⁷ Indeed, LFs are very diverse with respect to the semantic proximity they *inherently* imply – or not – between their argument and their value. For instance, Syn(L) = L', V₀(L) = L', A₁(L) = L', etc., obviously imply a strong *meaning relation* between L and L', as L' is a semantic derivative of L. By contrast, Magn(L) = L', Loc_{in}(L) = L', Oper₁(L) = L', etc., do not inherently imply any meaning relation between L and L'; such relation may exist “incidentally,” for a given case of f(L) = L', but not by virtue of the meaning ‘σ^f’ of f – e.g., Magn(*appetite*) = *ravenous* – where *appetite* means approximately ‘desire to eat’ and *ravenous*, when used freely (i.e., not as an intensifier), means ‘who has a strong desire / need to eat’.

In a Lexical System, each arc of the network that corresponds to an LF relation carries a semantic weight which is determined by the meaning ‘σ^f’ of the corresponding LF f. Prototypically, arcs corresponding to paradigmatic LFs (i.e., leading to semantic derivatives) tend to carry a stronger semantic weight than

¹⁵The lexicographic number of HEAVY_(Adj) 2 comes from the LDOCE.

¹⁶It is interesting to contrast two types of dictionaries of collocations in the light of the notion of relational profile: (i) those whose entries are collocation bases and that provide the list of collocates they select – see Benson et al. (2010) for English and Bosque (2006) for Spanish; (ii) those whose entries are collocates and that provide the list of corresponding bases in collocations – see Bosque (2010) for Spanish. The former describe the outgoing relational profile of bases, while the latter describe the incoming relational profile of collocates.

¹⁷The notion of semantic weight is further developed in Section 6.2.2, p. 194.

those corresponding to syntagmatic LFs. We will see shortly (Section 6.2.1.2) that the notion of semantic weight also applies to non-LF relations.

Before proceeding with the examination of lexical relations in Lexical Systems that are not based on LFs, it is important to note that a given node *n* of a Lexical System is a complex informational entity that encapsulates all the properties of the corresponding lexical unit *L* not accounted for by LF relations: grammatical characteristics, lexicographic definition, etc. That is, the lexical properties specified in *n* together with *n*'s relational profile are informationally equivalent to a lexical entry for *L*. Of course, in the context of the lexicography of lexical networks (Polguère 2014), special lexicographic tools are required to construct individual nodes with their individual properties and to weave the relations between nodes (see Section 6.3.2, p. 206).

6.2.1.2 Non-LF relations between lexical units

Three non-LF relations play an important role in Lexical Systems:

- copolysemy relation, holding between two lexical units of a polysemous vocable (Section 6.2.1.2.1);
- idiom-component relation, holding between an idiom and a lexeme that participates in its surface-syntactic structure (Section 6.2.1.2.2);
- meaning-inclusion relation, holding between a lexical unit and another lexical unit that is an element of its lexicographic definition (Section 6.2.1.2.3).

To make the presentation of these relations more straightforward and concrete, we base the following explanation mostly on data borrowed from the fr-LN.

The order of presentation of the three non-LF relations is determined by the expertise that has been gained from lexicographic work on the fr-LN: from the most extensively described relations to those whose encoding is still at an experimental stage.

Finally, note that the relational profile of a node / lexical unit, while primarily based on LF relations, also includes the three relations that are presented below.

6.2.1.2.1 Copolysemy relation

The fr-LN – as should be the case of any full-fledged Lexical System – contains the explicit modeling of the *polysemy structure* of vocables through the encoding of so-called *copolysemy relations* between lexical units of a vocable. A copolysemy

relation – that is, the corresponding arc of the lexical network – is oriented from the lexical unit that is the source of the “copolysemy derivation” toward the “derived” lexical unit. In other words, $L \vdash \text{copolysemy} \rightarrow L'$ reads as ‘lexical unit L has lexical unit L' as its copolysemy extension, metonymy, metaphor, ...’; this relation participates in the polysemy structure of a given vocable.

Some copolysemy relations have a strong semantic weight in the network: they imply that the two lexical units they connect belong to identical semantic fields. This is the case of the metonymy relation that links French CIGARETTE I ‘cigarette’ to French CIGARETTE II ‘habit of smoking *cigarettes*’¹⁸ – see Figure 6.5. Other copolysemy relations have weak (or, even, null) semantic weight – for instance, the metaphor relation (based on shape) that links French CIGARETTE I to the appositive noun French CIGARETTE III ‘[piece of clothing X] long and tight against the body, ‘as if’ it had the shape of a *cigarette*’ (*une robe cigarette* ‘a cigarette dress’). Such relations connect lexical units that are not expected to belong to the same semantic fields.



Figure 6.5: Polysemy of French CIGARETTE as represented in the fr-LN

A polysemous vocable is formally encoded in a Lexical System as a subnetwork consisting of nodes connected by copolysemy arcs, as shown in Figure 6.5 where two copolysemy arcs originate from the node CIGARETTE I. The *basic lexical unit* of a given vocable – i.e., the lexical unit that controls the polysemy structure of this vocable – corresponds to the unique node in the subnetwork that does not have an incoming copolysemy arc in its relational profile.

¹⁸Cf. *Julie veut arrêter la cigarette* ‘Julie wants to stop smoking cigarettes’. There is no corresponding lexical unit in English. The polysemy of French CIGARETTE is chosen as illustration for two reasons: (i) it is not identical to that of English CIGARETTE, which highlights the relevance of explicitly encoding polysemy structures, in particular, for the study of regular polysemy (Apresjan 1974, Peters & Peters 2000, Polguère 2022); (ii) though not too rich, it displays both metonymy and metaphor copolysemy relations, which we use later to illustrate the modeling of lexical proximity in lexical networks (Section 6.2.2.2, “Lexical vs. topological proximity,” p. 202).

6.2.1.2.2 Idiom-component relation

In a Lexical System, an *idiom-component relation* encodes the presence of a lexeme in the surface-syntactic structure [SSyntS] of an idiom (Pausé 2017a,b). This is illustrated in Figure 6.6, with the SSyntS of the French nominal idiom ‘BRAS CASSÉ’ lit. ‘arm broken’ – which means approximately ‘bungler’ – and the corresponding idiom-component relations encoded in the fr-LN.

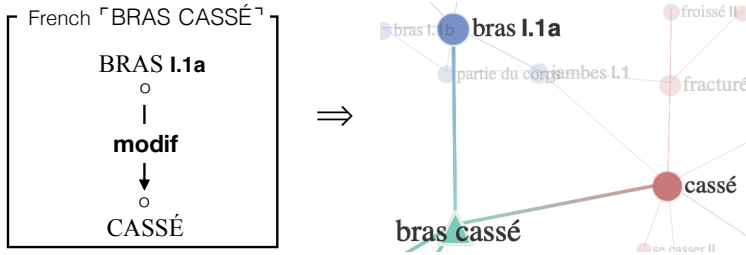


Figure 6.6: SSyntS of French ‘BRAS CASSÉ’ ‘bungler’ together with the two corresponding idiom-component relations encoded in the fr-LN¹⁹

Idiom-component relations belong to the outgoing relational profile of idioms and, conversely, to the incoming relational profile of the lexemes that compose the idiom. The list of all idioms in which a given lexeme participates is thus directly available in the network structure of a Lexical System and can be shown in the lexical entry of this lexeme, as it is the common practice in Explanatory Combinatorial Dictionaries; see the mention of the idiom ‘ANGER MANAGEMENT’ at the end of the lexical entry for ANGER_(N) (Appendix, p. 212).²⁰ We give in Table 6.2 the list of all idioms for which the lexeme BRAS I.1a ‘arm’ (cf. Figure 6.6) is a component in the fr-LN, at the time of writing. This list is automatically extracted from this lexeme’s incoming relational profile and displayed in textual visualizations of its lexical entry.

An idiom-component relation is *inherently* not a meaning relation: it has a null semantic weight. However, it is not necessarily devoid of *semantic relevance*: it

¹⁹For now, the SSyntS of an idiom is not encoded as such in the fr-LN. Each idiom is associated with a structural pattern that is simply a string of parts of speech slots – e.g., N+Adj, for ‘BRAS CASSÉ’. These slots are filled with actual lexical units in the description of each individual idiom by encoding idiom-component relations – for details, see Pausé (2017a).

²⁰In the lexicography of Lexical Systems, however, this information is encoded while building the description of the idiom, whereas it is usually done while writing the lexical entry of the embedded lexemes in the case of Explanatory Combinatorial Dictionaries and commercial dictionaries as well.

Table 6.2: Idioms with BRAS 1.1a ‘arm’ as component in the fr-LN

「À BRAS-LE-CORPS」 ¹ lit. ‘with arm-the-body’	‘with both arms around someone’s body’
「BAISSER LES BRAS」 ¹ lit. ‘lower the arms’	‘give up’
「BRAS CASSÉ」 ¹ lit. ‘broken arm’	‘bungler’
「BRAS DE FER」 ¹ lit. ‘arm of iron’	‘arm-wrestling’
「BRAS DE MORPHÉE」 ¹ lit. ‘arms of Morpheus’	‘state of sleeping / dreaming’
「BRAS DROIT」 ¹ lit. ‘right arm’	‘right-hand person’
「BRAS LONG」 ¹ lit. ‘long arm’	‘the long arm’
「COUPER BRAS ET JAMBES」 ¹ [à N _Y] lit. ‘cut.off arms and legs [to Y]’	‘take the steam out [of person Y]’
「LES BRAS EN TOMBENT」 ¹ [à N _X] lit. ‘The arms because.of.this fall [to X]’	‘X is completely flabbergasted’

can participate in establishing a semantically relevant path between two lexical nodes of the lexical network.²¹

Such is the case, obviously, for weak idioms (e.g., 「FRYING PAN」), whose definition contains the semantemes of all their lexical components: here, idiom-component relations are somehow doubling meaning-inclusion relations (cf. Section 6.2.1.2.3 below). More unexpectedly, strong idioms (e.g., 「PUSH [someone’s] BUTTONS」) can participate in establishing semantically relevant paths in a lexical network, especially when they are based on metaphors that are transparent and alive in the current state of the language (Alm-Arvius 2006). To illustrate this, let us consider the idiom 「AS SNOW」, that functions as Magn collocater for adjectives such as COLD_(Adj), PURE or WHITE_(Adj). One of the two idiom-component relations this idiom controls is part of a two-step path connecting the adjective WHITE_(Adj) and the noun SNOW_(N), where each step of the path has a null semantic weight due to the nature of the lexical relation it encodes, as shown in Table 6.3. This unsemantic two-step path is doubling in the lexical network a direct meaning-inclusion relation: WHITE_(Adj) is semantically included in SNOW_(N).²²

The explicit encoding of the above network path within a lexicon model helps account for the great proximity that exists in English between the semantic field of ‘snow’ and the semantic field of ‘whiteness [things that are white]’. This proximity is not a purely conceptual fact: it is linguistically determined. It results from existence in the English language of lexemes and phrasemes – such as the idiom 「AS SNOW」 – that are based on the inclusion of the semanteme ‘white’ in the definition of SNOW_(N).

²¹On the important notion of path in a graph / network, see Section 6.2.2.1, p. 197.

²²See ‘white flakes’ in the definition of SNOW_(N) in Chapter 4, Section 4.2, p. 82.

²³Remember that Magn(L) = L’ entails an inherently non-meaning relation between L and L’ – Section 6.2.1.1, p. 188.

Table 6.3: Two-step path from $\text{WHITE}_{(\text{Adj})}$ to $\text{SNOW}_{(\text{N})}$ based on inherently non-meaning relations²³

Steps	Relations	Comments
1.	$\text{WHITE}_{(\text{Adj})} \vdash \text{Magn} \rightarrow \ulcorner \text{AS SNOW} \urcorner$	LF relation – i.e., $\text{Magn}(\text{white}_{(\text{Adj})}) = \ulcorner \text{as snow} \urcorner$
2.	$\ulcorner \text{AS SNOW} \urcorner \vdash \text{idiom-comp.} \rightarrow \text{SNOW}_{(\text{N})}$	$\text{SNOW}_{(\text{N})}$ participates in the SSyntS of the idiom $\ulcorner \text{AS SNOW} \urcorner$

The above analysis illustrates the important role that idiom-component relations can play in the structuring of the Mental Lexicon. By encoding these relations in Lexical Systems, we account better for the multiplicity of ways the Speaker can navigate from one lexical unit to another during the activation of lexical knowledge; we also account for associations of ideas that take place in the Speaker’s mind and can influence lexical selection.

6.2.1.2.3 Meaning-inclusion relation

A lexicographic definition is a paraphrase of the defined lexical unit in terms of semantically simpler lexical units (Mel’čuk & Polguère 2018).²⁴ Consequently, each defining element in the definition of $\text{ANGER}_{(\text{N})}$ given in (2) – see also the Appendix (p. 209) – is a semanteme corresponding to a specific lexical unit of English.

- (2)
- X’s anger at Y for Z* : X’s unpleasant, unfavorable feeling directed at person Y and caused by Z – such that

Stimulus

 - X believes that Y is responsible for Z
 - Z is undesirable for X

Effect

 - X wants to do something hostile to Y (in order to stop Z)
 - X enters in a mental state such that X can easily lose self-control

From a lexical network viewpoint, this inclusion of lexical units (more precisely, the corresponding semantemes) in the definition translates into a set of

²⁴Lexical unit L’ is semantically simpler than lexical unit L if and only if L’ appears in the decomposition of the meaning of L, but not vice versa.

meaning-inclusion relations originating from $\text{ANGER}_{(N)}$ and leading to included lexical units.

Meaning-inclusion relations induced by lexicographic definitions are not all equally relevant as regards semantic weight. For instance, the three relations in (3), induced by the definition of $\text{ANGER}_{(N)}$, are listed by order of decreasing semantic weight.

- (3) $\text{ANGER}_{(N)} \vdash \text{meaning inclusion} \rightarrow \text{FEELING}_{(N)}$ – strong semantic weight
 $\text{ANGER}_{(N)} \vdash \text{meaning inclusion} \rightarrow \text{HOSTILE}_{(\text{Adj})}$ – average semantic weight
 $\text{ANGER}_{(N)} \vdash \text{meaning inclusion} \rightarrow \text{RESPONSIBLE}$ – weak semantic weight

In many instances, meaning-inclusion relations establish lexical connections that play a significant role in the lexicon and cannot be accounted for by LFs. An LF relation $L \vdash f \rightarrow L'$ (i.e., $f(L) = L'$) is, first of all, a *combinatorial* relation, that may or may not also be semantic: the meaning ‘ σ^f ’ of f is expressed in combination with L by the lexical entity L' . By contrast, an $L \vdash \text{meaning inclusion} \rightarrow L'$ relation is necessarily a purely meaning relation: the fact that the semanteme ‘ L ’ is included in the semanteme ‘ L' ’.

6.2.2 Formal properties of Lexical Systems

Lexical Systems have a formal structure known in graph theory as a small-world network. This characteristic has strong implications for Lexical Systems’ computability, so that it is essential to give some consideration to this topic. To present formal properties of Lexical Systems, we start with the introduction of some basic notions of graph theory, the knowledge of which is presupposed for the construction and study of lexical networks (Section 6.2.2.1). Next, we use these notions to explain how the informational proximity between lexical units in the Mental Lexicon can be modeled by exploiting the small-world structure of Lexical Systems (Section 6.2.2.2). We apologize in advance both:

- to readers who are unfamiliar with mathematical notions as they may find what follows too technical,
- to readers with background in graph theory for the simplifications and approximations we operate.

6.2.2.1 Some basics of graph theory

6.2.2.1.4 Networks and their mathematical counterparts: graphs

Until now, we have used the term *network* in a rather broad sense, which can be glossed as ‘system of individual entities connected by one-to-one relations’.

Actual networks can be *artificial* (designed) systems: they have been devised as such, to serve specific purposes – for instance, a city metro system, where metro stations are individual entities and tracks / tunnels connecting stations are relations between these entities.

Networks can also be *natural* systems: they happen to exist without having been consciously designed; we are specially interested here in networks constituted of a huge number of entities – for instance, the system formed in the brain by neurons and their synaptic connections. These very large networks that are natural systems found in our surrounding or remote environment are called *real-world networks*. Note that, in our use of the term, a real-world network can be the product of human activity – for instance, the network formed by hyperlinked webpages of the Internet. What matters is that this structure has emerged naturally, unlike the structure of a city metro system, which is relatively small and has been consciously designed.



Following Postulate 1 (p. 178), Mental Lexicons are natural real-world networks and, following Postulate 2 (p. 185), Lexical Systems are *human-made* networks whose structure is expected to be analogous to the *natural* structure of Mental Lexicons. Therefore, Lexical Systems are considered here as real-world networks as well.

Networks are studied as formal objects by *graph theory*, a branch of mathematics customarily traced back to Euler (1741): it deals with the formal representation and calculus of network systems. Let us provide some terminological clarification:

- we use the term *graph* to designate a mathematical object that underlies a network stripped of all (or most of) the attributes of its nodes and arcs;
- for the sake of simplicity, our presentation of graphs and their properties uses the terms *node* and *arc*, more habitual in linguistic works, even though

two other terms are generally favored in graph theory literature, respectively: *vertex* (pl. *vertices*) and *edge*.

6.2.2.1.5 Graph representations

Graphs are mathematical – therefore, informational – objects. They should not be equated with *graph representations*, which are used throughout the present chapter. Figure 6.7 gives three common modes of representation for a given *undirected graph* $\mathcal{G}$, i.e., a graph whose arcs are not oriented: geometric representation, set-theoretic representation and adjacency matrix.

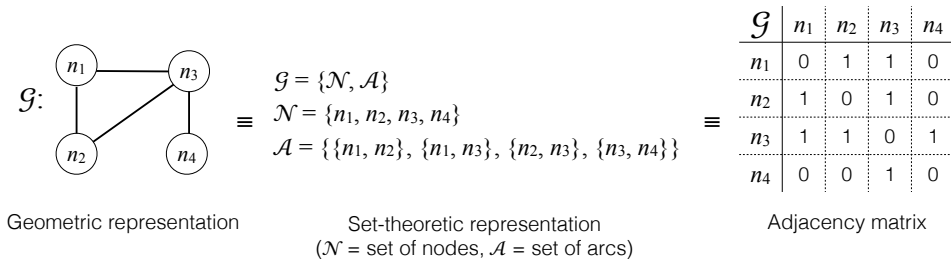


Figure 6.7: Three modes of representing a given undirected graph $\mathcal{G}$

The *geometric representation* is almost self-explanatory. In our figure, we put node names inside the circles representing nodes in order to highlight the fact that such names are only node IDs, and not attributes of the nodes.²⁵ A graph is entirely defined by its *topology*: its system of nodes and connecting arcs. Consequently, in a graph geometric representation, the spacial distribution of nodes and arcs is irrelevant from a formal point of view and only serves legibility purposes.²⁶ The relative positioning of nodes in the geometric representation of Figure 6.7 can be totally upended (n_4 at the very top, n_1 below n_3 , n_2 positioned much closer to n_3 , etc.); it will remain the representation of the same graph $\mathcal{G}$ as long as connections established by the arcs remain identical.

The *set-theoretic representation* defines a graph as a mathematical set comprised of two subsets: (i) a set of nodes and (ii) a set of arcs; this latter set contains unordered two-element sets encoding non-oriented binary relations between nodes.

²⁵This is in reference to what was said earlier about the distinction we maintain between the terms *network* and *graph* (network stripped of non-structural information).

²⁶We will show (Section 6.2.2.2, p. 201) that geometric distribution of nodes can nonetheless be used to encode additional information about the graph's overall topology.

The *adjacency matrix* is a table where each row and each column correspond to a node of the graph; each cell of the table indicates whether an arc connects the corresponding nodes: 1 means that the nodes are connected and 0 means that they are not. The adjacency matrix for graph $\mathcal{G}$ (Figure 6.7) is symmetrical relative to the table's top-left to bottom-right diagonal; this comes from the fact that, the arcs being non-oriented, any $n-n'$ arc is equally an $n'-n$ arc.

We exemplified the three modes of representation above with an undirected graph for pedagogical purposes. However, lexical relations examined earlier in Section 6.2.1 are all oriented and, therefore, the graphs underlying Lexical Systems are *directed graphs*. Figure 6.8 gives the three representations of a directed graph $\mathcal{G}'$.

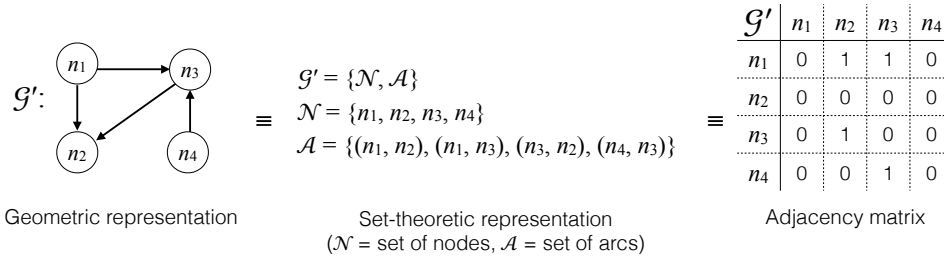


Figure 6.8: Three modes of representing a given directed graph $\mathcal{G}'$

Differences with the representations of Figure 6.7 (an undirected graph) are as follows:

- arcs of the geometric representations are *arrows* – they indicate the *starting node* and the *end node* for each relation;
- the set of arcs $\mathcal{A}$ in the set-theoretic representation contains (ordered) *couples* of nodes – instead of (unordered) two-element sets of nodes;
- 1s and 0s in the adjacency matrix take into account, for each cell, which node is the starting node and which node is the end node of the corresponding relation; note that the adjacency matrix of graph $\mathcal{G}'$ is non-symmetrical.

One of the advantages of the geometric representation of graphs over the two other representations is that it offers a clear visualization of possible paths between nodes. A *path* in a graph is a sequence of arcs that connects a sequence of nodes where neither arcs nor nodes are repeated. In an oriented graph, one

can consider *directed paths* – the sequence of arcs follows the orientation of the arcs – or *undirected paths* – the orientation of arcs is not relevant. For instance, the geometric representation of Figure 6.8 clearly shows:

- two directed paths from n_1 to n_2 , namely: n_1-n_2 and $n_1-n_3-n_2$;
- two undirected paths from n_1 to n_4 , namely: $n_1-n_3-n_4$ and $n_1-n_2-n_3-n_4$.

The notion of path is central in the definition of small-world networks (see Section 6.2.2.1.7, p. 199).

6.2.2.1.6 Structural properties of Lexical System graphs

Graphs are characterized by structural properties that pertain, primarily, (i) to the nature of their arcs – for instance, we have already established that we are concerned here with directed graphs only – and (ii) to the types of node-arc configurations they contain. We concentrate on *Lexical System graphs* – referred to as $\mathcal{G}_{LS}$ – and identify four properties (A–D) that characterize them as a specific type of directed graph. These properties are listed below, then summarized and exemplified in Figure 6.9 (p. 199).

- A) Labeled graph. Each arc of a $\mathcal{G}_{LS}$ is defined by a starting node, an end node and a *label* that identifies the lexical relation encoded by the arc: LF relation, copolysemy relation, etc. This characterizes a $\mathcal{G}_{LS}$ as an oriented *labeled graph*.
- B) Weighed graph. When presenting LF and non-LF relations encoded in Lexical Systems, we have indicated that these relations can be distinguished according to their semantic weight (Section 6.2.1.1, p. 188). This information is encoded in a $\mathcal{G}_{LS}$ for each arc of the graph, concurrently with the arc label, which characterizes a $\mathcal{G}_{LS}$ as a *weighed graph*.
- C) Multigraph. Two nodes of a $\mathcal{G}_{LS}$ can be connected by more than one arc – for instance, when two lexical units are connected simultaneously by an LF relation and a meaning-inclusion relation, as in (4). This results in *multiple arcs* in a $\mathcal{G}_{LS}$, which characterizes it as a *multigraph*.

- (4) $\text{Gener}(gas_{(N)}) = \text{substance}$
 $gas_{(N)} \vdash \text{meaning inclusion} \rightarrow \text{SUBSTANCE}$
 [Where ‘substance’ is the central component in the definition of $gas_{(N)}$]

D) Cyclic graph. A $\mathcal{G}_{LS}$ contains *cycles*, i.e., directed paths that originate and end at the same given node. This characterizes a $\mathcal{G}_{LS}$ as a *cyclic graph*. For instance, the set of LF relations in (5) is encoded as a two-step cycle in the corresponding $\mathcal{G}_{LS}$.

$$(5) \quad \begin{aligned} S_0(\text{present}_{(v)}) &= \text{presentation} \\ V_0(\text{presentation}) &= \text{present}_{(v)} \end{aligned}$$

The cycle in (5) also corresponds to a multigraph configuration – see property C) above.

The above four properties are recapitulated in Figure 6.9, with illustrations that apply to several properties wherever it is convenient to do so.

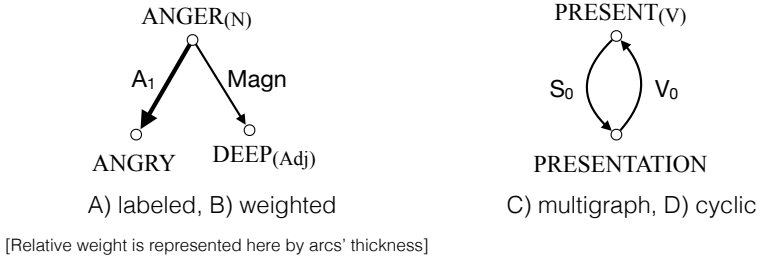


Figure 6.9: Four structural properties of a Lexical System graph $\mathcal{G}_{LS}$

6.2.2.1.7 Lexical Systems as small-world networks

The properties examined above characterize a $\mathcal{G}_{LS}$ at a *microscopic* level, i.e., at the level of individual nodes and relations they participate in. We are now taking a bird's-eye view of such graphs in order to contrast their microscopic and *macroscopic* structures. For this, we focus on a natural law that tends to apply to all real-world networks (as characterized in Section 6.2.2.1, p. 195).



Graph theory has determined that graphs of real-world networks have characteristic topological properties that identify them as *small-world networks* (Watts & Strogatz 1998).

The term *small-world network* refers to the small-world phenomenon in social networks,²⁷ which can be presented as follows: there is a high probability for any two randomly selected individuals i_1 and i_2 to be connected through a relatively short chain of acquaintances (Kleinberg 2000: 163) – hence the expression *It's a small world!* Though this phenomenon was originally conceptualized from the perspective of social and psychological studies (Milgram 1967), it is a characteristic feature of real-world networks of any kind. The topological structure of a small-world network has (at least) the four properties (A–D) described below; in order to anchor what follows more directly to our topic of interest (i.e., Lexical Systems), these properties are introduced as properties of a given $\mathcal{G}_{LS}$.²⁸

- A) Mainly connected graph. A $\mathcal{G}_{LS}$ may contain some isolated tiny subgraphs (possibly, isolated nodes); however, the bulk of its structure forms one huge *connected graph*: a graph in which there exists at least one path connecting any two nodes.
- B) Sparse graph. The arc-to-node ratio in a $\mathcal{G}_{LS}$ is low: there are more arcs than nodes, but the difference is not drastic. Consequently, there is an extremely low probability for any two nodes in a $\mathcal{G}_{LS}$ to be *directly* connected. This characterizes a $\mathcal{G}_{LS}$ as a *sparse graph*.²⁹
- C) It's a small world, nonetheless! In spite of its sparsity, a $\mathcal{G}_{LS}$ displays the small-world phenomenon in the sense that any two nodes of $\mathcal{G}_{LS}$, no matter how distant they may be in the topological structure of the graph, are connected by at least one relatively short path. This phenomenon is mainly made possible by two factors.

First, though most nodes of $\mathcal{G}_{LS}$ are weakly connected, there are nodes that function as lexical hubs from a relational viewpoint; for instance:

- generic terms that identify large lexical classes (ANIMAL, VEHICLE, FEELING, ...) and thus are ending nodes of many Gener arcs;

²⁷By *social network*, we mean here social structures made up of individuals having social interactions. We are not referring to social media platforms on the Internet.

²⁸We try to keep what follows as little mathematical as possible, at the expense of accuracy. See Gaume et al. (2010) for a purely mathematical characterization of lexical graphs as small-world networks; see Gader et al. (2014: Sec. 3) for the measurement of the *pedigree* of Lexical Systems as small-world networks.

²⁹The mathematical measurement used to evaluate this sparsity is the *average node degree* of the graph (the average number of arcs connected to each node) – for lexical networks, see Gaume et al. (2010: Sec. 2).

- collocative lexical units forming collocations with a large spectrum of potential bases – for instance, the Magn value $\text{HEAVY}_{(\text{Adj})}$ 2 mentioned in Section 6.2.1.1, p. 188.

Second, many copolysemy and idiom-component relations in a $\mathcal{G}_{LS}$ establish direct paths between semantic areas that are otherwise very far apart; for instance:

- $\text{CRANE}_{(\text{N})} \mid [\text{animals}] \vdash \text{copolysem. metaphor} \rightarrow \text{CRANE}_{(\text{N})} \parallel [\text{machines}];$
- $\text{「GO BANANAS」} [\text{feelings / behaviors}] \vdash \text{idiom-comp.} \rightarrow \text{BANANA} [\text{fruits}].$

D) High clustering rate. A $\mathcal{G}_{LS}$ is sparse at a macroscopic level but dense at a microscopic level, where a profusion of small node clusters are found. In mathematical terms, a $\mathcal{G}_{LS}$ has a high *clustering rate*: there is a much higher probability for two nodes n_1 and n_2 to be directly connected if both of them are directly connected to a node n_3 than if they are randomly selected. This is easily illustrated with LF relations, where various clustering *inference rules* such as (6) and (7) can be established. Figure 6.10 visualizes linguistic illustrations given in (6) and (7).

- (6) If $t(L) = L', L''$
 then there exists $\text{Syn} / \text{Syn}_{\sup} / \text{Syn}_{\subset} / \text{Syn}_{\cap}(L') = L''$,³⁰ for instance:
- $S_2(\text{married}) = \text{spouse}_{(\text{N})}$, 「better half」
 - $\text{Syn}_{\cap}(\text{spouse}_{(\text{N})}) = \text{「better half」}$
- (7) If $t_1(L) = L'$ and $t_2(L') = L''$
 then there is an above-average probability that there exists $t_3(L) = L''$;
 for instance:
- $\text{Conv}_{21}(\text{defeat}_{(\text{N})}) = \text{victory}$ and $A_1(\text{victory}) = \text{victorious}$
 - $A_2(\text{defeat}_{(\text{N})}) = \text{victorious}$

In Figure 6.10, solid arrows correspond to illustrations given in (6) and (7); dashed arrows are additional clustering LF relations.

6.2.2.2 Lexical vs. topological proximity

Formal properties presented above show that $\mathcal{G}_{LS}$ s are not *random graphs*, i.e., graphs in which nodes and connecting arcs are randomly distributed.³¹ They

³⁰This is true because all the elements of an LF value must be (quasi-)synonymous.

³¹In mathematical / statistical terms, a random graph is a graph in which the distribution of node degrees follows a Poisson law; by contrast, the degree distribution of a small-world graph

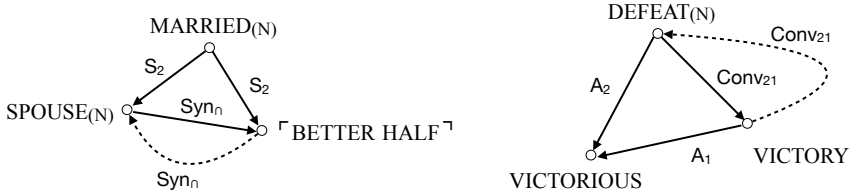


Figure 6.10: Two LF-based lexical clusters

are structured as small-world networks, whose topological properties can be exploited to access, study, and reason on the information they encode. The main advantage of small-world lexical networks over the better-known lexical taxonomies (Section 6.1, p.178) is that they do not adhere to one primary classifying / hierarchization principle; consequently, they allow for very flexible and varied types of computation.

To illustrate the computability of small-world networks, we focus here on *lexical proximity* (whether semantic or combinatorial) that can be inferred from the topology of $\mathcal{G}_{LS}$. This is well illustrated in visualizations performed by the Spiderlex graph browser (cf. Section 6.2.1.1, “Visualization of Lexical Systems,” p. 184), where clusters of lexical entities are identified by a stochastic algorithm that computes *topological proximity* between lexical nodes (Gaume 2004, 2008). This proximity is visually represented by two means:

- the geometric distribution of nodes in the visualization – the closer the nodes from the viewpoint of the graph topology, the closer they are positioned;
- the color coding of node clusters identified by the topological analysis – these clusters correspond to semantic-combinatorial classes of lexical entities.

Lexical spaces. We have established earlier that $\mathcal{G}_{LS}$ possess a high clustering rate (Section 6.2.2.1, p. 201): at the microscopic level, their nodes tend to form small clusters of lexical entities that are mutually interrelated. Such clusters are of course identified by the topological analysis of a $\mathcal{G}_{LS}$. However, at the macroscopic level, broader clusters are also identified; they bring together lexical entities that are more loosely related but form a distinct semantic-combinatorial *lexical space* in the graph and, hopefully, in the Mental Lexicon this graph encodes.

follows a power law (Gaume 2008: 849).

- Figure 6.11 below serves to illustrate the notion of lexical space and, more broadly, the outcome of a $\mathcal{G}_{LS}$'s topological analysis. It shows:

lexical space. These two spaces are located far apart, their connection holding by only one thread: the metaphor copolysemy relation between CIGARETTE_I and CIGARETTE_{III}. This reflects the fact that a metonymy belongs to the same broad conceptual domain as its source, while a metaphor can totally escape the gravitation field, so to speak, of the copolyseme it originates from.

Second, the two lexical spaces identified above are themselves constituted of clusters that display their own lexical cohesion. Such is the case, in the lexical space of CIGARETTE_I and CIGARETTE_{II} combined, of the small cluster constituted of FUMER_{III} ‘smoke_(V)’ – that is $\text{Real}_1(\text{cigarette}_I)$ – and three of its synonyms: CLOPER, GRILLER¹_{II} and SE GRILLER (see top left corner of Figure 6.11). This is an extremely tight cluster – and it is geometrically displayed as such – for two linguistic reasons: (i) the lexical units it contains are all values of Syn for each other³² and (ii) they are additionally all values of the same LF application $\text{Real}_1(\text{cigarette}_I)$.



Further reading

On the Lexical System approach to the modeling of polysemy, see Polguère (2018); on small-world lexical networks in lexicology / lexicography, see Loiseau et al. (2010), Polguère (2014, 2016); in psycholinguistics, see Gaume et al. (2002, 2008), De Deyne et al. (2017); in Natural Language Processing, see Biemann (2012).

6.3 Proof of concept: the French Lexical Network [fr-LN]

In lieu of conclusion, we present briefly the lexicographic enterprise that produced the first large-scale Lexical System: the *French Lexical Network* [fr-LN]. Bibliographic references on important aspects of the fr-LN project are provided at the end of the section.

³²Such cluster is called a *clique*: a (sub)graph where every two nodes are connected.

6.3.1 Organization and progression of the fr-LN project

The construction of the fr-LN is an ongoing long-term endeavor that started at the ATILF CNRS laboratory (Nancy, France) in 2011. An initial four-year funded project (called *RELIEF*, see Lux-Pogodalla & Polguère 2011) achieved the following tasks:

- 1.a) training of a team of lexicographers in the theory and practice of Explanatory Combinatorial Lexicology / Lexicography;
- 1.b) implementation – in parallel with 1.a – of a lexicographic editor tailor-made for the lexicography of Lexical Systems, whose capabilities were gradually incremented as the project proceeded with lexicographic description (Gader et al. 2012);
- 2) construction of a nucleus fr-LN for a core French vocabulary of around 3,700 vocables;
- 3) standard lexicographic work proceeding by important semantic fields (feelings, geographical elements, animals, ...).

This initial stage of the project resulted in a Lexical System that modeled a significant sample of the French lexicon and that later expanded into a full-fledged small-world network (as defined in Section 6.2.2, p. 194).

The fr-LN is in constant evolution both in terms of the number of lexical entities it describes and in terms of depth / accuracy of the description. At the time of writing, its coverage can be quantified as follows:

Lexical entities, i.e., network nodes	: 30,180
Vocables, i.e., lexicographic entries	: 19,084
Polysemy rate [= Lexical entities / Vocables]	: 1.58
Lexical relations, i.e., network arcs	: 85,941
LF relations	: 67,478
Copolysemy relations	: 9,960
Idiom-component relations	: 8,171
Meaning-inclusion relations	: 332

The above statistics show that the current fr-LN is a lexical network of significant proportions: approximately 30,000 lexical nodes for 85,900 lexical relations. In accordance with the properties of Lexical Systems (cf. Section 6.2.1.1, p. 184),

LF relations form the bulk of the relational structure of the model. Meaning-inclusion relations are underrepresented as the work on the writing of lexicographic definitions is still at an embryonic stage.

In the absence of a dedicated lexicographic team, the fr-LN now evolves mainly on demand, according to the needs of researchers, language teachers, etc. who are collaborators on the project or are making use of the resource. The fr-LN's lexicographic data is available in open access, to be used for Natural Language Processing or computational linguistic work – see ATILF (2024a), for the network itself, and ATILF (2024b), for the distribution of lexicographic examples embedded in the fr-LN.

6.3.2 New lexicography of lexical networks

The fr-LN project is mature enough – in terms of coverage and applications it has generated – to be seen as a proof of concept for Lexical Systems as plausible models of the Mental Lexicon. It demonstrates that network lexical models primarily structured by standard LF relations can offer new insights into the organization of lexical knowledge and, very importantly, that such models can indeed be lexicographically built.

As regards this last point, it is essential to stress the importance of the technological aspects of this new lexicography of lexical networks. Modern physics or astronomy could not exist without advanced technological means (cyclotrons, orbiting telescopes, etc.); similarly, though at a much more modest level, the lexicography of lexical networks has been made possible by the development of a sophisticated editor that allows lexicographers to literally jump from one lexical entity to another, encoding lexical relations while simultaneously creating new lexical nodes in the process, if the targeted lexical entity is not already present in the network.³³ This *lexicographic gesture* (Polguère 2012b) ensures that lexical entries always point to (the lexical entry of) actual lexical entities and forces lexicographers to encode a lexical relation as a link between two lexical entities that are (at least, minimally) modeled in the lexical network.

Anyone who has extensively used standard dictionaries (for language teaching, translation, etc.) knows that this is far from being the case in such models. The system of lexical relations present in a standard dictionary – that has been termed the dictionary's *mediostructure* (Wiegand 1996, Gouws & Prinsloo 1998) – is only modeled by cross-references in the text of the *microstructure*. At the

³³We do not illustrate the use of the Lexical System lexicographic editor (named *Dicet*, see Gader et al. 2012) as a next generation editor (*ItsyBitsy*) is under development at the time of writing.

scale of the whole lexicon, however, this cross-referencing is necessarily approximate for logistic reasons. Only by building a lexical model that *is*, so to speak, a mediostructure (i.e., a lexical network) can one ensure that the modeling of lexical relations – the essence of the lexicon’s structure (cf. Postulate 1, Section 6.1, p. 178) – is truly valid and unambiguous.

Finally, essential technology includes visualization and navigation tools such as Spiderlex (extensively used to produce illustrations in the present chapter). Their purpose is threefold: improve access to the content of Lexical Systems, visualize complex topological information and help lexicographers diagnose missing or wrongly encoded information.



Further reading

On the strategy adopted to organize the fr-LN’s initial growth based on the encoding of standard paradigmatic LF relations that are central in the system of LFs,³⁴ see Lux-Pogodalla & Polguère (2011) and Polguère & Sikora (2013); on the small-world structure of the fr-LN, see Gader et al. (2014); on the implementation of analogical reasoning in the fr-LN, see Ollinger (2014); on lexicographic edition of Lexical Systems, see Gader et al. (2012) and Polguère (2012b); on the modeling of terminologies within general language Lexical Systems, see Gotkova et al. (2024).

³⁴Roughly, those paradigmatic LFs whose ID is boxed in the table presenting the “System of standard simple lexical functions” (Chapter 3, p. 67).

Appendix: A sample lexical entry

The lexical entry of the noun $ANGER_{(N)}$ given below is intended to illustrate the use of lexical functions in lexicographic description. This is a reworked version of the entry presented in Iordanskaja & Mel'čuk (2022: Sec. 5, 42–46).

$ANGER_{(N)}$, noun, uncountable

Definition

X's anger at Y for Z : X's unpleasant, unfavorable feeling directed at person Y and caused by Z – such that

Stimulus

- X believes that Y is responsible for Z
- Z is undesirable for X

Effect

- X wants to do something hostile to Y (in order to stop Z)
- X enters in a mental state such that X can easily lose self-control

Government pattern

'X' $\Leftrightarrow$ I	'Y' $\Leftrightarrow$ II	'Z' $\Leftrightarrow$ III
1. –possess→ N's	1. –obl-obj1→ against N	1. –obl-obj2→ for N
2. –subj-adnom→ of N	2. –obl-obj1→ at N	2. –obl-obj2→ for V_{GER}
3. –modif→ $A_0(N)$	3. –obl-obj1→ toward N	3. –obl-obj2→ over N
		4. –obl-obj2→ over V_{GER}

Illustrations for the government pattern

- I.1 + II.1 *Judy's ⟨Her⟩ anger against co-workers is not helpful.*
- I.1 + II.2 + III.2 *They wanted to show their anger at the Government for signing the pact.*
- I.2 + II.2 + III.3 *The anger of seniors at the Government over the health care reform will impact the elections.*
- II.3 + III.2 *She felt anger toward her parents for leaving her there.*
- I.3 *Popular [= $A_0(\text{people})$] anger is rising.*

Appendix: A sample lexical entry

Lexical functions [LFs]

Paradigmatic LFs

Syn _⊃	: <i>fury; indignation; outrage; formal wrath; archaic ire</i>
Syn _∩	: <i>rage</i>
Anti _∩	: <i>gratefulness; gratitude</i>
Gener	: <i>feeling</i> [of ~] / [angry] <i>feeling; emotion</i> [of ~] / [angry] <i>emotion</i>
Magn+Figur	: <i>fire of ~, flame of ~ < firestorm of ~</i>
S ₂	: <i>object, subject, target</i> [of ~]
S ₃	: <i>cause, object, subject</i> [of ~], <i>reason</i> [for ~]
A ₁	: <i>angry, in ~</i>
Magn+A ₁	: <i>filled with ~, full of ~ < consumed by ~,</i> <i>colloquial</i> <i>‘blue in the face’, fuming,</i> <i>colloquial</i> <i>‘hot under the collar’</i>
AntiVer+Magn+A ₁	: <i>steamed-up</i> [I don’t see why you got steamed-up about this slip of the tongue.]
Able ₁	: <i>irascible; hot-tempered, short-tempered;</i> <i>short-fused</i>
PredAble ₁	: <i>‘be on a short fuse’, ‘have a short fuse’</i>
Adv ₁	: <i>angrily, in ~, with ~</i>
Sing	: <i>~ attack; burst of ~; fit of ~; flash of ~; gust of ~</i>

Syntagmatic LFs

Germ	: <i>seeds</i> [of ~]
Magn	: <i>deep, great < fierce, terrible < extreme; red < black; towering; unbridled, uncontrollable; a lot</i> [of ~]; <i>‘over the top’</i> anteposed; // <i>fury; // rage</i>
Magn ₁ ^{quant}	: <i>widespread</i>
AntiMagn	: <i>slight; ‘a little bit’</i> [of ~]; // <i>annoyance</i>
IncepPredPlusMagn	: <i>‘builds up’, grows, mounts, rises, ‘swells up’, ‘wells up’</i> [inside N _X]
CausPredPlusMagn	: <i>fuel, stir</i> [(ART) ~]
IncepPredMinusMagn	: <i>cools, fades, ‘goes away’, subsides</i>
CausPredMinusMagn	: <i>calm, soothe</i> [ART ~]
Ver	: <i>righteous</i>
Bon+Ver	: <i>noble</i>
Oper ₁	: <i>feel</i> [(ART) ~]
IncepOper ₁	: // <i>slang</i> <i>‘have a cow’, // slang</i> <i>‘have kittens’</i>
Magn+Oper ₁	: <i>blaze</i> [with ~]; <i>boil, burst, seethe</i> [with ~]

Magn+IncepOper ₁	: <i>erupt</i> [<i>in</i> ~], // <i>blow up</i> [<i>at</i> N _Y], // colloquial <i>blow a fuse</i> , // colloquial <i>blow a gasket</i> , // colloquial <i>blow</i> [N _X 's] <i>stack</i> , // colloquial <i>blow</i> [N _X 's] <i>top</i> , // <i>explode</i> , // colloquial <i>fly off the handle</i> , // colloquial <i>go ballistic</i> , // colloquial <i>go through the roof</i> , // colloquial <i>hit the ceiling</i> , // colloquial <i>see red</i>
CausOper ₁	: // <i>anger</i> _(V) [N _X]
IncepOper ₂ or Caus ₂ Oper ₂	: <i>attract</i> , <i>draw</i> [(ART) ~]
Caus ₍₂₎ Func ₀	: <i>provoke</i> [(ART) ~]; <i>arouse</i> , <i>spark</i> , <i>trigger</i> [(ART) ~]
Caus ₍₂₎ ContFunc ₀	: <i>feed</i> [ART ~]
Magn+Func ₁	: <i>boils</i> , <i>bubbles</i> , <i>seethes</i> [<i>in</i> N _X / <i>in</i> N _X 's gut / <i>in</i> N _X 's soul]; <i>fills</i> , <i>overcomes</i> , <i>seizes</i> [N _X]; <i>devours</i> [N _X] (<i>from inside</i>); // <i>Smoke</i> < <i>Steam</i> > <i>is coming</i> < <i>pouring</i> > <i>out</i> [<i>of</i> N _X 's ears]
Magn+IncepFunc ₁	: <i>explodes</i> [<i>in</i> N _X]
IncepFunc ₂	: <i>falls</i> [<i>on</i> N _Y]
Real ₁ , 'hostile action'	: <i>vent</i> [A _{(poss)X} ~ <i>on</i> N _Y]
Note. The semanteme configuration in subscript to the name of an LF identifies its target, i.e., the semantic component in the definition of the headword (in this case, ANGER _(N)) on which bears the meaning of the LF (see Chapter 3, Section 3.3.2, p. 49).	
AntiVerReal ₁ , 'hostile action'	: <i>take out</i> [A _{(poss)X} ~ <i>on</i> N _Ω], // <i>take it out</i> [<i>on</i> N _Ω] Ω is not responsible for Z
Adv ₁ Real ₁ , 'hostile action'	: <i>out of</i> ~
SingS ₀ Real ₁ , 'hostile action'	: <i>outburst</i> [<i>of</i> ~], [<i>angry</i>] <i>outburst</i>
Real ₁ , 'loss of self-control'	: <i>be beside oneself</i> [<i>with</i> ~]
Real ₂ , 'hostile action'	: <i>endure</i> [N _X 's ~]
Fact ₀ , 'loss of self-control'	: <i>explodes</i>
Perm ₁ Fact ₀	: <i>give vent</i> [<i>to</i> (ART) ~], <i>let out</i> , <i>unleash</i> , <i>vent</i> _(V) [(ART) ~]
NonPerm ₁ Fact ₀	: <i>contain</i> , <i>control</i> , <i>stifle</i> , <i>suppress</i> [A _{(poss)X} ~]
A ₂ NonPerm ₁ Fact ₀	: <i>pent-up</i> ; <i>suppressed</i>
Fact ₁ , 'loss of self-control'	: <i>blinds</i> [N _X]
Excess ^{motor} (<i>body</i>)—Sympt ₁₃	: [N _X] <i>is shaking</i> < <i>trembling</i> > [<i>with</i> ~]
Obstr(<i>breath</i>)—Sympt ₁₃	: [N _X] <i>is choking</i> [<i>of</i> ~]
Excess ^{motor} (<i>brows</i>)—Sympt ₁₂₃	: [N _X] <i>furrows</i> [A _{(poss)X} <i>brows</i> ((<i>in</i> ~))]
Excess ^{express} (<i>eyes</i>)—Sympt ₂₃	: [N _X 's <i>eyes</i>] <i>shine</i> [<i>with</i> ~]
Excess ^{express} (<i>eyes</i>)—Sympt ₁	: // [N _X] <i>looks daggers</i> [<i>at</i> N _Y]
Excess ^{express} (<i>eyes</i>)—Sympt ₃₁	: <i>seethes</i> [<i>in</i> N _X 's <i>eyes</i>]
Excess ^{motor} (<i>face</i>)—Sympt ₁	: // [N _X] <i>snarles</i> [<i>at</i> N _Y]
Excess ^{color} (<i>face</i>)—Sympt ₂₃	: [N _X 's <i>face</i>] <i>turns red</i> ([<i>with</i> ~])

Appendix: A sample lexical entry

Excess ^{color} (face)—Sympt ₁₃	: [N _X] <i>is flushed</i> ⟨ <i>is red</i> ⟩ ([<i>with</i> ~])
Excess ^{motor} (teeth)—Sympt ₁₂₃	: [N _X] <i>grinds</i> [A _{(poss)X}] <i>teeth</i> ([<i>in</i> ~])
Obstr ^{color} (face)—Sympt ₂₃	: [N _X 's face] <i>blanches</i> [<i>with</i> ~]; [N _X 's face] 「 <i>clouds over</i> 」 ([<i>with</i> ~])
Obstr ^{color} (face)—Sympt ₁₃	: [N _X] <i>is livid</i> [<i>with</i> ~]
Excess ^{motor} (feet)—Sympt ₁₂₃	: [N _X] <i>stamps</i> ⟨ <i>stomps</i> ⟩ [A _{(poss)X}] <i>feet</i> [<i>in</i> ~], // <i>thumps</i> [A _{(poss)X}] <i>feet</i>
Excess(voice)—Sympt ₁₃	: [N _X] <i>shouts</i> ⟨ <i>yells</i> ⟩ [<i>in</i> ~]

Examples

Examples are extracted from the COCA corpus (Davies 2008).

- (1) *Our anger generally tells us a lot more about ourselves than about the object of our anger.*
- (2) *Well, you're both anti-social misfits, filled with anger and rage.*
- (3) *Henrik looks at him with anger and, in an act of aggressive sublimation, knocks off the lamp and books from his father's table.*
- (4) *She came out publicly in a fit of anger when a fan harassed her online.*
- (5) *Their stories are outrageous and heartbreaking and fuel my anger.*
- (6) *This righteous anger was a relief; it masked his sense of shame.*
- (7) *Suddenly my suppressed anger explodes. How could he ever do this to me?*
- (8) *Even if protesters do not vent their anger at the troops, there are some who argue that any demonstrations will cause psychic wounds.*
- (9) *She had to endure the anger of friends and neighbors who saw her as a threat to their jobs in the mining industry.*
- (10) *Sam jumps up from the table, trembling with anger.*
- (11) *He was almost always on the verge of outright shouting in anger during the show.*

Idiom

「ANGER MANAGEMENT」

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Lexical functions: The naked truth

This book introduces **lexical functions**, a formal tool which ensures a systematic homogeneous description of two complementary families of relations between lexical units:

- **derivations** – paradigmatic lexical functions;
- **collocations**, which are constrained lexical combinations – syntagmatic lexical functions.

Each lexical function is associated with a specific meaning and once it is applied to a given lexical unit, which is seen as its argument, the function returns, as its value, the set of (near-)synonymous lexical entities which express this meaning depending on the argument. As an example, the paradigmatic lexical function S_1 is associated with the meaning ‘someone/something which does/is...’, and the syntagmatic lexical function *Magn* is associated with the meaning ‘very’/‘intense(ly)’ as illustrated below.

$S_1(\textit{felony})$	= <i>felon</i>
$S_1(\textit{gossip}_{(V)})$	= <i>gossiper; busybody</i>
$S_1(\textit{disappointing})$	= <i>disappointment; bummer</i>
<i>Magn</i> (<i>penalty</i>)	= <i>harsh, heavy, hefty, severe, stiff</i>
<i>Magn</i> (<i>fight</i> _(V))	= <i>fiercely</i> , ‘ <i>to the bitter end</i> ’, ‘ <i>tooth and nail</i> ’
<i>Magn</i> (<i>bored</i>)	= <i>stiff</i> [-], ‘ <i>to death</i> ’, ‘ <i>to tears</i> ’

The volume describes and illustrates (for English, French and Russian) the system of 66 simple standard lexical functions, which can be combined to form complex and configurational lexical functions. Standard lexical functions are language-universal: they seem sufficient for the description of standard derivations and collocations of any language. (For unique and deviating derivations and collocations non-standard lexical functions are assumed.) Balancing conceptual depth with pedagogical effort, *Lexical functions: The naked truth* guides readers from foundational notions to the full system of lexical functions, concluding with their role in modeling the network structure of natural language lexicons.